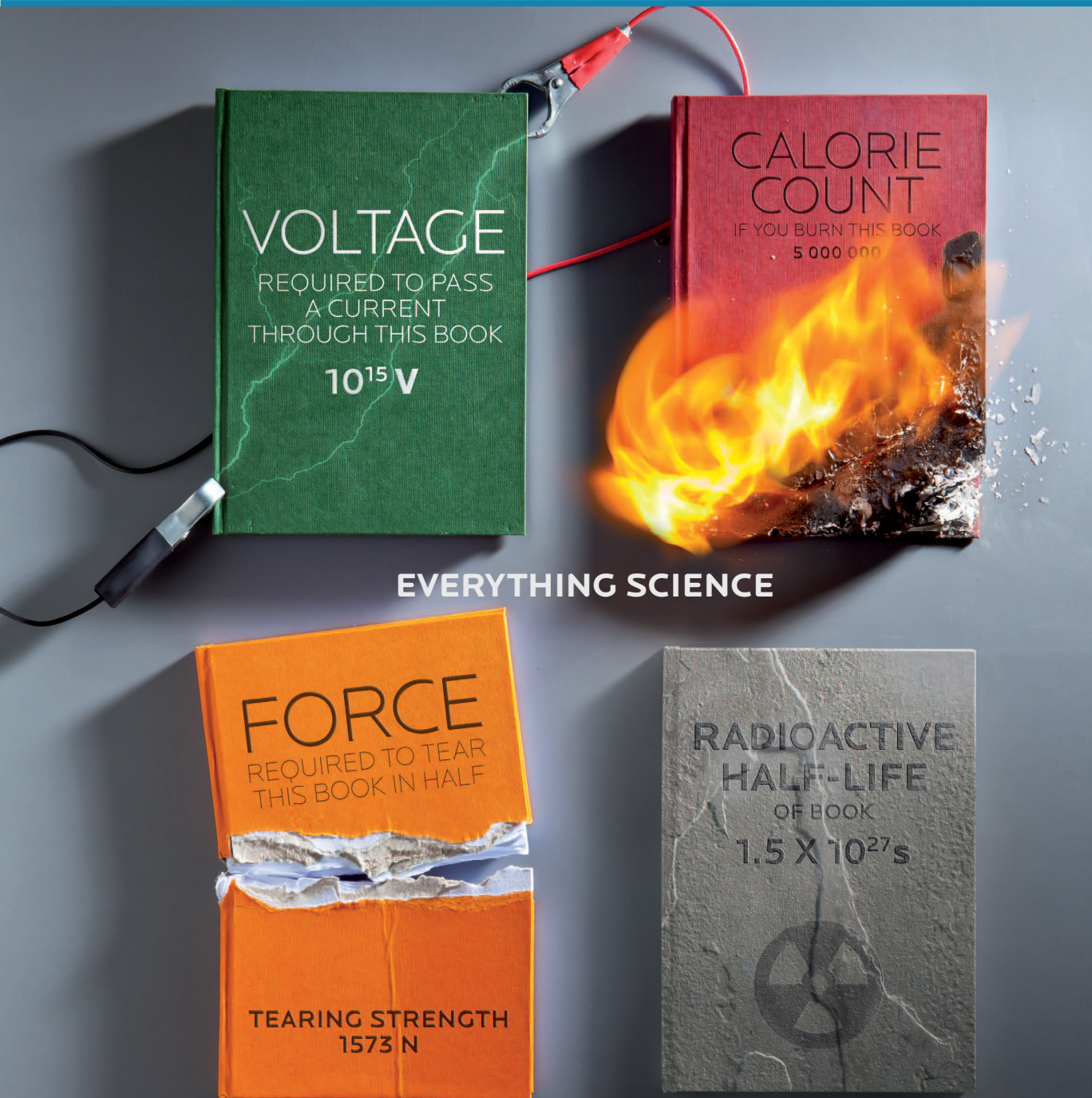


GRADE 12 PHYSICAL SCIENCES

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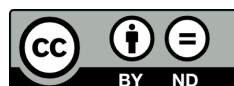
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EVERYTHING SCIENCE

When we look outside at everything in nature, look around us at everything manufactured or look up at everything in space we cannot but be struck by the incredible diversity and complexity of life; so many things, that look so different, operating in such unique ways. The physical universe really contains incredible complexity.

Yet, what is even more remarkable than this seeming complexity is the fact that things in the physical universe are knowable. We can investigate them, analyse them and understand them. It is this ability to understand the physical universe that allows us to transform elements and make technological progress possible.

If we look back at some of the things that developed over the last century – space travel, advances in medicine, wireless communication (from television to mobile phones) and materials a thousand times stronger than steel we see they are not the consequence of magic or some inexplicable phenomena. They were all developed through the study and systematic application of the physical sciences. So as we look forward at the 21st century and some of the problems of poverty, disease and pollution that face us, it is partly to the physical sciences we need to turn.

For however great these challenges seem, we know that the physical universe is knowable and that the dedicated study thereof can lead to the most remarkable advances. There can hardly be a more exciting challenge than laying bare the seeming complexity of the physical universe and working with the incredible diversity therein to develop products and services that add real quality to people's lives.

Physical sciences is far more wonderful, exciting and beautiful than magic! It is everywhere.

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The merger between Metropolitan and Momentum was lauded for the complementary fit between two companies. This complementary fit is also evident in the focus areas of CSI programmes where Metropolitan and Momentum together cover and support the most important sectors and where the greatest need is in terms of social participation.

HIV/AIDS is becoming a manageable disease in many developed countries but in a country such as ours, it remains a disease where people are still dying of this scourge unnecessarily. Metropolitan continues to make a difference in making sure that HIV AIDS moves away from being a death sentence to a manageable disease. Metropolitan's other focus area is education which remains the key to economic prosperity for our country.

Momentum's focus on persons with disabilities ensures that this community is included and allowed to make their contribution to society. Orphaned and vulnerable children are another focus area for Momentum and projects supported ensure that children are allowed to grow up safely, to assume their role along with other children in inheriting a prosperous future.

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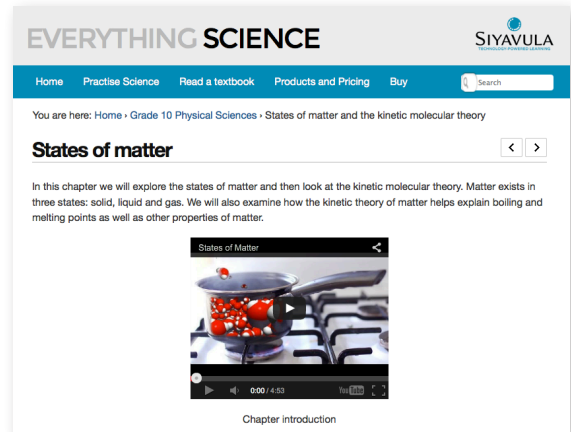
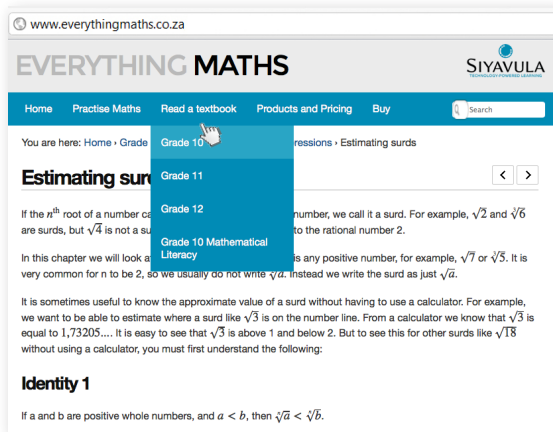


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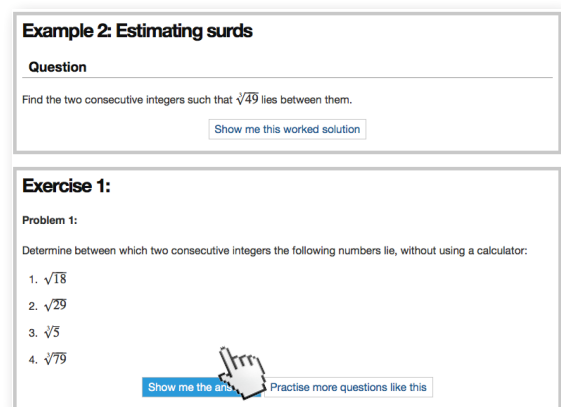
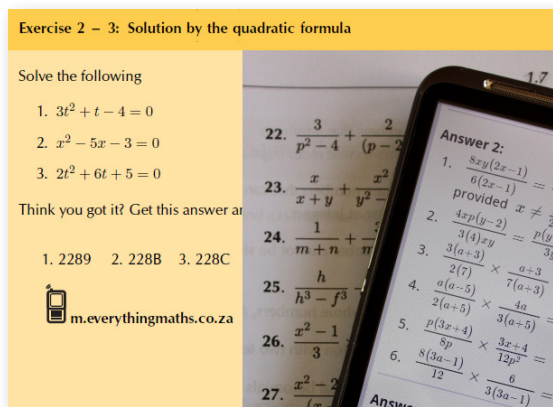
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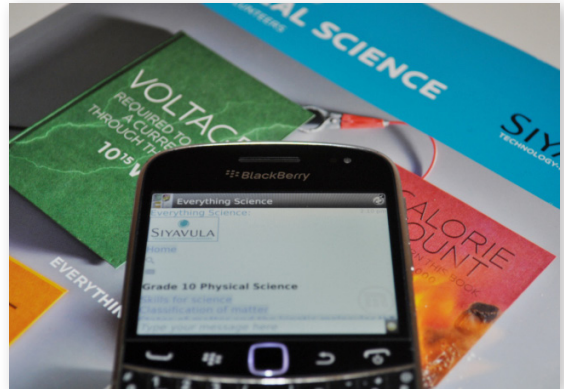


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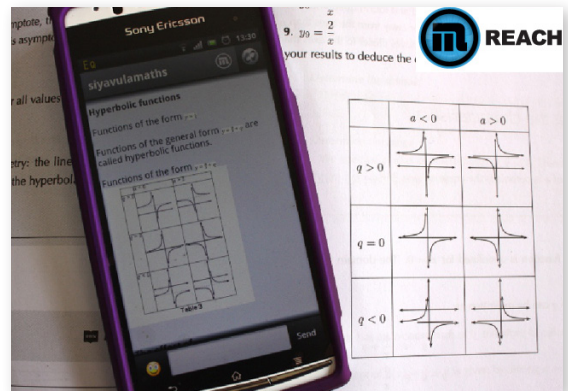
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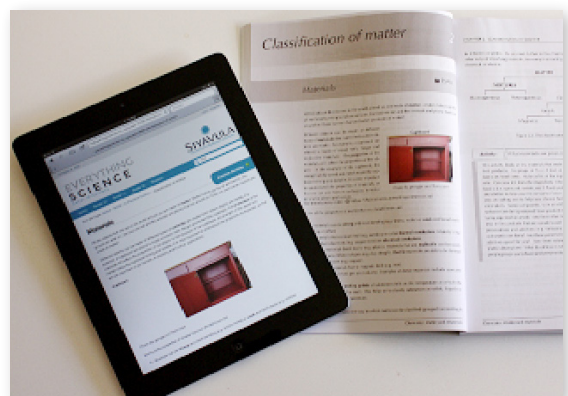
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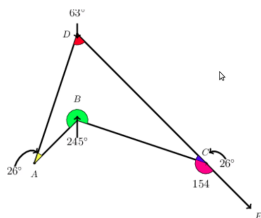
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Angles in quadrilaterals

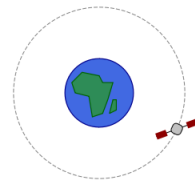
The diagram below represents quadrilateral ABCD with extended line $\overline{CE}$. Quadrilateral ABCD is a polygon with four sides and four angles. The sum of the interior angles in a quadrilateral = 360° . Angles on a straight line like $\overline{CE} = 180^\circ$.



Effect of mass on gravitational force

The International Space Station (ISS) has a mass M , as it orbits the Earth, it experiences a gravitational force of F . A space shuttle docks onto the ISS. The gravitational force the ISS experiences once the mass of the shuttle is added increases by a factor of 3.

By what factor does the mass of the ISS increase for it to experience this increase of gravitational force? Write your answer as a fraction of the original mass M_{ISS} of the ISS.

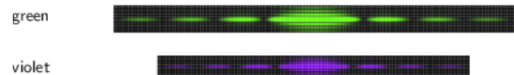


Answer: M_{ISS} [2 points] [Check answer](#)

[Help! How should I type my answer?](#)

Wavelength and diffraction

Two diffraction patterns are presented, determine which one has the longer wavelength based on the features of the diffraction pattern. The first pattern is for green light and the second pattern is for violet light:



The same diffraction grating is used to generate both diffraction patterns.

Answer: [Select an answer](#) [2 points] [Check answer](#)

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Chapters	Points	Mastery
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Classification of matter	22 / 34	☆☆☆
States of matter and the kinetic molecular theory		
The atom	66 / 77	☆☆☆☆
The periodic table	395 / 526	☆☆☆☆
Chemical bonding	71 / 128	☆☆☆☆
Transverse pulses	177 / 237	☆☆☆☆
Transverse waves		☆☆☆☆
Longitudinal waves		☆☆☆☆
Sound	100 / 139	☆☆☆☆
Electromagnetic radiation	453 / 598	☆☆☆☆
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Skills for science

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This book deals with the physical sciences - physics and chemistry. All the sciences are based in the use of experiment and testing to understand the world around us better. The scientific method requires us to constantly re-examine our understanding, by testing new evidence with our current theories and making changes to those theories if the evidence does not meet the test. The scientific method therefore is the powerful tool you will use throughout the physical sciences.

In this chapter you will learn how to gather evidence using the scientific method.

These skills will then be used throughout this textbook to test scientific theories and practices.

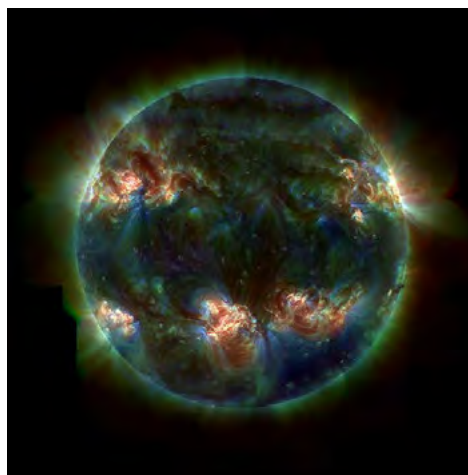


Figure 1.1: An ultraviolet image of the Sun.

1.1 The development of a scientific theory

ESCHQ

The most important, and most exciting, thing about science and scientific theories is that they are not fixed. Hypotheses are formed and carefully tested, leading to scientific theories that explain those observations and predict results. **The results are not made to fit the hypotheses.** If new information comes to light with the use of better equipment, or the results of other experiments, this new information is used to improve and expand current theories. If a theory is found to have been incorrect it is changed to fit this new information. The data should never be made to fit the theory, if the data does not fit the theory then the theory is reworked or discarded. Although this changing of opinion is often taken for inconsistency, it is this very willingness to adapt that makes science useful, and allows new discoveries to be made.

Remember that the term theory has a different meaning in science. A **scientific theory** is not like your theory of about why you can only ever find one sock. A scientific theory is one that has been tested and proven through repeated experiment and data. Scientists are constantly testing the data available, as well as commonly held beliefs, and it is this constant testing that allows progress, and improved theories.

Gravity

ESCHR

The theory of gravity has been slowly developing since the beginning of the 16th century. Galileo Galilei is credited with some of the earliest work. At the time it was widely believed that heavier objects accelerated faster toward the earth than light objects did. Galileo had a hypothesis that this was not true, and performed experiments to prove this.

Galileo's work allowed Sir Isaac Newton to hypothesise not only a theory of gravity on earth, but that gravity is what held the planets in their orbits. Newton's theory was used by John Couch Adams and Urbain Le Verrier to predict the planet Neptune in the solar system and this prediction was proved experimentally when Neptune was discovered by Johann Gottfried Galle.

Although a large majority of gravitational motion could be explained by Newton's theory of gravity, there were things that did not fit. But although a newer theory that better fits the facts was eventually proved by Albert Einstein, Newton's gravitational theory is still successfully used in many applications where the masses, speeds and energies are not too large.

FACT

Robert Boyle should be a familiar name to you. Boyle's law came about from his air pump experiments, where he discovered that pressure is inversely proportional to volume at a constant temperature ($p \propto \frac{1}{V}$ at constant T).

Thermodynamics

ESCHS

The principles of the three rules of thermodynamics describe how energy works, on all size levels (from the workings of the Earth's core, to a car engine). The basis for these three rules started as far back as 1650 with Otto von Guericke. He had a hypothesis that a vacuum pump could be made, and proved this by making one. In 1656 Robert Boyle and Robert Hooke used this information and built an air pump.

Over the next 150 years the theory was expanded on and improved. Denis Papin built a steam pressuriser and release valve, and designed a piston cylinder and engine, which Thomas Savery and Thomas Newcomen built. These engines inspired the study of heat capacity and latent heat. Joseph Black and James Watt increased the steam engine efficiency and it was their work that Sadi Carnot (considered the father of thermodynamics) studied before publishing a discourse on heat, power, energy and engine efficiency in 1824.

This work by Carnot was the beginning of modern thermodynamics as a science, with the first thermodynamics textbook written in 1859, and the first and second laws of thermodynamics being determined in the 1850s. Scientists such as Lord Kelvin, Max Planck, J. Willard Gibbs (all names you should recognise) among many many others studied thermodynamics. Over the course of 350 years thermodynamics has developed from the building of a vacuum pump, to some of the most important fundamental laws of energy.

1.2 Scientific method

ESCHT

The scientific method is the basic skill process in the world of science. Since the beginning of time humans have been curious as to why and how things happen in the world around us. The scientific method provides scientists with a well structured scientific platform to help find the answers to their questions. Using the scientific method there is no limit as to what we can investigate. The scientific method can be summarised as follows:

1. Ask a question about the world around you.
2. Do background research on your questions.
3. Make a hypothesis about the event that gives a sensible result. You must be able to test your hypothesis through experiment.
4. Design an experiment to test the hypothesis. These methods must be repeatable and follow a logical approach.
5. Collect data accurately and interpret the data. You must be able to take measurements, collect information, and present your data in a useful format (drawings, explanations, tables and graphs).
6. Draw conclusions from the results of the experiment. Your observations must be made objectively, never force the data to fit your hypothesis.
7. Decide whether your hypothesis explains the data collected accurately.
8. If the data fits your hypothesis, verify your results by repeating the experiment or getting someone else to repeat the experiment.
9. If your data does not fit your hypothesis perform more background research and

FACT

In science we never 'prove' a hypothesis through a single experiment because there is a chance that you made an error somewhere along the way. What you can say is that your results SUPPORT the original hypothesis.

make a new hypothesis.

Remember that in the development of both the gravitational theory and thermodynamics, scientists expanded on information from their predecessors or peers when developing their own theories. It is therefore very important to communicate findings to the public in the form of scientific publications, at conferences, in articles or TV or radio programmes. It is important to present your experimental data in a specific format, so that others can read your work, understand it, and repeat the experiment.

1. Aim: A brief sentence describing the purpose of the experiment.
2. Apparatus: A list of the apparatus.
3. Method: A list of the steps followed to carry out the experiment.
4. Results: Tables, graphs and observations about the experiment.
5. Discussion: What your results mean.
6. Conclusion: A brief sentence concluding whether or not the aim was met.

A hypothesis

A hypothesis should be specific and should relate directly to the question you are asking. For example if your question about the world was, why do rainbows form, your hypothesis could be: *Rainbows form because of light shining through water droplets.* After formulating a hypothesis, it needs to be tested through experiment. An incorrect prediction does not mean that you have failed. It means that the experiment has brought some new facts to light that you might not have thought of before.

Activity: Analysis of the scientific method.

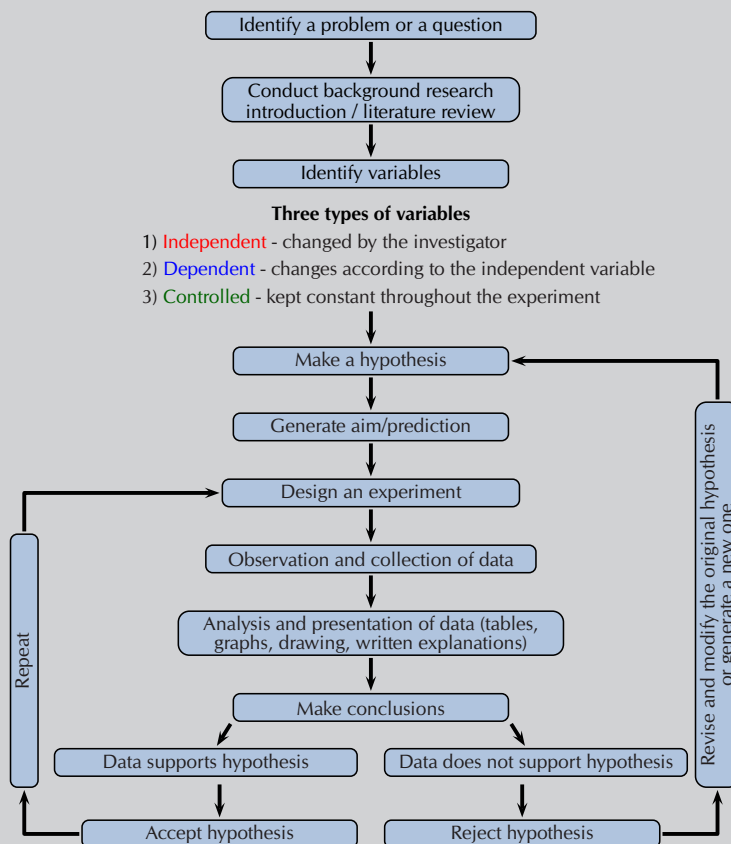


Figure 1.2: Overview of scientific method.

Break into groups of 3 or 4 and study the flow diagram provided, then discuss the questions that follow.

1. Once you have a problem you would like to study, why is it important to conduct background research before doing anything else?
2. What is the difference between a dependent, independent, and controlled variable and why is it important to identify them?
3. What is the difference between identifying a problem, a hypothesis, and a scientific theory?
4. Why is it important to repeat your experiment if the data fits the hypothesis?

Activity: Designing your own experiment

Recording and writing up an investigation is an integral part of the scientific method. In this activity you are required to design your own experiment. Use the information provided below, and the flow diagram in the previous experiment to help you design your experiment.

The experiment should be handed in as a 1 - 2 page report. Below are basic steps to follow when designing your own experiment.

1. Ask a question which you want to find an answer to.
2. Perform background research on your topic of choice.
3. Write down your hypothesis.
4. Identify variables important to your investigation: those that are relevant, those you can measure or observe.
5. Decide on the independent and dependent variables in your experiment, and those variables that must be kept constant.
6. Design the experiment you will use to test your hypothesis:
 - State the aim of the experiment.
 - List the apparatus (equipment) you will need to perform the experiment.
 - Write the method that will be used to test your hypothesis
 - in bullet format
 - in the correct sequence, with each step of the experiment numbered.
 - Indicate how the results should be presented, and what data is required.

Reading instruments

ESCHV

Before you perform an experiment you should be comfortable with certain apparatus that you will be using. The following pages give some commonly used apparatus and how to use them.

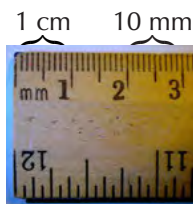


Figure 1.3: The end of a ruler.

Most rulers you find have two sets of lines on them. You can ignore those with the numbers spaced further apart. We only work in the metric system and those are for the imperial system. The closest together lines are for millimetres, the thicker lines are for 5 mm and the thicker, longer lines with numbers next to them mark off every 10 mm (1 cm).



Figure 1.4: Reading a thermometer.

A thermometer can have one, or two sets of numbers on it. If it has two sets of numbers one will be in Celsius, and one will be in Fahrenheit. We use Celsius, so you can ignore the side with a larger temperature range. In Figure 1.4 you can ignore the right-hand side. Looking on the left you can see that the red line (coloured ethanol here) is next to the fourth line above 0 °C. Each small line is 1 °C, so the temperature is 4 °C.



Figure 1.5: A laboratory style thermometer.

Laboratory thermometers will go to much higher temperatures than those used for measuring the temperature outside, or your body temperature. It is important to make sure that the thermometer you are using can handle the temperature you will be measuring too. If not, do not use that thermometer as you will break it. Make sure your thermometer is upright whenever you use it in an experiment, to avoid incorrect results.

📺 See video: [27HM](#) at www.everythingscience.co.za



Figure 1.6: A scale (also referred to as a balance).

Different scales have different functions. However, a basic function of all scales is a **tare** button. This zeros the balance. It is important that you zero the balance before you take any measurements. If you are weighing something on a piece of paper you should tare the balance with the piece of paper on it, and weigh the substance. Make sure you check the units that your scale is weighing in. If you want your value to be accurate to ,00 g then the scale must measure to that accuracy. A scale that measures in mg would be best.



Figure 1.7: The meniscus of water in a burette.

The surface of the water (the *meniscus*) is slightly higher at the edges of a container than in the middle. This is due to surface tension and the interaction between the water and the edge of the container (Figure 1.7). When measuring the volume in a burette or measuring cylinder or pipette you should look at the bottom of the meniscus. Where that lies is where you measure the volume. So in this example the meniscus is on the fifth line below the large line that represents 1 ml. Therefore the volume is 1,5 ml.

It is also possible that the liquid being measured has greater internal forces than those between it and the container. Then the meniscus would be higher in the middle than at the sides, and you would use the top of the meniscus to measure your volume.

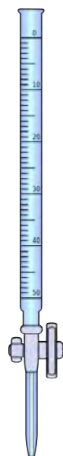


Figure 1.8: A burette.

A measuring cylinder is used to measure volumes that you want accurate to the nearest millilitre or so. It is not a highly accurate way of measuring volumes. The volume in a measuring cylinder is measured in the same way as for a burette, the difference is that in a measuring cylinder the smallest volume would be at the bottom, while the largest would be at the top.



Figure 1.9: A measuring cylinder with water.

There are two types of pipettes you might encounter this year. A volumetric pipette has a large bulb, marked with the set volume it can measure. Above the bulb on these pipettes there is a line. For a 5 ml volumetric pipette, when the meniscus of your liquid sits on the line, then the volume in that pipette is 5 ml.



Figure 1.10: A 5 ml volumetric pipette.

A graduated pipette has the same type of marking you see on a burette. The top is 0 ml, and the volume increases as you move down the pipette. In this pipette you should fill the pipette to near the 0 ml line and make a note of the volume. You can then add the desired volume, stopping when the volume in the pipette has decreased by the required amount.



Figure 1.11: A graduated pipette.

A learner wondered whether the rate of evaporation of a substance was related to the boiling point of the substance. Having done background research they realised that the boiling point of a substance is linked to the intermolecular forces within the substance. They know that greater intermolecular forces require more energy to overcome. This led them to form the following hypothesis:

The larger the intermolecular forces of a substance the higher the boiling point. Therefore, if a substance has higher boiling point it will have a slower rate of evaporation.

Perform the following experiment that the learner designed to test that hypothesis.

Experiment: Boiling points and rate of evaporation: Part 1

Aim:

To determine whether the rate of evaporation of a substance is related to its boiling point.

Apparatus:

You will need the following items for this experiment:

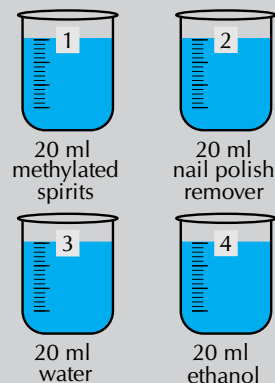
- 220 ml water, 20 ml methylated spirits, 20 ml nail polish remover, 20 ml water, 20 ml ethanol
- One 250 ml beaker, four 20 ml beakers, a thermometer, a stopwatch or clock

Method:

WARNING!

All alcohols are toxic, methanol is particularly toxic and can cause blindness, coma or death. Handle all chemicals with care.

1. Place 200 ml of water into the 250 ml beaker and move the beaker to sunny spot. Place the thermometer in the water.
2. Label the four 20 ml beakers **1 - 4**. These beakers should be marked.
3. Place 20 ml methylated spirits into beaker 1, 20 ml nail polish remover into beaker 2, 20 ml water into beaker 3 and 20 ml ethanol into beaker 4.
4. Carefully move each beaker to the warm (sunny) spot.
5. Observe each dish every two minutes. Note the volume in the beaker each time.
6. Continue making observations for 20 minutes. Record the volumes in a table.



Results:

- Record your observations from the investigation in a table like the one below.

Substance	Methylated spirits	Nail polish remover	Water	Ethanol
Boiling point (°C)	78,5	56,5	100	78,4
Initial volume (ml)	20	20	20	20
2 min				
4 min				
6 min				

In order to analyse the data obtained during experiments it is often necessary to convert that data into different representations. One type of representation is a graph. A few examples are given here.

How to draw graphs in science

ESCHY

For all graphs plotted from experimental data it is important to remember that you should never *connect the dots*. Data will never follow a line or curve perfectly. By obtaining multiple experimental data points any discrepancies in each data point can be removed. The line added after the points are plotted should be a **best fit line** that can then be used to more accurately determine further information.

Features of graphs you plot:

- An appropriate scale is used for each axis so that the plotted points use most of the axis/space (work out the range of the data and the highest and lowest points).
- The scale must **remain the same** along the entire axis and should use easy intervals such as 10's, 20's, 50's. Use graph paper for accuracy.
- Each axis must be labelled with what is shown on the axis and must include the appropriate units in brackets, e.g. Temperature ($^{\circ}\text{C}$), time (seconds), height (cm).
- The independent variable is generally plotted along the x-axis, while the dependent variable is generally plotted along the y-axis.
- Each point has an x and y co-ordinate and is plotted with a symbol which is big enough to see, e.g. a cross or circle.
- A best fit line is then added to the graph.
- **Do not** start the graph at the origin unless there is a data point for (0,0), or if the best fit line runs through the origin.
- The graph must have a clear, descriptive title which outlines the relationship between the dependent and independent variable.
- If there is more than one set of data drawn on a graph, a different symbol (and/or colour) must be used for each set and a key or legend must define the symbols.
- Use line graphs when the relationship between the dependent and independent variables is continuous.
- For a line graph, you can draw a line of best fit with a ruler. The number of points are distributed fairly evenly on each side of the line (see Figure 1.12).
- With an exponential graph (when the points appear to be following a curve) you can draw a best fit line freehand (see 1.13).

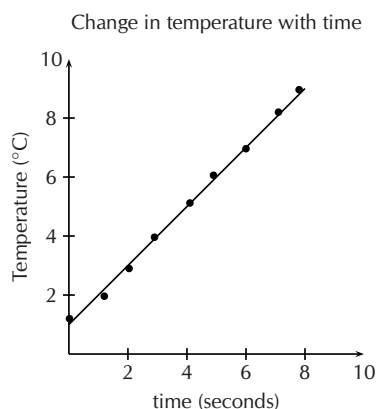


Figure 1.12: A straight line graph of the change in temperature with time.

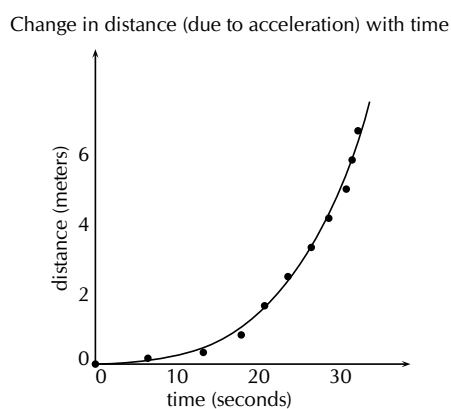


Figure 1.13: A graph with an exponential best fit line.

Remember that without units much of our work as scientists would be meaningless. We need to express our thoughts clearly and units give meaning to the numbers we measure and calculate. Depending on which units we use, the numbers are different. For example if you have 12 water, it means nothing. You could have 12 ml of water, 12 litres of water, or even 12 bottles of water. Units are an essential part of the language we use. Units must be specified when expressing physical quantities.

Qualitative and quantitative analysis

ESCHZ

Qualitative analysis

In **qualitative** research you look at the quality of a substance. At how it looks compared to other cases. You do an in-depth analysis of a specific case, and then make an informed decision about similar cases.

For example, a qualitative study of the study habits of university students could include only a few people, or over twenty. Each person would be asked in-depth questions about how they study, and what works for them, and a general, informed assertion can then be made about these study habits.

Quantitative analysis

In **quantitative** research you look at specific numbers. You study a large group (data sample) and do statistical analyses of the group, with experimental controls, manipulation of variables, and the modelling and analysis of your data.

For example, a quantitative study of those same study habits of university students would include a large number of people, for statistical relevance. The questions asked would include raw data of actual studying hours, and the most productive study times. These data points would then be analysed using graphical models.

Experiment: Boiling points and rate of evaporation: Part 2

Results:

Now that you remember how to plot graphs, go back to the data you obtained during the previous experiment.

- On the same set of axes, plot a graph of the volume (ml) versus the time (min) for each substance.

Analysis of results or discussion:

- Analyse the results plotted on the graphs and the table.
- Which substance has the fastest decrease in volume? Which has the slowest decrease?
- Discuss if there are any relationships between your independent (time) and dependent (volume) variables (what type of graph did you plot?).
- It is important to look for patterns/trends in your graphs or tables and describe these clearly in words.
- Compare the different graphs and the different rates of evaporation to the boiling points of the substances.

Evaluation of results:

- This is where you answer the question *Is the rate of evaporation of a substance related to its boiling point?*
- You need to carefully consider the results:

- Were there any unusual results? If so then these should be discussed and possible reasons given for them.
- Discuss how you ensured the validity and reliability of the investigation. Was it a fair test (validity) and if the experiment were to be repeated would the results obtained be similar (reliability)? The best way to ensure reliability is to repeat the experiment several times and obtain an average.
- Discuss any experimental errors that may have occurred during the experiment. These can include errors in the methods and apparatus. Make suggestions on what could be done differently next time.
- Did this experiment yield qualitative or quantitative results?

Conclusions based on scientific evidence

ESCJ2

Experiment: Boiling points and rate of evaporation: Part 3

Conclusion:

You have your results, and the analysis of your results. Now you need to look back at your hypothesis.

The conclusion needs to link the results to the aim and hypothesis. In a short paragraph, write down if what was observed supports or reject the hypothesis. If your original hypothesis does not match up with the final results of your experiment, do not change the hypothesis. Instead, try to explain what might have been wrong with your original hypothesis. What information did you not have originally that caused you to be wrong in your prediction.

Activity: Conclusions and bias

Read the following extract on bias taken from radiology.rsna.org (01/09/13), and answer the questions that follow:

Bias is a form of systematic error that can affect scientific investigations and distort the measurement process. A biased study loses validity in relation to the degree of the bias. While some study designs are more prone to bias, its presence is universal. It is difficult or even impossible to completely eliminate bias. In the process of attempting to do so, new bias may be introduced or a study may be rendered less generalizable. Therefore, the goals are to minimize bias and for both investigators and readers to comprehend its residual effects, limiting misinterpretation and misuse of data.

- In the light of the above quotation, why is it important for you to clearly state all your experimental parameters?
- Why is it important never to try and make the data fit your hypothesis?
- Look again at the conclusions you drew during your experiment.
 - Are they biased?
 - Did you make any assumptions based on preconceived ideas?
 - Is the data presented in a clear way that does not force a reader to agree with your conclusions?

Activity: Different explanations for the same set of experimental data.

To prevent bias, it is important to be able to look at the same information in different ways, to make sure your conclusion is the most logical one.

Divide into groups of three or four and compare the conclusions you drew from the boiling points and rate of evaporation experiment, and answer the questions that follow:

- Did anyone in the group draw a different conclusion from the one you drew? If yes, discuss the merits and short-falls of the different conclusions.
- Did your conclusion match what you were expecting to find from the hypothesis?
- Can you think of any other explanation that would explain your data?

Activity: Methods of knowing used by non-scientists.

Research one of the following topics and report your findings to the class in a five minute oral presentation:

- Traditional medicines.
- Navigation and knowledge of the seasons, from the stars.

Designing a model

ESCJ3

Sometimes a system is too large to be studied, or too difficult to recreate experimentally. In these cases it is possible to design a model based on a smaller system, that fits the data observed for the larger system. Here are some key points to remember when designing a model:

- A model is a testable idea that describes a large system that is not easily testable.
- The model should be able to explain as many observations of the large system as possible, and yet be relatively simple.

An example of a model was the spherical model of the Earth, rather than a flat one.

- Many educated people of the day (in the late 1400s) knew that the Earth could not be flat due to observations that did not fit. A spherical Earth model was proposed, which was testable on a small scale.
- The model explained many previously unexplained phenomenon (such as that ships appeared to *sink* over the horizon, regardless of the direction of travel).
- The model was further verified by the shape of the Earth's shadow on the moon during lunar eclipses.

1.4 Laboratory safety procedures

ESCJ4

Laboratories have rules that are enforced as safety precautions. These rules are:

- Before doing any scientific experiment make sure that you know where the fire extinguishers are in your laboratory and there should also be a bucket of sand to extinguish fires.
- You are responsible for your own safety as well as the safety of others in the laboratory. Never perform experiments alone.

- Do not eat or drink in the laboratory. Do not use laboratory glassware to eat or drink from.
- Ensure that you are dressed appropriately whenever you are near chemicals:
 - hair tied back
 - no loose or flammable clothing
 - closed shoes
 - gloves
 - safety glasses
- Always behave responsibly in the laboratory. Do not run around or play practical jokes. Always check the safety data of any chemicals you are going to use. Never smell, taste or touch chemicals unless instructed to do so. Do not take chemicals from the laboratory.
- Only perform the experiments that your teacher instructs you to. Never mix chemicals for fun. Follow the given instructions exactly. Do not mix up steps or try things in a different order.
- Care needs to be taken when pouring liquids or powders from one container to another. When spillages occur you need to call the teacher immediately to assist in cleaning up the spillage.
- Care needs to be taken when using strong acids and bases. A good safety precaution is to have a solution of sodium bicarbonate in the vicinity to neutralise any spills as quickly as possible. If you spill on yourself wash the area with lots of water and seek medical attention. Never add water to acid. Always add the acid to water.
- When working with chemicals and gases that are hazardous, a fume cupboard should be used. Always work in a well ventilated area.
- When you are instructed to smell chemicals, place the container on a laboratory bench and use your hand to gently waft (fan) the vapours towards you.
- Be alert and careful when handling chemicals, hot glassware, etc. Never heat thick glassware as it will break. (For example, do not heat measuring cylinders).
- When lighting a Bunsen burner the correct procedure needs to be followed: securely connect the rubber tubing to the gas pipe, have your matches ready, turn on the gas, then light a match and the bunsen burner. Do not leave Bunsen burners and flames unattended.
- When heating substances in a test tube do not overheat the solution. Remember to face the mouth of the test tube away from you and members of your group when heating a test tube.
- Always check with your teacher how to dispose of waste. Chemicals should not be disposed of down the sink.
- Ensure all Bunsen burners are turned off at the end of the practical and all chemical containers are sealed.

Hazard data

ESCJ5

Before starting any experiment, research the chemicals and materials you will be using in that experiment. Laboratory chemicals can be dangerous, and you should study the safety data sheets should before working with a chemical. The data sheets can be found at <http://www.msds.com/>.

Before working with a chemical, you should also make sure that you know how to, and have the facilities available to, dispose of those chemicals correctly and safely. Many chemicals cannot simply be washed down the sink. If you follow these few simple guidelines you can safely carry out experiments in the laboratory without endangering yourself or others around you.

Momentum and impulse

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2.1 Introduction

ESCJ6

In Grade 10 we studied motion but not what caused the motion, in Grade 11 we learnt about forces and how they can alter the motion of an object. In this chapter we will focus on what happens when two bodies undergo a contact interaction and how their motion is affected. We learn more about how force and motion are related. We are introduced to two new concepts, momentum and impulse.

We can begin by considering some scenarios to set the context. Most people have some intuition for physics based on their everyday experiences but they haven't formalised it. We can use our intuitive answers to lead into more structured thinking about physical events.

Everyone has experienced a mosquito landing on their arm and it can happen quite unnoticed. Consider the case of a falcon landing on your arm (ignore the sharp claws for now). You would definitely notice, why? What makes a mosquito different to a falcon? Would you still notice if the mosquito flew the same way as a falcon, or if the falcon copied the flight of a mosquito before landing? You probably would still notice, but try to think about what makes them so different.



Look at a motorcycle, motorcar and truck. Which of them is more likely to result in less damage in a collision situation, why? What factors would you change to reduce potential damage.



The factors that come up in these considerations are how fast things are moving and how massive they are. A falcon moving at the same speed as a mosquito still has a much larger mass. Even if a mosquito moved as fast as a falcon it wouldn't bother us because the mass of a mosquito is so small.

If a motorcycle, motorcar and truck were all moving at the same speed then it would be much safer to be in a collision with the motorcycle but a truck doesn't have to be moving as fast as a motorcycle to have a huge impact in a collision because of its large mass.

Why is the Moon's orbit largely unaffected when it is hit by asteroid?

There is an interplay between mass and speed (velocity to be precise) that governs what would happen if these objects came into contact with another object. There are two quantities that depend on mass and velocity, kinetic energy and momentum. Kinetic energy is something that we learnt about previously. Momentum is different to kinetic energy and is what we will focus on in the first part of this chapter. In the second part we will cover some of the differences between kinetic energy and momentum.



TIP

Momentum transfer doesn't require a contact interaction but we won't consider any non-contact scenarios in this chapter.

Key linked concepts

- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Techniques of vector addition — Physical Sciences, Grade 10, Vectors and scalars
- Newton's laws — Physical Sciences, Grade 11, Forces

2.2 Momentum

ESCJ7

Momentum is a physical quantity which is closely related to forces. Momentum is a property which applies to moving objects, in fact it is mass in motion. If something has mass and it is moving then it has momentum.

DEFINITION: Momentum

The linear momentum of a particle (object) is a vector quantity equal to the product of the mass of the particle (object) and its velocity.

The momentum (symbol $\vec{p}$) of an object of mass m moving at velocity $\vec{v}$ is: $\vec{p} = m\vec{v}$

Momentum is directly proportional to both the mass and velocity of an object. A small car travelling at the same velocity as a big truck will have a smaller momentum than the truck. The smaller the mass, the smaller the momentum for a fixed velocity. If the mass is constant then the greater the velocity the greater the momentum. The momentum will always be in the same direction as the velocity because mass is a scalar not a vector.

Vector nature of momentum

ESCJ8

A car travelling at $120 \text{ km}\cdot\text{hr}^{-1}$ will have a larger momentum than the same car travelling at $60 \text{ km}\cdot\text{hr}^{-1}$. Momentum is also related to velocity; the smaller the velocity, the smaller the momentum.

Different objects can also have the same momentum, for example a car travelling slowly can have the same momentum as a motorcycle travelling relatively fast. We can easily demonstrate this.

TIP

A vector multiplied by a scalar has the same direction as the original vector but a magnitude that is scaled by the multiplicative factor.

Consider a car of mass 1000 kg with a velocity of $8 \text{ m}\cdot\text{s}^{-1}$ (about $30 \text{ km}\cdot\text{hr}^{-1}$) East. The momentum of the car is therefore:

$$\begin{aligned}\vec{p} &= m\vec{v} \\ &= (1000) (8) \\ &= 8000 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ East}\end{aligned}$$

Now consider a motorcycle, also travelling East, of mass 250 kg travelling at $32 \text{ m}\cdot\text{s}^{-1}$ (about $115 \text{ km}\cdot\text{hr}^{-1}$). The momentum of the motorcycle is:

$$\begin{aligned}\vec{p} &= m\vec{v} \\ &= (250 \text{ kg}) (32 \text{ m}\cdot\text{s}^{-1}) \\ &= 8000 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ East}\end{aligned}$$

Even though the motorcycle is considerably lighter than the car, the fact that the motorcycle is travelling much faster than the car means that the momentum of both vehicles is the same.

From the calculations above, you are able to derive the unit for momentum as $\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$.

Momentum is also vector quantity, because it is the product of a scalar (m) with a vector ($\vec{v}$).

This means that whenever we calculate the momentum of an object, we should include the direction of the momentum.

▶ See video: 27HN at www.everythingscience.co.za

Worked example 1: Momentum of a soccer ball

QUESTION

A soccer ball of mass 420 g is kicked at $20 \text{ m}\cdot\text{s}^{-1}$ towards the goal post. Calculate the momentum of the ball.



SOLUTION

Step 1: Identify what information is given and what is asked for

The question explicitly gives:

- the mass of the ball, and
- the velocity of the ball.

The mass of the ball must be converted to SI units. $420 \text{ g} = 0,42 \text{ kg}$

We are asked to calculate the momentum of the ball. From the definition of momentum, $\vec{p} = m\vec{v}$ we see that we need the mass and velocity of the ball, which we are given.

Step 2: Do the calculation

We calculate the magnitude of the momentum of the ball,

$$\begin{aligned}\vec{p} &= m\vec{v} \\ &= (0,42) (20) \\ &= 8,40 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}\end{aligned}$$

Step 3: Quote the final answer

We quote the answer with the direction of motion included, $\vec{p} = 8,40 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ in the direction of the goal post.

Worked example 2: Momentum of a cricket ball**QUESTION**

A cricket ball of mass 160 g is bowled at $40 \text{ m}\cdot\text{s}^{-1}$ towards a batsman. Calculate the momentum of the cricket ball.

**SOLUTION****Step 1: Identify what information is given and what is asked for**

The question explicitly gives

- the mass of the ball ($m = 160 \text{ g} = 0,16 \text{ kg}$), and
- the velocity of the ball ($\vec{v} = 40 \text{ m}\cdot\text{s}^{-1}$ towards the batsman)

To calculate the momentum we will use $\vec{p} = m\vec{v}$.

Step 2: Do the calculation

$$\begin{aligned}\vec{p} &= m\vec{v} \\ &= (0,16) (40) \\ &= 6,4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \\ &= 6,4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ in the direction of the batsman}\end{aligned}$$

Step 3: Quote the final answer

The momentum of the cricket ball is $6,4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ in the direction of the batsman.

Worked example 3: Momentum of the Moon**QUESTION**

The centre of the Moon is approximately 384 400 km away from the centre of the Earth and orbits the Earth in 27,3 days. If the Moon has a mass of $7,35 \times 10^{22} \text{ kg}$, what is the magnitude of its momentum (using the definition given in this chapter) if we assume a circular orbit? The actual momentum of the Moon is more complex but we do not cover that in this chapter.

**SOLUTION****Step 1: Identify what information is given and what is asked for**

The question explicitly gives

- the mass of the Moon ($m = 7,35 \times 10^{22} \text{ kg}$)
- the distance to the Moon ($384\,400 \text{ km} = 384\,400\,000 \text{ m} = 3,844 \times 10^8 \text{ m}$)
- the time for one orbit of the Moon ($27,3 \text{ days} = 27,3 \times 24 \times 60 \times 60 = 2,36 \times 10^6 \text{ s}$)

We are asked to calculate only the magnitude of the momentum of the Moon (i.e. we do not need to specify a direction). In order to do this we require the mass and the magnitude of the velocity of the Moon, since $\vec{p} = m\vec{v}$.

Step 2: Find the magnitude of the velocity of the Moon

The magnitude of the average velocity is the same as the speed. Therefore: $v = \frac{\Delta x}{\Delta t}$

We are given the time the Moon takes for one orbit but not how far it travels in that time. However, we can work this out from the distance to the Moon and the fact that the Moon has a circular orbit.

Using the equation for the circumference, C , of a circle in terms of its radius, we can determine the distance travelled by the Moon in one orbit:

$$\begin{aligned} C &= 2\pi r \\ &= 2\pi (3,844 \times 10^8) \\ &= 2,42 \times 10^9 \text{ m} \end{aligned}$$

Combining the distance travelled by the Moon in an orbit and the time taken by the Moon to complete one orbit, we can determine the magnitude of the Moon's velocity or speed,

$$\begin{aligned} v &= \frac{\Delta x}{\Delta t} \\ &= \frac{C}{T} \\ &= \frac{2,42 \times 10^9 \text{ m}}{2,36 \times 10^6 \text{ s}} \\ &= 1,02 \times 10^3 \text{ m}\cdot\text{s}^{-1}. \end{aligned}$$

Step 3: Finally calculate the momentum and quote the answer

The magnitude of the Moon's momentum is:

$$\begin{aligned} \vec{p} &= m\vec{v} \\ p &= mv \\ &= (7,35 \times 10^{22}) (1,02 \times 10^3) \\ &= 7,50 \times 10^{25} \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \end{aligned}$$

The magnitude of the momentum of the Moon is $7,50 \times 10^{25} \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$.

As we have said, momentum is a vector quantity. Since momentum is a vector, the techniques of vector addition discussed in [Vectors and scalars in Grade 10](#) must be used when dealing with momentum.

Change in momentum

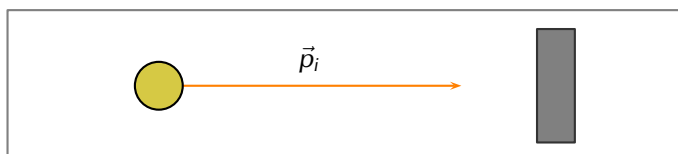
ESCJ9

Particles or objects can collide with other particles or objects, we know that this will often change their velocity (and maybe their mass) so their momentum is likely to change as well. We will deal with collisions in detail a little bit later but we are going to start by looking at the details of the change in momentum for a single particle or object.

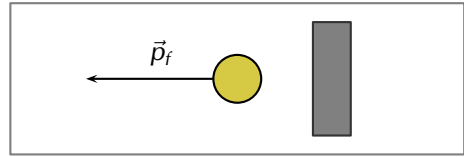
Case 1: Object bouncing off a wall

Lets start with a simple picture, a ball of mass, m , moving with initial velocity, $\vec{v}_i$, to the right towards a wall.

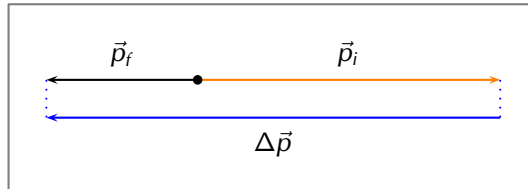
It will have momentum $\vec{p}_i = m\vec{v}_i$ to the right as shown in this picture:



The ball bounces off the wall. It will now be moving to the left, with the same mass, but a different velocity, $\vec{v}_f$ and therefore, a different momentum, $\vec{p}_f = m\vec{v}_f$, as shown in this picture:



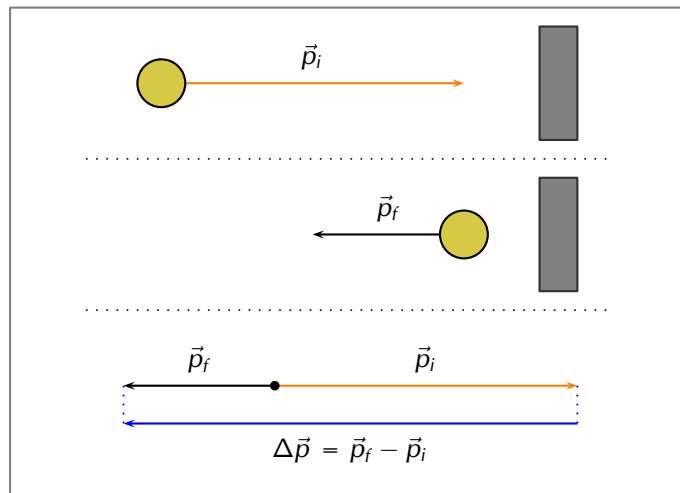
We know that the final momentum vector must be the sum of the initial momentum vector and the change in momentum vector, $\Delta\vec{p} = m\Delta\vec{v}$. This means that, using tail-to-head vector addition, $\Delta\vec{p}$, must be the vector that starts at the head of $\vec{p}_i$ and ends on the head of $\vec{p}_f$ as shown in this picture:



We also know from algebraic addition of vectors that:

$$\begin{aligned}\vec{p}_f &= \vec{p}_i + \Delta\vec{p} \\ \vec{p}_f - \vec{p}_i &= \Delta\vec{p} \\ \Delta\vec{p} &= \vec{p}_f - \vec{p}_i\end{aligned}$$

If we put this all together we can show the sequence and the change in momentum in one diagram:



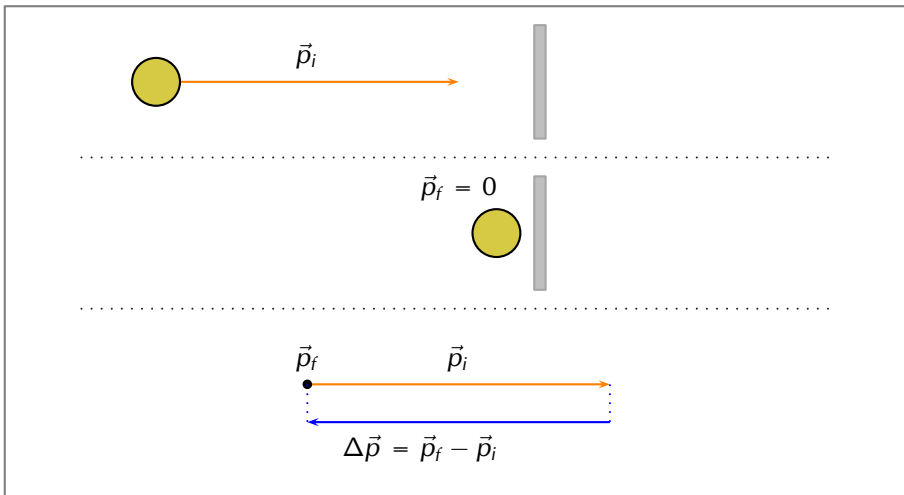
We have just shown the case for a rebounding object. There are a few other cases we can use to illustrate the basic features but they are all built up in the same way.

Case 2: Object stops



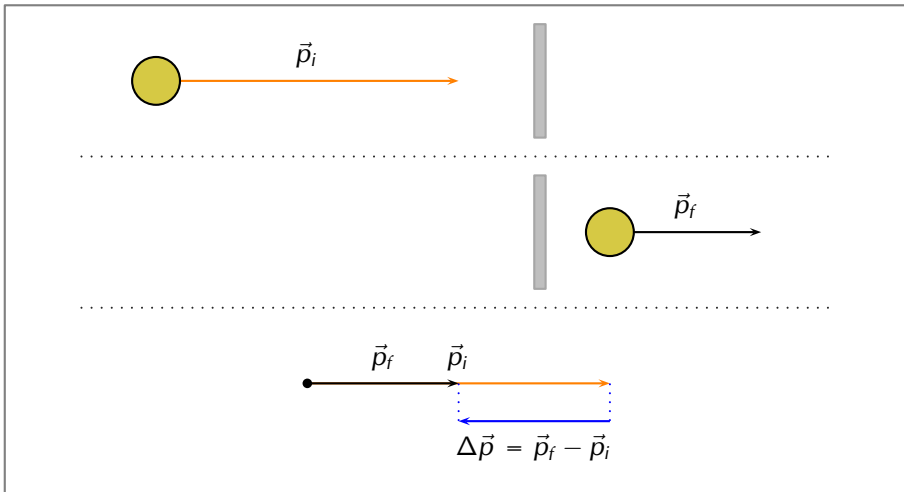
In some scenarios the object may come to a standstill (rest). An example of such a case is a tennis ball hitting the net. The net stops the ball but doesn't cause it to bounce back. At the instant before it falls to the ground its velocity is zero.

This scenario is described in this image:



Case 3: Object continues more slowly

In this case, the object continues in the same direction but more slowly. To give this some context, this could happen when a ball hits a glass window and goes through it or an object sliding on a frictionless surface encounters a small rough patch before carrying on along the frictionless surface.



Important: note that even though the momentum remains in the same direction the change in momentum is in the opposite direction because the magnitude of the final momentum is less than the magnitude of the initial momentum.

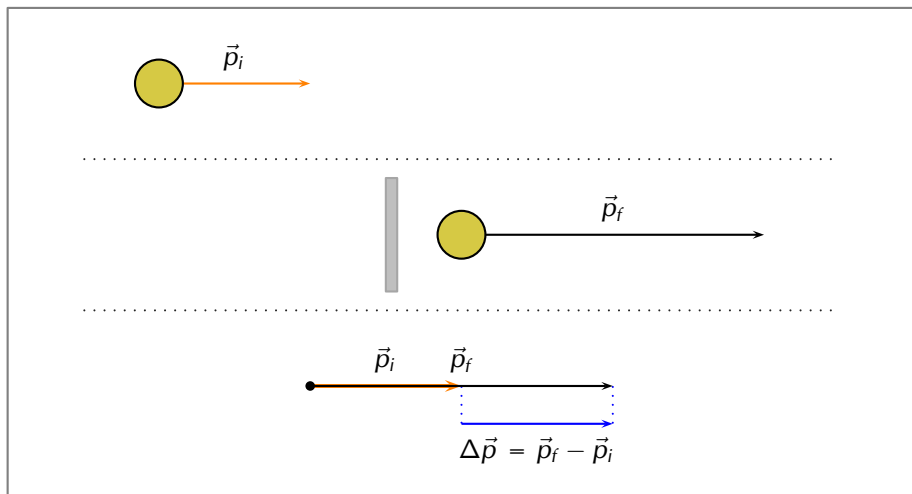
Case 4: Object gets a boost

In this case the object interacts with something that increases the velocity it has without changing its direction. For example, in squash the ball can bounce off a back wall

towards the front wall and a player can hit it with a racquet in the same direction, increasing its velocity.



If we analyse this scenario in the same way as the first 3 cases, it will look like this:



Case 5: Vertical bounce

IMPORTANT!

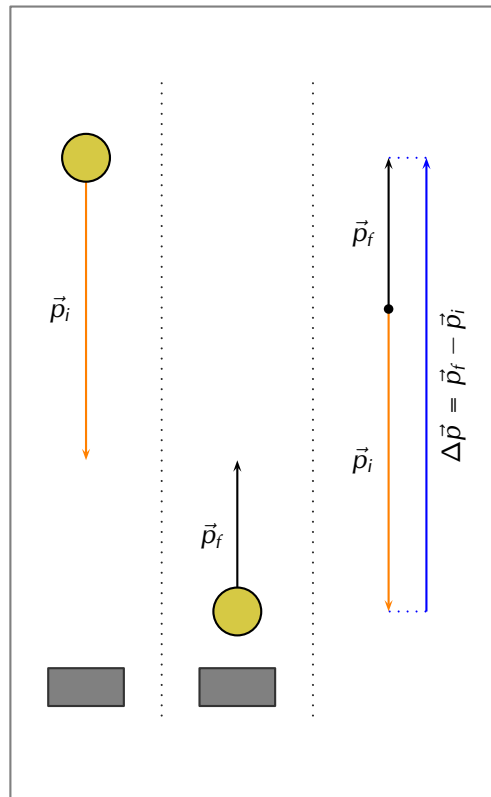
For this explanation we are ignoring any effect of gravity. This isn't accurate but we will learn more about the role of gravity in this scenario in the next chapter.

All of the examples that we've shown so far have been in the horizontal direction. That is just a coincidence, this approach applies for vertical or horizontal cases. In fact, it applies to any scenario where the initial and final vectors fall on the same line, any 1-dimensional (1D) problem. We will only deal with 1D scenarios in this chapter.

For example, a stationary basketball player bouncing a ball.



To illustrate the point, here is what the analysis would look like for a ball bouncing off the floor:



Exercise 2 – 1:

1.
 - a) The fastest recorded delivery for a cricket ball is $161,3 \text{ km}\cdot\text{hr}^{-1}$, bowled by Shoaib Akhtar of Pakistan during a match against England in the 2003 Cricket World Cup, held in South Africa. Calculate the ball's momentum if it has a mass of 160 g.
 - b) The fastest tennis service by a man is $246,2 \text{ km}\cdot\text{hr}^{-1}$ by Andy Roddick of the United States of America during a match in London in 2004. Calculate the ball's momentum if it has a mass of 58 g.
 - c) The fastest server in the women's game is Venus Williams of the United States of America, who recorded a serve of $205 \text{ km}\cdot\text{hr}^{-1}$ during a match in Switzerland in 1998. Calculate the ball's momentum if it has a mass of 58 g.
 - d) If you had a choice of facing Shoaib, Andy or Venus and didn't want to get hurt, who would you choose based on the momentum of each ball?
2. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27HP



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In the previous section we considered a number of scenarios where the momentum of an object changed but we didn't look at the details of what caused the momentum to change. In each case it interacted with something which we know would have exerted a force on the object and we've learnt a lot about forces in Grade 11 so now we can tie the two together.

You have learnt about Newton's Laws of motion in Grade 11. We know that an object will continue in its state of motion unless acted on by a force so unless a force acts the momentum will not change.

In its most general form Newton's Second Law of motion is defined in terms of momentum which actually allows for the mass and the velocity to vary. We will not deal with the case of changing mass as well as changing velocity.

DEFINITION: *Newton's Second Law of Motion (N2)*

The net or resultant force acting on an object is equal to the rate of change of momentum.

Mathematically, Newton's Second Law can be stated as: $\vec{F}_{\text{net}} = \frac{\Delta \vec{p}}{\Delta t}$

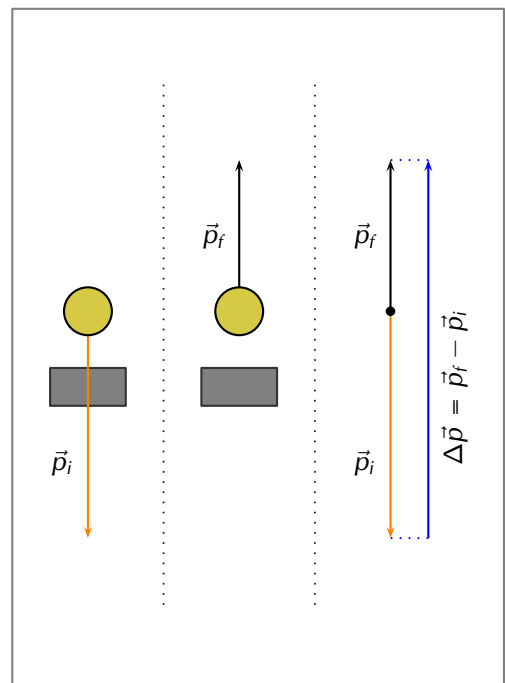
If a force is acting on an object whose mass is not changing, then Newton's Second Law describes the relationship between the motion of an object and the net force on the object through:

$$\vec{F}_{\text{net}} = m\vec{a}_{\text{net}}$$

We can therefore say that because a net force causes an object to change its motion, it also causes its momentum to change.

$$\begin{aligned}\vec{F}_{\text{net}} &= m\vec{a}_{\text{net}} \\ \vec{F}_{\text{net}} &= m \frac{\Delta \vec{v}}{\Delta t} \\ \vec{F}_{\text{net}} &= \frac{m\Delta \vec{v}}{\Delta t} \\ \vec{F}_{\text{net}} &= \frac{\Delta \vec{p}}{\Delta t}\end{aligned}$$

Let us apply this to the last case from the previous section, consider a tennis ball (mass = 0,1 kg) that is thrown and strikes the floor with a velocity of $5 \text{ m}\cdot\text{s}^{-1}$ downwards and bounces back at a final velocity of $3 \text{ m}\cdot\text{s}^{-1}$ upwards. As the ball approaches the floor it has an initial momentum $\vec{p}_i$. When it moves away from the floor it has a final momentum $\vec{p}_f$.



The bounce on the floor can be thought of as a collision taking place where the floor exerts a force on the tennis ball to change its momentum.

Remember: momentum and velocity are vectors so we have to choose a direction as positive. For this example we choose the initial direction of motion as positive, in other words, downwards is positive.

$$\begin{aligned}\vec{p}_i &= m\vec{v}_i \\ &= (0,1)(+5) \\ &= 0,5 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ downwards}\end{aligned}$$

When the tennis ball bounces back it changes direction. The final velocity will thus have a negative value because it is in the negative direction. The momentum after the bounce can be calculated as follows:

$$\begin{aligned}\vec{p}_f &= m\vec{v}_f \\ &= (0,1)(-3) \\ &= -0,3 \\ &= 0,3 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ upwards}\end{aligned}$$

Now let us look at what happens to the momentum of the tennis ball. The momentum changes during this bounce.

We keep our initial choice of downwards as positive. This means that the final momentum will have a negative number.

$$\begin{aligned}\Delta\vec{p} &= \vec{p}_f - \vec{p}_i \\ &= m\vec{v}_f - m\vec{v}_i \\ &= (-0,3) - (0,5) \\ &= -0,8 \\ &= 0,8 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ upwards}\end{aligned}$$

You will notice that this number is bigger than the previous momenta calculated. This should be the case as the change has to cancel out the initial momentum and then still be as large as the final momentum over and above the initial momentum.

Worked example 4: Change in Momentum

QUESTION

A tennis ball of mass 58 g strikes a wall perpendicularly with a velocity of $10 \text{ m}\cdot\text{s}^{-1}$. It rebounds at a velocity of $8 \text{ m}\cdot\text{s}^{-1}$. Calculate the change in the momentum of the tennis ball caused by the wall.

SOLUTION

Step 1: Identify the information given and what is asked

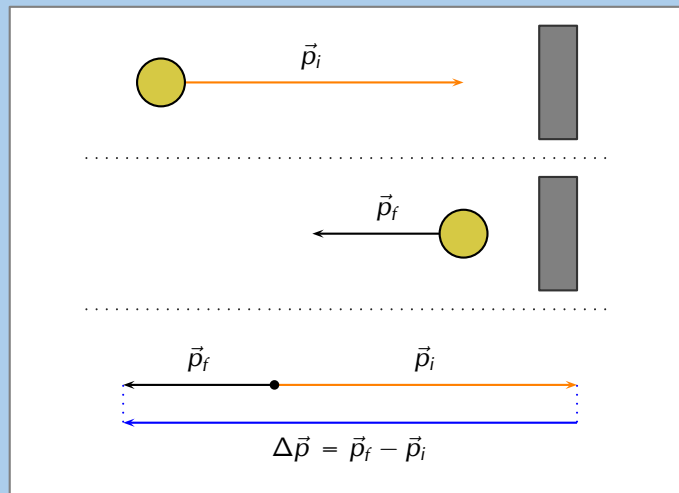
The question explicitly gives a number of values which we identify and convert into SI units:

- the ball's mass ($m = 58 \text{ g} = 0,058 \text{ kg}$),
- the ball's initial velocity ($\vec{v}_i = 10 \text{ m}\cdot\text{s}^{-1}$) towards the wall, and

- the ball's final velocity ($\vec{v}_f = 8 \text{ m}\cdot\text{s}^{-1}$) away from the wall

We are asked to calculate the change in momentum of the ball,

$$\Delta\vec{p} = m\vec{v}_f - m\vec{v}_i$$



We have everything we need to find $\Delta\vec{p}$. Since the initial momentum is directed towards the wall and the final momentum is away from the wall, we can use the algebraic method of subtraction discussed in Vectors in Grade 10.

Step 2: Choose a frame of reference

Let us choose towards the wall as the positive direction.

Step 3: Do the calculation

$$\begin{aligned}\Delta\vec{p} &= m\vec{v}_f - m\vec{v}_i \\ &= (0,058)(-8) - (0,058)(+10) \\ &= (-0,46) - (0,58) \\ &= -1,04 \\ &= 1,04 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ away from the wall}\end{aligned}$$

Step 4: Quote the final answer

The change in momentum is $1,04 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ away from the wall.

Worked example 5: Change in momentum

QUESTION

A rubber ball of mass $0,8 \text{ kg}$ is dropped and strikes the floor with an initial velocity of $6 \text{ m}\cdot\text{s}^{-1}$. It bounces back with a final velocity of $4 \text{ m}\cdot\text{s}^{-1}$. Calculate the change in the momentum of the rubber ball caused by the floor.

SOLUTION

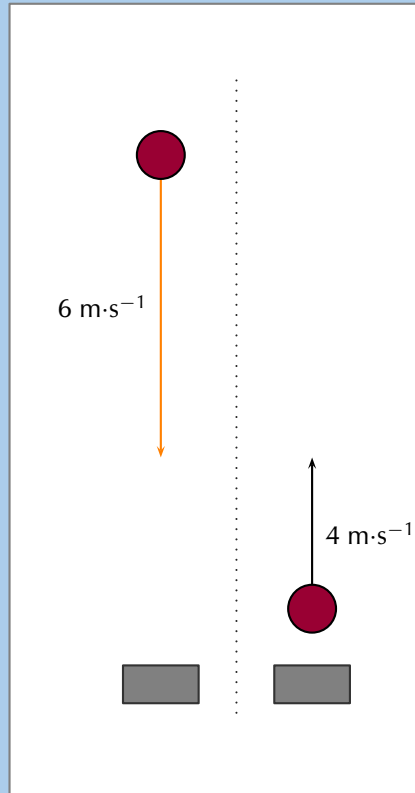
Step 1: Identify the information given and what is asked

The question explicitly gives a number of values which we identify and convert into SI units:

- the ball's mass ($m = 0,8 \text{ kg}$),
- the ball's initial velocity ($\vec{v}_i = 6 \text{ m}\cdot\text{s}^{-1}$) downwards, and
- the ball's final velocity ($\vec{v}_f = 4 \text{ m}\cdot\text{s}^{-1}$) upwards

We are asked to calculate the change in momentum of the ball,

$$\Delta \vec{p} = m\vec{v}_f - m\vec{v}_i$$



Step 2: Do the calculation

Let us choose down as the positive direction.

$$\begin{aligned}\Delta \vec{p} &= m\vec{v}_f - m\vec{v}_i \\ &= (0,8)(-4) - (0,8)(+6) \\ &= (-3,2) - (4,8) \\ &= -8 \\ &= 8,0 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ upwards}\end{aligned}$$

Step 3: Quote the final answer

The change in momentum is $8,0 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ upwards.

Worked example 6: Change in momentum

QUESTION

A regulation squash ball weighs 24 g. In a squash match a ball bounces off the back wall in the direction of the front wall at $1 \text{ m}\cdot\text{s}^{-1}$ before a player hits it with a racquet. After being struck towards the front wall the ball is moving at $20 \text{ m}\cdot\text{s}^{-1}$. What is the change in momentum?

SOLUTION

Step 1: Identify the information given and what is asked

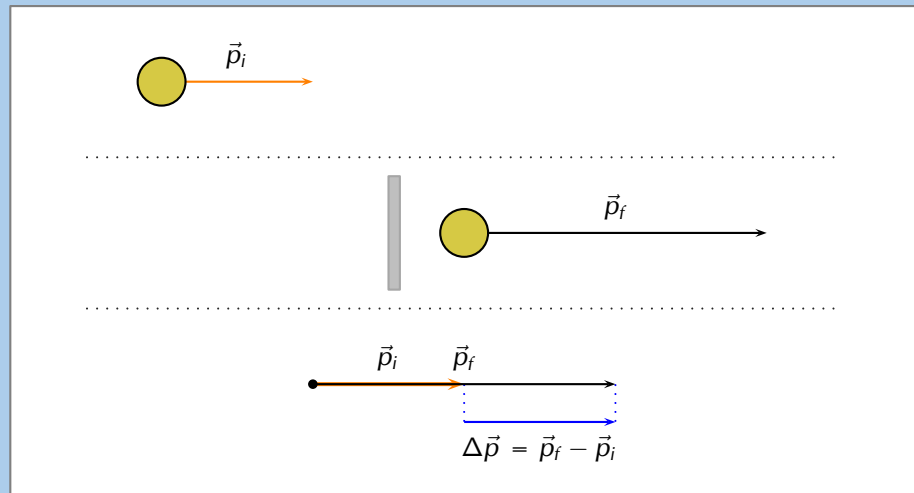
The question explicitly gives a number of values which we identify and convert into SI

units:

- the ball's mass ($m = 24 \text{ g} = 0,024 \text{ kg}$),
- the ball's initial velocity ($\vec{v}_i = 1 \text{ m}\cdot\text{s}^{-1}$) towards the front wall, and
- the ball's final velocity ($\vec{v}_f = 20 \text{ m}\cdot\text{s}^{-1}$) towards the front wall

We are asked to calculate the change in momentum of the ball,

$$\Delta\vec{p} = m\vec{v}_f - m\vec{v}_i$$



We have everything we need to find $\Delta\vec{p}$.

Step 2: Choose a frame of reference

Let us choose towards the front wall as the positive direction.

Step 3: Do the calculation

$$\begin{aligned}\Delta\vec{p} &= m\vec{v}_f - m\vec{v}_i \\ &= (0,024)(20) - (0,024)(+1) \\ &= (0,48) - (0,024) \\ &= 0,456 \\ &= 0,46 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ towards the front wall}\end{aligned}$$

Step 4: Quote the final answer

The change in momentum is $0,46 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ towards the front wall.

Exercise 2 – 2:

- Which expression accurately describes the change of momentum of an object?
 - $\frac{\vec{F}}{m}$
 - $\frac{\vec{F}}{\Delta t}$
 - $\vec{F} \cdot m$
 - $\vec{F} \cdot \Delta t$
- A child drops a ball of mass 100 g. The ball strikes the ground with a velocity of $5 \text{ m}\cdot\text{s}^{-1}$ and rebounds with a velocity of $4 \text{ m}\cdot\text{s}^{-1}$. Calculate the change of momentum of the ball.
- More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [27HQ](#) 2. [27HR](#)



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2.4 Conservation of momentum

ESCJC

In this section we are going to look at momentum when two objects interact with each other and, specifically, treat both objects as one system. To do this properly we first need to define what we mean we talk about a system, then we need to look at what happens to momentum overall and we will explore the applications of momentum in these interactions.

Systems

ESCJD

DEFINITION: *System*

A system is a physical configuration of particles and or objects that we study.

For example, earlier we looked at what happens when a ball bounces off a wall. The system that we were studying was just the wall and the ball. The wall must be connected to the Earth and something must have thrown or hit the ball but we ignore those. A system is a subset of the physical world that we are studying. The system exists in some larger environment.

In the problems that we are solving we actually treat our system as being isolated from the environment. That means that we can completely ignore the environment. In reality, the environment can affect the system but we ignore that for isolated systems. We try to choose isolated systems when it makes sense to ignore the surrounding

environment.

DEFINITION: *Isolated system*

An isolated system is a physical configuration of particles and or objects that we study that doesn't exchange any matter with its surroundings and is not subject to any force whose source is external to the system.

An external force is a force acting on the pieces of the system that we are studying that is not caused by a component of the system.

It is a choice we make to treat objects as an isolated system but we can only do this if we think it really make sense, if the results we are going to get will still be reasonable. In reality, no system is completely isolated except for the whole universe (we think). When we look at a ball hitting a wall it makes sense to ignore the force of gravity. The effect isn't exactly zero but it will be so small that it will not make any real difference to our results.

Conservation of momentum

ESCJF

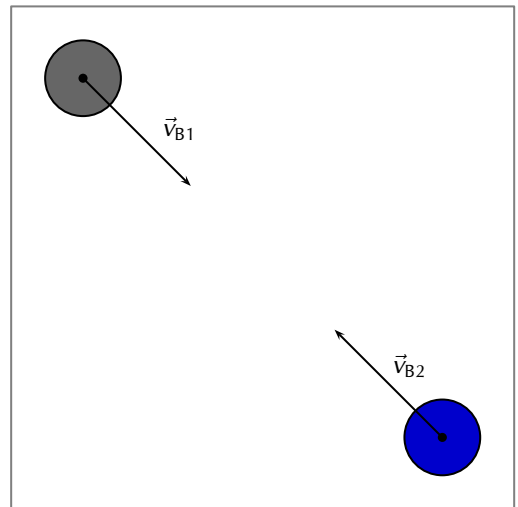
There is a very useful property of isolated systems, total momentum is conserved. Lets use a practical example to show why this is the case, let us consider two billiard balls moving towards each other. Here is a sketch alongside (not to scale). When they come into contact, ball 1 exerts a contact force on the ball 2, $\vec{F}_{B1}$, and the ball 2 exerts a force on ball 1, $\vec{F}_{B2}$. We also know that the force will result in a change in momentum:

$$\vec{F}_{net} = \frac{\Delta \vec{p}}{\Delta t}$$

We also know from Newton's third law that:

$$\begin{aligned}\vec{F}_{B1} &= -\vec{F}_{B2} \\ \frac{\Delta \vec{p}_{B1}}{\Delta t} &= -\frac{\Delta \vec{p}_{B2}}{\Delta t} \\ \Delta \vec{p}_{B1} &= -\Delta \vec{p}_{B2} \\ \Delta \vec{p}_{B1} + \Delta \vec{p}_{B2} &= 0\end{aligned}$$

This says that if you add up all the changes in momentum for an isolated system the net result will be zero. If we add up all the momenta in the system the total momentum won't change because the net change is zero.

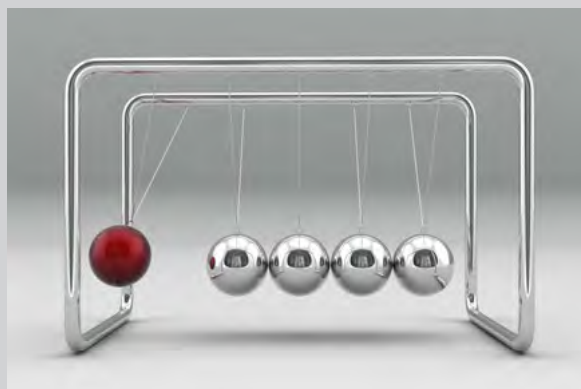


Important: note that this is because the forces are internal forces and Newton's third law applies. An external force would not necessarily allow momentum to be conserved. In the absence of an external force acting on a system, momentum is conserved.

Activity: Newton's cradle demonstration

Momentum conservation:

A Newton's cradle demonstrates a series of collisions in which momentum is conserved.



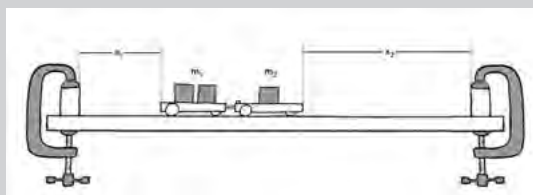
Informal experiment: Conservation of momentum

Aim:

To investigate the changes in momentum when two bodies are separated by an explosive force.

Apparatus:

- two spring-loaded trolleys
- stopwatch
- meter-stick
- two barriers



Method:

- Clamp the barriers one metre apart onto a flat surface.
- Find the mass of each trolley and place a known mass on one of the trolleys.
- Place the two trolleys between the barriers end to end so that the spring on the one trolley is in contact with the flat surface of the other trolley
- Release the spring by hitting the release knob and observe how the trolleys push each other apart.
- Repeat the explosions with the trolleys at a different position until they strike the barriers simultaneously. Each trolley now has the same time of travel t .
- Measure the distances x_1 and x_2 . These can be taken as measures of the respective velocities.

- Repeat the experiment for a different combination of masses.

Results:

Record your results in a table, using the following headings for each trolley. Total mass in kg, distance travelled in m, momentum in $\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$. What is the relationship between the total momentum after the explosion and the total momentum before the explosion?

DEFINITION: Conservation of Momentum

The total momentum of an isolated system is constant.

The total momentum of a system is calculated by the vector sum of the momenta of all the objects or particles in the system. For a system with n objects the total momentum is:

$$\vec{p}_T = \vec{p}_1 + \vec{p}_2 + \dots + \vec{p}_n$$

Worked example 7: Calculating the total momentum of a system

QUESTION

Two billiard balls roll towards each other. They each have a mass of 0,3 kg. Ball 1 is moving at $\vec{v}_1 = 1 \text{ m}\cdot\text{s}^{-1}$ to the right, while ball 2 is moving at $\vec{v}_2 = 0,8 \text{ m}\cdot\text{s}^{-1}$ to the left. Calculate the total momentum of the system.

SOLUTION

Step 1: Identify what information is given and what is asked for

The question explicitly gives

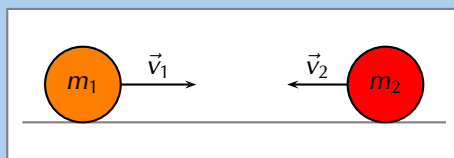
- the mass of each ball,
- the velocity of ball 1, $\vec{v}_1$, and
- the velocity of ball 2, $\vec{v}_2$,

all in the correct units.

We are asked to calculate the **total momentum of the system**. In this example our system consists of two balls. To find the total momentum we must determine the momentum of each ball and add them.

$$\vec{p}_T = \vec{p}_1 + \vec{p}_2$$

Since ball 1 is moving to the right, its momentum is in this direction, while the second ball's momentum is directed towards the left.



Thus, we are required to find the sum of two vectors acting along the same straight line.

Step 2: Choose a frame of reference

Let us choose moving to the right as the positive direction.

Step 3: Calculate the momentum

The total momentum of the system is then the sum of the two momenta taking the directions of the velocities into account. Ball 1 is travelling at $1 \text{ m}\cdot\text{s}^{-1}$ to the right or $+1 \text{ m}\cdot\text{s}^{-1}$. Ball 2 is travelling at $0,8 \text{ m}\cdot\text{s}^{-1}$ to the left or $-0,8 \text{ m}\cdot\text{s}^{-1}$. Thus,

$$\begin{aligned}\vec{p}_T &= m_1 \vec{v}_1 + m_2 \vec{v}_2 \\ &= (0,3)(+1) + (0,3)(-0,8) \\ &= (+0,3) + (-0,24) \\ &= +0,06 \\ &= 0,06 \text{ m}\cdot\text{s}^{-1} \text{ to the right}\end{aligned}$$

In the last step the direction was added in words. Since the result in the second last line is positive, the total momentum of the system is in the positive direction (i.e. to the right).

Exercise 2 – 3:

- Two golf balls roll towards each other. They each have a mass of 100 g. Ball 1 is moving at $\vec{v}_1 = 2,4 \text{ m}\cdot\text{s}^{-1}$ to the right, while ball 2 is moving at $\vec{v}_2 = 3 \text{ m}\cdot\text{s}^{-1}$ to the left. Calculate the total momentum of the system.
- Two motorcycles are involved in a head on collision. Motorcycle A has a mass of 200 kg and was travelling at $120 \text{ km}\cdot\text{hr}^{-1}$ south. Motor cycle B has a mass of 250 kg and was travelling north at $100 \text{ km}\cdot\text{hr}^{-1}$. A and B are about to collide. Calculate the momentum of the system before the collision takes place.
- A 700 kg truck is travelling north at a velocity of $40 \text{ km}\cdot\text{hr}^{-1}$ when it is approached by a 500 kg car travelling south at a velocity of $100 \text{ km}\cdot\text{hr}^{-1}$. Calculate the total momentum of the system.
- More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27HS 2. 27HT 3. 27HV



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We have shown that the net change in momentum is zero for an isolated system. The momenta of the **individual objects can change** but the total momentum of the **system remains constant**.

This means that it makes sense to define the total momentum at the beginning of the problem as the initial total momentum, $\vec{p}_{Ti}$, and the final total momentum $\vec{p}_{Tf}$. Momentum conservation implies that, no matter what happens inside an isolated system:

$$\vec{p}_{Ti} = \vec{p}_{Tf}$$

This means that in an isolated system the total momentum before a **collision** or **explosion** is equal to the total momentum after the collision or explosion.

Consider a simple collision of two billiard balls. The balls are rolling on a frictionless, horizontal surface and the system is isolated. We can apply conservation of momentum. The first ball has a mass m_1 and an initial velocity $\vec{v}_{i1}$. The second ball has a mass m_2 and moves first ball with an initial velocity $\vec{v}_{i2}$. This situation is shown in Figure 2.1.

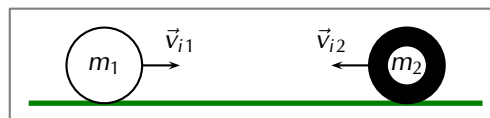


Figure 2.1: Before the collision.

The total momentum of the system before the collision, $\vec{p}_{Ti}$ is: $\vec{p}_{Ti} = m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2}$

After the two balls collide and move away they each have a different momentum. If the first ball has a final velocity of $\vec{v}_{f1}$ and the second ball has a final velocity of $\vec{v}_{f2}$ then we have the situation shown in 2.2.



Figure 2.2: After the collision.

The total momentum of the system after the collision, $\vec{p}_{Tf}$ is: $\vec{p}_{Tf} = m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2}$

This system of two balls is isolated since there are no external forces acting on the balls. Therefore, by the principle of conservation of linear momentum, the total momentum before the collision is equal to the total momentum after the collision. This gives the equation for the conservation of momentum in a collision of two objects,

$\vec{p}_i = \vec{p}_f$	$m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} = m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2}$
m_1	: mass of object 1 (kg)
m_2	: mass of object 2 (kg)
$\vec{v}_{i1}$	: initial velocity of object 1 ($\text{m}\cdot\text{s}^{-1}$ + direction)
$\vec{v}_{i2}$	: initial velocity of object 2 ($\text{m}\cdot\text{s}^{-1}$ - direction)
$\vec{v}_{f1}$	: final velocity of object 1 ($\text{m}\cdot\text{s}^{-1}$ - direction)
$\vec{v}_{f2}$	: final velocity of object 2 ($\text{m}\cdot\text{s}^{-1}$ + direction)

This equation is always true - momentum is always conserved in collisions.

Worked example 8: Conservation of momentum

QUESTION

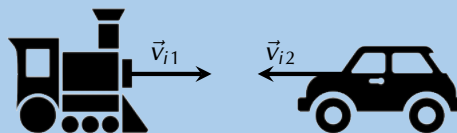
A toy car of mass 1 kg moves westwards with a speed of $2 \text{ m}\cdot\text{s}^{-1}$. It collides head-on with a toy train. The train has a mass of 1,5 kg and is moving at a speed of $1,5 \text{ m}\cdot\text{s}^{-1}$ eastwards. If the car rebounds at $2,05 \text{ m}\cdot\text{s}^{-1}$, calculate the final velocity of the train.

SOLUTION

Step 1: Analyse what you are given and draw a sketch

The question explicitly gives a number of values which we identify and convert into SI units if necessary:

- the train's mass ($m_1 = 1,50 \text{ kg}$),
- the train's initial velocity ($\vec{v}_{1i} = 1,5 \text{ m}\cdot\text{s}^{-1}$ eastward),
- the car's mass ($m_2 = 1,00 \text{ kg}$),
- the car's initial velocity ($\vec{v}_{2i} = 2,00 \text{ m}\cdot\text{s}^{-1}$ westward), and
- the car's final velocity ($\vec{v}_{2f} = 2,05 \text{ m}\cdot\text{s}^{-1}$ eastward).



We are asked to find the final velocity of the train.

Step 2: Choose a frame of reference

We will choose to the East as positive.

Step 3: Apply the Law of Conservation of momentum

$$\begin{aligned}\vec{p}_{Ti} &= \vec{p}_{Tf} \\ m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} &= m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2} \\ (1,5)(+1,5) + (2)(-2) &= (1,5)(\vec{v}_{f1}) + (2)(2,05) \\ 2,25 - 4 - 4,1 &= (1,5) \vec{v}_{f1} \\ 5,85 &= (1,5) \vec{v}_{f1} \\ \vec{v}_{f1} &= 3,9 \text{ m} \cdot \text{s}^{-1} \text{ eastwards}\end{aligned}$$

Worked example 9: Conservation of momentum

QUESTION

A jet flies at a speed of $275 \text{ m} \cdot \text{s}^{-1}$. The pilot fires a missile forward off a mounting at a speed of $700 \text{ m} \cdot \text{s}^{-1}$ relative to the ground. The respective masses of the jet and the missile are 5000 kg and 50 kg . Treating the system as an isolated system, calculate the new speed of the jet immediately after the missile had been fired.



SOLUTION

Step 1: Analyse what you are given and draw a sketch

The question explicitly gives a number of values which we identify and convert into SI units if necessary:

- the mass of the jet ($m_1 = 5000 \text{ kg}$)
- the mass of the rocket ($m_2 = 50 \text{ kg}$)
- the initial velocity of the jet and rocket ($\vec{v}_{i1} = \vec{v}_{i2} = 275 \text{ m} \cdot \text{s}^{-1}$ to the left)
- the final velocity of the rocket ($\vec{v}_{f2} = 700 \text{ m} \cdot \text{s}^{-1}$ to the left)

We need to find the final speed of the jet and we can use momentum conservation because we can treat it as an isolated system.

We choose the original direction that the jet was flying in as the positive direction, to the left.



After the missile is launched we need to take both into account:



Figure 2.3: System is jet and missile

Step 2: Apply the Law of Conservation of momentum

The jet and missile are connected initially and move at the same velocity. We will therefore combine their masses and change the momentum equation as follows:

$$\begin{aligned}\vec{p}_i &= \vec{p}_f \\ (m_1 + m_2) \vec{v}_i &= m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2} \\ (5000 + 50)(275) &= (5000)(\vec{v}_{f1}) + (50)(700) \\ 1\,388\,750 - 35\,000 &= (5000)(\vec{v}_{f1}) \\ \vec{v}_{f1} &= 270,75 \text{ m}\cdot\text{s}^{-1} \text{ in the original direction}\end{aligned}$$

Step 3: Quote the final answer

The speed of the jet is the magnitude of the final velocity, $270,75 \text{ m}\cdot\text{s}^{-1}$.

Worked example 10: Conservation of momentum

QUESTION

A bullet of mass 50 g travelling horizontally to the right at $500 \text{ m}\cdot\text{s}^{-1}$ strikes a stationary wooden block of mass 2 kg resting on a smooth horizontal surface. The bullet goes through the block and comes out on the other side at $200 \text{ m}\cdot\text{s}^{-1}$. Calculate the speed of the block after the bullet has come out the other side.

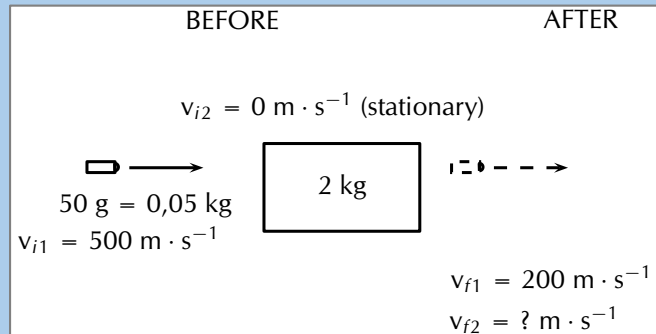
SOLUTION

Step 1: Analyse what you are given and draw a sketch

The question explicitly gives a number of values which we identify and convert into SI units if necessary:

- the mass of the bullet ($m_1 = 50 \text{ g} = 0,050 \text{ kg}$)
- the mass of the block ($m_2 = 2,00 \text{ kg}$)
- the initial velocity of the bullet ($\vec{v}_{i1} = 500 \text{ m}\cdot\text{s}^{-1}$ to the right)
- the final velocity of the bullet ($\vec{v}_{f1} = 200 \text{ m}\cdot\text{s}^{-1}$ to the right)

We need to find the velocity of the wood block. We choose the direction in which the bullet was travelling to be the positive direction, to the right.



Step 2: Apply the Law of Conservation of momentum

$$\vec{p}_i = p_f$$

$$m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} = m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2}$$

$$(0,05)(+500) + (2)(0) = (0,05)(+200) + (2)(\vec{v}_{f2})$$

$$25 + 0 - 10 = 2 \vec{v}_{f2}$$

$$\vec{v}_{f2} = 7,5 \text{ m} \cdot \text{s}^{-1} \text{ in the same direction as the bullet}$$

Step 3: Apply the Law of Conservation of momentum

The block is travelling at $7,5 \text{ m} \cdot \text{s}^{-1}$.

We have been applying conservation of momentum to collisions and explosion which is valid but there are actually two different types of collisions and they have different properties.

Two types of collisions are of interest:

- elastic collisions
- inelastic collisions

In both types of collision, total momentum is *a/ways* conserved. Kinetic energy is conserved for elastic collisions, but not for inelastic collisions.

Initially in a collision the objects have kinetic energy. In some collisions that energy is transformed through processes like deformation. In a car crash the car gets all mangled which requires the permanent transfer of energy.

DEFINITION: *Elastic Collisions*

An elastic collision is a collision where total momentum and total kinetic energy are both conserved.

This means that in an elastic collision the total momentum *and* the total kinetic energy before the collision is the same as after the collision. For these kinds of collisions, the kinetic energy is not transformed permanently through work or deformation of the objects. During the collision the energy is going to be transferred (for example as a ball compresses) but will be recovered during the elastic response of the system (for example the ball then expanding again).

Before the collision

2.4 shows two balls rolling toward each other, about to collide:

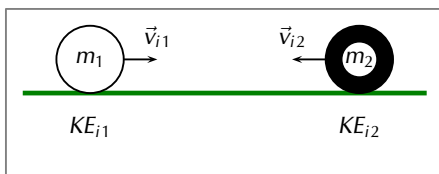


Figure 2.4: Two balls before they collide.

After the collision

2.5 shows two balls after they have collided:

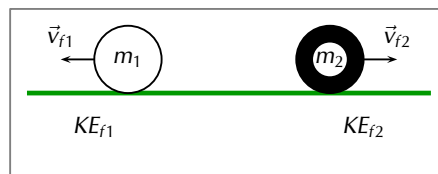


Figure 2.5: Two balls after they collide.

We have calculated the total initial momentum previously, now we calculate the total kinetic energy of the system in the same way. Ball 1 has a kinetic energy which we call KE_{i1} and ball 2 has a kinetic energy which we call KE_{i2} , the total kinetic energy before the collision is: $KE_{Ti} = KE_{i1} + KE_{i2}$

Ball 1 now has a kinetic energy which we call KE_{f1} and ball 2 now has a kinetic energy which we call KE_{f2} , it means that the total kinetic energy after the collision is: $KE_{Tf} = KE_{f1} + KE_{f2}$

Since this is an *elastic* collision, the total momentum before the collision equals the total momentum after the collision **and** the total kinetic energy before the collision equals the total kinetic energy after the collision. Therefore:

$$\begin{aligned}\vec{p}_{Ti} &= \vec{p}_{Tf} \\ \vec{p}_{i1} + \vec{p}_{i2} &= \vec{p}_{f1} + \vec{p}_{f2} \\ \text{and} \\ KE_{Ti} &= KE_{Tf} \\ KE_{i1} + KE_{i2} &= KE_{f1} + KE_{f2}\end{aligned}$$

Worked example 11: An elastic collision

QUESTION

Consider a collision between two pool balls. Ball 1 is at rest and ball 2 is moving towards it with a speed of $2 \text{ m}\cdot\text{s}^{-1}$. The mass of each ball is $0,3 \text{ kg}$. After the balls collide *elastically*, ball 2 comes to an immediate stop and ball 1 moves off. What is the final velocity of ball 1?

SOLUTION

Step 1: Choose a frame of reference

Choose to the right as positive and we assume that ball 2 is moving towards the left approaching ball 1.

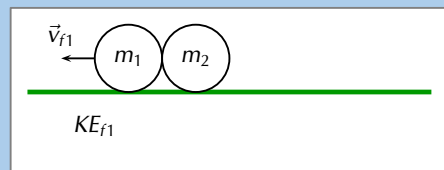
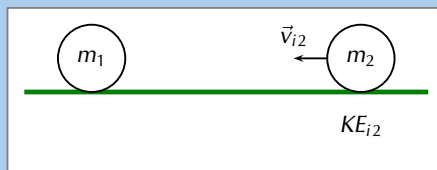
Step 2: Determine what is given and what is needed

- Mass of ball 1, $m_1 = 0,3 \text{ kg}$.
- Mass of ball 2, $m_2 = 0,3 \text{ kg}$.
- Initial velocity of ball 1, $\vec{v}_{i1} = 0 \text{ m}\cdot\text{s}^{-1}$.
- Initial velocity of ball 2, $\vec{v}_{i2} = 2 \text{ m}\cdot\text{s}^{-1}$ to the left.
- Final velocity of ball 2, $\vec{v}_{f2} = 0 \text{ m}\cdot\text{s}^{-1}$.
- The collision is elastic.

All quantities are in SI units. We need to find the final velocity of ball 1, v_{f1} . Since the collision is elastic, we know that

- momentum is conserved, $m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} = m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2}$
- energy is conserved, $\frac{1}{2} (m_1 v_{i1}^2 + m_2 v_{i2}^2) = \frac{1}{2} (m_1 v_{f1}^2 + m_2 v_{f2}^2)$

Step 3: Draw a rough sketch of the situation



Step 4: Solve the problem

Momentum is conserved in all collisions so it makes sense to begin with momentum. Therefore:

$$\begin{aligned}\vec{p}_{Ti} &= \vec{p}_{Tf} \\ m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} &= m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2} \\ (0,3)(0) + (0,3)(-2) &= (0,3) \vec{v}_{f1} + 0 \\ \vec{v}_{f1} &= -2,00 \text{ m}\cdot\text{s}^{-1}\end{aligned}$$

Step 5: Quote your final answer

The final velocity of ball 1 is $2 \text{ m}\cdot\text{s}^{-1}$ to the left.

Worked example 12: Another elastic collision

QUESTION

Consider 2 marbles. Marble 1 has mass 50 g and marble 2 has mass 100 g. Edward rolls marble 2 along the ground towards marble 1 in the positive x-direction. Marble 1 is initially at rest and marble 2 has a velocity of $3 \text{ m}\cdot\text{s}^{-1}$ in the positive x-direction. After they collide *elastically*, both marbles are moving. What is the final velocity of each marble?

SOLUTION

Step 1: Decide how to approach the problem

We are given:

- mass of marble 1, $m_1 = 50 \text{ g}$
- mass of marble 2, $m_2 = 100 \text{ g}$
- initial velocity of marble 1, $\vec{v}_{i1} = 0 \text{ m}\cdot\text{s}^{-1}$
- initial velocity of marble 2, $\vec{v}_{i2} = 3 \text{ m}\cdot\text{s}^{-1}$ to the right
- the collision is elastic

The masses need to be converted to SI units.

$$m_1 = 0,05\text{kg}$$

$$m_2 = 0,1\text{kg}$$

We are required to determine the final velocities:

- final velocity of marble 1, $\vec{v}_{f1}$
- final velocity of marble 2, $\vec{v}_{f2}$

Since the collision is elastic, we know that

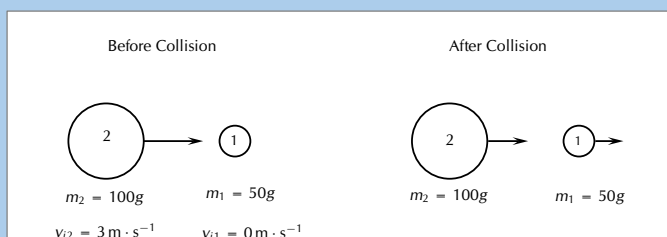
- momentum is conserved, $\vec{p}_{Ti} = \vec{p}_{Tf}$.
- energy is conserved, $KE_{Ti} = KE_{Tf}$.

We have two equations and two unknowns ($\vec{v}_1$, $\vec{v}_2$) so it is a simple case of solving a set of simultaneous equations.

Step 2: Choose a frame of reference

Choose to the right as positive.

Step 3: Draw a rough sketch of the situation



Step 4: Solve the problem

Momentum is conserved. Therefore:

$$\begin{aligned}\vec{p}_i &= \vec{p}_f \\ \vec{p}_{i1} + \vec{p}_{i2} &= \vec{p}_{f1} + \vec{p}_{f2} \\ m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} &= m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2} \\ m_2 \vec{v}_{i2} &= m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2} \\ m_1 \vec{v}_{f1} &= m_2 \vec{v}_{i2} - m_2 \vec{v}_{f2} \\ \vec{v}_{f1} &= \frac{m_2}{m_1} (\vec{v}_{i2} - \vec{v}_{f2})\end{aligned}$$

Energy is also conserved. Therefore:

$$\begin{aligned}KE_i &= KE_f \\ KE_{i1} + KE_{i2} &= KE_{f1} + KE_{f2} \\ \frac{1}{2} m_1 v_{i1}^2 + \frac{1}{2} m_2 v_{i2}^2 &= \frac{1}{2} m_1 v_{f1}^2 + \frac{1}{2} m_2 v_{f2}^2 \\ \frac{1}{2} m_2 v_{i2}^2 &= \frac{1}{2} m_1 v_{f1}^2 + \frac{1}{2} m_2 v_{f2}^2 \\ m_2 v_{i2}^2 &= m_1 v_{f1}^2 + m_2 v_{f2}^2\end{aligned}$$

Substitute the expression we derived from conservation of momentum into the expression derived from conservation of kinetic energy and solve for v_{f2} .

$$\begin{aligned}m_2 v_{i2}^2 &= m_1 v_{f1}^2 + m_2 v_{f2}^2 \\ &= m_1 \left(\frac{m_2}{m_1} (v_{i2} - v_{f2}) \right)^2 + m_2 v_{f2}^2 \\ &= m_1 \frac{m_2^2}{m_1^2} (v_{i2} - v_{f2})^2 + m_2 v_{f2}^2 \\ &= \frac{m_2^2}{m_1} (v_{i2} - v_{f2})^2 + m_2 v_{f2}^2 \\ v_{i2}^2 &= \frac{m_2}{m_1} (v_{i2} - v_{f2})^2 + v_{f2}^2 \\ &= \frac{m_2}{m_1} (v_{i2}^2 - 2 \cdot v_{i2} \cdot v_{f2} + v_{f2}^2) + v_{f2}^2 \\ 0 &= \left(\frac{m_2}{m_1} - 1 \right) v_{i2}^2 - 2 \frac{m_2}{m_1} v_{i2} \cdot v_{f2} + \left(\frac{m_2}{m_1} + 1 \right) v_{f2}^2 \\ &= \left(\frac{0.1}{0.05} - 1 \right) (3)^2 - 2 \frac{0.1}{0.05} (3) \cdot v_{f2} + \left(\frac{0.1}{0.05} + 1 \right) v_{f2}^2 \\ &= (2 - 1) (3)^2 - 2 \cdot 2 (3) \cdot v_{f2} + (2 + 1) v_{f2}^2 \\ &= 9 - 12 v_{f2} + 3 v_{f2}^2 \\ &= 3 - 4 v_{f2} + v_{f2}^2 \\ &= (v_{f2} - 3) (v_{f2} - 1)\end{aligned}$$

Therefore $v_{f2} = 1$ or $v_{f2} = 3$

Substituting back into the expression from conservation of momentum, we get:

$$\begin{aligned}v_{f1} &= \frac{m_2}{m_1} (v_{i2} - v_{f2}) \\&= \frac{0.1}{0.05} (3 - 3) \\&= 0\end{aligned}$$

or

$$\begin{aligned}v_{f1} &= \frac{m_2}{m_1} (v_{i2} - v_{f2}) \\&= \frac{0.1}{0.05} (3 - 1) \\&= 4 \text{ m}\cdot\text{s}^{-1}\end{aligned}$$

But according to the question, marble 1 is moving after the collision, therefore marble 1 moves to the right at $4 \text{ m}\cdot\text{s}^{-1}$.

Therefore marble 2 moves with a velocity of $1 \text{ m}\cdot\text{s}^{-1}$ to the right.

Worked example 13: Colliding billiard balls

QUESTION

Two billiard balls each with a mass of 150 g collide head-on in an elastic collision. Ball 1 was travelling at a speed of $2 \text{ m}\cdot\text{s}^{-1}$ and ball 2 at a speed of $1,5 \text{ m}\cdot\text{s}^{-1}$. After the collision, ball 1 travels away from ball 2 at a velocity of $1,5 \text{ m}\cdot\text{s}^{-1}$.

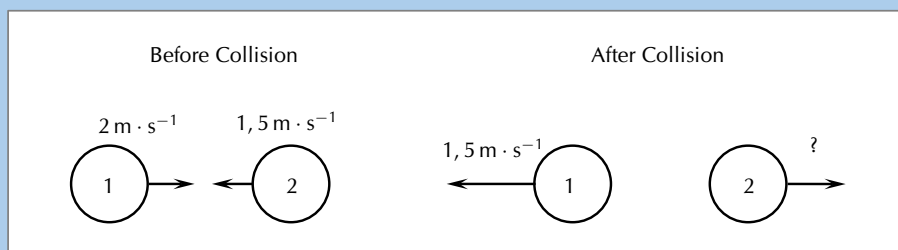
1. Calculate the velocity of ball 2 after the collision.
2. Prove that the collision was elastic. Show calculations.

SOLUTION

Step 1: Choose a frame of reference

Choose to the right as positive.

Step 2: Draw a rough sketch of the situation



Step 3: Decide how to approach the problem

Since momentum is conserved in all kinds of collisions, we can use conservation of momentum to solve for the velocity of ball 2 after the collision.

Step 4: Solve problem

$$\begin{aligned}\vec{p}_{Ti} &= \vec{p}_{Tf} \\ m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} &= m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2} \\ \left(\frac{150}{1000}\right)(2) + \left(\frac{150}{1000}\right)(-1, 5) &= \left(\frac{150}{1000}\right)(-1, 5) + \left(\frac{150}{1000}\right)(\vec{v}_{f2}) \\ 0,3 - 0,225 &= -0,225 + 0,15\vec{v}_{f2} \\ \vec{v}_{f2} &= 3 \text{ m}\cdot\text{s}^{-1}\end{aligned}$$

So after the collision, ball 2 moves with a velocity of $3 \text{ m}\cdot\text{s}^{-1}$ to the right.

Step 5: Elastic collisions

The fact that characterises an elastic collision is that the total kinetic energy of the particles before the collision is the same as the total kinetic energy of the particles after the collision. This means that if we can show that the initial kinetic energy is equal to the final kinetic energy, we have shown that the collision is elastic.

Step 6: Calculating the initial total kinetic energy

$$\begin{aligned}EK_{\text{before}} &= \frac{1}{2}m_1 v_{i1}^2 + \frac{1}{2}m_2 v_{i2}^2 \\ &= \left(\frac{1}{2}\right)(0,15)(2)^2 + \left(\frac{1}{2}\right)(0,15)(-1,5)^2 \\ &= 0,469\dots\text{J}\end{aligned}$$

Step 7: Calculating the final total kinetic energy

$$\begin{aligned}EK_{\text{after}} &= \frac{1}{2}m_1 v_{f1}^2 + \frac{1}{2}m_2 v_{f2}^2 \\ &= \left(\frac{1}{2}\right)(0,15)(-1,5)^2 + \left(\frac{1}{2}\right)(0,15)(2)^2 \\ &= 0,469\dots\text{J}\end{aligned}$$

So $EK_{Ti} = EK_{Tf}$ and hence the collision is elastic.

Inelastic collisions

ESCJJ

DEFINITION: Inelastic Collisions

An inelastic collision is a collision in which **total momentum** is conserved but **total kinetic energy** is *not* conserved. The kinetic energy is *transformed* from or into other kinds of energy.

So the total momentum before an inelastic collisions is the same as after the collision. But the total *kinetic* energy before and after the inelastic collision is *different*. Of course this does not mean that total energy has not been conserved, rather the energy has been *transformed* into another type of energy.

As a rule of thumb, inelastic collisions happen when the colliding objects are distorted in some way. Usually they change their shape. The modification of the shape of an object requires energy and this is where the “missing” kinetic energy goes. A classic example of an inelastic collision is a motor car accident. The cars change shape and there is a noticeable change in the kinetic energy of the cars before and after the collision. This energy was used to bend the metal and deform the cars. Another example of an inelastic collision is shown in 2.6.

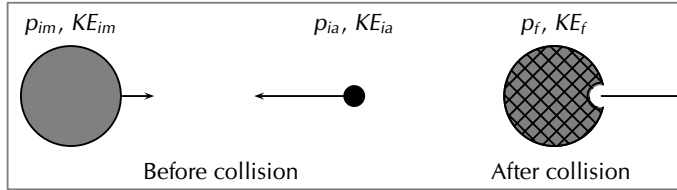


Figure 2.6: Asteroid moving towards the Moon.

An asteroid is moving through space towards the Moon. Before the asteroid crashes into the Moon, the total momentum of the system is:

$$\vec{p}_{Ti} = \vec{p}_{iM} + \vec{p}_{ia}$$

The total kinetic energy of the system is:

$$KE_i = KE_{iM} + KE_{ia}$$

When the asteroid collides **inelastically** with the Moon, its kinetic energy is transformed mostly into heat energy. If this heat energy is large enough, it can cause the asteroid and the area of the Moon’s surface that it hits, to melt into liquid rock. From the force of impact of the asteroid, the molten rock flows outwards to form a crater on the Moon.

After the collision, the total momentum of the system will be the same as before. But since this collision is *inelastic*, (and you can see that a change in the shape of objects has taken place), total kinetic energy is **not** the same as before the collision.

Momentum is conserved:

$$\vec{p}_{Ti} = \vec{p}_{Tf}$$

But the total kinetic energy of the system is not conserved:

$$KE_i \neq KE_f$$

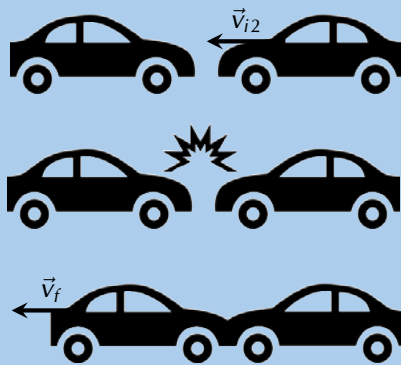
Worked example 14: An inelastic collision

QUESTION

Consider the collision of two cars. Car 1 is at rest and Car 2 is moving at a speed of $2 \text{ m}\cdot\text{s}^{-1}$ to the left. Both cars each have a mass of 500 kg . The cars collide *inelastically* and stick together. What is the resulting velocity of the resulting mass of metal?

SOLUTION

Step 1: Draw a rough sketch of the situation



Step 2: Determine how to approach the problem

We are given:

- mass of car 1, $m_1 = 500 \text{ kg}$
- mass of car 2, $m_2 = 500 \text{ kg}$
- initial velocity of car 1, $\vec{v}_{i1} = 0 \text{ m}\cdot\text{s}^{-1}$
- initial velocity of car 2, $\vec{v}_{i2} = 2 \text{ m}\cdot\text{s}^{-1}$ to the left
- the collision is inelastic

All quantities are in SI units. We are required to determine the final velocity of the resulting mass, $\vec{v}_f$.

Since the collision is inelastic, we know that

- momentum is conserved,
 $m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} = m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2} = (m_1 + m_2) \vec{v}_f$
- kinetic energy is not conserved

Step 3: Choose a frame of reference

Choose to the left as positive.

Step 4: Solve problem

So we must use conservation of momentum to solve this problem.

$$\begin{aligned}\vec{p}_{Ti} &= \vec{p}_{Tf} \\ \vec{p}_{i1} + \vec{p}_{i2} &= \vec{p}_f \\ m_1 \vec{v}_{i1} + m_2 \vec{v}_{i2} &= (m_1 + m_2) \vec{v}_f \\ (500)(0) + (500)(2) &= (500 + 500) \vec{v}_f \\ 1000 &= 1000 \vec{v}_f \\ \vec{v}_f &= 1 \text{ m}\cdot\text{s}^{-1}\end{aligned}$$

Therefore, the final velocity of the resulting mass of cars is $1 \text{ m}\cdot\text{s}^{-1}$ to the left.

Exercise 2 – 4: Collisions

1. A truck of mass 4500 kg travelling at $20 \text{ m}\cdot\text{s}^{-1}$ hits a car from behind. The car (mass 1000 kg) was travelling at $15 \text{ m}\cdot\text{s}^{-1}$. The two vehicles, now connected carry on moving in the same direction.
 - a) Calculate the final velocity of the truck-car combination after the collision.
 - b) Determine the kinetic energy of the system before and after the collision.
 - c) Explain the difference in your answers for b).
 - d) Was this an example of an elastic or inelastic collision? Give reasons for your answer.
2. Two cars of mass 900 kg each collide head-on and stick together. Determine the final velocity of the cars if car 1 was travelling at $15 \text{ m}\cdot\text{s}^{-1}$ and car 2 was travelling at $20 \text{ m}\cdot\text{s}^{-1}$.
3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [27HW](#) 2. [27HX](#)



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2.5 Impulse

ESCJK

When a net force acts on a body it will result in an acceleration which alters the motion of the body. A large net force will cause a larger acceleration than a small net force. The total change in motion of the object can be the same if the large and small forces act for different time intervals. The combination of the force and time that it acts is a useful quantity which leads us to define impulse.

DEFINITION: Impulse

Impulse is the product of the net force and the time interval for which the force acts.

$$\text{Impulse} = \vec{F}_{\text{net}} \cdot \Delta t$$

However, from Newton's Second Law, we know that

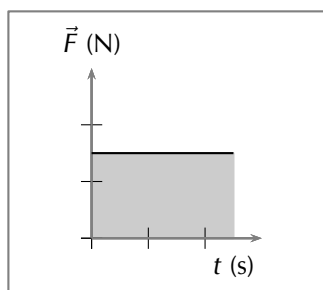
$$\begin{aligned}\vec{F}_{\text{net}} &= \frac{\Delta \vec{p}}{\Delta t} \\ \therefore \vec{F}_{\text{net}} \cdot \Delta t &= \Delta \vec{p} \\ &= \text{Impulse}\end{aligned}$$

Therefore we can define the impulse-momentum theorem:

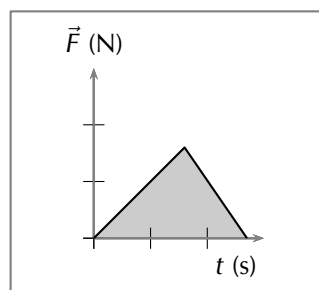
$$\text{Impulse} = \Delta \vec{p}$$

Impulse is equal to the change in momentum of an object. From this equation we see, that for a given change in momentum, $\vec{F}_{\text{net}}\Delta t$ is fixed. Thus, if $\vec{F}_{\text{net}}$ is reduced, Δt must be increased (i.e. a smaller resultant force must be applied for longer to bring about the same change in momentum). Alternatively if Δt is reduced (i.e. the resultant force is applied for a shorter period) then the resultant force must be increased to bring about the same change in momentum.

The graphs below show how the force acting on a body changes with time.



The area under the graph, shaded in, represents the impulse of the body.



Worked example 15: Impulse and change in momentum

QUESTION

A 150 N resultant force acts on a 300 kg trailer. Calculate how long it takes this force to change the trailer's velocity from $2 \text{ m}\cdot\text{s}^{-1}$ to $6 \text{ m}\cdot\text{s}^{-1}$ in the same direction. Assume that the forces acts to the right which is the direction of motion of the trailer.

SOLUTION

Step 1: Identify what information is given and what is asked for

The question explicitly gives

- the trailer's mass as 300 kg,
- the trailer's initial velocity as $2 \text{ m}\cdot\text{s}^{-1}$ to the right,
- the trailer's final velocity as $6 \text{ m}\cdot\text{s}^{-1}$ to the right, and
- the resultant force acting on the object.

We are asked to calculate the time taken Δt to accelerate the trailer from the 2 to $6 \text{ m}\cdot\text{s}^{-1}$. From the Newton's second law,

$$\begin{aligned}\vec{F}_{net}\Delta t &= \Delta\vec{p} \\ &= m\vec{v}_f - m\vec{v}_i \\ \Delta t &= \frac{m}{\vec{F}_{net}} (\vec{v}_f - \vec{v}_i).\end{aligned}$$

Thus we have everything we need to find Δt !

Step 2: Choose a frame of reference

Choose right as the positive direction.

Step 3: Do the calculation and quote the final answer

$$\begin{aligned}\Delta t &= \frac{m}{\vec{F}_{net}} (\vec{v}_f - \vec{v}_i) \\ \Delta t &= \left(\frac{300}{+150} \right) ((+6) - (+2)) \\ \Delta t &= \left(\frac{300}{150} \right) (4) \\ \Delta t &= \frac{(300)(+4)}{150} \\ \Delta t &= 8 \text{ s}\end{aligned}$$

It takes 8 s for the force to change the object's velocity from $2 \text{ m}\cdot\text{s}^{-1}$ to the right to $6 \text{ m}\cdot\text{s}^{-1}$ to the right.

QUESTION

A cricket ball weighing 156 g is moving at $54 \text{ km}\cdot\text{hr}^{-1}$ towards a batsman. It is hit by the batsman back towards the bowler at $36 \text{ km}\cdot\text{hr}^{-1}$. Calculate

1. the ball's impulse, and
2. the average force exerted by the bat if the ball is in contact with the bat for 0,13 s.

SOLUTION
Step 1: Identify what information is given and what is asked for

The question explicitly gives

- the ball's mass,
- the ball's initial velocity,
- the ball's final velocity, and
- the time of contact between bat and ball

We are asked to calculate the impulse:

$$\text{Impulse} = \Delta \vec{p} = \vec{F}_{\text{net}} \Delta t$$

Since we do not have the force exerted by the bat on the ball ($\vec{F}_{\text{net}}$), we have to calculate the impulse from the change in momentum of the ball. Now, since

$$\begin{aligned} \Delta \vec{p} &= \vec{p}_f - \vec{p}_i \\ &= m\vec{v}_f - m\vec{v}_i, \end{aligned}$$

we need the ball's mass, initial velocity and final velocity, which we are given.

Step 3: Choose a frame of reference

Let us choose the direction from the batsman to the bowler as the positive direction. Then the initial velocity of the ball is $\vec{v}_i = -15 \text{ m}\cdot\text{s}^{-1}$, while the final velocity of the ball is $\vec{v}_f = +10 \text{ m}\cdot\text{s}^{-1}$.

Step 4: Calculate the momentum

Now we calculate the change in momentum,

$$\begin{aligned} \Delta \vec{p} &= \vec{p}_f - \vec{p}_i \\ &= m\vec{v}_f - m\vec{v}_i \\ &= m(\vec{v}_f - \vec{v}_i) \\ &= (0,156)((+10) - (-15)) \\ &= +3,9 \\ &= 3,9 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ in the direction from the batsman to the bowler} \end{aligned}$$

Step 2: Convert to S.I. units

Firstly let us change units for the mass

$$\begin{aligned} 1000 \text{ g} &= 1 \text{ kg} \\ \text{So, } 1 \text{ g} &= \frac{1}{1000} \text{ kg} \\ \therefore 156 \times 1 \text{ g} &= 156 \times \frac{1}{1000} \text{ kg} \\ &= 0,156 \text{ kg} \end{aligned}$$

Next we change units for the velocity

$$\begin{aligned} 1 \text{ km}\cdot\text{h}^{-1} &= \frac{1000 \text{ m}}{3600 \text{ s}} \\ \therefore 54 \times 1 \text{ km}\cdot\text{h}^{-1} &= 54 \times \frac{1000 \text{ m}}{3600 \text{ s}} \\ &= 15 \text{ m}\cdot\text{s}^{-1} \end{aligned}$$

Similarly, $36 \text{ km}\cdot\text{hr}^{-1} = 10 \text{ m}\cdot\text{s}^{-1}$.

Step 5: Determine the impulse

Finally since impulse is just the change in momentum of the ball,

$$\begin{aligned}\text{Impulse} &= \Delta \vec{p} \\ &= 3,9 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ in the direction from the batsman to the bowler}\end{aligned}$$

Step 6: Determine the average force exerted by the bat

$$\text{Impulse} = \vec{F}_{\text{net}} \Delta t = \Delta \vec{p}$$

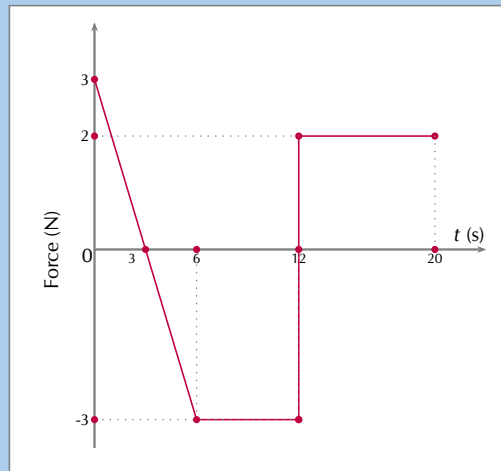
We are given Δt and we have calculated the impulse of the ball.

$$\begin{aligned}\vec{F}_{\text{net}} \Delta t &= \text{Impulse} \\ \vec{F}_{\text{net}} (0,13) &= +3,9 \\ \vec{F}_{\text{net}} &= \frac{+3,9}{0,13} \\ &= +30 \\ &= 30 \text{ N in the direction from the batsman to the bowler}\end{aligned}$$

Worked example 17: Analysing a force graph**QUESTION**

Analyse the Force vs. time graph provided and answer the following questions:

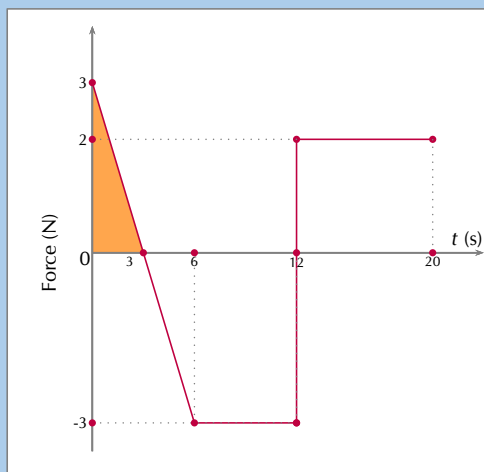
- What is the impulse for the interval 0 s to 3 s?
- What is the impulse for the interval 3 s to 6 s?
- What is the change in momentum for the interval 0 s to 6 s?
- What is the impulse for the interval 6 s to 20 s?
- What is the impulse for the interval 0 s to 20 s?

**SOLUTION****Step 1: Identify what information is given and what is being asked for**

A graph of force versus time is provided. We are asked to determine both impulse and change in momentum from it. We know that the area under the graph is the impulse and we can relate impulse to change in momentum with the impulse-momentum theorem.

We need to calculate the area under the graph for the various intervals to determine impulse and then work from there.

Step 2: Impulse for interval 0 s to 3 s



We need to calculate the area of the shaded portion under the graph. This is a triangle with a base of 3 s and a height of 3 N therefore:

$$\begin{aligned}\text{Impulse} &= \frac{1}{2}bh \\ &= \frac{1}{2}(3)(3) \\ &= 4,5 \text{ N}\cdot\text{s}\end{aligned}$$

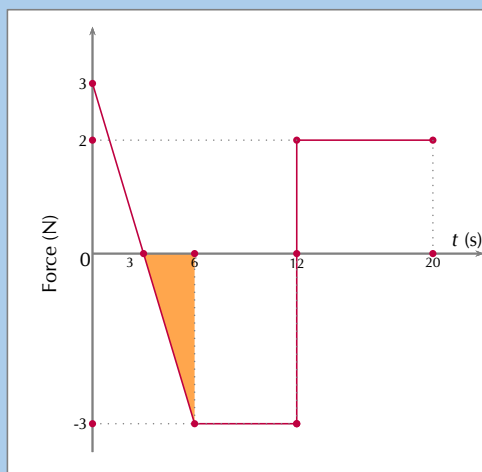
The impulse is 4,5 N·s in the positive direction.

Step 3: Impulse for interval 3 s to 6 s

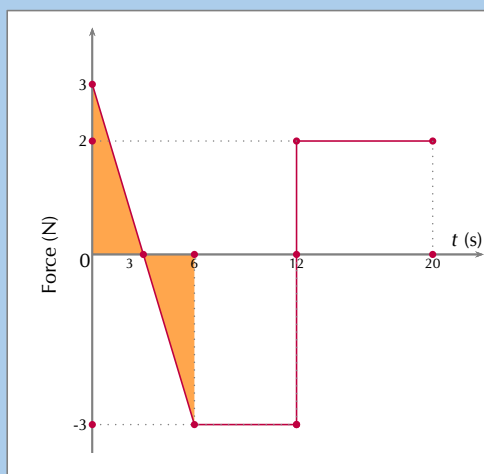
We need to calculate the area of the shaded portion under the graph. This is a triangle with a base of 3 s and a height of -3 N. Note that the force has a negative value so is pointing in the negative direction.

$$\begin{aligned}\text{Impulse} &= \frac{1}{2}bh \\ &= \frac{1}{2}(3)(-3) \\ &= -4,5 \text{ N}\cdot\text{s}\end{aligned}$$

The impulse is 4,5 N·s in the negative direction.



Step 4: What is the change in momentum for the interval 0 s to 6 s



From the impulse-momentum theorem we know that that impulse is equal to the change in momentum. We have worked out the impulse for the two sub-intervals making up 0 s to 6 s. We can sum them to find the impulse for the total interval:

$$\begin{aligned}\text{impulse}_{0-6} &= \text{impulse}_{0-3} + \text{impulse}_{3-6} \\ &= (4,5) + (-4,5) \\ &= 0 \text{ N}\cdot\text{s}\end{aligned}$$

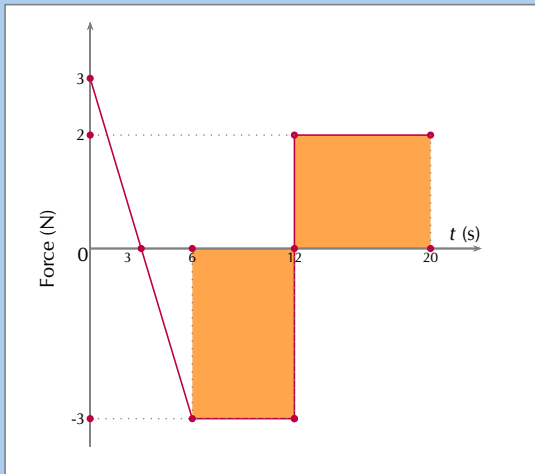
The positive impulse in the first 3 seconds is exactly opposite to the impulse in the second 3 second interval making the total impulse for the first 6 seconds zero:

$$\text{impulse}_{0-6} = 0 \text{ N}\cdot\text{s}$$

From the impulse-momentum theorem we know that:

$$\Delta \vec{p} = \text{impulse} = 0 \text{ N}\cdot\text{s}$$

Step 5: What is the impulse for the interval 6 s to 20 s



We need to calculate the area of the shaded portion under the graph. This is divided into two areas, 6 s to 12 s and 12 s to 20 s, which we need to sum to get the total impulse.

$$\begin{aligned} \text{Impulse}_{6-12} &= (6)(-3) \\ &= -18 \text{ N}\cdot\text{s} \end{aligned}$$

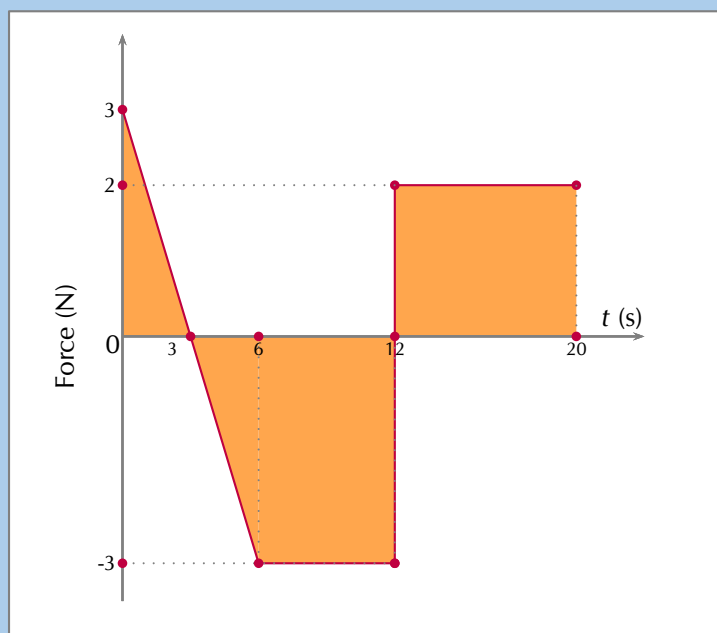
$$\begin{aligned} \text{Impulse}_{12-20} &= (8)(2) \\ &= +16 \text{ N}\cdot\text{s} \end{aligned}$$

The total impulse is the sum of the two:

$$\begin{aligned} \text{Impulse}_{6-20} &= \text{Impulse}_{6-12} + \text{Impulse}_{12-20} \\ &= (-18) + (16) \\ &= -2 \text{ N}\cdot\text{s} \end{aligned}$$

The impulse is 2 N·s in the negative direction.

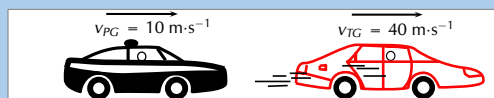
Step 6: What is the impulse of the entire period



The impulse is 2 N·s in the negative direction.

QUESTION

A patrol car is moving on a straight horizontal road at a velocity of $10 \text{ m}\cdot\text{s}^{-1}$ east. At the same time a thief in a car ahead of him is driving at a velocity of $40 \text{ m}\cdot\text{s}^{-1}$ in the same direction.



v_{PG} : velocity of the **patrol** car relative to the **ground**

v_{TG} : velocity of the **thief's** car relative to the **ground**

Questions 1 and 2 from the original version in 2011 Paper 1 are no longer part of the curriculum.

1. While travelling at $40 \text{ m}\cdot\text{s}^{-1}$, the thief's car of mass 1000 kg , collides head-on with a truck of mass 5000 kg moving at $20 \text{ m}\cdot\text{s}^{-1}$. After the collision, the car and the truck move together. Ignore the effects of friction.



State the *law of conservation of linear momentum* in words.

(2 marks)

2. Calculate the velocity of the thief's car immediately after the collision.
(6 marks)
3. Research has shown that forces greater than $85\,000 \text{ N}$ during collisions may cause fatal injuries. The collision described above lasts for $0,5 \text{ s}$.
Determine, by means of a calculation, whether the collision above could result in a fatal injury.
(5 marks)

[TOTAL: 13 marks]

SOLUTION

Question 1

The total (linear) momentum remains constant (**OR** is conserved **OR** does not change) in an isolated (**OR** in a closed system **OR** in the absence of external forces).

(2 marks)

Question 2

Option 1:

Taking to the right as positive

$$\begin{aligned}\sum p_{\text{before}} &= \sum p_{\text{after}} \\ (1000)(40) + (5000)(-20) &= (1000 + 5000)v_f \\ v_f &= -10 \text{ m} \cdot \text{s}^{-1} \\ &= 10 \text{ m} \cdot \text{s}^{-1} \quad \text{left OR west}\end{aligned}$$

Option 2:

Taking to the right as positive

$$\begin{aligned}\Delta p_{\text{car}} &= -\Delta p_{\text{truck}} \\ m_{\text{car}}(v_f - v_{i,\text{car}}) &= -m_{\text{truck}}(v_f - v_{i,\text{truck}}) \\ (1000)(v_f - (40)) &= -(5000)(v_f - (-20)) \\ 6000v_f &= -60\,000 \\ \therefore v_f &= -10 \text{ m} \cdot \text{s}^{-1} \\ \therefore v_f &= 10 \text{ m} \cdot \text{s}^{-1} \quad \text{left OR west}\end{aligned}$$

(6 marks)

Question 3

Option 1:

Force on the car: (Taking to the right as positive)

$$\begin{aligned}F_{\text{net}}\Delta t &= \Delta p = mv_f - mv_i \\ F_{\text{net}}(0,5) &= (1000)(-10 - 40) \\ \therefore F_{\text{net}} &= -10^5 \text{ N} \\ &\text{OR} \\ \therefore F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\ \therefore F_{\text{net}} &> 85\,000 \text{ N}\end{aligned}$$

Yes, the collision is fatal.

OR

Force on the car: (Taking to the left as positive)

$$\begin{aligned}F_{\text{net}}\Delta t &= \Delta p = mv_f - mv_i \\ F_{\text{net}}(0,5) &= (1000)(10 - (-40)) \\ \therefore F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\ \therefore F_{\text{net}} &> 85\,000 \text{ N}\end{aligned}$$

Yes, the collision is fatal.

Option 2:

Force on the truck: (Taking to the right as positive)

$$\begin{aligned}F_{\text{net}}\Delta t &= \Delta p = mv_f - mv_i \\ F_{\text{net}}(0,5) &= (5000)(-10 - (-20)) \\ \therefore F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\ \therefore F_{\text{net}} &> 85\,000 \text{ N}\end{aligned}$$

Yes, the collision is fatal.

OR

Force on the truck: (Taking to the left as positive)

$$\begin{aligned}F_{\text{net}}\Delta t &= \Delta p = mv_f - mv_i \\ F_{\text{net}}(0,5) &= (1000)(10 - 20) \\ \therefore F_{\text{net}} &= -10^5 \text{ N} \\ &\text{OR} \\ \therefore F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\ \therefore F_{\text{net}} &> 85\,000 \text{ N}\end{aligned}$$

Yes, the collision is fatal.

Option 3:

Force on the car: (Taking to the right as positive)

$$\begin{aligned}
 v_f &= v_i + a\Delta t \\
 -10 &= 40 + a(0,5) \\
 \therefore a &= -100 \text{ m} \cdot \text{s}^{-2} \\
 F_{\text{net}} &= ma \\
 &= (1000)(-100) \\
 F_{\text{net}} &= -10^5 \text{ N} \quad (-100\,000 \text{ N}) \\
 F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\
 \therefore F_{\text{net}} &> 85\,000 \text{ N}
 \end{aligned}$$

Yes, the collision is fatal.

OR

Force on the car: (Taking to the left as positive)

$$\begin{aligned}
 v_f &= v_i + a\Delta t \\
 10 &= -40 + a(0,5) \\
 \therefore a &= 100 \text{ m} \cdot \text{s}^{-2} \\
 F_{\text{net}} &= ma \\
 &= (1000)(100) \\
 F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\
 \therefore F_{\text{net}} &> 85\,000 \text{ N}
 \end{aligned}$$

Yes, the collision is fatal.

[TOTAL: 13 marks]

Option 4:

Force on the truck: (Taking to the right as positive)

$$\begin{aligned}
 v_f &= v_i + a\Delta t \\
 -10 &= -20 + a(0,5) \\
 \therefore a &= 20 \text{ m} \cdot \text{s}^{-2} \\
 F_{\text{net}} &= ma \\
 &= (5000)(20) \\
 F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\
 \therefore F_{\text{net}} &> 85\,000 \text{ N}
 \end{aligned}$$

Yes, the collision is fatal.

OR

Force on the truck: (Taking to the left as positive)

$$\begin{aligned}
 v_f &= v_i + a\Delta t \\
 10 &= 20 + a(0,5) \\
 \therefore a &= -20 \text{ m} \cdot \text{s}^{-2} \\
 F_{\text{net}} &= ma \\
 &= (5000)(-20) \\
 F_{\text{net}} &= -10^5 \text{ N} \quad (-100\,000 \text{ N}) \\
 F_{\text{net}} &= 10^5 \text{ N} \quad (100\,000 \text{ N}) \\
 \therefore F_{\text{net}} &> 85\,000 \text{ N}
 \end{aligned}$$

Yes, the collision is fatal.

(5 marks)

Exercise 2 – 5:

1. Which one of the following is **not** a unit of impulse?

- a) N·s
- b) kg·m·s⁻¹
- c) J·m·s⁻¹
- d) J·m⁻¹·s

2. A toy car of mass 1 kg moves eastwards with a speed of 2 m·s⁻¹. It collides head-on with a toy train. The train has a mass of 2 kg and is moving at a speed of 1,5 m·s⁻¹ westwards. The car rebounds (bounces back) at 3,4 m·s⁻¹ and the train rebounds at 1,2 m·s⁻¹.

- a) Calculate the change in momentum for each toy.
- b) Determine the impulse for each toy.

- c) Determine the duration of the collision if the magnitude of the force exerted by each toy is 8 N.
3. A bullet of mass 20 g strikes a target at $300 \text{ m}\cdot\text{s}^{-1}$ and exits at $200 \text{ m}\cdot\text{s}^{-1}$. The tip of the bullet takes 0,0001 s to pass through the target. Determine:
- the change of momentum of the bullet.
 - the impulse of the bullet.
 - the magnitude of the force experienced by the bullet.
4. A bullet of mass 20 g strikes a target at $300 \text{ m}\cdot\text{s}^{-1}$. Determine under which circumstances the bullet experiences the greatest change in momentum, and hence impulse:
- When the bullet exits the target at $200 \text{ m}\cdot\text{s}^{-1}$.
 - When the bullet stops in the target.
 - When the bullet rebounds at $200 \text{ m}\cdot\text{s}^{-1}$.
5. A ball with a mass of 200 g strikes a wall at right angles at a velocity of $12 \text{ m}\cdot\text{s}^{-1}$ and rebounds at a velocity of $9 \text{ m}\cdot\text{s}^{-1}$.
- Calculate the change in the momentum of the ball.
 - What is the impulse of the wall on the ball?
 - Calculate the magnitude of the force exerted by the wall on the ball if the collision takes 0,02 s.
6. If the ball in the previous problem is replaced with a piece of clay of 200 g which is thrown against the wall with the same velocity, but then sticks to the wall, calculate:
- The impulse of the clay on the wall.
 - The force exerted by the clay on the wall if it is in contact with the wall for 0,5 s before it comes to rest.
7. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27HY 2. 27HZ 3. 27J2 4. 27J3 5. 27J4 6. 27J5



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2.6 Physics in action: Impulse

ESCJM

A very important application of impulse is improving safety and reducing injuries. In many cases, an object needs to be brought to rest from a certain initial velocity. This means there is a certain specified change in momentum. If the time during which the momentum changes can be increased then the force that must be applied will be less and so it will cause less damage. This is the principle behind arrestor beds for trucks, airbags, and bending your knees when you jump off a chair and land on the ground.

Air bags are used in motor vehicles because they are able to reduce the effect of the force experienced by a person during an accident. Air bags extend the time required to stop the momentum of the driver and passenger. During a collision, the motion of the driver and passenger carries them towards the windshield. If they are stopped by a collision with the windshield, it would result in a large force exerted over a short time in order to bring them to a stop. If instead of hitting the windshield, the driver and passenger hit an air bag, then the time of the impact is increased. Increasing the time of the impact results in a decrease in the force.

$$F = \frac{\Delta p}{\Delta t}$$

Therefore if t is increased, for a constant change in momentum, force on body is reduced

The same principle explains why wicket keepers in cricket use padded gloves or why there are padded mats in gymnastics. In cricket, when the wicket keeper catches the ball, the padding is slightly compressible, thus reducing the effect of the force on the wicket keepers hands. Similarly, if a gymnast falls, the padding compresses and reduces the effect of the force on the gymnast's body.

An arrestor bed is a patch of ground that is softer than the road. Trucks use these when they have to make an emergency stop. When a truck reaches an arrestor bed the time interval over which the momentum is changed is increased. This decreases the force and causes the truck to slow down.

In sports where rackets and bats are used, like tennis, cricket, squash, badminton and baseball, the hitter is often encouraged to follow-through when striking the ball. High speed films of the collisions between bats/rackets and balls have shown that following through increases the time over which the collision between the racket/bat and ball occurs. This increase in the time of the collision causes an increase in the velocity change of the ball. This means that a hitter can cause the ball to leave the racket/bat faster by following through. In these sports, returning the ball with a higher velocity often increases the chances of success.

Another safety application of trying to reduce the force experienced is in crumple zones in cars. When two cars have a collision, two things can happen:

1. the cars bounce off each other, or
2. the cars crumple together.

Which situation is more dangerous for the occupants of the cars? When cars bounce off each other, or rebound, there is a larger change in momentum and therefore a larger impulse. A larger impulse means that a greater force is experienced by the occupants of the cars. When cars crumple together, there is a smaller change in momentum and therefore a smaller impulse. The smaller impulse means that the occupants of the cars experience a smaller force. Car manufacturers use this idea and design crumple zones into cars, such that the car has a greater chance of crumpling than rebounding in a collision. Also, when the car crumples, the change in the car's momentum happens over a longer time. Both these effects result in a smaller force on the occupants of the car, thereby increasing their chances of survival.

Activity: Egg Throw

This activity demonstrates the effect of impulse and how it is used to improve safety. Have two learners hold up a bed sheet or large piece of fabric. Then toss an egg at the sheet. The egg should not break, because the collision between the egg and the bed sheet lasts over an extended period of time since the bed sheet has some give in it. By increasing the time of the collision, the force of the impact is minimised. Take care to aim at the sheet, because if you miss the sheet, you will definitely break the egg and have to clean up the mess!

Exercise 2 – 6:

1. A canon, mass 500 kg, fires a shell, mass 1 kg, horizontally to the right at $500 \text{ m}\cdot\text{s}^{-1}$. What is the magnitude and direction of the initial recoil velocity of the canon?
2. A trolley of mass 1 kg is moving with a speed of $3 \text{ m}\cdot\text{s}^{-1}$. A block of wood, mass 0,5 kg, is dropped vertically into the trolley. Immediately after the collision, the speed of the trolley and block is $2 \text{ m}\cdot\text{s}^{-1}$. By way of calculation, show whether momentum is conserved in the collision.
3. A 7200 kg empty railway truck is stationary. A fertiliser firm loads 10 800 kg fertiliser into the truck. A second, identical, empty truck is moving at $10 \text{ m}\cdot\text{s}^{-1}$ when it collides with the loaded truck.
 - a) If the empty truck stops completely immediately after the collision, use a conservation law to calculate the velocity of the loaded truck immediately after the collision.

- b) Calculate the distance that the loaded truck moves after collision, if a constant frictional force of 24 kN acts on the truck.
4. A child drops a squash ball of mass 0,05 kg. The ball strikes the ground with a velocity of $4 \text{ m}\cdot\text{s}^{-1}$ and rebounds with a velocity of $3 \text{ m}\cdot\text{s}^{-1}$. Considering only the squash ball, does the law of conservation of momentum apply to this situation? Explain.
5. A bullet of mass 50 g travelling horizontally at $600 \text{ m}\cdot\text{s}^{-1}$ strikes a stationary wooden block of mass 2 kg resting on a smooth horizontal surface. The bullet gets stuck in the block.
- Name and state the principle which can be applied to find the speed of the block-and-bullet system after the bullet entered the block.
 - Calculate the speed of the bullet-and-block system immediately after impact.
 - If the time of impact was $5 \times 10^{-4} \text{ s}$, calculate the force that the bullet exerts on the block during impact.
6. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27J6 2. 27J7 3. 27J8 4. 27J9 5. 27JB



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2.7 Chapter summary

ESCJT

► See presentation: 27JC at www.everythingscience.co.za

- **Newton's Second Law:** The resultant force acting on a body will cause the body to accelerate in the direction of the resultant force. The acceleration of the body is directly proportional to the magnitude of the resultant force and inversely proportional to the mass of the object.
- **Newton's Third Law:** If body A exerts a force on body B then body B will exert an equal but opposite force on body A.
- **Momentum:** The *momentum* of an object is defined as its mass multiplied by its velocity.
- **Momentum of a System:** The *total momentum of a system* is the sum of the momenta of each of the objects in the system.
- **Principle of Conservation of Linear Momentum:** 'The total linear momentum of an isolated system is constant' or 'In an isolated system the total momentum before a collision (or explosion) is equal to the total momentum after the collision (or explosion)'.
- **Elastic collision:** both total momentum and total kinetic energy are conserved.
- **Inelastic collision:** only total momentum is conserved, total kinetic energy is not conserved.
- **Impulse:** the product of the net force and the time interval for which the force acts.
- **Law of Momentum:** The applied resultant force acting on an object is equal to

the rate of change of the object's momentum and this force is in the direction of the change in momentum.

- **Impulse-momentum theorem:** the impulse is equal to the change in momentum.

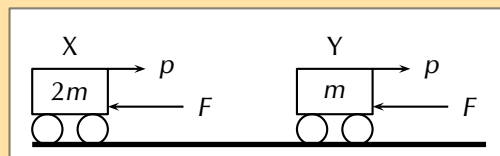
Physical quantities			
Quantity	Vector	Unit name	Unit symbol
Mass (m)	-	kilogram	kg
Velocity ($\vec{v}$)	✓	metre per second	$\text{m}\cdot\text{s}^{-1}$
Momentum ($\vec{p}$)	✓	kilogram metres per second	$\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$
Kinetic energy (KE)	-	joule	J
Impulse	✓	newton seconds	$\text{N}\cdot\text{s}$

Table 2.1: Units used in **momentum and impulse**.

Exercise 2 – 7: Momentum

- [SC 2003/11] A projectile is fired vertically upwards from the ground. At the highest point of its motion, the projectile explodes and separates into two pieces of equal mass. If one of the pieces is projected vertically upwards after the explosion, the second piece will...
 - drop to the ground at zero initial speed.
 - be projected downwards at the same initial speed as the first piece.
 - be projected upwards at the same initial speed as the first piece.
 - be projected downwards at twice the initial speed as the first piece.
- [IEB 2004/11 HG1] A ball hits a wall horizontally with a speed of $15 \text{ m}\cdot\text{s}^{-1}$. It rebounds horizontally with a speed of $8 \text{ m}\cdot\text{s}^{-1}$. Which of the following statements about the system of the ball and the wall is **true**?
 - The total linear momentum of the system is not conserved during this collision.
 - The law of conservation of energy does not apply to this system.
 - The change in momentum of the wall is equal to the change in momentum of the ball.
 - Energy is transferred from the ball to the wall.
- [IEB 2001/11 HG1] A block of mass M collides with a stationary block of mass $2M$. The two blocks move off together with a velocity of $\vec{v}$. What is the velocity of the block of mass M immediately **before** it collides with the block of mass $2M$?
 - $\vec{v}$
 - $2\vec{v}$
 - $3\vec{v}$
 - $4\vec{v}$
- [IEB 2003/11 HG1] A cricket ball and a tennis ball move horizontally towards you with the same momentum. A cricket ball has greater mass than a tennis ball. You apply the same force in stopping each ball.
How does the time taken to stop each ball compare?

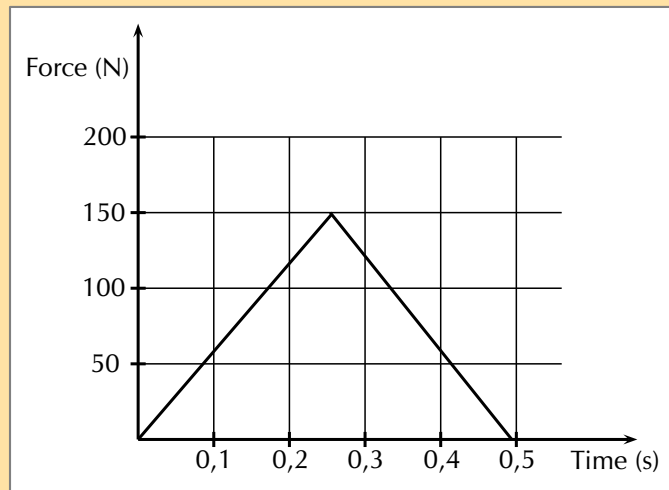
- a) It will take longer to stop the cricket ball.
 - b) It will take longer to stop the tennis ball.
 - c) It will take the same time to stop each of the balls.
 - d) One cannot say how long without knowing the kind of collision the ball has when stopping.
5. [IEB 2004/11 HG1] Two identical billiard balls collide head-on with each other. The first ball hits the second ball with a speed of V , and the second ball hits the first ball with a speed of $2V$. After the collision, the first ball moves off in the opposite direction with a speed of $2V$. Which expression correctly gives the speed of the second ball after the collision?
- a) V
 - b) $2V$
 - c) $3V$
 - d) $4V$
6. [SC 2002/11 HG1] Which one of the following physical quantities is the same as the rate of change of momentum?
- a) resultant force
 - b) work
 - c) power
 - d) impulse
7. [IEB 2005/11 HG] Cart X moves along a smooth track with momentum p . A resultant force F applied to the cart stops it in time t . Another cart Y has only half the mass of X, but it has the same momentum p .



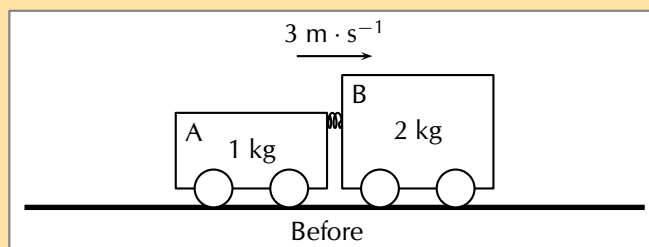
- In what time will cart Y be brought to rest when the same resultant force F acts on it?
- a) $\frac{1}{2}t$
 - b) t
 - c) $2t$
 - d) $4t$
8. [SC 2002/03 HG1] A ball with mass m strikes a wall perpendicularly with a speed, v . If it rebounds in the opposite direction with the same speed, v , the magnitude of the change in momentum will be ...
- a) $2mv$
 - b) mv
 - c) $\frac{1}{2}mv$
 - d) $0mv$
9. Show that impulse and momentum have the same units.
10. A golf club exerts an average force of 3 kN on a ball of mass $0,06 \text{ kg}$. If the golf club is in contact with the golf ball for 5×10^{-4} seconds, calculate

- a) the change in the momentum of the golf ball.
- b) the velocity of the golf ball as it leaves the club.

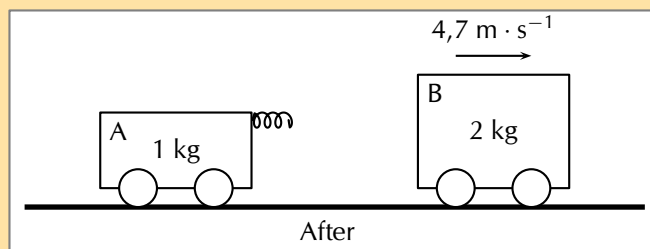
11. During a game of hockey, a player strikes a stationary ball of mass 150 g . The graph below shows how the force of the ball varies with the time.



- a) What does the area under this graph represent?
 - b) Calculate the speed at which the ball leaves the hockey stick.
 - c) The same player hits a practice ball of the same mass, but which is made from a softer material. The hit is such that the ball moves off with the same speed as before. How will the **area**, the **height** and the **base** of the triangle that forms the graph, compare with that of the original ball?
12. The fronts of modern cars are deliberately designed in such a way that in case of a head-on collision, the front would crumple. Why is it desirable that the front of the car should crumple?
13. [SC 2002/11 HG1] In a railway shunting yard, a locomotive of mass 4000 kg, travelling due east at a velocity of $1,5 \text{ m}\cdot\text{s}^{-1}$, collides with a stationary goods wagon of mass 3000 kg in an attempt to couple with it. The coupling fails and instead the goods wagon moves due east with a velocity of $2,8 \text{ m}\cdot\text{s}^{-1}$.
- a) Calculate the magnitude and direction of the velocity of the locomotive immediately after collision.
 - b) Name and state in words the law you used to answer the previous question
14. [SC 2005/11 SG1] A combination of trolley A (fitted with a spring) of mass 1 kg, and trolley B of mass 2 kg, moves to the right at $3 \text{ m}\cdot\text{s}^{-1}$ along a frictionless, horizontal surface. The spring is kept compressed between the two trolleys.



While the combination of the two trolleys is moving at $3 \text{ m}\cdot\text{s}^{-1}$, the spring is released and when it has expanded completely, the 2 kg trolley is then moving to the right at $4,7 \text{ m}\cdot\text{s}^{-1}$ as shown below.



- a) State, in words, the principle of *conservation of linear momentum*.
 - b) Calculate the magnitude and direction of the velocity of the 1 kg trolley immediately after the spring has expanded completely.
15. [IEB 2002/11 HG1] A ball bounces back from the ground. Which of the following statements is true of this event?
- a) The magnitude of the change in momentum of the ball is equal to the magnitude of the change in momentum of the Earth.
 - b) The magnitude of the impulse experienced by the ball is greater than the magnitude of the impulse experienced by the Earth.
 - c) The speed of the ball before the collision will always be equal to the speed of the ball after the collision.
 - d) Only the ball experiences a change in momentum during this event.
16. [SC 2002/11 SG] A boy is standing in a small stationary boat. He throws his schoolbag, mass 2 kg, horizontally towards the jetty with a velocity of $5 \text{ m} \cdot \text{s}^{-1}$. The *combined mass* of the boy and the boat is 50 kg.
- a) Calculate the magnitude of the horizontal momentum of the bag immediately after the boy has thrown it.
 - b) Calculate the velocity (magnitude and direction) of the *boat-and-boy* immediately after the bag is thrown.
17. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

- | | | | | | |
|----------|----------|----------|----------|----------|----------|
| 1. 27JD | 2. 27JF | 3. 27JG | 4. 27JH | 5. 27JJ | 6. 27JK |
| 7. 27JM | 8. 27JN | 9. 27JP | 10. 27JQ | 11. 27JR | 12. 27JS |
| 13. 27JT | 14. 27JV | 15. 27JW | 16. 27JX | | |



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Vertical projectile motion in one dimension

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3.1 Introduction

ESCJV

In this chapter we are going to look at the motion of objects that are either projected, thrown, or shot directly into the air, be it vertically upwards, downwards or when objects are dropped. This is something that everyone has experienced while playing games and we all have a strong intuition that ‘what goes up, must come down’, but now we’ll study why that is the case.

▶ See video: 27JY at www.everythingscience.co.za

Key linked concepts

- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Rectilinear motion — Physical Sciences, Grade 10, Motion in one dimension
- Linear equations — Mathematics, Grade 10, Equations and inequalities
- Quadratic equations — Mathematics, Grade 10, Functions

3.2 Vertical projectile motion

ESCJW

In Grade 11, we studied the motion of objects in free fall and saw that such an object has a constant gravitational acceleration of $\vec{g}$. We now study the motion of objects that are moving upwards or downwards while experiencing a force due to gravity. We call this projectile motion. We will only consider the case where objects move vertically upwards and/or downwards – meaning that there is no horizontal displacement of the object, only vertical displacement.

When we talk about a projectile we mean a body or particle that is project or launched and then moves under the influence of **only gravity**.

Motion in a gravitational field

ESCJX

We know from Newton’s Law of Universal Gravitation that an object in the Earth’s gravitational field experiences a force pulling it towards the centre of the Earth. If this is the only force acting on the object then the object will accelerate towards the centre of the Earth.

A person standing on the Earth’s surface will interpret this acceleration as objects always falling downwards. In Grade 11, we showed that the value of the acceleration due to gravity can be treated as a having a constant magnitude of $g = 9,8 \text{ m}\cdot\text{s}^{-2}$.

IMPORTANT!

In reality, if you go very far away from the Earth’s surface, the magnitude of $\vec{g}$ would change, but, for everyday problems, we can safely treat it as constant. We also ignore any effects that air resistance (drag) might have.

For the rest of this chapter we will deal only with the case where the force due to gravity is the only force acting on the projectile that is falling. Any projectile can be described as falling, even if its motion is upwards initially.

The initial velocity, $\vec{v}_i$, that an object has and the acceleration that it experiences are two different quantities. It is very important to remember that the gravitational acceleration is always towards the centre of the Earth and constant, regardless of the direction or magnitude of the velocity.

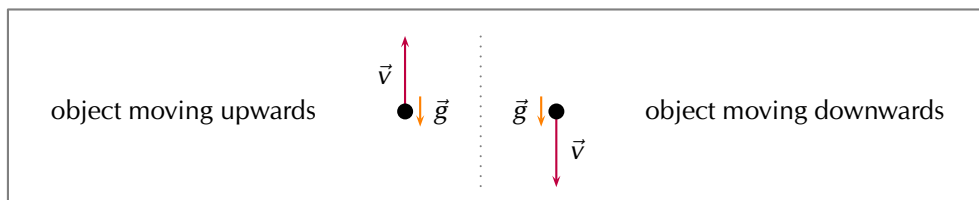


Figure 3.1: Objects moving upwards or downwards in the Earth's gravitational field always accelerate towards the centre of the Earth. This looks like a downwards acceleration to someone standing on the Earth's surface. We don't draw both vectors on the object because we would be mixing velocity and acceleration which are two *different* physical quantities.

► See video: 27JZ at www.everythingscience.co.za

This means that if an object is moving upwards with some initial velocity in the vertical direction, the magnitude of the velocity in the vertical direction decreases until it stops ($v = 0 \text{ m}\cdot\text{s}^{-1}$) for an instant. The point at which the velocity is reduced to zero corresponds with the maximum height, $h_{\max}$, that the object reaches. After this, the object starts to fall. It is very important to remember that the acceleration is constant but the velocity vector has changed in magnitude and direction. At the maximum height where the velocity is zero the acceleration is still $\vec{g}$.

TIP

Projectiles moving upwards or downwards in the Earth's gravitational field always accelerate downwards with a constant acceleration $\vec{g}$. Note: non-zero acceleration means that the velocity is *changing*.

TIP

Projectiles that have an initial velocity upwards will have zero velocity at their greatest height, $h_{\max}$. The acceleration is still $\vec{g}$.

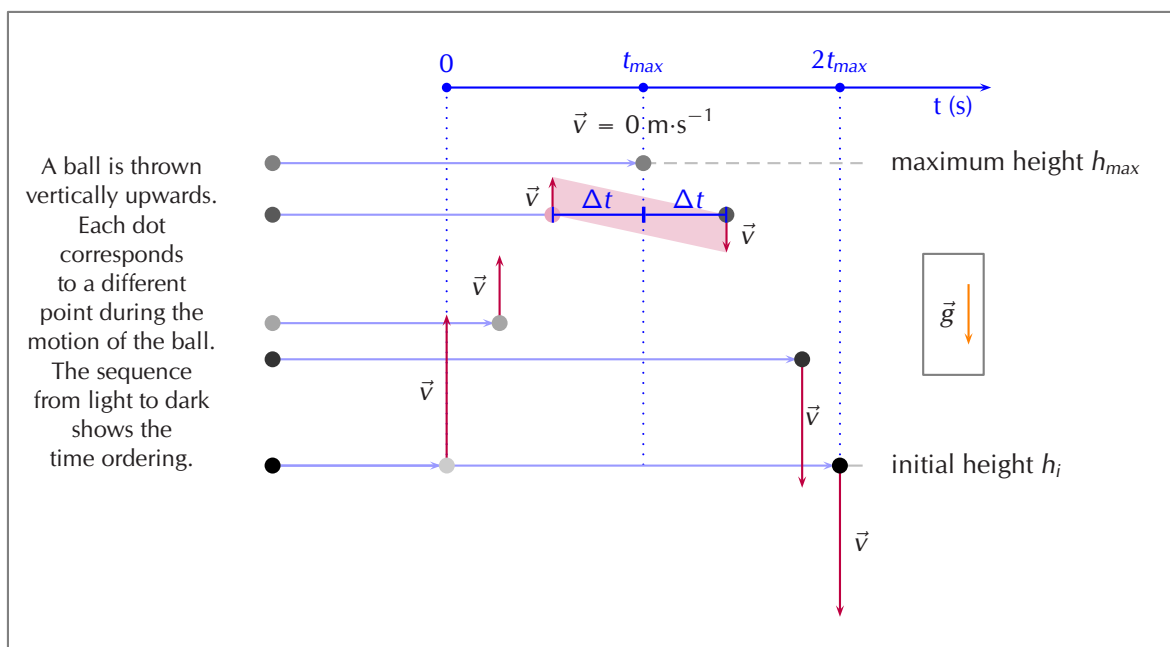


Figure 3.2: (a) An object is thrown upwards from height h_i . After time t_m , the object reaches its maximum height, and starts to drop downwards. After a time $2t_m$ the object returns to height h_i .

Consider an object thrown upwards from a vertical height h_i . We have said that the object will travel upwards with decreasing vertical velocity until it stops, and then starts moving vertically downwards. The time that it takes for the object to fall back down

to height h_i is the same as the time taken for the object to reach its maximum height from height h_i . This is known as *time symmetry*. This is a consequence of the uniform acceleration the projectile experiences.

This has two implications for projectiles that pass through a point on both the upward and downward part of their motion when in free-fall:

- **time symmetry:** the time intervals during the upward motion and the downward motion are the same, for example it will take the same time to rise from initial position to maximum height as it will to drop back to the initial position. This applies to any point as shown by the pink dot in Figure 3.2. The time interval between the projectile passing the point and being at maximum height is the same, Δt .
- **magnitude of velocity:** the magnitude of the velocity at the same point on the upward and downward motion will be the same, the direction will be reversed, as indicated by the shaded region in Figure 3.2.

This is very useful when solving problems because if you have any information about the upward motion you can learn something about the downward motion as well and vice versa.

Formal experiment: Investigate the motion of a falling body

Aim:

To measure the position and time during motion and to use that data to plot a “Position vs Time” and “Velocity vs Time” graph from which we can calculate the magnitude for gravitational acceleration, $\vec{g}$.

Apparatus:

- ticker tape apparatus
- ticker timer
- tape
- graph paper
- ruler
- foam cushion

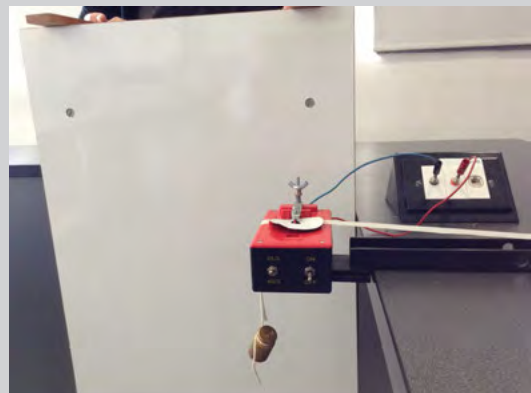




Figure 3.3: Motion at constant velocity

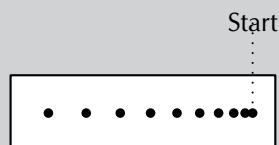


Figure 3.4: Motion with increasing velocity

Method:

1. Work with a friend. Copy the table below into your workbook.
2. Clamp the timer to the table to allow the tape to fall through the staples on the timer such that the ticker tape will be perpendicular to the floor. Plug in the timer.
3. Thread a piece of ticker tape through the staples. Make sure the tape is under the carbon disc. The length of the tape should be the distance from the timer to the floor.
4. Using masking tape attach a 100 g mass to the end of the tape closest to the floor.
5. Position the foam cushion directly below the mass. (The cushion will protect your masses and your floor)
6. Hold the tape so that it is vertical and will easily pass through the staples.
7. Turn on the timer.
8. Let go of the tape with the mass, allowing it to drop to the floor.
9. Turn off the timer.
10. Repeat.
11. Take each tape and find the first clear dot nearest the mass end of the tape. Circle this first clear dot and label it as zero. Then count five dots and label the fifth dot as number 1. Count another 5 dots and label the fifth dot number 2. Continue this procedure.
12. On each piece of tape, measure the distance between successive dots. Note these distances in the table below.
13. Use the frequency of the ticker timer to work out the time intervals between successive dots. Note these times in the table below.
14. Work out the average of the two distance values between successive numbered dots.

15. The time interval between two dots is 0,02 s so the time between numbered dots is 0,1 s.
16. Use the average distance between numbered dots and the known time interval to calculate velocity.
17. Use the magnitude of the velocity and time values to plot a graph of “Velocity vs Time” *onto graph paper*. Stick the graph paper into your workbook.
18. Draw the best straight line through your data points.
19. Determine the slope of the straight line. This is the magnitude of the acceleration.

Results:

Change in distance (m)	Change in time (s)	Magnitude of velocity ($\text{m}\cdot\text{s}^{-1}$)

Discussion:

Describe the motion of the mass using the graphs.

Equations of motion

ESCJY

The equations of rectilinear motion that you learnt about in Grade 10 can be used for vertical projectile motion, with acceleration from gravity: $\vec{a} = \vec{g}$. Why, because the equations of rectilinear motion can be applied to any motion in a straight line with constant acceleration. If we are only considering vertical motion (up and down) it is motion in a straight line and the acceleration, $\vec{g}$, is constant and along the same line. We use the magnitude of $g = 9,8 \text{ m}\cdot\text{s}^{-2}$ for our calculations.

IMPORTANT!

Now would be a good time to revise both rectilinear motion and quadratic equations from Grade 10.

Remember that when you use these equations, you are dealing with vectors which have magnitude *and* direction. Therefore, you need to decide which direction will be the positive direction so that your vectors have the correct signs.

The equations of motion that you learnt about in Grade 10 are:

$$\begin{aligned}\vec{v}_f &= \vec{v}_i + \vec{a}t \\ \Delta\vec{x} &= \frac{(\vec{v}_i + \vec{v}_f)}{2}t \\ \Delta\vec{x} &= \vec{v}_i t + \frac{1}{2}\vec{a}t^2 \\ \vec{v}_f^2 &= \vec{v}_i^2 + 2\vec{a}\Delta\vec{x}\end{aligned}$$

In the case of vertical projectile motion, we know that the only force we will consider is gravity and therefore the acceleration will be $\vec{a} = \vec{g}$.

$$\begin{aligned}\vec{v}_f &= \vec{v}_i + \vec{g}t \\ \Delta\vec{x} &= \frac{(\vec{v}_i + \vec{v}_f)}{2}t \\ \Delta\vec{x} &= \vec{v}_i t + \frac{1}{2}\vec{g}t^2 \\ \vec{v}_f^2 &= \vec{v}_i^2 + 2\vec{g}\Delta\vec{x}\end{aligned}$$

where

$$\begin{aligned}\vec{v}_i &= \text{initial velocity (m} \cdot \text{s}^{-1}\text{) at start} \\ \vec{v}_f &= \text{final velocity (m} \cdot \text{s}^{-1}\text{) at time } t \\ \Delta\vec{x} &= \text{change in vertical position (m)} \\ t &= \text{time (s)} \\ \Delta t &= \text{time interval (s)} \\ \vec{a} = \vec{g} &= \text{acceleration due to gravity (m} \cdot \text{s}^{-2}\text{)}\end{aligned}$$

IMPORTANT!

$\Delta\vec{x}$ is displacement, the change in position. If the coordinate system you choose isn't centred at the point from which the motion starts (i.e. you don't have $\vec{x}_i = 0$) then you must remember that the displacement and the final position are not the same. The position is given by $\vec{x}_f = \vec{x}_i + \Delta\vec{x}$. If $\vec{x}_i = 0$ then $\vec{x}_f = (0) + \Delta\vec{x}$.

🔗 See simulation: 27K2 at www.everythingscience.co.za

Now we can apply this mathematical description of vertical motion to solving problems.

Worked example 1: Projectile motion

QUESTION

A ball is thrown vertically upwards with an initial velocity of $10 \text{ m} \cdot \text{s}^{-1}$. A photograph (shown) is taken when the ball is 1,5 m above the point of release.

1. Determine the maximum height above the thrower's hand reached by the ball.
2. Determine the time it takes the ball to reach its maximum height.
3. Determine the time(s) when the ball will be at the position shown in the photograph.

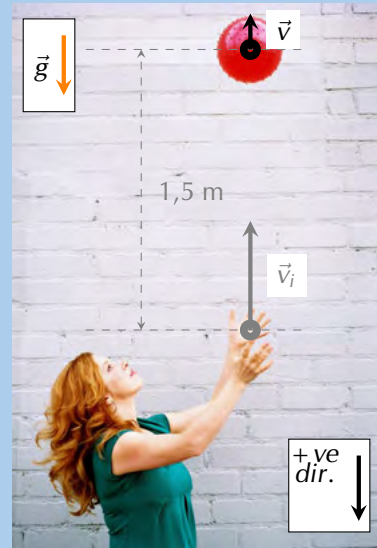


SOLUTION

Step 1: Identify what is required and what is given

We are required to determine the maximum height reached by the ball and how long it takes to reach this height. We are given the initial velocity $\vec{v}_i = 10 \text{ m}\cdot\text{s}^{-1}$ upwards and the acceleration due to gravity $\vec{g} = 9,8 \text{ m}\cdot\text{s}^{-2}$ downwards. We also need to determine when the ball is at the position shown in the photograph.

The ball is thrown straight up and will reach a maximum height, at which point its velocity will be zero, before beginning to fall downwards. The height of the ball in the photograph is given as 1,5 m above the initial point. Adding the information to the picture helps us understand what is going on more effectively:



Step 2: Determine how to approach the problem

Even though there are a number of questions to answer, the best approach is to work through them one at a time.

First, as with rectilinear motion in Grade 10, select a positive direction. We choose **down as positive** and remember to **keep this sign convention** for the whole problem. This will allow us to analyse the information given and determine directions and signs. We know that at the maximum height the velocity of the ball is $0 \text{ m}\cdot\text{s}^{-1}$. We therefore have the following:

- $\vec{v}_i = -10 \text{ m}\cdot\text{s}^{-1}$ (it is negative because we chose downwards as positive)
- $\vec{v}_f = 0 \text{ m}\cdot\text{s}^{-1}$
- $\vec{g} = 9,8 \text{ m}\cdot\text{s}^{-2}$

We know that the position shown (1,5 m above the point of release) will be passed as the ball rises to its maximum height and then again as the ball falls downwards.

Step 3: Identify the appropriate equation to determine the maximum height

We can use $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{g}\Delta\vec{x}$ to solve for the height.

Step 4: Substitute the values into the equation and find the maximum height

$$\begin{aligned}\vec{v}_f^2 &= \vec{v}_i^2 + 2\vec{g}\Delta\vec{x} \\ (0)^2 &= (-10)^2 + (2)(9,8)(\Delta x) \\ -100 &= 19,6\Delta x \\ \Delta x &= -5,102 \text{ m}\end{aligned}$$

The value for the displacement will be negative because the displacement is upwards and we have chosen downward as positive (and upward as negative). The maximum height will be a positive number, $h_m = 5,10 \text{ m}$.

Step 5: Identify the appropriate equation to determine the time to reach maximum height

We know the values of $\vec{v}_i$, $\vec{v}_f$ and $\vec{g}$ so we can use: $\vec{v}_f = \vec{v}_i + \vec{g}t$ to solve for the time.

Step 6: Substitute the values in and find the time to reach maximum height

$$\begin{aligned}\vec{v}_f &= \vec{v}_i + \vec{g}t \\ 0 &= -10 + 9,8t \\ 10 &= 9,8t \\ t &= 1,02 \text{ s}\end{aligned}$$

Step 7: Determine the times at which the ball is in the position in the photograph

We know the displacement, initial velocity and the acceleration of the ball. We also expect to get two values as we know the ball will pass through the point twice.

We substitute our values into the equation $\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{g}t^2$, which is a quadratic equation in time, so we expect to find two different solutions if we solve for time.

It is important to notice that we can use the value $-1,5 \text{ m}$ as the displacement above the initial position.

$$\begin{aligned}\Delta\vec{x} &= v_i t + \frac{1}{2}\vec{g}t^2 \\ \frac{1}{2}\vec{g}t^2 + \vec{v}_i t - \Delta\vec{x} &= 0 \\ \frac{1}{2}(9,8)t^2 + (-10)t - (-1,5) &= 0 \\ \underbrace{(4,9)}_a \underbrace{t^2}_{x^2} + \underbrace{(-10)}_b \underbrace{t}_x + \underbrace{(1,5)}_c &= 0 \\ t &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ t &= \frac{-(-10) \pm \sqrt{(-10)^2 - 4(4,9)(1,5)}}{2(4,9)} \\ t &= \frac{10 \pm \sqrt{70,6}}{9,8} \\ t &= \frac{10 \pm 8,40238}{9,8} \\ t &= 1,02041 \pm 0,857386 \\ t &= 0,16 \text{ s or } t = 1,88 \text{ s}\end{aligned}$$

Step 8: Write the final answer

The ball:

1. reaches a maximum height of $5,10 \text{ m}$;
2. takes $1,02 \text{ s}$ to reach the top; and
3. passes through the point in the photograph at $t = 0,16 \text{ s}$ on the way up and $t = 1,88 \text{ s}$ on the way down.

The use of a sign convention should not affect your result. As an exercise, repeat the

previous worked example using the opposite sign convention to verify that you do get the same result.

Worked example 2: Height of a projectile

QUESTION

A cricketer hits a cricket ball so that it goes vertically upwards. If the ball takes 10 s to return to the initial height, determine its maximum height above the initial position.



SOLUTION

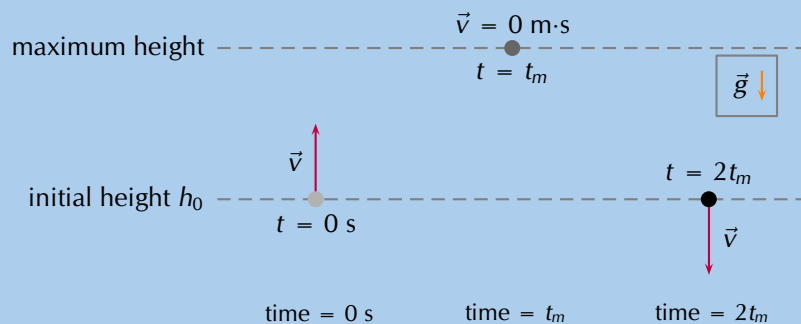
Step 1: Identify what is required and what is given

We need to find how high the ball goes. We know that it takes 10 s to go up and down. We do not know what the initial velocity ($\vec{v}_i$) of the ball is. After the batsman strikes the ball it goes directly upwards and the only force that will be acting on the ball will be the gravitational force. This information is represented in the photograph.



Step 2: Determine how to approach the problem

For a problem like this it is useful to divide the motion into two parts. The first part of the motion consists of the upward motion of the ball with an initial velocity ($\vec{v}_i$) and a final velocity ($\vec{v}_f = 0$) at the maximum height. The second part of the motion consists of the downward motion of the ball with an initial velocity ($\vec{v}_i = 0$) and final velocity ($\vec{v}_f$) which is not yet known.



Choose down as positive. We know that at the maximum height, the velocity of the ball is $0 \text{ m}\cdot\text{s}^{-1}$. We also know that the ball takes the same time to reach its maximum height as it takes to travel from its maximum height to the initial position, due to time symmetry. The time taken is half the total time. Therefore, we have the following information for the second (downward) part of the motion of the ball:

- $t = 5 \text{ s}$ (half of the total time)
- $\vec{v}_{\text{top}} = 0 \text{ m}\cdot\text{s}^{-1}$
- $\vec{g} = 9,8 \text{ m}\cdot\text{s}^{-2}$ downwards
- $\Delta\vec{x} = ?$

Step 3: Find an appropriate equation to use

We do not know the final velocity of the ball coming down. We need to choose an equation that does not have $\vec{v}_f$ in it. We can use the following equation to solve for $\Delta\vec{x}$:

$$\Delta\vec{x} = \vec{v}_i t + \frac{1}{2} \vec{g} t^2$$

Step 4: Substitute values and find the height

$$\Delta\vec{x} = (0)(5) + \frac{1}{2}(9,8)(5)^2$$

$$\Delta\vec{x} = 122,5 \text{ m downwards}$$

In the second motion, the displacement of the ball is 122,5 m downwards. This means that the height was $h = 122,5 \text{ m}$.

Step 5: Write the final answer

The ball reaches a maximum height of 122,5 m.

Worked example 3: Equal displacements [attribution: Sunil Kumar Singh]

QUESTION

A sequence of balls are successively dropped from the top of a tower. The time interval between the release of consecutive balls is equal. At the precise instant that the ninth ball (ball 9) is released, the first ball hits the ground. Which of the balls in the sequence is at $\frac{3}{4}$ of the height, h , of the tower when the first ball hits the ground?

SOLUTION

Step 1: Understand what is happening

A ball is dropped from a tower. After a time, t_i , a second ball is dropped. After another time interval, t_i , a third ball is dropped. This means that when ball 3 is dropped the total time that has passed is $2t_i$.

After a third time interval, t_i , a fourth ball is dropped. This means that at the instant that ball 4 is dropped the total time that has passed is $3t_i$. This process continues until the ninth ball is dropped. At the same instant that the ninth ball is dropped, the first ball that was dropped strikes the ground. The ninth ball would have been released after 8 time intervals, therefore it took the first ball a time interval of $8t_i$ to hit the ground.

Step 2: Describe the events mathematically

As this is a vertical motion problem where everything is falling towards the ground, it is simplest to choose downwards as the positive direction.

The first ball hits the ground when the ninth ball is dropped. This means that the first

ball has fallen for a total time of $(9 - 1)t_i = 8t_i$. The displacement of the first ball is also described by the equation $\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{g} t^2$. The balls are dropped so their initial velocity is zero, $\vec{v}_i = 0$ and we know the time for the first ball to fall the height of the tower therefore, in the case of the first ball, we have:

$$\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{g} t^2$$

$$h = \frac{1}{2} (9,8) (8t_i)^2$$

Let the n th ball be the ball that is at $\frac{3}{4}$ of the height, h , of the tower when the first ball strikes the ground. The n th ball will have fallen for a time of $(n - 1)t_i$. If it is $\frac{3}{4}$ of the way up the tower it will have fallen a distance of $\frac{1}{4}h$. This motion is also described by $\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{g} t^2$, and therefore we know, in the case of the n th ball, we have:

$$\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{g} t^2$$

$$\frac{1}{4}h = \frac{1}{2} (9,8) ((n - 1)t_i)^2$$

$$h = 2(9,8) ((n - 1)t_i)^2$$

We can't solve for n from this final equation because it will contain h and t_i . We can use the two equations we have to eliminate h . Both equations are written in the form $h = \dots$ therefore we can equate them eliminating h :

$$\frac{1}{2} (9,8) (8t_i)^2 = 2(9,8) ((n - 1)t_i)^2$$

$$\frac{1}{2} (9,8) 8^2 t_i^2 = 2(9,8) (n - 1)^2 t_i^2$$

$$\left(\frac{2}{(9,8)t_i^2} \right) \times \frac{1}{2} (9,8) 8^2 t_i^2 = \left(\frac{2}{(9,8)t_i^2} \right) \times 2(9,8) (n - 1)^2 t_i^2$$

$$8^2 = 4(n - 1)^2$$

$$8 = 2(n - 1)$$

$$4 = (n - 1)$$

$$n = 5$$

IMPORTANT!

We know that n must be a positive integer which is why we use the positive roots when we take the square root above.

Step 3: Write the final answer

The fifth ball is at $\frac{3}{4}$ of the height of the tower when the first ball strikes the ground.

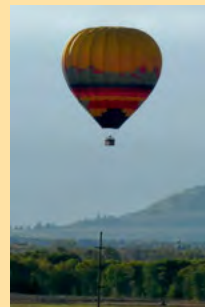
Exercise 3 – 1: Equations of motion

1. A cricketer hits a cricket ball straight up into the air. The cricket ball has an initial velocity of $20 \text{ m} \cdot \text{s}^{-1}$ vertically upwards.

- a) What height does the ball reach before it stops to fall back to the ground.
 - b) How long has the ball been in the air for?
2. Zingi throws a tennis ball straight up into the air. It reaches a height of 80 cm.
- a) Determine the initial velocity of the tennis ball.
 - b) How long does the ball take to reach its maximum height?

3. A tourist takes a trip in a hot air balloon. The hot air balloon is ascending (moving up) at a velocity of $4 \text{ m}\cdot\text{s}^{-1}$ vertically upwards. He accidentally drops his camera over the side of the balloon's basket, at a height of 20 m. Calculate the velocity with which the camera hits the ground.

A hot air balloon.



4. A ball is dropped vertically from a tower. If the vertical distance covered in the last second is equal to the distance covered in first three seconds, find the height of the tower.
5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27K3 2. 27K4 3. 27K5 4. 27K6



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Graphs of vertical projectile motion

ESCJZ

We can describe vertical projectile motion through a series of graphs: position, velocity and acceleration versus time graphs. This isn't new, in Grade 10 you learnt about the graphs for describing rectilinear motion with constant acceleration. In the previous section we motivated for using the same equations of motion as in rectilinear motion. Therefore, everything you learnt about the graphs describing rectilinear motion is applicable when describing vertical projectile motion.

IMPORTANT!

Now would be a good time to revise the section on graphs in rectilinear motion from Grade 10.

Remember that: $\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$ and $\vec{v} = \frac{\Delta \vec{x}}{\Delta t}$. For the graphs of these quantities the slope and the area under a graph can tell us about changes in other quantities. As a reminder, here is a summary table, from Grade 10 rectilinear motion, about which information we can derive from the slope and the area of various graphs:

Graph	Slope	Area
Displacement vs time	$\vec{v}$	-
Velocity vs time	$\vec{a}$	$\Delta\vec{x}$
Acceleration vs time	-	$\vec{v}$

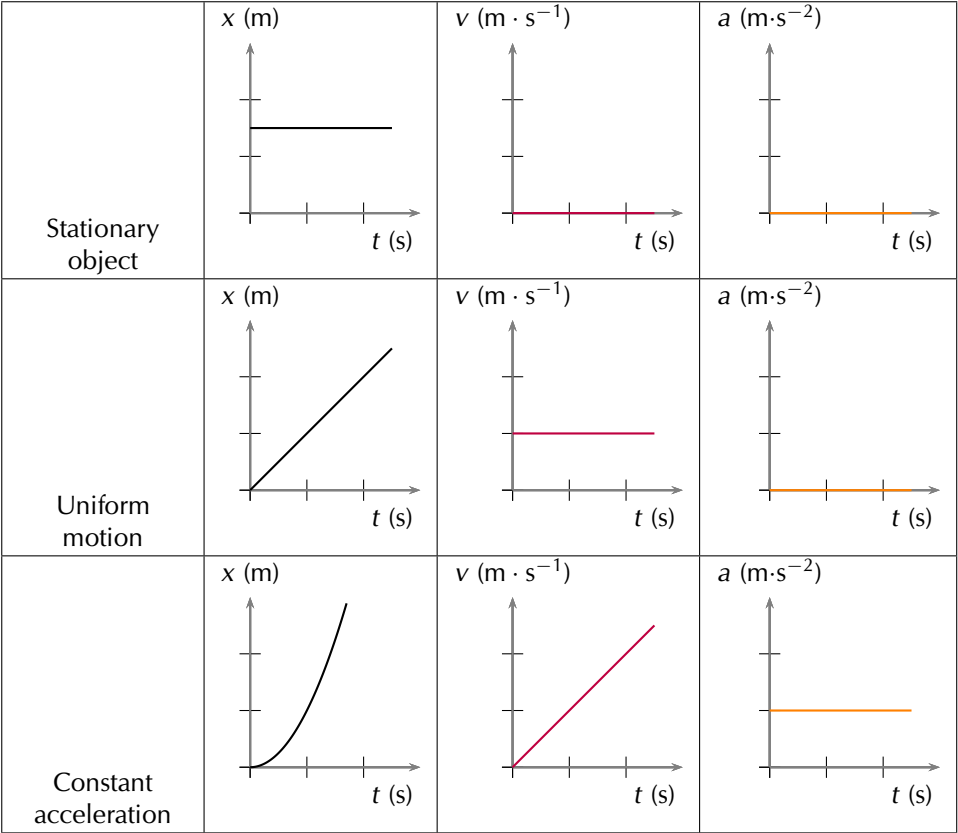


Figure 3.5: Position vs time, velocity vs time and acceleration vs time graphs.

The graphs are the graphical representations of the equations of motion. This means that if you have the graph for one of position, velocity or acceleration you should be able to write down the corresponding equation and vice versa.

The type of graphs is related to the equation of motion: position is a parabola, velocity is a straight line and acceleration is a constant. The characteristic features then depend on the sign convention and the specific values of the problem.

To illustrate the variation, we consider three separate cases: (1) the object has an initial velocity in the upward direction (opposite direction to $\vec{g}$), (2) the object falls from rest (no initial velocity), and (3) the object has an initial velocity in the downward direction (same direction as $\vec{g}$).

Important: the horizontal axis in these graphs is time, not space.

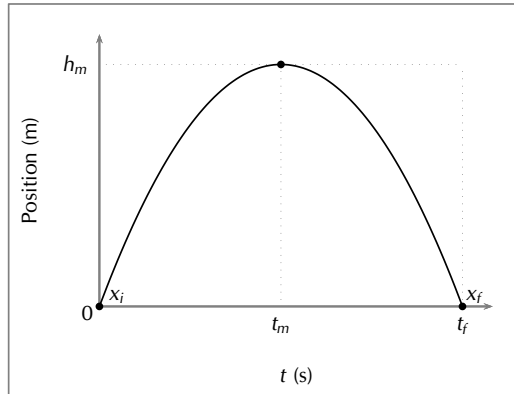
Case 1: Initial velocity upwards

Consider an object undergoing the following motion:

1. at time $t = 0$ s the object has an initial position, $\vec{x}_i = 0$, and an initial upward velocity of magnitude v_i ,
2. the object reaches a maximum height, h_m , above the ground,
3. the object then falls down to a final position, $\vec{x}_f$, with a final velocity v_f downward at time $= t_f$.

The equation for position is given by: $\vec{x} = \vec{x}_i + \vec{v}_i t + \frac{1}{2}\vec{g}t^2$. We have to choose a positive direction as before.

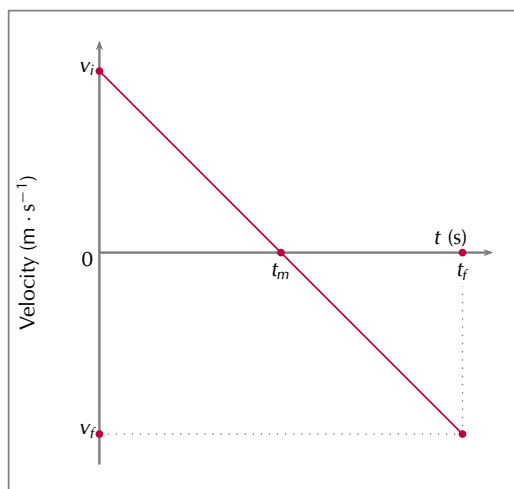
We choose upwards as positive. The position versus time graph for the motion described is shown below:

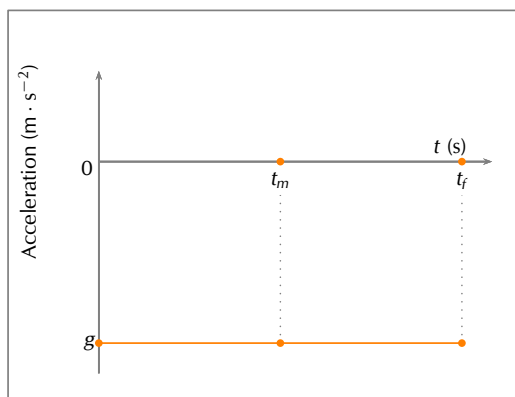


IMPORTANT!

It is important to notice that we are plotting position versus time and the initial position, $\vec{x}_i$, is not necessarily zero. The position as a function of time will be given by $\vec{x}_f = \vec{x}_i + \vec{v}_i t + \frac{1}{2}\vec{g}t^2$. The initial position, $\vec{x}_i$, will shift the graph of $\vec{x}$ up or down by a constant value. Whether it is an upward or downward shift will depend on which direction was chosen as the positive direction. The magnitude of the shift will depend on the choice of origin for the coordinate system (frame of reference).

The second graph is a linear function showing the velocity as a function of time: $\vec{v}_f = \vec{v}_i + \vec{g}t$. Note that we chose upwards as our positive direction therefore the acceleration due to gravity will be negative. The slope (i.e. the coefficient of t) of the velocity versus time graph is the acceleration and the slope is negative which is consistent with our expectations after choosing upwards as the positive direction.



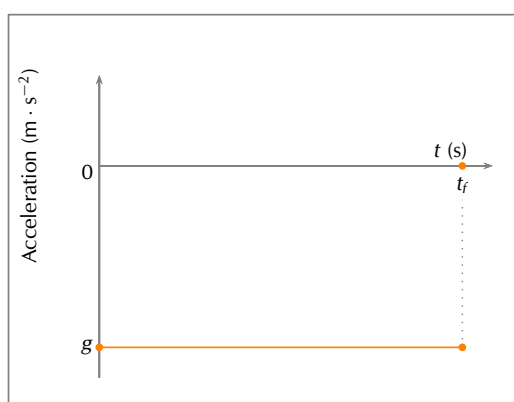


The final plot is a graph of acceleration versus time, and is constant because the magnitude and direction of the acceleration due to gravity are constant.

Cases 2 & 3: Initial velocity zero or downwards

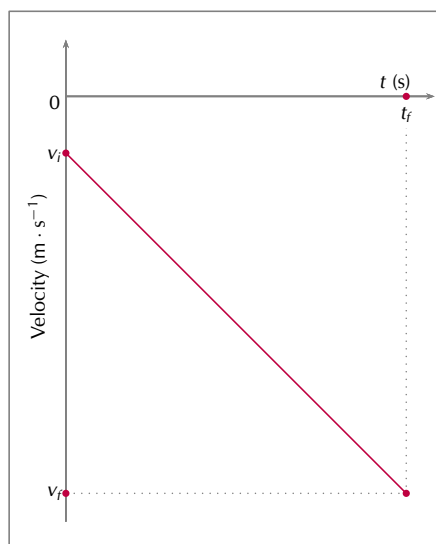
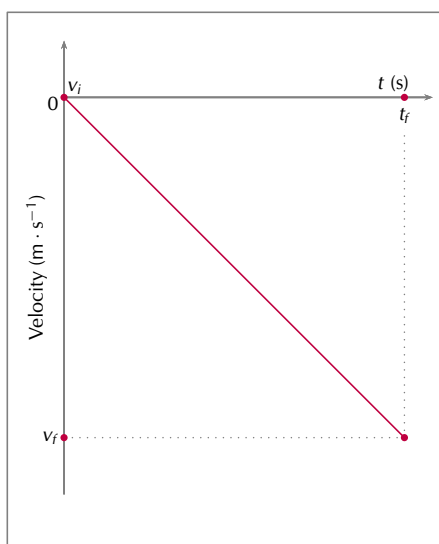
We start by choosing a direction as the positive direction. We want to compare the graphs with the first case, so we will choose the same direction (upwards) to be the positive direction.

The one thing that remains the same is the acceleration, $\vec{g}$, due to gravity. Therefore:



If an object starts from rest (for example is dropped), then the initial velocity is zero. We know that there is constant acceleration and hence the graph of velocity versus time looks like this:

If the object starts with an initial velocity downwards the velocity graph looks like:



IMPORTANT!

It is important to notice that the slope of all the velocity versus time graphs has the same magnitude. If the same direction is chosen as positive then the slope has the same magnitude AND the same sign (direction). The difference in these graphs is related to the initial velocity, $\vec{v}_i$. The magnitude of the slope of the graph is the magnitude of the acceleration.

For the case where the object falls from rest, the equation for position is $\vec{x} = \vec{x}_i + \underbrace{\vec{v}_i t}_0 + \frac{1}{2}\vec{g}t^2$ and the graph

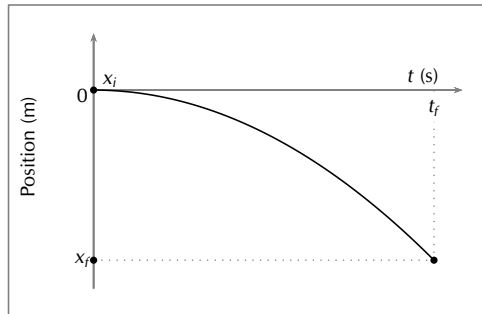
of the position versus time looks as shown on the right.

For motion with an initial velocity in the negative direction, the graph of displacement versus time will be a narrower parabola as compared to the case where the object falls from rest.

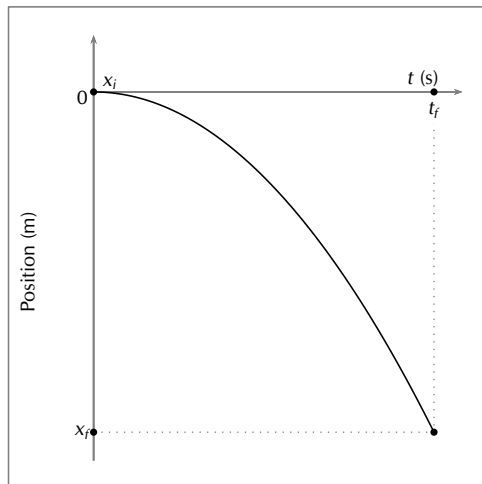
The primary difference in features between the cases we've looked at is that for Case 1, the position graph has a maximum positive displacement (it has a peak), while in Cases 2 and 3, where the initial velocity is negative, the displacement is always negative.

You may be wondering if the final position is always the same as the initial position, as we had for our Case 1 example. That is not necessarily true. If an object is thrown up from the edge of a cliff, the initial position of the object will be the height of the cliff but the final position will be at the base of the cliff.

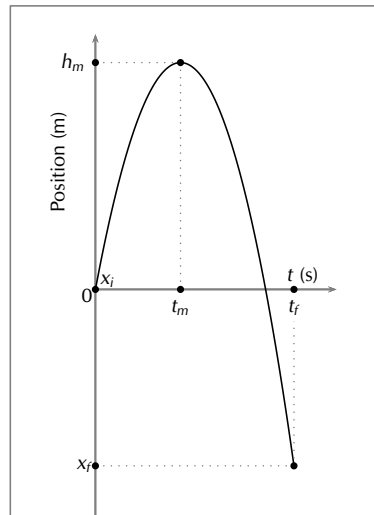
In this graph the initial velocity is zero:



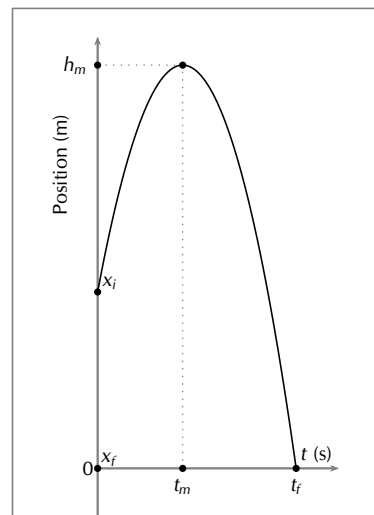
In this graph the initial velocity is in the negative direction:



If we choose the point from which the object is thrown as the origin of our coordinate system then the graph will look like this:



If we choose to centre our coordinate system at the base of the cliff then the graph will look like this, because the initial displacement is the height of the cliff:

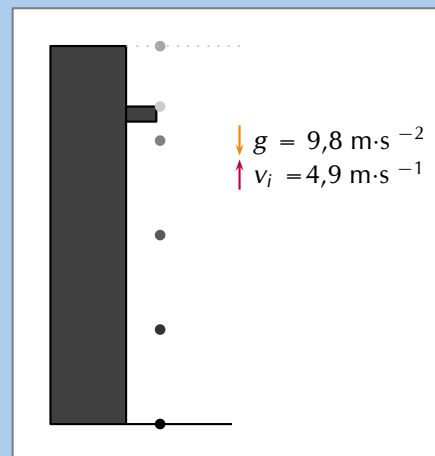


Now let's look at some examples involving graphs of projectile motion.

Worked example 4: Drawing graphs of projectile motion

QUESTION

Stanley is standing on a balcony 20 m above the ground. Stanley tosses a rubber ball upwards with an initial velocity of $4,9 \text{ m}\cdot\text{s}^{-1}$. The ball travels upwards to a maximum height, and then falls to the ground. Draw graphs of position vs time, velocity vs time and acceleration vs time. Choose upwards as the positive direction.



SOLUTION

Step 1: Determine what is required

We are required to draw graphs of:

1. $\vec{x}$ vs t
2. $\vec{v}$ vs t
3. $\vec{a}$ vs t

Step 2: Determine how to approach the problem

There are two parts to the motion of the ball:

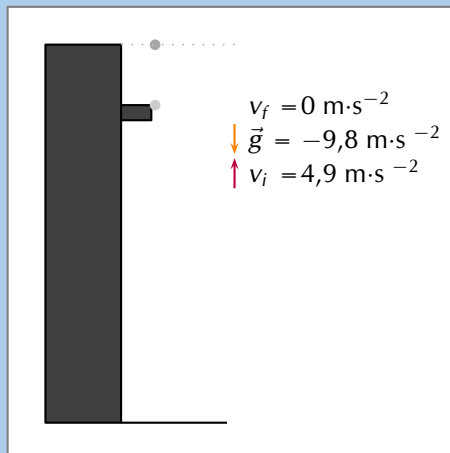
1. ball travelling upwards from the building
2. ball falling to the ground

We examine each of these parts separately. To be able to draw the graphs, we need to determine the time taken and displacement for each of the motions.

Step 3: Find the height and the time taken for the first motion

For the first part of the motion we have:

- $\vec{v}_i = +4,9 \text{ m}\cdot\text{s}^{-1}$
- $\vec{v}_f = 0 \text{ m}\cdot\text{s}^{-1}$
- $\vec{g} = -9,8 \text{ m}\cdot\text{s}^{-2}$



Therefore we can use $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{g}\Delta\vec{x}$ to solve for the height and $\vec{v}_f = \vec{v}_i + \vec{g}t$ to solve for the time.

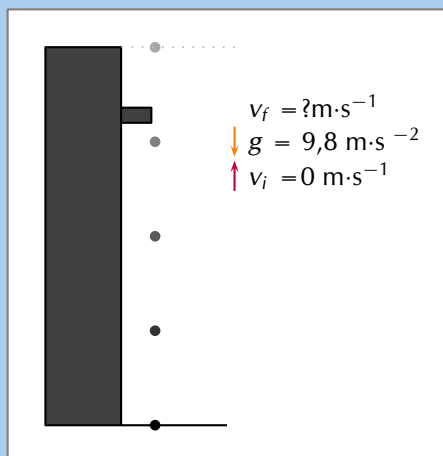
$$\begin{aligned}\vec{v}_f^2 &= \vec{v}_i^2 + 2\vec{g}\Delta\vec{x} \\ (0)^2 &= (4,9)^2 + 2 \times (-9,8) \times \Delta\vec{x} \\ 19,6\Delta x &= (4,9)^2 \\ \Delta\vec{x} &= +1,225 \text{ m} \\ h &= \Delta x = 1,225 \text{ m}\end{aligned}$$

$$\begin{aligned}\vec{v}_f &= \vec{v}_i + \vec{g}t \\ 0 &= 4,9 + (-9,8) \times t \\ 9,8t &= 4,9 \\ t &= 0,5 \text{ s}\end{aligned}$$

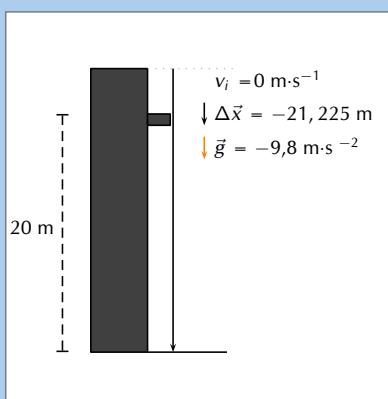
Step 4: Find the height and the time taken for the second motion

For the second part of the motion we have: Therefore we can use $\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{g}t^2$ to solve for the time.

- $\vec{v}_i = 0 \text{ m}\cdot\text{s}^{-1}$
- $\Delta\vec{x} = -(20 + 1,225) \text{ m}$
- $\vec{g} = -9,8 \text{ m}\cdot\text{s}^{-2}$

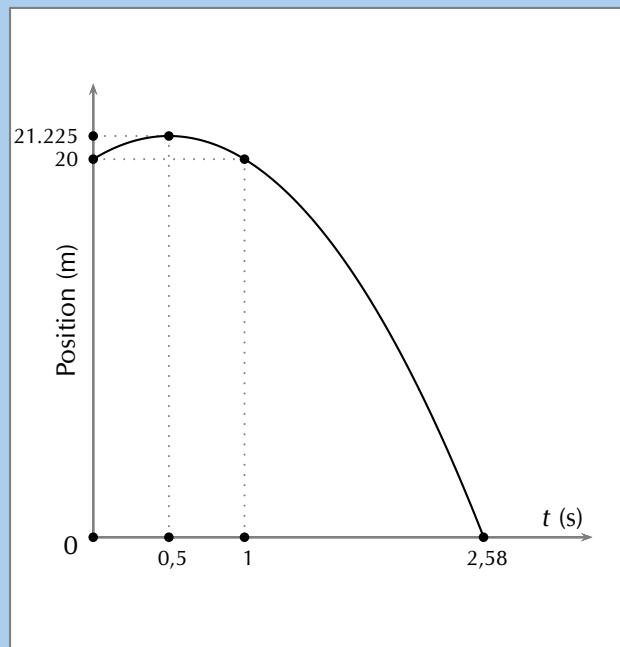


$$\begin{aligned}\Delta\vec{x} &= \vec{v}_i t + \frac{1}{2}\vec{g}t^2 \\ -(20 + 1,225) &= (0) \times t + \frac{1}{2} \times (-9,8) \times t^2 \\ -21,225 &= 0 - 4,9t^2 \\ t^2 &= 4,33163... \\ t &= 2,08125 \text{ s}\end{aligned}$$



Step 5: Graph of position vs time

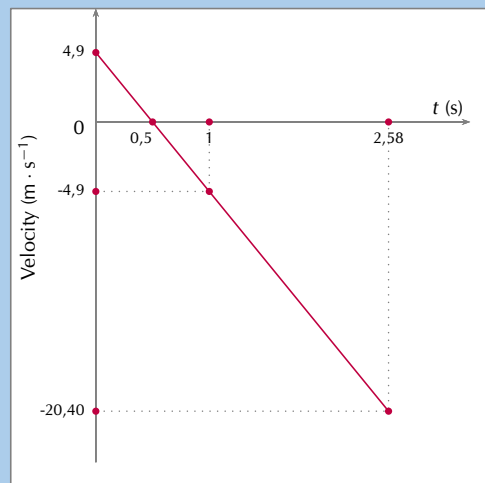
The ball starts from a position of 20 m (at $t = 0$ s) from the ground and moves upwards until it reaches $20 + 1,225$ m (at $t = 0,5$ s). It then falls back to 20 m (at $t = 0,5 + 0,5 = 1,0$ s) and then falls to the ground, $\Delta \vec{x} = 0$ m (at $t = 0,5 + 2,08 = 2,58$ s).



Step 6: Graph of velocity vs time

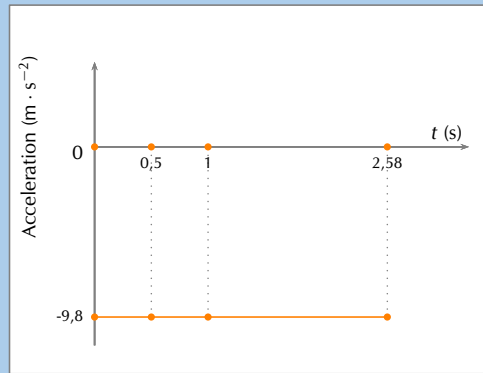
The ball starts off with a velocity of $+4,9 \text{ m} \cdot \text{s}^{-1}$ at $t = 0$ s, it then reaches a velocity of $0 \text{ m} \cdot \text{s}^{-1}$ at $t = 0,5$ s. It stops and falls back to the Earth. At $t = 1,0$ s (i.e. after a further 0,5 s) it has a velocity of $-4,9 \text{ m} \cdot \text{s}^{-1}$. This is the same as the initial upwards velocity but it is downwards. It carries on at constant acceleration until $t = 2,58$ s. In other words, the velocity graph will be a straight line. The final velocity of the ball can be calculated as follows:

$$\begin{aligned}\vec{v}_f &= \vec{v}_i + \vec{g}t \\ &= 0 + (-9,8)(2,08...) \\ &= -20,396... \text{ m} \cdot \text{s}^{-1}\end{aligned}$$



Step 7: Graph of a vs t

We chose upwards to be positive. The acceleration of the ball is downward, $\vec{g} = 9,8 \text{ m} \cdot \text{s}^{-2}$ downwards. Because the acceleration is constant throughout the motion, the graph looks like this:



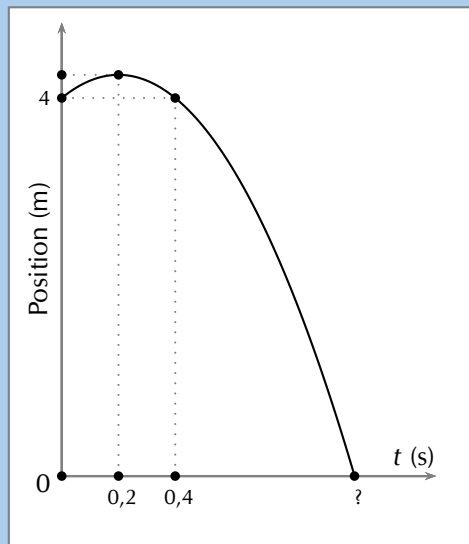
Worked example 5: Analysing graphs of projectile motion

QUESTION

The graph below (not drawn to scale) shows the motion of a tennis ball that was thrown vertically upwards from an open window some distance from the ground. It takes the ball 0,2 s to reach its highest point before falling back to the ground.

Study the graph given and calculate:

1. how high the window is above the ground.
2. the time it takes the ball to reach the maximum height.
3. the initial velocity of the ball.
4. the maximum height that the ball reaches.
5. the final velocity of the ball when it reaches the ground.



SOLUTION

Step 1: Find the height of the window

The initial position of the ball will tell us how high the window is. From the y-axis on the graph we can see that the ball is 4 m from the ground.

The window is therefore 4 m above the ground.

Step 2: Find the time taken to reach the maximum height

The maximum height is where the position vs time graph reaches its maximum position. This is when $t = 0,2$ s.

It takes the ball 0,2 s to reach the maximum height.

Step 3: Find the initial velocity ($\vec{v}_i$) of the ball

To find the initial velocity we only look at the first part of the motion of the ball. That is from when the ball is released until it reaches its maximum height. We have the following for this:

In this case, let's choose upwards as positive.

To calculate the initial velocity of the ball ($\vec{v}_i$), we use:

$$t = 0,2 \text{ s}$$

$$\vec{g} = 9,8 \text{ m} \cdot \text{s}^{-2}$$

$$\vec{v}_f = 0 \text{ m} \cdot \text{s}^{-1} \text{ (because the ball stops)}$$

$$\vec{v}_f = \vec{v}_i + \vec{g}t$$

$$0 = \vec{v}_i + (-9,8)(0,2)$$

$$v_i = 1,96 \text{ m} \cdot \text{s}^{-1}$$

The initial velocity of the ball is $1,96 \text{ m} \cdot \text{s}^{-1}$ upwards.

Step 4: Find the maximum height of the ball

To find the maximum height we look at the initial motion of the ball.

We have the following:

To calculate the displacement from the window to the maximum height (Δx) we use:

$$t = 0,2 \text{ s}$$

$$\vec{g} = 9,8 \text{ m} \cdot \text{s}^{-2}$$

$$\vec{v}_f = 0 \text{ m} \cdot \text{s}^{-1} \text{ (because the ball stops)}$$

$$\vec{v}_i = +1,96 \text{ m} \cdot \text{s}^{-1} \text{ (calculated above)}$$

$$\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{g} t^2$$

$$\Delta \vec{x} = (1,96)(0,2) + \frac{1}{2}(-9,8)(0,2)^2$$

$$\Delta \vec{x} = 0,196 \text{ m}$$

The maximum height of the ball is $(4 + 0,196) = 4,196 \text{ m}$ above the ground.

Step 5: Find the final velocity ($\vec{v}_f$) of the ball

To find the final velocity of the ball we look at the second part of the motion. For this we have:

$$\Delta \vec{x} = -4,196 \text{ m (because upwards is positive)}$$

$$\vec{g} = -9,8 \text{ m} \cdot \text{s}^{-2}$$

$$\vec{v}_i = 0 \text{ m} \cdot \text{s}^{-1}$$

We can use $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{g}\Delta\vec{x}$ to calculate the final velocity of the ball.

$$\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{g}\Delta\vec{x}$$

$$\vec{v}_f^2 = (0)^2 + 2(-9,8)(-4,196)$$

$$\vec{v}_f^2 = 82,2416$$

$$\vec{v}_f = 9,0687 \text{ m} \cdot \text{s}^{-1} \text{ downwards}$$

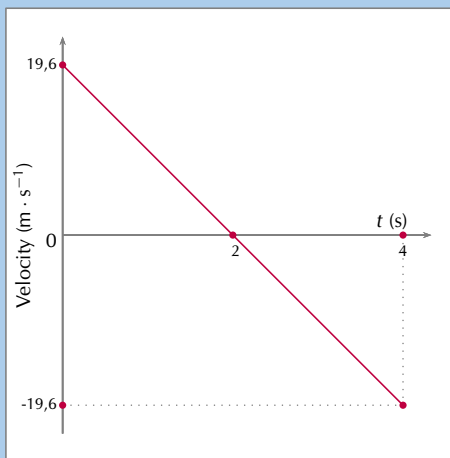
The final velocity of the ball is $9,07 \text{ m}\cdot\text{s}^{-1}$ downwards.

Worked example 6: Describing projectile motion

QUESTION

A cricketer hits a cricket ball from the ground and the following graph of velocity vs time was drawn. Upwards was taken as positive. Study the graph and follow the instructions below:

1. Describe the motion of the ball according to the graph.
2. Draw a sketch graph of the corresponding position-time graph. Label the axes.
3. Draw a sketch graph of the corresponding acceleration-time graph. Label the axes.



SOLUTION

Step 1: Describe the motion of the ball

We need to study the velocity-time graph to answer this question. We will break the motion of the ball up into two time zones: $t = 0 \text{ s}$ to $t = 2 \text{ s}$ and $t = 2 \text{ s}$ to $t = 4 \text{ s}$.

From $t = 0 \text{ s}$ to $t = 2 \text{ s}$ the following happens:

The ball starts to move at an initial velocity of $19,6 \text{ m}\cdot\text{s}^{-1}$ and decreases its velocity to $0 \text{ m}\cdot\text{s}^{-1}$ at $t = 2 \text{ s}$. At $t = 2 \text{ s}$ the velocity of the ball is $0 \text{ m}\cdot\text{s}^{-1}$ and therefore it stops.

From $t = 2 \text{ s}$ to $t = 4 \text{ s}$ the following happens:

The ball moves from a velocity of $0 \text{ m}\cdot\text{s}^{-1}$ to $19,6 \text{ m}\cdot\text{s}^{-1}$ in the opposite direction to the original motion.

If we assume that the ball is hit straight up in the air (and we take upwards as positive), it reaches its maximum height at $t = 2 \text{ s}$, stops, and falls back to the Earth to reach the ground at $t = 4 \text{ s}$.

Step 2: Draw the position-time graph

To draw this graph, we need to determine the displacements at $t = 2 \text{ s}$ and $t = 4 \text{ s}$.

At $t = 2 \text{ s}$:

The displacement is equal to the area under the graph:

Area under graph = Area of triangle

$$\text{Area} = \frac{1}{2}bh$$

$$\text{Area} = \frac{1}{2} \times 2 \times 19,6$$

$$\text{Position} = 19,6 \text{ m}$$

At $t = 4 \text{ s}$:

The displacement is equal to the area under the whole graph (top and bottom). Remember that an area under the time line must be subtracted:

Area under graph = Area of triangle 1 + Area of triangle 2

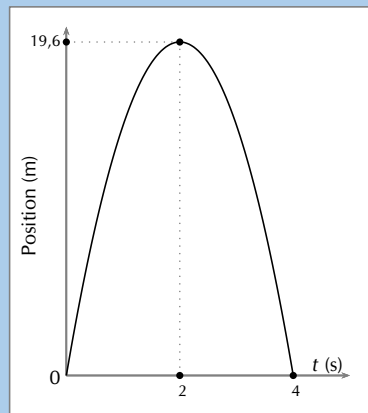
$$\text{Area} = \frac{1}{2}bh + \frac{1}{2}bh$$

$$\text{Area} = \left(\frac{1}{2} \times 2 \times 19,6 \right) + \left(\frac{1}{2} \times 2 \times (-19,6) \right)$$

$$\text{Area} = 19,6 - 19,6$$

$$\text{Displacement} = 0 \text{ m}$$

The position vs time graph for motion at constant acceleration is a curve. The graph will look like this:



Step 3: Draw the acceleration versus time graph

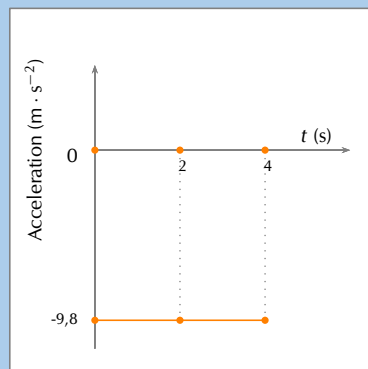
To draw the acceleration vs time graph, we need to know what the acceleration is. The velocity versus time graph is a straight line which means that the acceleration is constant. The gradient of the line will give the acceleration.

The line has a negative slope (goes down towards the left) which means that the acceleration has a negative value.

Calculate the gradient of the line:

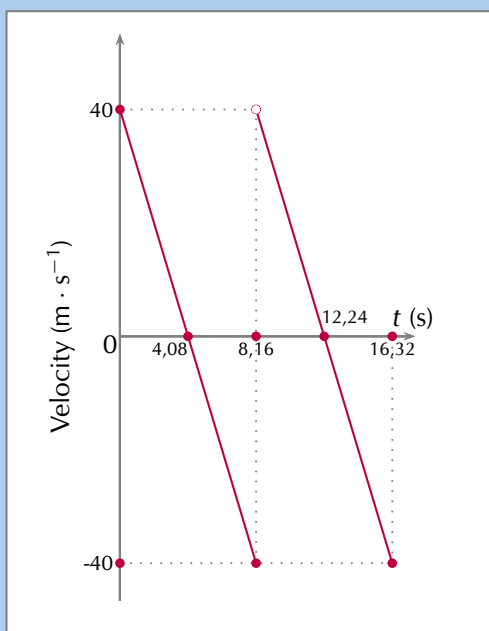
$$\begin{aligned} \text{gradient} &= \frac{\Delta v}{\Delta t} \\ \text{gradient} &= \frac{0 - 19,6}{2 - 0} \\ \text{gradient} &= \frac{-19,6}{2} \\ \text{gradient} &= -9,8 \end{aligned}$$

Therefore the acceleration = $9,8 \text{ m} \cdot \text{s}^{-2}$ downwards.



QUESTION

Describe the motion corresponding to the following graph and plot the position vs time and acceleration vs time assuming the initial position is 0:



SOLUTION

Step 1: Determine what is required

The velocity vs time graph corresponding to the motion of an object is given, using the graph we need to describe the actual motion. Notice that the graph has discontinuities, this naturally breaks the graph into phases. We should process each phase separately and then try to infer (figure out) what probably happened to change from one phase to the next.

The first thing that you normally do in a vertical projectile motion problem is choose a positive direction. In this case you can't do that because it has already been decided. We need to use the information given to figure out which direction was chosen as positive.

Step 2: Determine the direction chosen as positive

We are given a velocity vs time graph. We know that the slope of the velocity time graph is the acceleration and we know that in these problems we are only dealing with gravitational acceleration. We can use what we know about gravitational acceleration and the slope of the graph to determine which direction was chosen as positive.

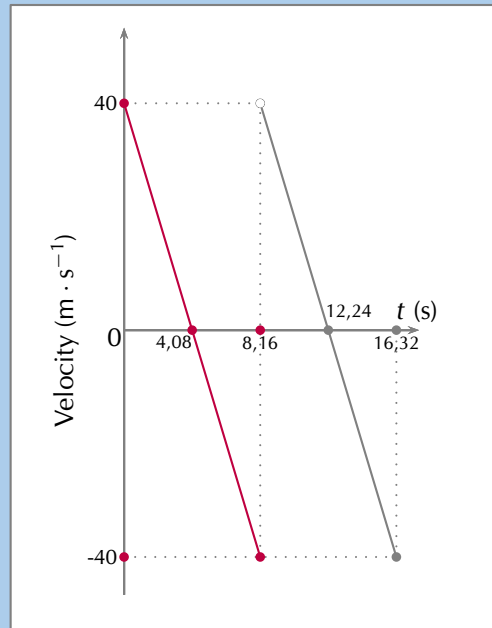
Gravitational acceleration is always directed towards the centre of the Earth. If we choose:

- **upwards as the positive** direction then the slope of the velocity vs time graph will be negative
- **downwards as the positive** direction then the slope of the velocity vs time graph will be positive

In this problem the slope of the velocity vs time graph is negative and so we know that upwards was chosen as the positive direction.

Step 3: First phase

In the plot below we highlight only the first phase of interest.

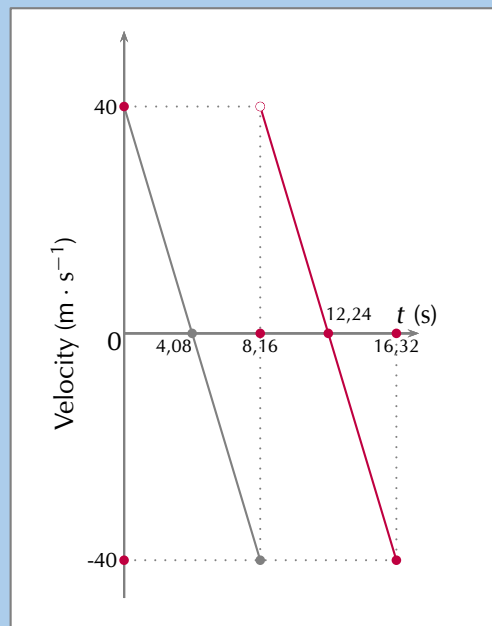


The initial velocity is positive and then decreases linearly, passes through zero and reaches the same magnitude in the negative direction. The positive starting value means the object was going upwards, the linear decrease means a constant acceleration in the negative direction, passing through zero means that the object reaches a maximum height before beginning to fall in the negative direction. The fact that the magnitude of the initial and final velocities is the same means that there is time symmetry.

This is just the plot of an object being thrown/shot/projected upwards, peaking and falling back to the same level as it was thrown from.

Step 4: Second phase

In the plot below we highlight only the second phase.



Just as in the first phase, the initial velocity is positive and then decreases linearly, passes through zero and reaches the same magnitude in the negative direction. The positive starting value means the object was going upwards, the linear decrease means a constant acceleration in the negative direction, passing through zero means that the object reaches a maximum height before beginning to fall in the negative direction. The fact that the magnitude of the initial and final velocities is the same means that there is time symmetry.

This is just the plot of an object being thrown upwards, peaking and falling back to the same level as it was thrown from.

Step 5: Combining phases

We know that the two phases look the same, the object starts off moving upwards, peaks, falls down to the initial position and then repeats the process. This could be a description of a ball bouncing as an example.

Step 6: Position vs time

To draw an accurate position graph we know that we are dealing with the Case 1 situation from earlier. We need to determine the height that the object rises to and then we can plot the position vs time. We know the timing information from the velocity vs time plot.

We can verify that the acceleration is in fact from gravity by using $\vec{v}_f = \vec{v}_i + \vec{g}t$:

$$\begin{aligned}\vec{v}_f &= \vec{v}_i + \vec{g}t \\ (-40) &= (40) + \vec{g}(8,16) \\ (-80) &= \vec{g}(8,16) \\ \vec{g} &= \frac{-80}{8,16} \\ \vec{g} &= -9,8 \text{ m} \cdot \text{s}^{-2}\end{aligned}$$

We can work out the height using the equation for displacement:

$$\begin{aligned}\vec{v}_f^2 &= \vec{v}_i^2 + 2\vec{g}\Delta\vec{x} \\ 0 &= (40)^2 + 2(-9,8)\Delta\vec{x} \\ \Delta\vec{x} &= \frac{-(40)^2}{2(-9,8)} \\ \Delta\vec{x} &= 81,63 \text{ m}\end{aligned}$$

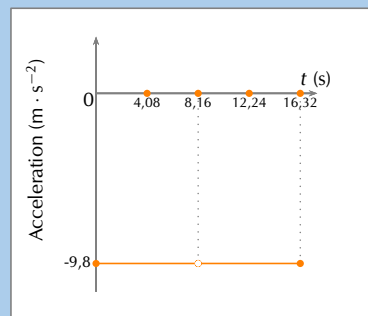
So we can now draw the displacement vs time plot:



At that point the acceleration must be some other, very large, value in the positive direction but we don't have enough information to determine what the value must be.

We explicitly note that at that point the acceleration is NOT $-9,8 \text{ m} \cdot \text{s}^{-2}$.

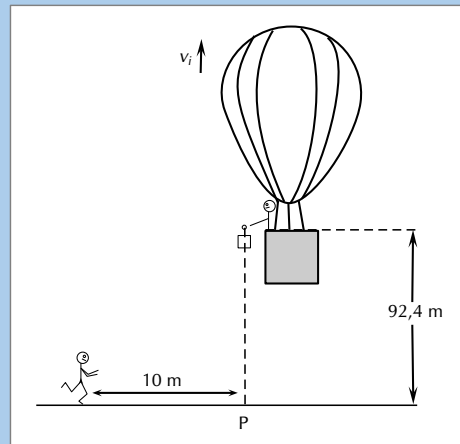
Step 7: Acceleration vs time



The acceleration is due to gravity except at one point, when the object bounces.

QUESTION

A hot-air balloon is moving vertically upwards at a constant speed. A camera is accidentally dropped from the balloon at a height of 92,4 m as shown in the diagram below. The camera strikes the ground after 6 s. Ignore the effects of friction.



- At the instant the camera is dropped, it moves upwards. Give a reason for this observation. (1 mark)
- Calculate the speed v_i at which the balloon is rising when the camera is dropped. (4 marks)
- Draw a sketch graph of velocity versus time for the entire motion of the camera. Indicate the following on the graph:
 - Initial velocity
 - Time at which it reaches the ground
 (4 marks)
- If a jogger, 10 m away from point **P** as shown in the above diagram and running at a constant speed of $2 \text{ m} \cdot \text{s}^{-1}$, sees the camera at the same instant it starts falling from the balloon, will he be able to catch the camera before it strikes the ground? Use a calculation to show how you arrived at the answer. (5 marks)

[TOTAL: 14 marks]

SOLUTION

Question 1 The initial velocity / speed of the camera is the same (as that of the balloon). (1 marks)

Question 2

Taking downwards as the positive direction:

$$\begin{aligned}\Delta y &= v_i \Delta t + \frac{1}{2} a \Delta t^2 \\ \therefore 92,4 &= v_i(6) + \frac{1}{2}(9,8)(6)^2 \\ \therefore v_i &= -14 \text{ m} \cdot \text{s}^{-1} \\ \therefore v_i &= 14 \text{ m} \cdot \text{s}^{-1}\end{aligned}$$

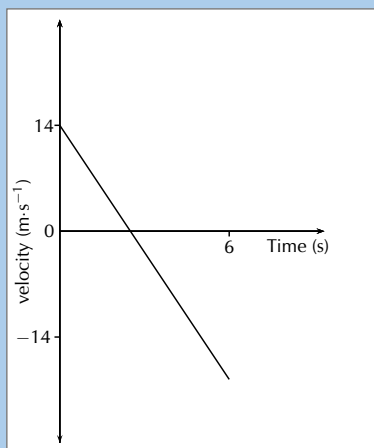
Taking downwards as the negative direction:

$$\begin{aligned}\Delta y &= v_i \Delta t + \frac{1}{2} a \Delta t^2 \\ \therefore -92,4 &= v_i(6) + \frac{1}{2}(-9,8)(6)^2 \\ \therefore v_i &= 14 \text{ m} \cdot \text{s}^{-1}\end{aligned}$$

(4 marks)

Question 3

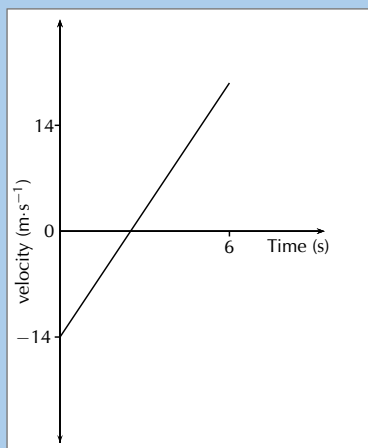
Taking downwards as the positive direction:



Criteria for graph:

- Correct shape as shown. (Straight line with gradient.)
- Graph starts at $v = 14 \text{ m} \cdot \text{s}^{-1} / v_i$ at $t = 0 \text{ s}$.
- Graph extends below t axis until $t = 6 \text{ s}$.
- Section of graph below t axis is longer than section above t axis.

Taking downwards as the negative direction:



Criteria for graph:

- Correct shape as shown. (Straight line with gradient.)
- Graph starts at $v = -14 \text{ m} \cdot \text{s}^{-1} / v_i$ at $t = 0 \text{ s}$.
- Graph extends above t axis until $t = 6 \text{ s}$.
- Section of graph above t axis is longer than section below t axis.

(4 marks)

Question 4

Option 1:

$$\begin{aligned}\Delta x &= v\Delta t \\ \therefore 10 &= (2)\Delta t \\ \therefore \Delta t &= 5 \text{ s}\end{aligned}$$

Yes, he will catch the camera since the time is less than 6 s.

Option 2:

$$\begin{aligned}\Delta x &= v\Delta t \\ \therefore &= (2)(6) \\ \therefore &= 12 \text{ m}\end{aligned}$$

Yes, he will catch the camera since the distance covered is greater than 10 m.

Option 3:

$$\begin{aligned}\Delta x &= v_i\Delta t + \frac{1}{2}a\Delta t^2 \\ \therefore 10 &= (2)\Delta t + \frac{1}{2}(0)\Delta t^2 \\ \therefore \Delta t &= 5 \text{ s}\end{aligned}$$

Yes, he will catch the camera since the time is less than 6 s.

Option 4:

$$\begin{aligned}\Delta x &= \left(\frac{v_i + v_f}{2}\right)\Delta t \\ \therefore 10 &= \left(\frac{2 + 2}{2}\right)\Delta t \\ \therefore \Delta t &= 5 \text{ s}\end{aligned}$$

Yes, he will catch the camera since the time is less than 6 s.

Option 5:

$$\begin{aligned}\Delta x &= \left(\frac{v_i + v_f}{2}\right)\Delta t \\ &= \left(\frac{2 + 2}{2}\right)(6) \\ \therefore \Delta x &= 12 \text{ m}\end{aligned}$$

Yes, he will catch the camera since the distance covered is greater than 10 m. (5 marks)

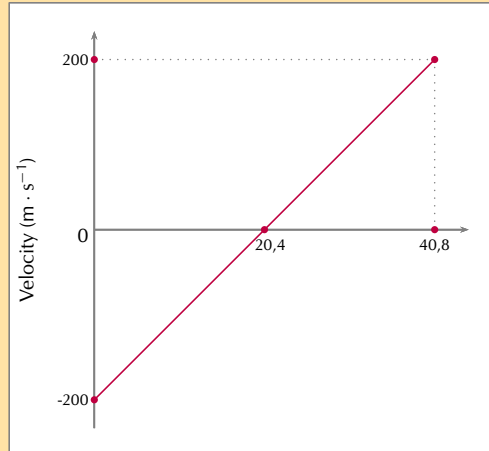
[TOTAL: 14 marks]

Exercise 3 – 2: Graphs of vertical projectile motion

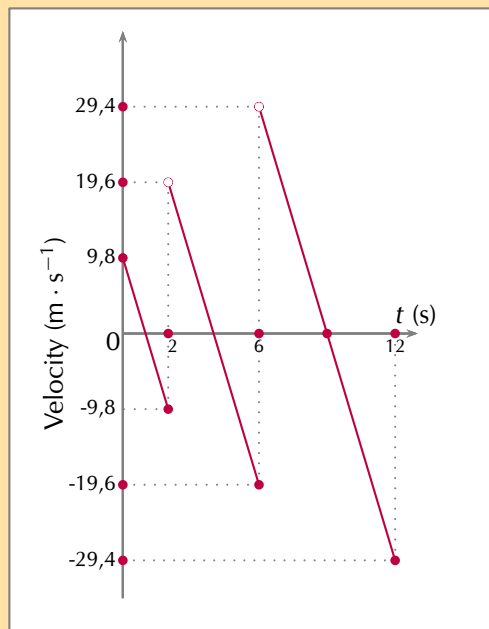
1. Amanda throws a tennis ball from a height of 1,5 m straight up into the air and then lets it fall to the ground. Draw graphs of x vs t ; v vs t and a vs t for the motion of the ball. The initial velocity of the tennis ball is $2\text{ m} \cdot \text{s}^{-1}$. Choose upwards as positive.
2. A bullet is shot straight upwards from a gun.

The following graph is drawn. Downwards was chosen as positive

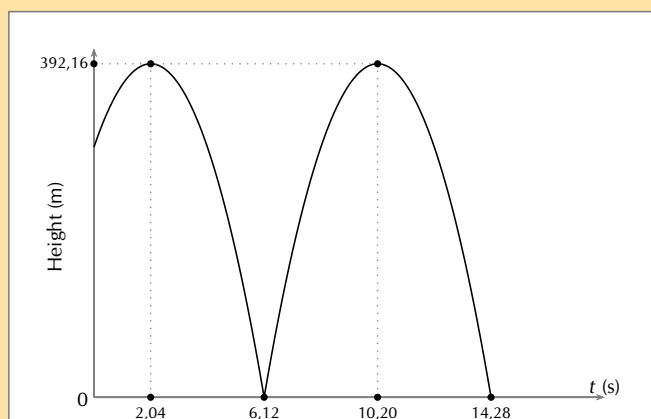
- a) Describe the motion of the bullet.
- b) Draw a displacement - time graph.
- c) Draw an acceleration - time graph.



3. Describe the motion resulting from the velocity vs time graph shown below, assuming an initial height of 0 m, and draw a displacement vs time graph and an acceleration vs time graph:



4. What is the initial height? Draw a velocity vs time graph and an acceleration vs time graph that would result in this displacement vs time graph:



5. A ball that always bounces off the floor with the same magnitude of the velocity as it hit the floor is dropped from 3 m above a stack of 4 crates. Each crate is 30 cm high. After the ball bounces off a crate it is quickly removed so that the next time the ball bounces it bounces off the next crate in the pile. This is repeated until all the crates have been removed. Draw the following graphs for the situation:

- Displacement vs time
- Velocity vs time
- Acceleration vs time

6. A ball is dropped from 4 m above a cushioned mat. Each time the ball bounces the magnitude of the velocity in the upwards direction is half the magnitude of the velocity with which it hit the floor. The ball is allowed to bounce 3 times, draw the following graphs for the situation and the equation describing each section of the graph:

- Displacement vs time
- Velocity vs time
- Acceleration vs time

7. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27K7 2. 27K8 3. 27K9 4. 27KB 5. 27KC 6. 27KD



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3.3 Chapter summary

ESCK2

► See presentation: 27KF at www.everythingscience.co.za

- Projectiles are objects that move through the air. In vertical projectile motion we deal with objects that fall under the influence of gravity and only vertically.
- Objects that move up and down (vertical projectiles) on the Earth accelerate with a constant acceleration $\vec{g}$ which is approximately equal to $9,8 \text{ m}\cdot\text{s}^{-2}$ directed

downwards towards the centre of the earth.

- The time it takes an object to rise to its maximum height, is the same as the time it will take to fall back to its initial height. The magnitude of the velocity will also be the same but the direction will be reversed. This is known as time symmetry and is a consequence of uniform gravitational acceleration.
- The equations of motion can be used to solve vertical projectile problems.

$$\vec{v}_f = \vec{v}_i + \vec{g}t$$

$$\Delta \vec{x} = \frac{(\vec{v}_i + \vec{v}_f)}{2} t$$

$$\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{g} t^2$$

$$\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{g}\Delta \vec{x}$$

- Graphs for vertical projectile motion are similar to graphs for motion at constant acceleration.

Physical Quantities			
Quantity	Vector	Unit name	Unit symbol
Position (x)	-	metre	m
Displacement ($\Delta \vec{x}$)	✓	metre	m
Instantaneous velocity ($\vec{v}$)	✓	metre per second	m·s ⁻¹
Instantaneous speed (v)	-	metre per second	m·s ⁻¹
Instantaneous acceleration ($\vec{a}$)	✓	metre per second per second	m·s ⁻²
Magnitude of acceleration (a)	-	metre per second per second	m·s ⁻²

Table 3.1: Units used in **vertical projectile motion**.

Exercise 3 – 3:

- [SC 2001/11 HG1] A bullet fired vertically upwards reaches a maximum height and falls back to the ground.
Which **one** of the following statements is **true** with reference to the acceleration of the bullet during its motion, if air resistance is ignored? The acceleration:
 - a) is always downwards
 - b) is first upwards and then downwards
 - c) is first downwards and then upwards
 - d) decreases first and then increases
- [IEB 2002/11 HG1] Two balls, P and Q, are simultaneously thrown into the air from the same height above the ground. P is thrown vertically upwards and Q vertically downwards with the same initial speed. Which of the following is true of both balls just before they hit the ground? (Ignore any air resistance. Take downwards as the positive direction.)

	Velocity	Acceleration
A	The same	The same
B	P has a greater velocity than Q	P has a negative acceleration; Q has a positive acceleration
C	P has a greater velocity than Q	The same
D	The same	P has a negative acceleration; Q has a positive acceleration

3. [SC 2003/11] A ball X of mass m is projected vertically upwards at a speed u_x from a bridge 20 m high. A ball Y of mass $2m$ is projected vertically downwards from the same bridge at a speed of u_y . The two balls reach the water at the same speed. Air friction can be ignored.

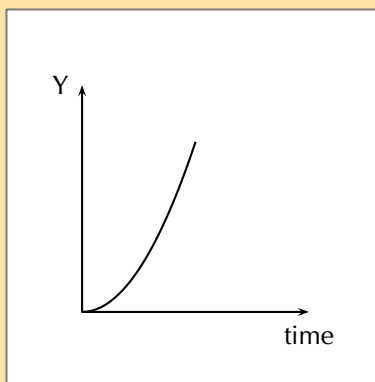
Which of the following is true with reference to the speeds with which the balls are projected?

- a) $u_x = \frac{1}{2} u_y$
- b) $u_x = u_y$
- c) $u_x = 2u_y$
- d) $u_x = 4u_y$

4. [SC 2002/03 HG1]

A stone falls freely from rest from a certain height. Which one of the following quantities could be represented on the y-axis of the graph below?

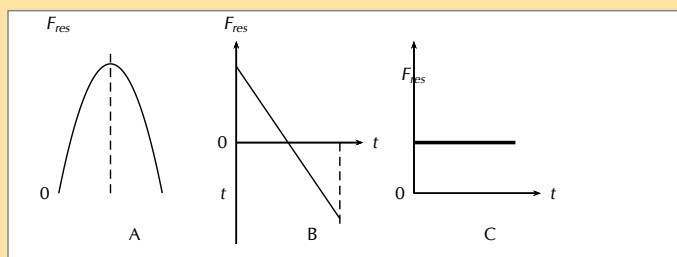
- a) velocity
- b) acceleration
- c) momentum
- d) displacement



5. A stone is thrown vertically upwards and it returns to the ground. If friction is ignored, its acceleration as it reaches the highest point of its motion is

- a) greater than just after it left the thrower's hand.
- b) less than just before it hits the ground.
- c) the same as when it left the thrower's hand.
- d) less than it will be when it strikes the ground.

6. [IEB 2004/11 HG1] A stone is thrown vertically up into the air. Which of the following graphs best shows the resultant force exerted on the stone against time while it is in the air? (Air resistance is negligible.)



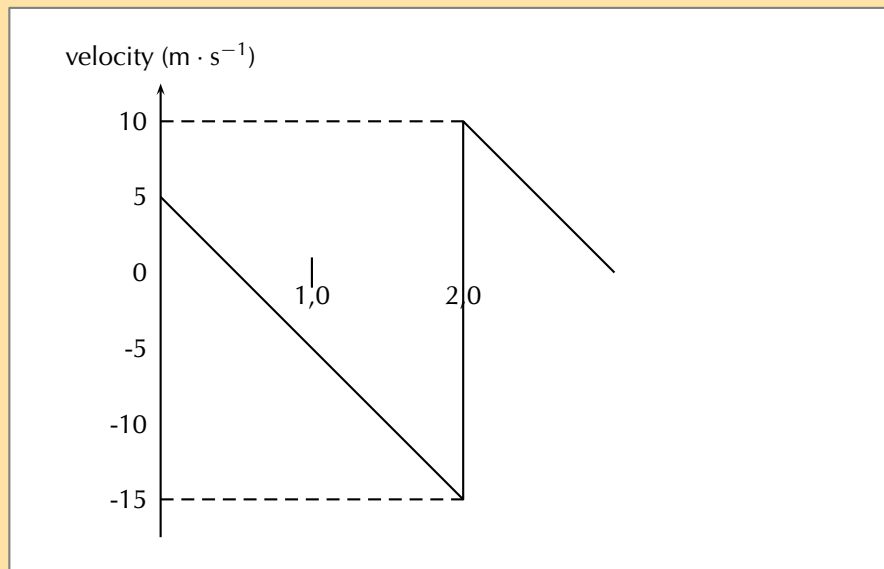
7. What is the velocity of a ball just as it hits the ground if it is thrown upward at $10 \text{ m}\cdot\text{s}^{-1}$ from a height 5 meters above the ground?

8. [IEB 2002/11 HG1 - Bouncing Ball]

A ball bounces vertically on a hard surface after being thrown vertically up into the air by a boy standing on the ledge of a building.

Just before the ball hits the ground for the first time, it has a velocity of magnitude $15 \text{ m}\cdot\text{s}^{-1}$. Immediately, after bouncing, it has a velocity of magnitude $10 \text{ m}\cdot\text{s}^{-1}$.

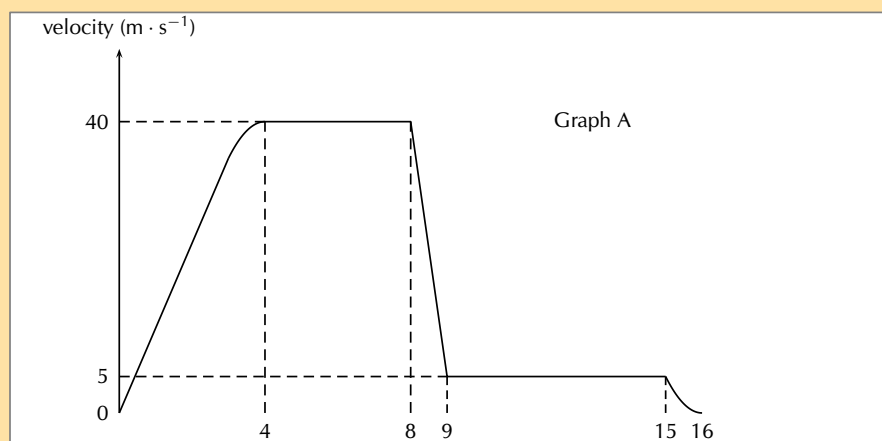
The graph below shows the velocity of the ball as a function of time from the moment it is thrown upwards into the air until it reaches its maximum height after bouncing once.



- At what velocity does the boy throw the ball into the air?
- What can be determined by calculating the gradient of the graph during the first two seconds?
- Determine the gradient of the graph over the first two seconds. State its units.
- How far below the boy's hand does the ball hit the ground?
- Use an equation of motion to calculate how long it takes, from the time the ball was thrown, for the ball to reach its maximum height after bouncing.
- What is the position of the ball, measured from the boy's hand, when it reaches its maximum height after bouncing?

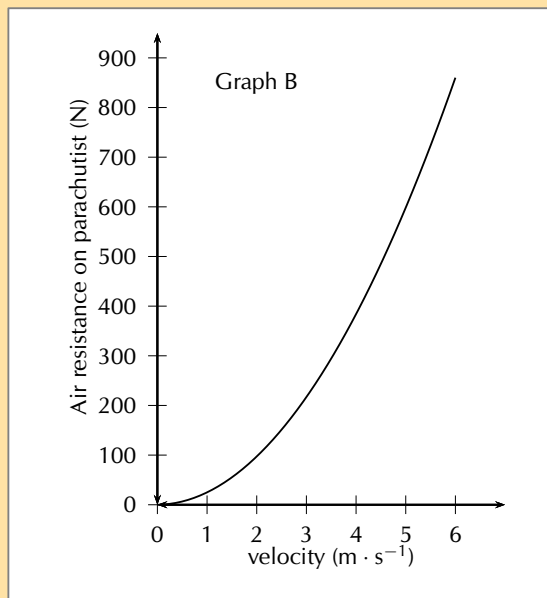
9. [IEB 2001/11 HG1] - **Free Falling?**

A parachutist steps out of an aircraft, flying high above the ground. She falls for the first 8 seconds before opening her parachute. A graph of her velocity is shown in Graph A below.

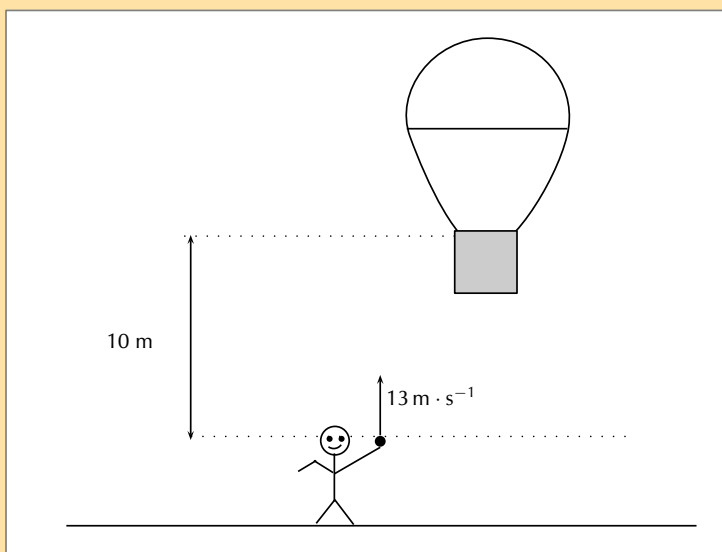


- a) Use the information from the graph to calculate an approximate height of the aircraft when she stepped out of it (to the nearest 10 m).
- b) What is the magnitude of her velocity during her descent with the parachute fully open?

The air resistance acting on the parachute is related to the speed at which the parachutist descends. Graph B shows the relationship between air resistance and velocity of the parachutist descending with the parachute open.



- c) Use Graph B to find the magnitude of the air resistance on her parachute when she was descending with the parachute open.
 - d) Assume that the mass of the parachute is negligible. Calculate the mass of the parachutist showing your reasoning clearly.
10. Niko, in the basket of a hot-air balloon, is stationary at a height of 10 m above the level from where his friend, Bonggi, will throw a ball. Bonggi intends throwing the ball upwards and Niko, in the basket, needs to **descend** (move downwards) to catch the ball at its maximum height.



Bongi throws the ball upwards with a velocity of $13 \text{ m}\cdot\text{s}^{-1}$. Niko starts his descent at the same instant the ball is thrown upwards, by letting air escape from the balloon, causing it to accelerate downwards. Ignore the effect of air friction on the ball.

- a) Calculate the maximum height reached by the ball.
- b) Calculate the magnitude of the minimum average acceleration the balloon must have in order for Niko to catch the ball at its maximum height.

11. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [27KG](#) 2. [27KH](#) 3. [27KJ](#) 4. [27KK](#) 5. [27KM](#) 6. [27KN](#)
7. [27KP](#) 8. [27KQ](#) 9. [27KR](#) 10. [27KS](#)



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Organic molecules

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4.1 What are organic molecules?

ESCK3

Organic chemistry is the branch of chemistry that deals with **organic molecules**. An organic molecule is one which contains **carbon**, although not all compounds that contain carbon are organic molecules. Noticeable exceptions are carbon monoxide (CO), carbon dioxide (CO₂), carbonates (e.g. calcium carbonate), carbides (e.g. calcium carbide) and cyanides (e.g. sodium cyanide). Pure carbon compounds such as diamond and graphite are also **not** organic compounds. Organic molecules can range in size from simple molecules to complex structures containing thousands of atoms!

Although carbon is present in all organic compounds, other elements such as hydrogen (H), oxygen (O), nitrogen (N), sulfur (S) and phosphorus (P) are also common in these molecules.

DEFINITION: *Organic molecule*

An organic molecule is a molecule that contains carbon atoms (generally bonded to other carbon atoms as well as hydrogen atoms).

Organic compounds are very important in daily life and they range from simple to extremely complex (Figure 4.1).

Organic molecules make up a big part of our own bodies, they are in the food we eat and in the clothes we wear. Organic compounds are also used to make products such as medicines, plastics, washing powders, dyes, along with a long list of other items. There are millions organic compounds found in nature, as well as millions of synthetic (man-made) organic compounds.



Figure 4.1: A simple organic molecule, propane, can be used in a gas lamp (left). The complex organic molecule DNA carries the genetic code of a person and can be used to identify them.

4.2 Organic molecular structures

ESCK4

Special properties of carbon

ESCK5

Carbon has a number of unique properties which influence how it behaves and how it bonds with other atoms:

- Carbon (Figure 4.2) has *four valence electrons* which means that each carbon atom can form a maximum of four bonds with other atoms. Because of the number of bonds that carbon can form with other atoms, organic compounds can be very complex.
 - Carbon can form bonds with other carbon atoms to form single, double or triple covalent bonds.
 - Carbon can also form bonds with other atoms like hydrogen, oxygen, nitrogen and the halogens.
 - Carbon can bond to form straight chain, branched, and cyclic molecules.

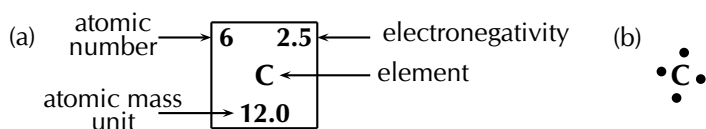


Figure 4.2: Carbon (a) as seen on the periodic table and (b) a Lewis dot representation.

TIP

Do not confuse organic compounds with *naturally produced* food. Organic compounds are often produced in a laboratory.

- Because of this, long *chain structures* can form. This is known as **catenation** - the bonding of atoms of the same element into longer chains. These chains can either be *unbranched* (Figure 4.3) or *branched* (have a branched group, Figure 4.4) and can contain single carbon-carbon bonds only, or double and triple carbon-carbon bonds as well.

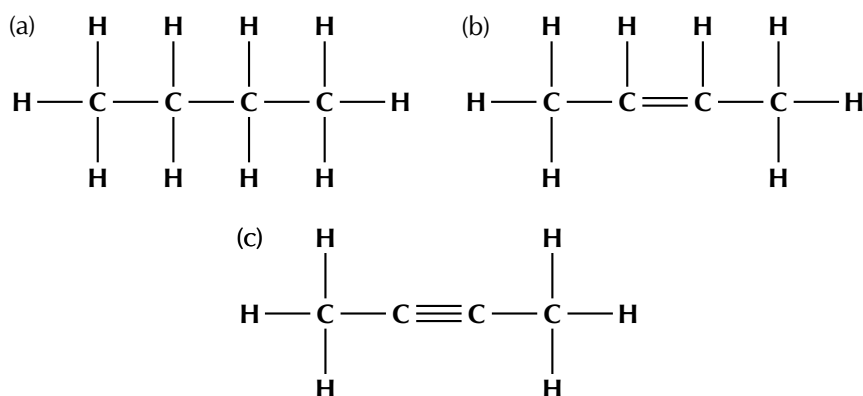


Figure 4.3: Unbranched carbon chains with (a) single carbon-carbon bonds, (b) single and double carbon-carbon bonds and (c) single and triple carbon-carbon bonds.

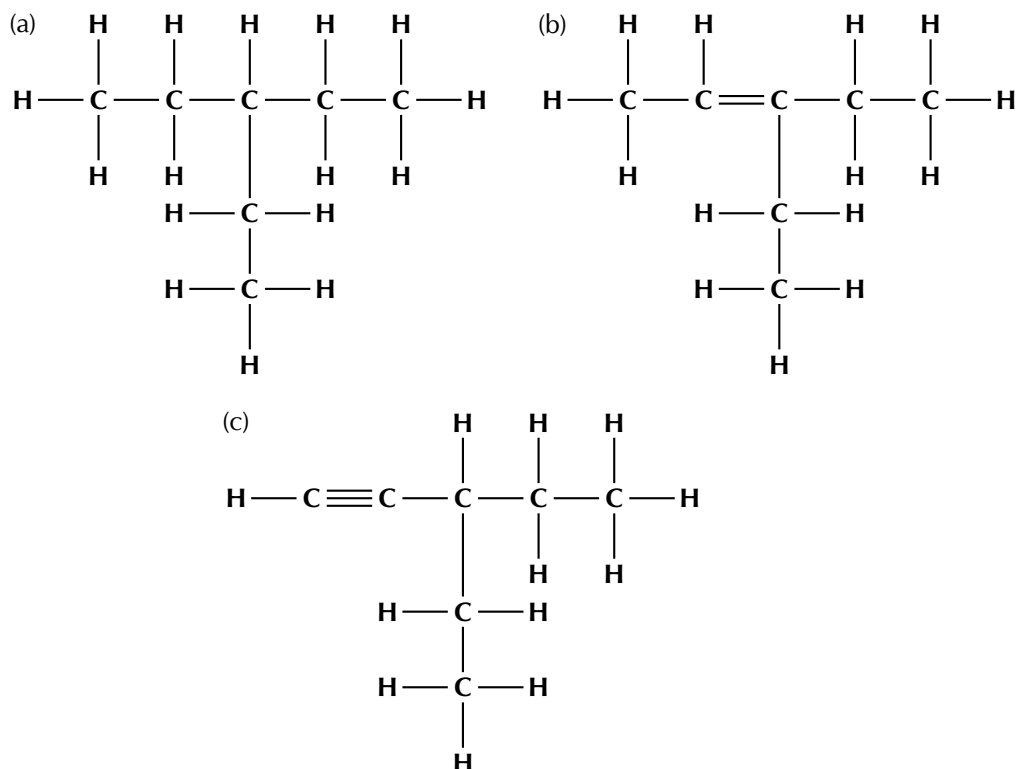
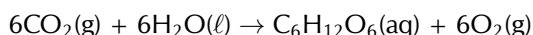


Figure 4.4: Branched carbon chains with (a) single carbon-carbon bonds, (b) single and double carbon-carbon bonds and (c) single and triple carbon-carbon bonds.

- Because of its position on the periodic table, most of the bonds that carbon forms with other atoms are *covalent*. Think for example of a C – C bond. The difference in electronegativity between the two atoms is zero, so this is a pure covalent bond. In the case of a C – H bond, the difference in electronegativity between carbon (2,5) and hydrogen (2,2) is so small that C – H bonds are almost purely covalent. The result of this is that most organic compounds are non-polar. This affects some of the properties of organic compounds.

Sources of carbon

The main source of the carbon in organic compounds is **carbon dioxide** in the atmosphere. Plants use sunlight to convert carbon dioxide and water (inorganic compounds) into sugar (an organic compound) through the process of **photosynthesis**.



Plants are therefore able to make their own organic compounds through photosynthesis, while animals feed on plants or plant products in order to gain the organic compounds that they need to survive.

Other important sources of carbon are **fossil fuels** such as coal, petroleum and natural gas. This is because fossil fuels are themselves formed from the decaying remains of dead organisms (refer to Grade 11 for more information on fossil fuels).

Representing organic molecules

ESCK6

There are a number of ways to represent organic compounds. It is useful to know all of these so that you can recognise a molecule regardless of how it is shown. There are four main ways of representing a compound in two dimensions (on your page). We will use the examples of two molecules called 2-methylpropane and butane to help explain the difference between each.

• Structural formula

The structural formula of an organic compound shows every bond between every atom in the molecule. Each bond is represented by a line. The structural formulae of 2-methylpropane and butane are shown in Figure 4.5.

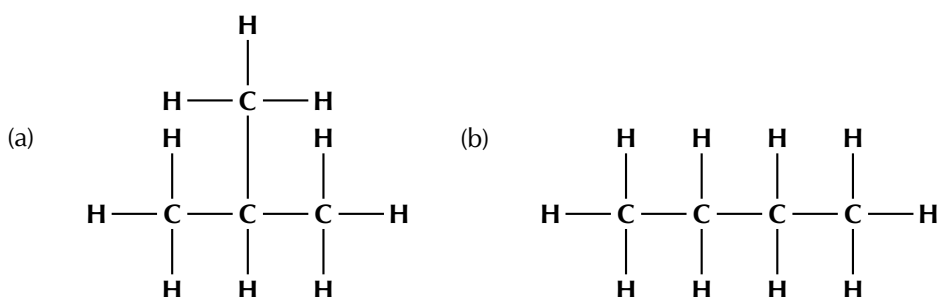


Figure 4.5: The structural formula of (a) 2-methylpropane and (b) butane.

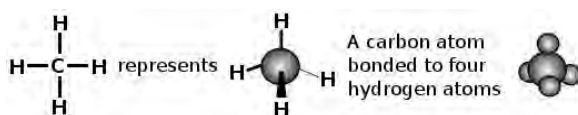


Figure 4.6: Different ways of representing a carbon atom bonding to four hydrogen atoms.

- **Semi-structural formula**

It is possible to understand the structure of an organic molecule without writing out all the carbon-hydrogen bonds. This way of writing a structure is called a semi-structural formula and is shown in Figure 4.7.

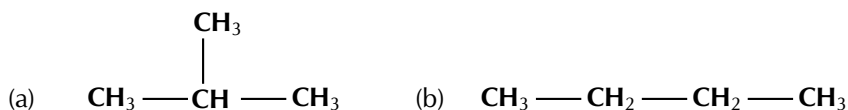


Figure 4.7: The semi-structural formulae of (a) 2-methylpropane and (b) butane.

Compare these semi-structural representations with the structural representations shown in Figure 4.5.

- **Condensed structural formula**

It is also possible to represent a molecule without showing any bonds between atoms at all. This is called a **condensed structural formula** (Figure 4.8). As for a semi-structural representation, the carbon atoms are grouped with the hydrogen atoms bonded directly to it. The bonds between these groups are not shown. Branched or substituent groups are shown in brackets after the carbon atom to which they are bonded.



Figure 4.8: The condensed structural formulae of (a) 2-methylpropane and (b) butane.

Note that in Figure 4.8 (b) the two CH_2 groups can be abbreviated to $(\text{CH}_2)_2$. Compare these condensed structural representations with the structural (Figure 4.5) and the semi-structural representations (Figure 4.7).

- **Molecular formula**

The molecular formula of a compound shows how many atoms of each type are in a molecule. The number of each atom is written as a subscript after the atomic symbol. The molecular formula of 2-methylpropane is:



This means that each molecule of 2-methylpropane consists of four carbon atoms and ten hydrogen atoms. The molecular formula of butane is also C_4H_{10} . Molecular formula gives no structural information about the compound.

Of course molecules are not two-dimensional so shown below are a few examples of different ways to represent methane (CH_4 , Figure 4.9) and ethane (C_2H_6 , Figure 4.10).

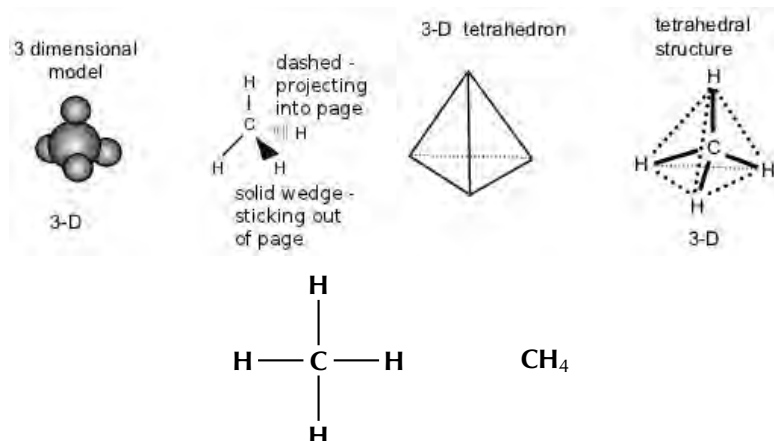


Figure 4.9: Different ways of representing methane.

TIP

A substituent is an atom or group of atoms that replaces a hydrogen atom on the main chain of an organic molecule. Therefore a branched group is a substituent. A halogen atom can also be a substituent.

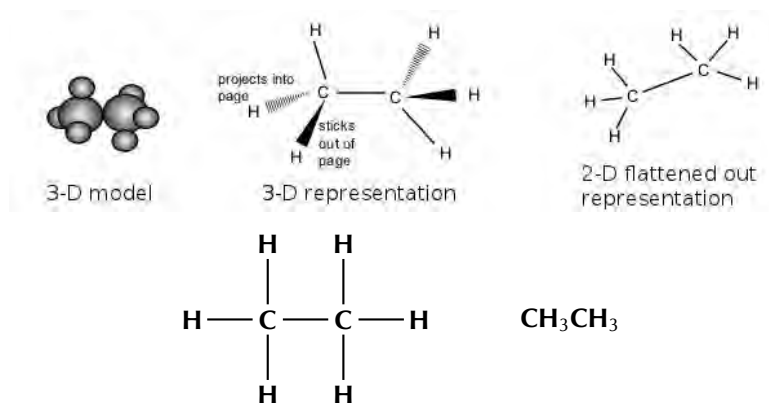


Figure 4.10: Different ways of representing ethane.

This means that butane can be represented in two dimensions as shown in Figure 4.11 (a) but it actually looks more like the three-dimensional representation given in Figure 4.11 (b).

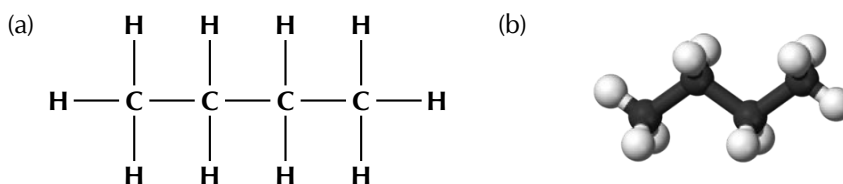


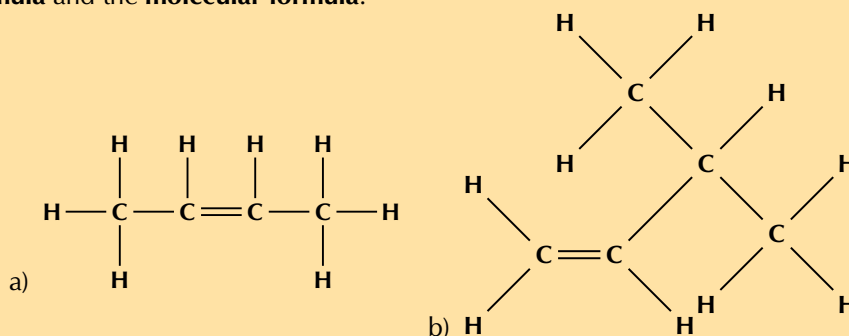
Figure 4.11: (a) Two-dimensional and (b) three-dimensional representations of butane.

Exercise 4 – 1: Representing organic compounds

- For each of the following, give the **structural formula** and the **molecular formula**.

a) CH₃CH₂CH₃ b) CH₃CH₂CH(CH₃)CH₃ c) CH₃CH₃

- For each of the following organic compounds, give the **condensed structural formula** and the **molecular formula**.



- Give two possible structural formulae for the compound with a molecular formula of C₄H₁₀.
- More questions. Sign in at Everything Science online and click 'Practise Science'. Check answers online with the exercise code below or click on 'show me the answer'.

1. 27KT 2. 27KV 3. 27KW



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The way in which a compound will react is determined by a particular characteristic of a group of atoms and the way they are bonded (e.g. double C–C bond, C–OH group). This is called the **functional group**. This group is important in determining how a compound will react. The same functional group will undergo the same or similar chemical reaction(s) regardless of the size of the molecule it is a part of. Molecules can have more than one functional group.

DEFINITION: *Functional group*

In organic chemistry a functional group is a specific group of atoms (and the bonds between them) that are responsible for the characteristic chemical reactions of those molecules.

In one group of organic compounds, called the **hydrocarbons**, the single, double and triple bonds between carbon atoms give rise to the alkanes, alkenes and alkynes, respectively. The double carbon-carbon bonds (in the alkenes) and triple carbon-carbon bonds (in the alkynes) are examples of functional groups.

In another group of organic compounds, called the **alcohols**, an oxygen and a hydrogen atom are bonded to each other to form the functional group (in other words an alcohol has an OH in it). All alcohols will contain an oxygen and a hydrogen atom bonded together in some part of the molecule. Table 4.1 summarises some of the common functional groups. We will look at these in more detail later in this chapter.

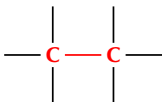
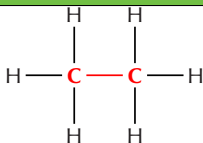
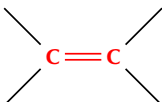
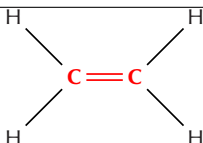
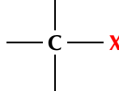
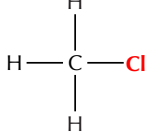
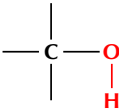
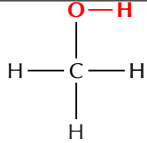
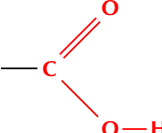
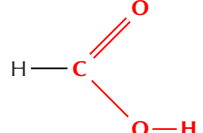
Name of group	Functional group	Example	Structural Formula
Alkane		Ethane	
Alkene		Ethene	
Alkyne		Ethyne	
Haloalkane/alkyl halide	 (X = F, Cl, Br, I)	Chloromethane	
Alcohol / alkanol		Methanol	
Carboxylic acid		Methanoic acid	

Table 4.1: Some functional groups of organic compounds.

There are some important points to note as we discuss functional groups:

- The beginning of a compound name (prefix) comes from the number of carbons in the longest chain:

meth-	1 carbon atom
eth-	2 carbon atoms
prop-	3 carbon atoms
but-	4 carbon atoms

- The end of a compound name (suffix) comes from the functional group, e.g. an alkane has the suffix -ane. Refer to the examples in Table 4.1.

For more information on naming organic molecules see Section 4.3.

Saturated and unsaturated structures

ESCK8

Hydrocarbons that contain only single bonds are called **saturated** hydrocarbons because each carbon atom is bonded to as many hydrogen atoms as possible. Figure 4.12 shows a molecule of ethane, which is a saturated hydrocarbon.

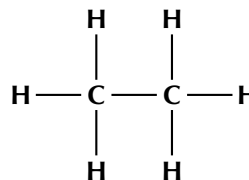


Figure 4.12: A saturated hydrocarbon, ethane.

DEFINITION: *Saturated compounds*

A **saturated** compound has no double or triple bonds (i.e. they have single bonds only). All carbon atoms are bonded to four other atoms.

Hydrocarbons that contain double or triple bonds are called **unsaturated** hydrocarbons because they don't contain as many hydrogen atoms as possible.

DEFINITION: *Unsaturated compounds*

An **unsaturated** compound contains double or triple bonds. A carbon atom may therefore be bonded to only two or three other atoms.

Figure 4.13 shows molecules of ethene and ethyne which are unsaturated hydrocarbons. If you compare the number of carbon and hydrogen atoms in a molecule of ethane and a molecule of ethene, you will see that the number of hydrogen atoms in ethene is *less* than the number of hydrogen atoms in ethane despite the fact that they both contain two carbon atoms. In order for an unsaturated hydrocarbon compound to become saturated, one of the two (or three) bonds in a double (or triple) bond has to be broken, and additional atoms added.

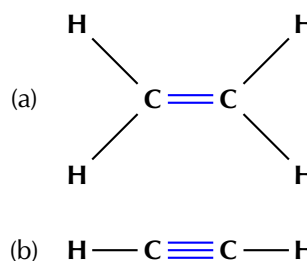


Figure 4.13: Unsaturated hydrocarbons: (a) ethene and (b) ethyne.

▶ See video: 27KX at www.everythingscience.co.za

Let us first look at a group of organic compounds known as the **hydrocarbons**.

DEFINITION: *Hydrocarbon*

An organic molecule which contains only carbon and hydrogen atoms with no other functional groups besides single, double or triple carbon-carbon bonds.

The hydrocarbons that we are going to look at are called **aliphatic compounds**. The aliphatic compounds are divided into *acyclic compounds* (chain structures) and *cyclic compounds* (ring structures). The chain structures are further divided into structures that contain only *single bonds* (**alkanes**), those that contain at least one *double bond* (**alkenes**) and those that contain at least one *triple bond* (**alkynes**).

Cyclic compounds (which will not be covered in this book) include structures such as a *cyclopentane ring*, which is found in insulating foam and in appliances such as fridges and freezers. Figure 4.14 summarises the classification of the hydrocarbons.

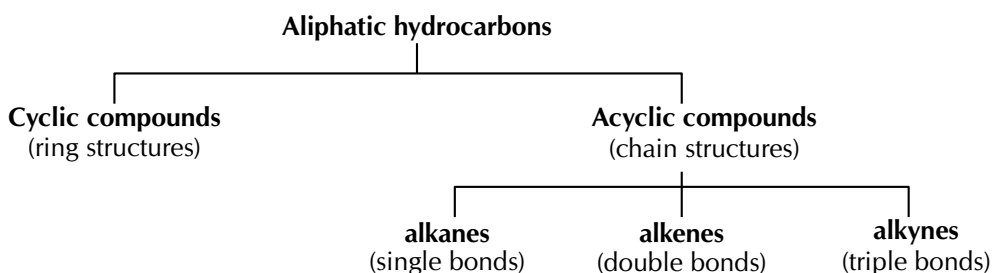


Figure 4.14: The classification of the aliphatic hydrocarbons.

We will now look at each of the acyclic, aliphatic hydrocarbon groups in more detail.

The alkanes

The alkanes are hydrocarbons that only contain *single covalent bonds* between their carbon atoms. This means that they are *saturated* compounds and are quite unreactive. The simplest alkane has only one carbon atom and is called **methane**. This molecule is shown in Figure 4.15.

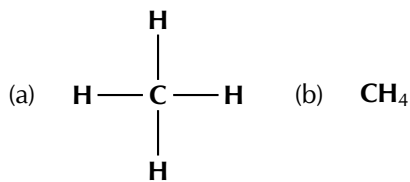


Figure 4.15: The (a) structural and (b) molecular formula representations of methane.

The second alkane in the series has two carbon atoms and is called **ethane**. This is shown in Figure 4.16.

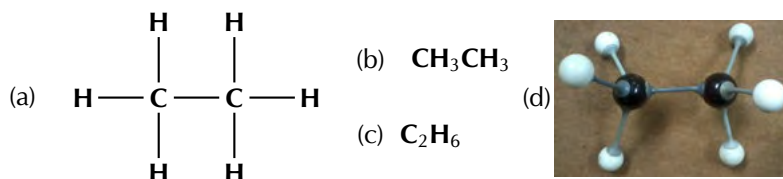
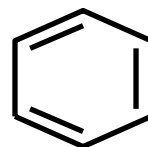


Figure 4.16: The (a) structural, (b) condensed structural and (c) molecular formula representations of ethane. (d) An atomic model of ethane.

FACT

An aliphatic compound is one that does not contain an aromatic ring:



benzene

The simplest aromatic compound is benzene. There are aliphatic cyclic compounds, but if a compound contains an aromatic ring it is an aromatic compound, not an aliphatic one.

FACT

Some fungi use alkanes as a source of carbon and energy. One fungus *amorphotheca resinae* (also known as kerosene fungus) prefers the alkanes used in aviation fuel, and this can cause problems for aircraft in tropical areas.



The third alkane in the series has three carbon atoms and is called **propane** (Figure 4.2).

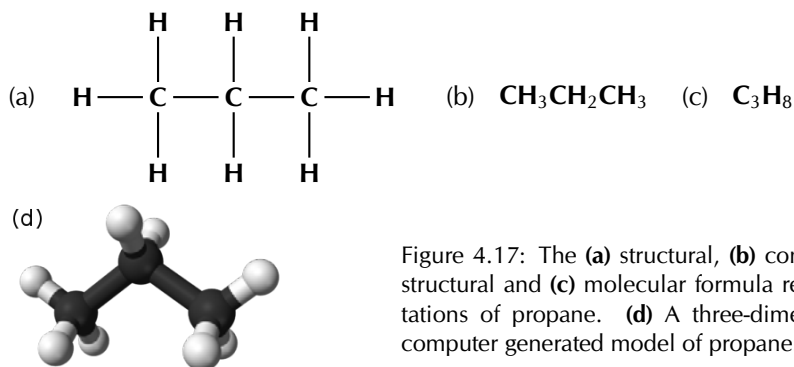


Figure 4.17: The (a) structural, (b) condensed structural and (c) molecular formula representations of propane. (d) A three-dimensional computer generated model of propane.

When you look at the molecular formula for each of the alkanes, you should notice a pattern developing. For each carbon atom that is added to the molecule, two hydrogen atoms are added. In other words, each molecule differs from the one before it by CH_2 . This is called a *homologous series*.

DEFINITION: *Homologous series*

A homologous series is a series of compounds with the same general formula. All molecules in this series will contain the same functional groups.

The general formula is similar to both the molecular formula and the condensed structural formula. The functional group is written as it would be in the condensed structural formula (to make it more obvious), while the rest of the atoms in the compound are written in the same style as the molecular formula. The alkanes have the **general formula: $\text{C}_n\text{H}_{2n+2}$** .

- The alkanes are the most important source of fuel in the world and are used extensively in the chemical industry.
- Alkanes that contain four or less carbon atoms are gases (e.g. methane and ethane).
- Others are liquid fuels (e.g. octane, an important component of petrol).

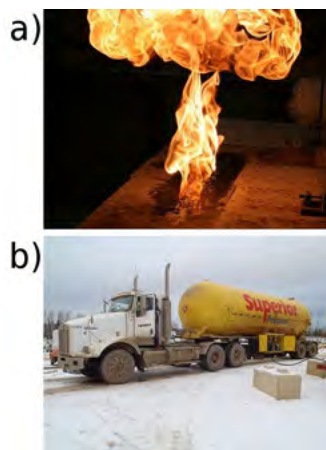


Figure 4.18: (a) Methane gas bubbles burning and (b) propane (under high pressure) being transported by truck.



Figure 4.19: Liquid fuels that contain octane are kept in tanks at petrol stations.

The alkenes

In the alkenes there must be at least one double bond between two carbon atoms. This means that they are *unsaturated* and are *more reactive* than the alkanes. The simplest alkene is ethene (also known as ethylene), which is shown in Figure 4.20.

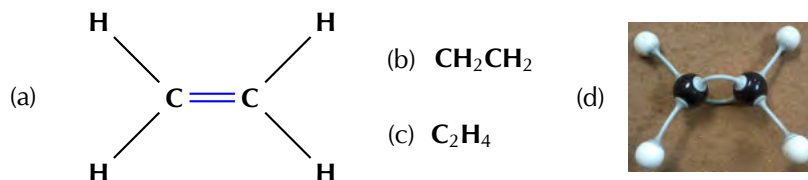


Figure 4.20: The (a) structural, (b) condensed structural and (c) molecular formula representations of ethene. (d) An atomic model of ethene.

As with the alkanes, the alkenes also form a homologous series. They have the **general formula**: C_nH_{2n} . The second alkene in the series would therefore be C_3H_6 . This molecule is known as propene (Figure 4.21).

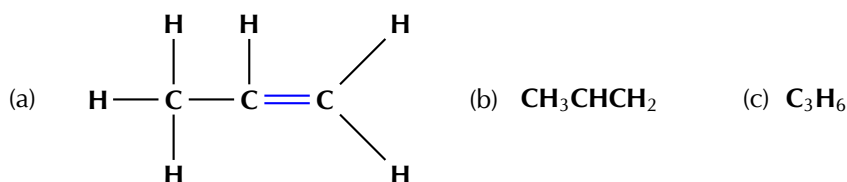


Figure 4.21: The (a) structural, (b) condensed structural and (c) molecular formula representations of propene.

There can be more than one double bond in an alkene as shown in Figure 4.22. The naming of these compounds is covered in Section 4.3, IUPAC naming and formulae.

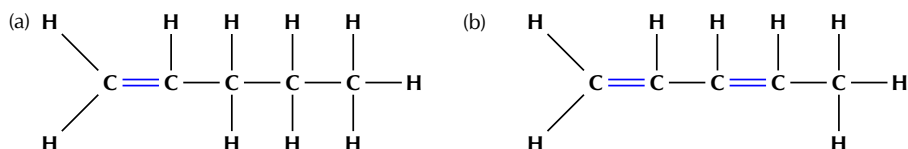


Figure 4.22: The structural representations of (a) pent-1-ene and (b) pent-1,3-diene.

The alkenes are more reactive than the alkanes because they are unsaturated. As with the alkanes, compounds that have four or less carbon atoms are gases at room temperature. Those with five or more carbon atoms are liquids.

The alkenes have a variety of uses:

- For example, ethene is a chemical compound used in plants to stimulate the ripening of fruits and the opening of flowers.

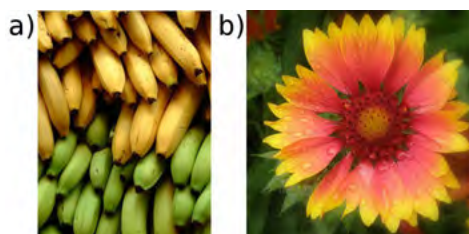


Figure 4.23: (a) Unripe (green) and ripe (yellow) bananas and (b) a flowering plant.

- Propene is an important compound in the petrochemicals industry. It is used to make polypropylene (see Section 4.7 for more information) and is also used as a fuel gas for other industrial processes.

TIP

Note that if an alkene has two double bonds, it is called a **diene**. If you don't understand the names of compounds, don't worry. We will go into more detail on this later in the chapter.

FACT

Acetylene is the industrial name for the organic compound ethyne. The raw materials that are needed to make acetylene are calcium carbonate and coal. An important use of acetylene is in oxyacetylene gas welding. The fuel gas burns with oxygen in a torch. Because the combustion of alkenes and alkynes is exothermic an incredibly high heat is produced, which is hot enough to melt metal.

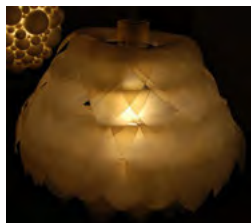


Figure 4.24: A lamp made of polypropylene. Propene is used to make polypropylene.

The alkynes

In the alkynes there must be at least one triple bond between two of the carbon atoms. They are unsaturated compounds and are therefore more reactive than alkanes. Their **general formula** is C_nH_{2n-2} . For example **but-1-yne** has the molecular formula C_4H_6 . The simplest alkyne is ethyne (Figure 4.25), also known as acetylene. Many of the alkynes are used to synthesise other chemical products.

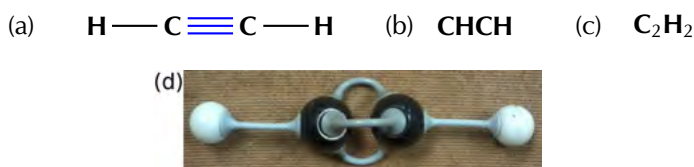


Figure 4.25: The (a) structural, (b) condensed structural and (c) molecular representations of ethyne (acetylene). (d) An atomic model of ethyne.

Remember that organic molecules do not need to be straight chains. They can have branched groups as well, as shown in Figure 4.26.

A summary of the relative reactivity and the homologous series that occur in the hydrocarbons is given in Table 4.2.

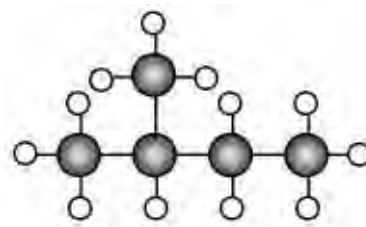


Figure 4.26: A methyl branched group on carbon 2 of butane (2-methylbutane).

Functional group	Homologous series	Reactivity
alkane	C_nH_{2n+2}	low reactivity
alkene	C_nH_{2n}	high reactivity
alkyne	C_nH_{2n-2}	high reactivity

Table 4.2: A summary of the homologous series of the hydrocarbons.

FACT

Liquid bromine is highly corrosive and toxic. Handle with extreme care!



Experiment: Saturated vs. unsaturated compounds

Aim:

To study the effect of bromine water and potassium permanganate on saturated and unsaturated compounds.

Apparatus:

WARNING!

Liquid bromine (required to make bromine water) is a highly volatile, corrosive and toxic compound. Please handle with care: wear the appropriate safety clothing including gloves, labcoat, safety glasses and mask. Work in a fumehood. If you do not have the apparatus to handle liquid bromine safely, use potassium permanganate only.

- cyclohexane, cyclohexene, bromine water ($\text{Br}_2(\text{aq})$), potassium permanganate (KMnO_4) in an acidic solution
- 4 glass containers (test tubes/beakers/shallow basins), two A4 sheets of paper
- 2 plastic pipettes

Method:

1. Label one piece of paper A and the other piece of paper B.
2. Place 20 ml of cyclohexane into a container and place the container on paper A.
3. Place 20 ml of cyclohexane into a container and place the container on paper B.
4. Repeat steps 2 and 3 with cyclohexene.
5. Take 12 ml of bromine water and add it to the beaker of cyclohexane on paper A. Observe any colour changes.
6. Repeat step 5 with the beaker of cyclohexene on paper A.
7. Take 12 ml of KMnO_4 and add it to the beaker of cyclohexane on paper B. Observe any colour changes.
8. Repeat step 7 with the beaker of cyclohexene on paper B.

Results:

Record your results in the table.

Cyclohexane is an *alkane*, cyclohexene is an *alkene*.

Compound	Initial colour	Solution added	Final colour
cyclohexane		bromine water	
cyclohexane		KMnO_4	
cyclohexene		bromine water	
cyclohexene		KMnO_4	

Questions:

- Which of these compounds (cyclohexane, cyclohexene) is saturated and which is unsaturated?
- What colour changes did you observe with the alkane compound?
- What colour changes did you observe with the alkene compound?
- Can you suggest a reason for the differences?

Discussion and conclusion:

Bromine water and KMnO_4 both have intense colours. Cyclohexane is a saturated, colourless liquid. When bromine water and KMnO_4 are added to the cyclohexane there is no reaction and the solution becomes the colour of the bromine water or KMnO_4 .

Cyclohexene is also a colourless liquid, but it is unsaturated. This results in a reaction with bromine water and with KMnO_4 . Cyclohexene will form a bromoalkane with bromine water. Bromoalkanes are colourless liquids and the solution will be colourless - liquid bromine is decolourised by cyclohexene. Similarly KMnO_4 will be decolourised by the cyclohexene.

Exercise 4 – 2: The hydrocarbons

1. Answer these questions on the hydrocarbons.
 - a) What is the difference between the alkanes, alkenes and alkynes?
 - b) Give the general formula for the alkynes

carbon atom. When the hydroxyl is bonded to a tertiary carbon atom it is a **tertiary alcohol**. Examples are given below.

- (a) **Primary Alcohols** (C atom at the end of a chain)
-OH (hydroxyl group) is bonded to a carbon atom that is bonded to only one other carbon atom

butan-1-ol

- (b) **Secondary Alcohols** (C atom inside a chain)
-OH group is bonded to a carbon atom that is bonded to two other carbon atoms

butan-2-ol

- (c) **Tertiary Alcohols** (needs a branched chain)
-OH group is bonded to a carbon atom that is bonded to three other carbon atoms

2-methylpropan-2-ol

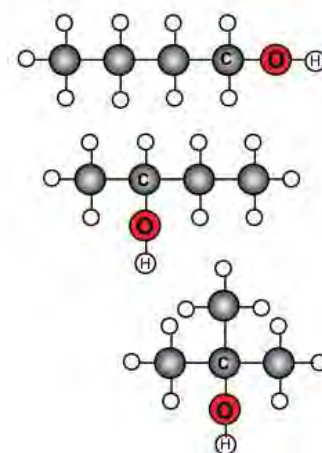


Figure 4.29: A (a) primary (butan-1-ol), (b) secondary (butan-2-ol) and (c) tertiary (2-methylpropan-2-ol) alcohol.

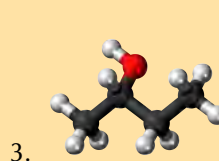
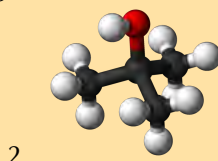
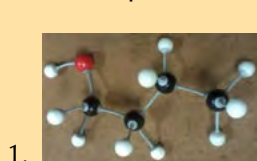
The alcohols have a number of different uses:

- methylated spirits is ethanol with methanol added
- all alcohols are toxic, but in low concentrations ethanol can be used in alcoholic drinks
- ethanol is the only alcohol used in alcoholic drinks
- ethanol is used as an industrial solvent
- methanol and ethanol can both be used as a fuel and they burn more cleanly than petrol or diesel.
- ethanol is used as a solvent in medical drugs, perfumes and plant essences
- ethanol is an antiseptic

Exercise 4 – 3: The alcohols

Give the structural and condensed structural formula for the following alcohols. State, with reasons, whether the compound is a primary, secondary, or tertiary alcohol.

(Note: a black ball represents a carbon atom, a white ball represents a hydrogen atom, and a red ball represents an oxygen atom)



4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27M3 2. 27M4 3. 27M5



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FACT



Methanol is toxic. If ingested it forms formic acid or formate salts, which damage the central nervous system and can cause *blindness, coma or death*.

FACT

CFC stands for *chlorofluorocarbons*. Due to their low toxicity and low reactivity, CFCs were widely used in refrigeration and as propellants in aerosols. However, the low reactivity means that CFCs can get into the upper atmosphere where they are degraded by UV light and damage the ozone layer.

**FACT**

Chloroform (CHCl_3) was used as an anesthetic for years. However, aside from causing dizziness, fatigue and headaches, it was discovered to be toxic, often fatally so. Even non-fatal doses can cause damage to the kidneys and liver. Chloroform can sometimes be found in cough syrups, although not often anymore.



Alkyl halides

ESCKC

Alkyl halides are hydrocarbons with one hydrogen atom replaced by a halogen atom (F, Cl, Br, I). The **alkyl** is due to the fact that a hydrocarbon branched group has the suffix *-yl* and is one of the three hydrocarbons: alkanes, alkenes or alkynes. These alkyl groups contain one or more halogen atoms, which leads to the name alkyl halides. Our focus will be on the alkane alkyl halides also known as the haloalkanes (or halogenoalkanes) (see Table 4.1).

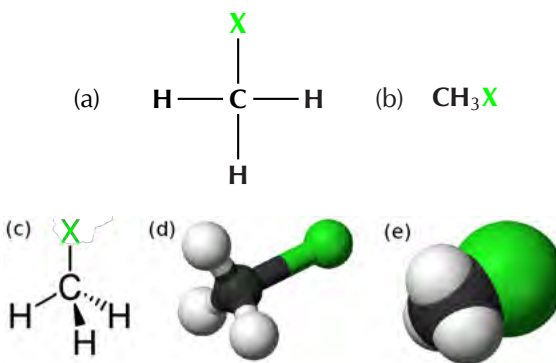


Figure 4.30: Representations of a halomethane where X can be F, Cl, Br or I: (a) structural, (b) molecular formula, (c) 3-D line drawing, (d) 3-D ball and stick model and (e) 3-D space-filling model.

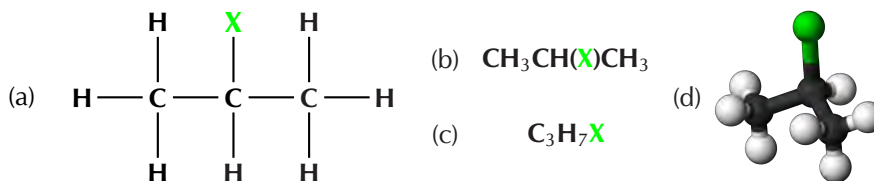


Figure 4.31: Representations of 2-halopropane where X can be F, Cl, Br or I: (a) structural, (b) condensed structural, (c) molecular formula and (d) a ball and stick model.

Note that the halogen atom is called a *substituent*.

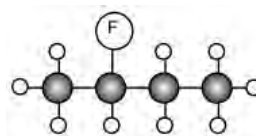


Figure 4.32: A fluorine atom as a substituent on carbon 2 of butane (2-fluorobutane). Remember the branched chain shown in Figure 4.26. That branched chain is also called a substituent.

DEFINITION: *Substituent*

A substituent is an *atom or group of atoms* bonded to a carbon chain. This can be an inorganic atom (e.g. halogen) or an alkyl group that is shorter than the main group.

An organic compound is always named in accordance with the longest chain of carbon atoms that contains the functional group. If the substituent is an alkyl group it is known as a branched chain.

Some uses of haloalkanes include:

- in fire extinguishers
- as aerosol propellants
- in refrigeration
- generating foamed plastics
- solvents in dry cleaning processes (not actually *dry*, but no water is required)

Extension

Chloroform

Haloalkanes can contain more than one halogen atom. Chloromethanes are substances that can be used as anaesthetics during operations. One example is trichloromethane, also known as *chloroform* (Figure 4.33).

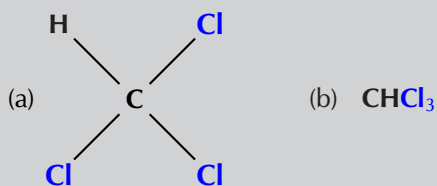


Figure 4.33: The (a) structural and (b) molecular formula representations of trichloromethane (chloroform).

Exercise 4 – 4: Haloalkanes

- Answer these questions on the haloalkanes.
 - Give the general formula for the haloalkanes with only one halogen atom
 - Are haloalkanes saturated compounds?
- Draw the structural formulae for:
 - $\text{CH}_2(\text{Br})\text{CH}_2\text{CH}_3$
 - $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{CH}_3$
 - $\text{CH}_2(\text{F})\text{CH}_3$
- More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27M6 2. 27M7



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Carbonyl-containing compounds

ESCKD

The carbonyl group consists of a carbon atom that is joined to an oxygen by a double bond (see Figure 4.34).

In Figure 4.34 R' and R are used to represent the rest of the atoms in the molecule. For example R could represent an alkyl chain, or a hydrogen atom.

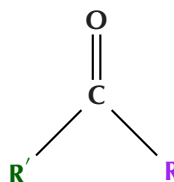


Figure 4.34: A compound containing a carbonyl group.

Aldehydes and ketones

If the functional group is on the *end* of the carbon chain, the organic compound is called an **aldehyde** (Figure 4.35 (a)). Being at the *end* of the chain means that R' or R represents a hydrogen atom. The simplest aldehyde is *methanal*.

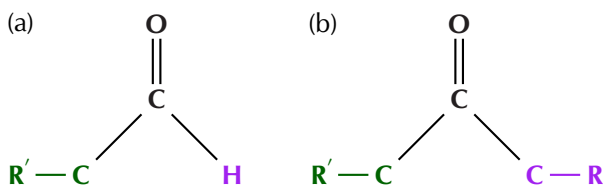


Figure 4.35: (a) An aldehyde and (b) a ketone.

TIP

Note that the condensed structural formula for an aldehyde ends in **CHO** not **COH**. This is because **COH** could be confused with the hydroxyl (**–OH**) group of an alcohol.

FACT

The molecular formulae representations for butanal and butanone are identical (**C₄H₈O**). This is why structural and condensed structural representations are necessary.

The aldehyde containing 4 carbon atoms, butanal, is illustrated in Figure 4.2. In this example **R** represents **H** and **R'** represents **CH₃CH₂CH₂**.

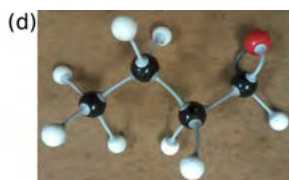
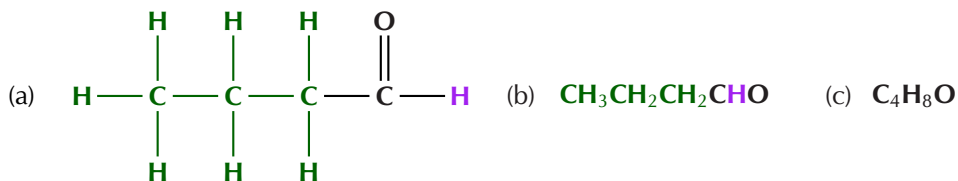


Figure 4.36: The (a) structural, (b) condensed structural and (c) molecular formula representations of butanal. (d) An atomic model of butanal.

Some uses of aldehydes include:

- in resins (over 6 million tons of formaldehyde are produced per year)
- in the production of plasticisers and alcohols used in detergents
- in perfumes and flavourants

If the carbonyl group is in the middle of the carbon chain, the compound is called a **ketone** (Figure 4.35 (b)). Being in the *middle* of the chain means that **R'** and **R** *cannot* represent **H**. The simplest ketone is *propanone* (also known as acetone, the compound in nail varnish remover), which contains three carbon atoms. The ketone containing 4 carbon atoms, butanone, is illustrated in Figure 4.2.

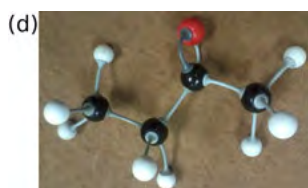
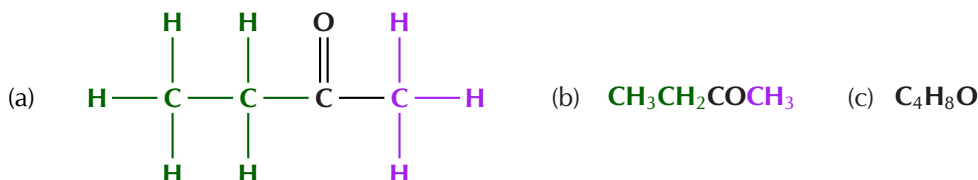


Figure 4.37: The (a) structural, (b) condensed structural and (c) molecular formula representations of butanone. (d) An atomic model of butanone.

Some uses of ketones include:

- as solvents
- in the production of polymers
- in the production of pharmaceuticals

The **general formula** for both the aldehydes and ketones can be written as: **C_nH_{2n}O**. This means that they cannot be told apart from their general formula alone. There are more complex general formulas that allow aldehydes and ketones to be distinguished, but they are not covered in this book.

Carboxylic acids

Carboxylic acids are organic acids that are characterised by having a carboxyl group, written as **–COOH**. In a carboxyl group a carbon atom is double-bonded to an oxygen atom (carbonyl group), and it is also bonded to a hydroxyl group (**R**). The simplest carboxylic acid, methanoic acid, is shown in Figure 4.38 and ethanoic acid is shown in Figure 4.2.

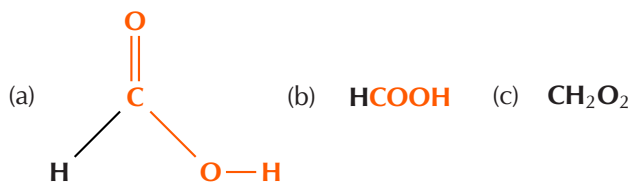


Figure 4.38: The (a) structural, (b) condensed structural and (c) molecular formula representations of methanoic acid.

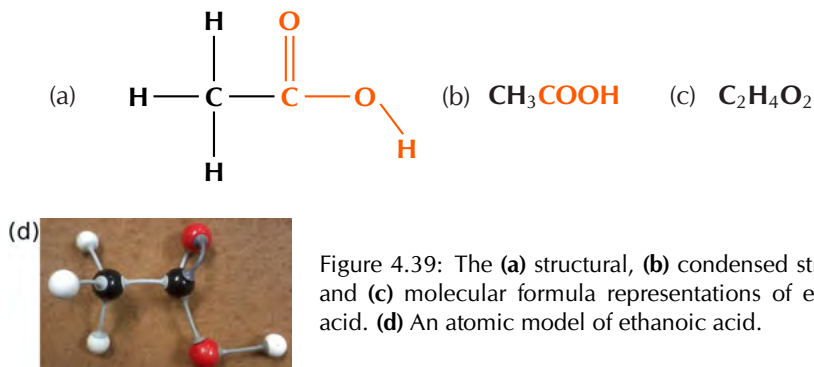


Figure 4.39: The (a) structural, (b) condensed structural and (c) molecular formula representations of ethanoic acid. (d) An atomic model of ethanoic acid.

FACT

A certain type of ant, called formicine ants, manufacture and secrete formic acid, which is used to defend themselves against other organisms that might try to eat them.



Carboxylic acids are widespread in nature. Methanoic acid (also known as *formic acid*) has the formula HCOOH and is found in insect stings. Ethanoic acid (CH_3COOH), or *acetic acid*, is the main component of vinegar. More complex organic acids also have a variety of different functions. Benzoic acid for example, is used as a food preservative. Carboxylic acids have the **general formula**: $\text{C}_n\text{H}_{2n+1}\text{COOH}$.

Ethanoic acid can be produced through the oxidation of ethanol upon exposure to the oxygen in air. This is why wine that is left too long can taste acidic. Wine can easily go sour if exposed to the oxygen molecules (O_2) in the air, especially if the weather is warm.

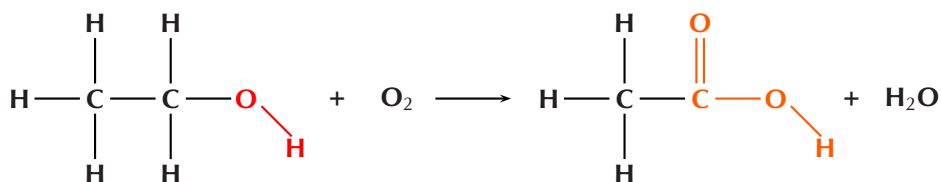
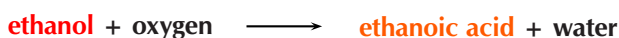
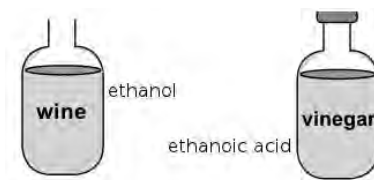
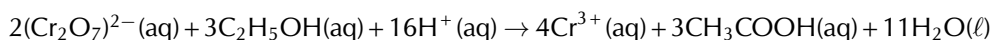


Figure 4.40: The oxidation of wine.

The oxidation of ethanol to ethanoic acid can also be seen in the reaction of ethanol with potassium dichromate:



The colour change that occurs is shown in the image below and the following video:



▶ See video: 27M8 at www.everythingscience.co.za

Figure 4.41: The different colours of potassium dichromate (left) and potassium dichromate and ethanol (right). (Screenshot taken from video by beggar098 on YouTube)

TIP

Esters will be dealt with in much greater detail in Section 4.7.

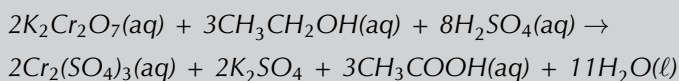
Case study: Breathalysers

Read the following extract taken from HowStuffWorks (12/08/13):

The **Breathalyzer** device contains:

- A system to sample the breath of the suspect
- Two glass vials containing the chemical reaction mixture
- A system of photocells connected to a meter to measure the color change associated with the chemical reaction

To measure alcohol, a suspect breathes into the device. The breath sample is bubbled in one vial through a mixture of sulfuric acid, potassium dichromate, silver nitrate and water. The principle of the measurement is based on the following chemical reaction:



In this reaction:

1. The **sulfuric acid removes the alcohol from the air** into a liquid solution.
2. The **alcohol reacts with potassium dichromate** to produce: chromium sulfate, potassium sulfate, acetic acid, water

The silver nitrate is a **catalyst**, a substance that makes a reaction go faster without participating in it. The sulfuric acid, in addition to removing the alcohol from the air, also might provide the acidic condition needed for this reaction.

During this reaction, the reddish-orange dichromate ion **changes color** to the green chromium ion when it reacts with the alcohol; the degree of the color change is directly related to the level of alcohol in the expelled air. To determine the amount of alcohol in that air, the reacted mixture is compared to a vial of unreacted mixture in the **photocell system**, which produces an **electric current** that causes the needle in the meter to move from its resting place. The operator then rotates a knob to bring the needle back to the resting place and reads the level of alcohol from the knob – the more the operator must turn the knob to return it to rest, the greater the level of alcohol.

Break into groups of three or four. Research breathalysers and then report your information to the class.

Make sure to cover the following areas:

- The effect of alcohol on the body
- The effect of alcohol on reaction times
- The origins of the breathalyser
- The term *mouth alcohol* and its effect on breathalyser tests.

Esters

When an alcohol reacts with a carboxylic acid, an **ester** is formed. Most esters have a characteristic smell. In the reaction a molecule of water is removed from the two compounds and a new bond is formed between what remains of the alcohol and the carboxylic acid. A **catalyst** is required in this reaction, in this case it must be an inorganic acid (e.g. H_2SO_4). An example is shown in Figure 4.42.

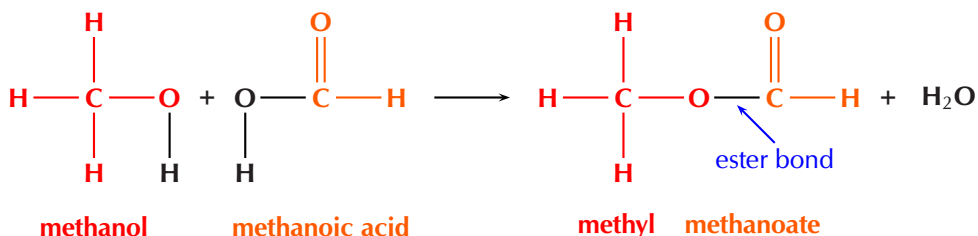


Figure 4.42: The formation of an ester and water from an alcohol and carboxylic acid.

FACT

The esterification process is reversible with large quantities of water (although it can be slow). In an acidic environment the reaction speeds up. Reversible reactions are covered in greater detail in Chapter 8.

The esterification process with methanol and methanoic acid is shown with atomic models in Figure 4.43. Esters have the **general formula: $\text{C}_n\text{H}_{2n}\text{O}_2$** . This general formula can also be applied to carboxylic acids, but the more complex general formula for esters alone is not covered in this book.

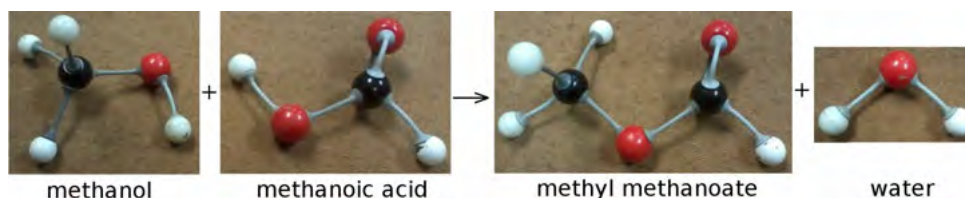


Figure 4.43: The esterification process of methanol and methanoic acid to methyl methanoate and water, shown with three-dimensional model kits.

Some common uses for esters are:

- in cosmetics and beauty products because they typically have a fruity smell, making them good as artificial flavourants and scents
- in nail varnish remover and model plane glue
- as solvents for non-water soluble compounds (e.g. oils, resins) because the ester of a specific carboxylic acid will be less water soluble than the carboxylic acid
- as plasticisers because esters can make a compound less brittle, and more flexible

Exercise 4 – 5: Carbonyl compounds

- Answer these questions on carbonyl compounds.
 - What other functional group does a carboxylic acid have in addition to a carbonyl group?
 - What is the main difference between aldehydes and ketones
 - What two reactants are required to make an ester?
 - How is ethanoic acid produced?
- Draw the structural formulae for each of the following compounds. What series does each compound belong to?
 - CH_3COCH_3
 - $\text{CH}_3\text{CH}_2\text{COOH}$
 - $\text{CH}_3\text{CH}_2\text{CHO}$
 - $\text{CH}_3\text{COOCH}_3$
- More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 27M9
- 27MB



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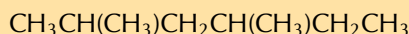


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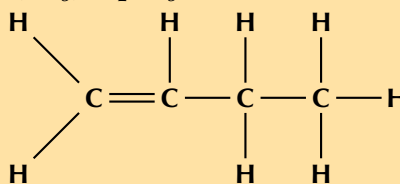
Exercise 4 – 6: Functional groups

1. Write down the:

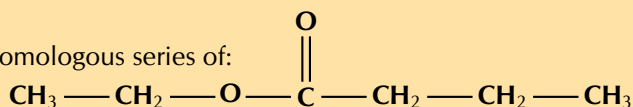
a) name of the homologous series of:



b) general formula of:



c) name of the homologous series of:



d) structural formula of: $\text{CH}_3\text{CH}_2\text{COOH}$

2. Look at the list of organic compounds in the table below:

Organic compound	Homologous series
$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	
$\begin{array}{ccccccc} & \text{H} & & \text{O} & & \text{H} & \text{H} & \text{H} \\ & & & & & & & \\ \text{H} - & \text{C} - & \text{C} - & \text{O} - & \text{C} - & \text{C} - & \text{C} - & \text{H} \\ & & & & & & \\ & \text{H} & & & \text{H} & \text{H} & \text{H} \end{array}$	
CH_3CHO	
$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} \\ & & & & & \\ \text{H} - & \text{C} - & \text{C} - & \text{C} - & \text{H} \\ & & & & & \\ & \text{H} & & \text{H} & & \text{H} \end{array}$	
$\text{CH}_3\text{CCCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	

a) Complete the table by identifying the functional group of each compound.

b) Give the structural representation of the compounds represented by condensed structural formulae.

3. A chemical reaction takes place and ethyl methanoate is formed.

a) Identify the homologous series to which ethyl methanoate belongs?

b) Name the two types of reactants used to produce this compound in a chemical reaction.

c) Give the structural formula of ethyl methanoate ($\text{HCOOCH}_2\text{CH}_3$).

4. The following reaction takes place: $\text{CH}_3\text{CHCH}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CH}_3\text{CH}_2\text{CH}_3(\text{g})$

a) Give the name of the homologous series of the organic compound in the reactants.

b) What is the name of the homologous series of the product?

c) Which compound in the reaction is a saturated hydrocarbon?

5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27MC 2. 27MD 3. 27MF 4. 27MG



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It is possible for two organic compounds to have the same *molecular formula* but a different *structural formula*. Look at the two organic compounds that are shown in Figure 4.44 for example.

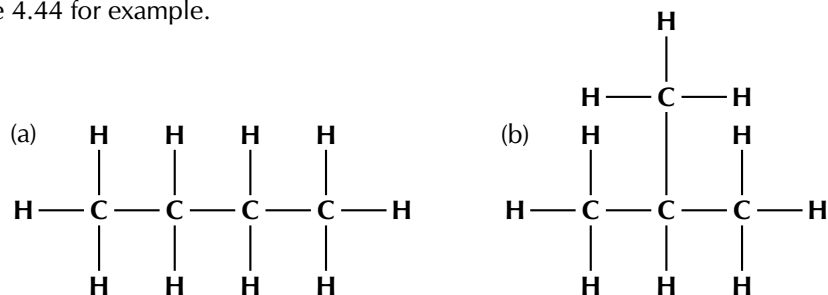


Figure 4.44: Isomers of a 4-carbon organic compound with the molecular formula C_4H_{10} . (a) butane and (b) 2-methylpropane.

If you were to count the number of carbon and hydrogen atoms in each compound, you would find that they are the same. They both have the same molecular formula C_4H_{10} , but their structure is different and so are their properties. Such compounds are called **isomers**.

DEFINITION: *Isomer*

In chemistry, isomers are molecules with the same molecular formula but different structural formula.

Isomers are molecules with the same molecular formula and often (though not always) with the same kinds of chemical bonds between atoms, but with the atoms arranged differently.

The isomers shown in Figure 4.44 differ only in the location of the carbon atoms. The functional groups are the same but butane has all four carbons in one chain, while 2-methylpropane has three carbons in the longest chain and a methyl group attached to the second carbon in the chain. It is also possible to have *positional* isomers (Figure 4.45). In this case the $-OH$ functional group can be on different carbon atoms, for example carbon 1 for pentan-1-ol, on carbon 2 for pentan-2-ol or on carbon 3 for pentan-3-ol.

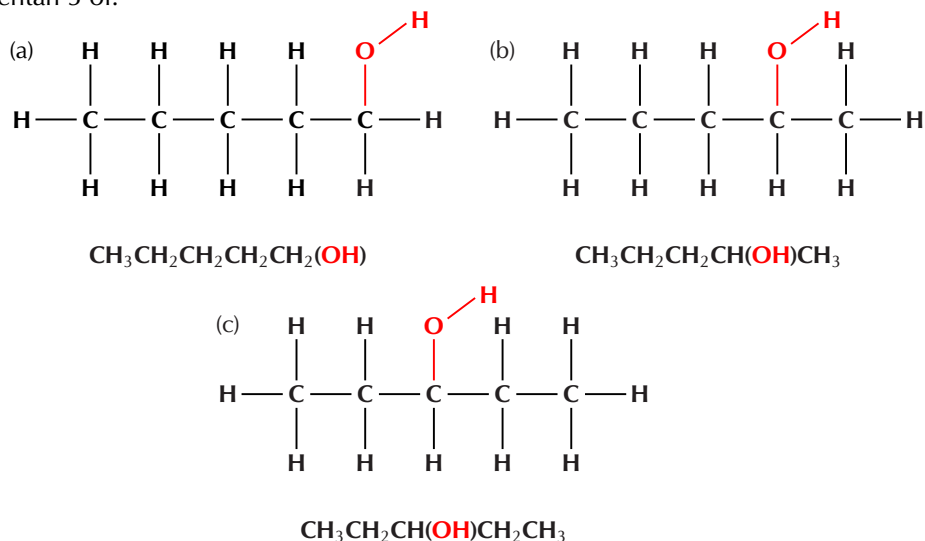


Figure 4.45: The structural and condensed structural representations of the isomers (a) pentan-1-ol, (b) pentan-2-ol and (c) pentan-3-ol.

FACT

Both butane and 2-methylpropane (isobutane) are used in camping stoves and lighters.



Positional isomers are also found in esters. If one ester was made from ethanol and hexanoic acid and another was made from hexanol and ethanoic acid the two esters produced are isomers (see Figure 4.46).

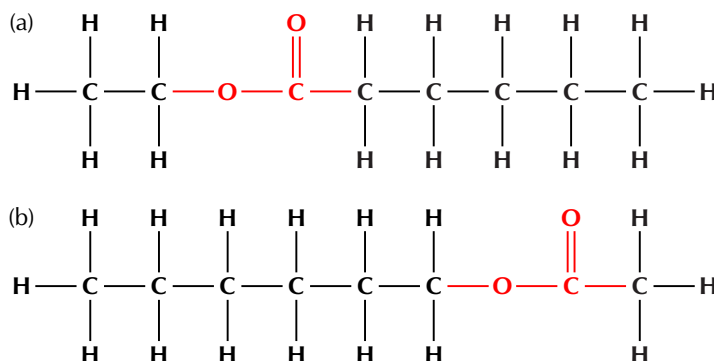


Figure 4.46: The structural representations of the isomers (a) ethyl hexanoate and (b) hexyl ethanoate.

It is important to note that molecules need not have the same functional groups to be isomers. For example propanone (commonly known as acetone) and propanal (Figure 4.47) have the same molecular formula (C_3H_6O) but different functional groups and properties (Table 4.3). These types of isomers are **functional isomers**.

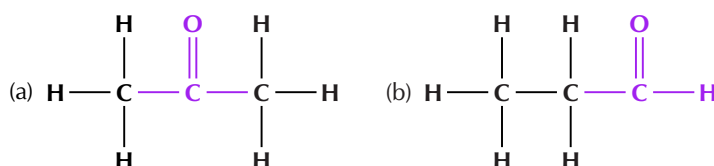


Figure 4.47: Two C_3H_6O isomers (a) propanone and (b) propanal.

Name	Functional Group	Melting point ($^{\circ}C$)	Boiling point ($^{\circ}C$)	Reactivity
propanone	ketone	-95	56	less reactive
propanal	aldehyde	-81	48	more reactive

Table 4.3: Some properties of the isomers propanone and propanal.

Heptanoic acid and butyl propanoate (Figure 4.48) are another example of *functional isomers* (containing different functional groups). They both have the same molecular formula of $C_7H_{14}O_2$ but have different functional groups and different properties (Table 4.4).

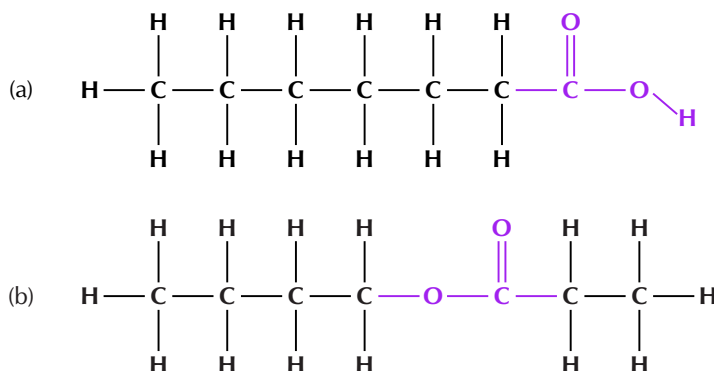


Figure 4.48: The structural representations of the isomers (a) heptanoic acid and (b) butyl propanoate.

Name	Functional group	Melting point (°C)	Boiling point (°C)
heptanoic acid	carboxylic acid	-7,5	223
butyl propanoate	ester	-89,5	145

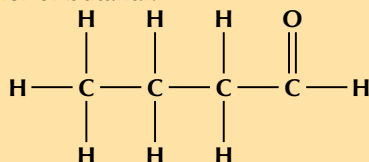
Table 4.4: The different physical properties of the isomers heptanoic acid and butyl propanoate.

Exercise 4 – 7: Isomers

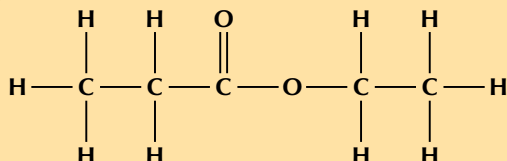
1. Match the organic compound in Column A with its isomer in Column B:

	Column A	Column B
1	$\text{CH}_3\text{CH}_2\text{CH}_2(\text{OH})$	$\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$
2	<pre> H H H H H — C — C — C — C — H H H H H </pre>	<pre> O CH₃ — CH₂ — CH₂ — C — CH₃ </pre>
3	<pre> H O H H — C — — C — — C — H H — C — H H — C — H H H </pre>	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$

2. a) Give the ketone isomer of butanal:



b) Give a carboxylic acid that is an isomer of:



3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [27MH](#) 2a. [27MJ](#) 2b. [27MK](#)



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4.3 IUPAC naming and formulae

ESCKG

What is IUPAC naming?

ESCKH

In order to give compounds a name, certain rules must be followed. When naming organic compounds, the IUPAC (International Union of Pure and Applied Chemistry) nomenclature (naming scheme) is used. This is to give consistency to the names. It also enables every compound to have a unique name, which is not possible with the common names used (for example in industry). We will first look at some of the steps

TIP

Molecules can contain both double or triple bonds and other functional groups (e.g. an alkene and an alcohol functional group in one molecule - propenol). However, all molecules explored in this book will contain only single carbon-carbon bonds when combined with other functional groups.

that need to be followed when naming a compound, and then try to apply these rules to some specific examples.

A good general rule to follow is to start at the **end** (the suffix) and work backwards (from right to left) in the name.

1. Recognise the *functional group* in the compound. This will determine the suffix of the name (see Table 4.5).
2. Find the longest continuous carbon chain that contains the functional group (it won't always be a *straight* chain) and count the number of carbon atoms in this chain.
This number will determine the prefix (the *beginning*) of the compound's name (see Table 4.6).

3. Number the carbons in the longest carbon chain (**Important:** If the molecule is not an alkane (i.e. has a functional group) you need to start numbering so that the *functional group* is on the carbon with the *lowest possible number*). Start with the carbon at the end closest to the functional group.

4. Look for any branched groups:

- Name them by counting the number of carbon atoms in the branched group and referring to Table 4.6, these groups will all end in -yl.
- Note the position of the group on the main carbon chain. If there is more than one of the same type of branched group then both numbers must be listed (e.g. 2,4 -) and one of the prefixes listed in Table 4.7 must be used. **Important:** If the molecule is an *alkane* the branched group must be on the carbon with the *lowest possible number*.
- The branched groups must be listed before the name of the main chain in alphabetical order (ignoring di/tri/tetra).

If there are no branched groups this step can be ignored.

Number	prefix
2	di-
3	tri-
4	tetra-

Table 4.7: Prefixes for multiple substituents with the same name. These apply to multiple functional groups as well.

5. For the alkyl halides the halogen atom is treated in much the same way as branched groups:
 - To name them take the name of the halogen atom (e.g. iodine) and replace the -ine with -o (e.g. iodo).

Functional group	suffix
alkane	-ane
alkene	-ene
alkyne	-yne
alcohol	-ol
aldehyde	-al
ketone	-one
carboxylic acid	-oic acid
ester	-oate

Table 4.5: The suffix associated with various functional groups.

Carbon atoms	prefix
1	meth-
2	eth-
3	prop-
4	but-
5	pent-
6	hex-
7	hept-
8	oct-
9	non-
10	dec-

Table 4.6: The prefix of a compound's name is determined by the number of carbon atoms in the longest chain that contains the functional group.

- Give the halogen atom a number to show its position on the carbon chain. If there is more than one halogen atom the numbers should be listed and a prefix should be used (e.g. 3,4-diiodo- or 1,2,2-trichloro-). See Table 4.7 for a list of the prefixes.

Halogen	name
fluorine	fluoro
chlorine	chloro
bromine	bromo
iodine	iodo

Table 4.8: Naming halogen atoms in organic molecules.

- The halogen atoms must be listed before the name of the main chain in alphabetical order (ignore di/tri/tetra).

If there are no halogen atoms this step can be ignored.

6. Combine the elements of the name into a single word in the following order:

- branched groups/halogen atoms in alphabetical order (ignoring prefixes)
- prefix of main chain
- name ending according to the functional group and its position on the longest carbon chain.

Naming hydrocarbons

ESCKJ

Naming alkanes

The suffix for an alkane is -ane.

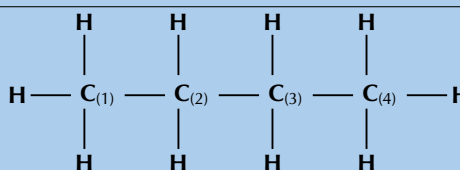
Worked example 1: Naming the alkanes

QUESTION

Give the IUPAC name for the following compound:

Note: The numbers attached to the carbon atoms would not normally be shown.

The carbon atoms have been numbered to help you to name the compound.



SOLUTION

Step 1: Identify the functional group

The compound is a hydrocarbon with single bonds between the carbon atoms. It is an alkane and will have a suffix of -ane.

Step 2: Find the longest carbon chain

There are four carbon atoms in the longest chain. The prefix of the compound will be but-.

Step 3: Number the carbon atoms in the longest chain

The numbering has been done for you here.

Step 4: Look for any branched group, name them and give their position on the carbon chain

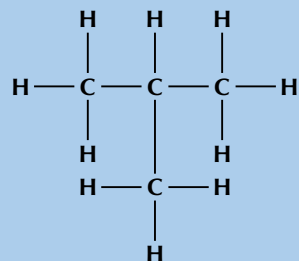
There are no branched groups in this compound.

Step 5: Combine the elements of the name into a single word

The name of the compound is **butane**.

Worked example 2: Naming the alkanes**QUESTION**

Give the IUPAC name for the following compound:

**SOLUTION****Step 1: Identify the functional group**

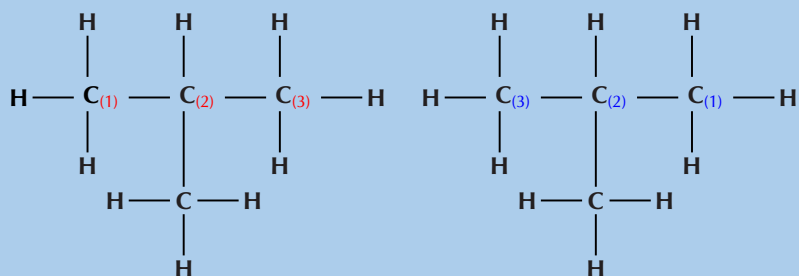
The compound is a hydrocarbon with single bonds between the carbon atoms. It is an alkane and will have the suffix -ane.

Step 2: Find the longest carbon chain

There are three carbon atoms in the longest chain. The prefix for this compound is prop-.

Step 3: Number the carbons in the carbon chain

If we start at the carbon on the left, we can number the atoms as shown in red (left). If we start at the carbon on the right, we can number the atoms as shown in blue (right).

**Step 4: Look for any branched groups, name them and give their position on the carbon chain**

There is a branched group attached to the second carbon atom. In this case the methyl group is on carbon 2 regardless of which side you number the longest chain from.

This group has the formula CH_3 , which is methane without a hydrogen atom. However, because it is not part of the main chain, it is given the suffix -yl (i.e. methyl). The position of the methyl group comes just before its name (see the next step).

Step 5: Combine the elements of the compound's name into a single word in the order of branched group; prefix; name ending according to the functional group

The compound's name is **2-methylpropane**.

Worked example 3: Naming the alkanes

QUESTION

Draw the semi-structural structural and condensed structural formula for the organic compound **2,2,4-trimethylhexane**

SOLUTION

Step 1: Identify the functional group

The name ends in -ane therefore the compound is an alkane.

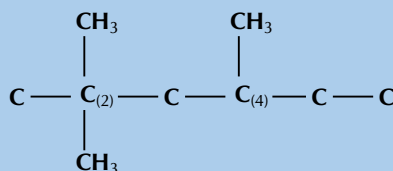
Step 2: Determine the number of carbon atoms in the longest chain

The longest chain has the prefix hex-.
There are therefore 6 carbon atoms in the longest chain.



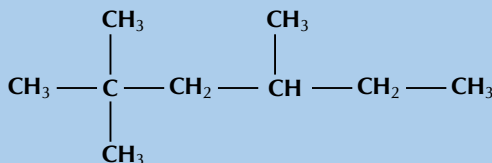
Step 3: Look for any branched groups and place them on the structure

The compound is **2,2,4-trimethylhexane**.
Therefore there are three branched groups.
Two on carbon 2 and one on carbon 4.



Step 4: Combine this information and add the hydrogen atoms

Carbon atoms can have four single bonds.
Therefore wherever a carbon atom has less than four bonds draw in hydrogen atoms until there are four bonds.



Step 5: Condense the structural formula

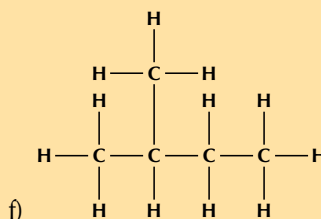
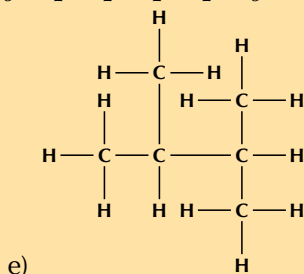
First condense the main chain: $\text{CH}_3\text{CCH}_2\text{CHCH}_2\text{CH}_3$

Then add the side chains (in brackets) on the relevant carbon atoms:



Exercise 4 – 8: Naming alkanes

- Give the structural formula for each of the following alkanes
 - Octane
 - Propane
 - 2-methylpropane
 - 3-ethylpentane
- Give the IUPAC name for each of the following alkanes:
 - $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
 - $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 - CH_3CH_3



3. More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.

1. 27MM 2. 27MN



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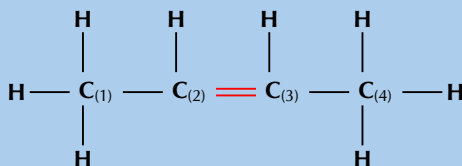
Naming alkenes

The suffix for an alkene is -ene.

Worked example 4: Naming the alkenes

QUESTION

Give the IUPAC name for the following compound:



SOLUTION

Step 1: Identify the functional group

The compound has a double carbon-carbon bond and is an alkene. It will have the suffix -ene.

Step 2: Find the longest carbon chain containing the functional group

The functional group is a double bond, so the longest chain must contain the double bond. There are four carbon atoms in the longest chain and so the prefix for this compound will be but-.

Step 3: Number the carbon atoms

Remember that the carbon atoms must be numbered so that the functional group is at the lowest numbered carbon atom possible. In this case, it doesn't matter whether we number the carbons from the left to right, or from the right to left. The double bond will still fall between the second and third carbon atoms.

Step 4: Look for any branched groups, name them and give their position on the carbon chain

There are no branched groups in this molecule.

Step 5: Combine the elements of the name into a single word in the following order: branched groups; prefix; name ending according to the functional group and its position along the longest carbon chain

The name of this compound is **but-2-ene** or **2-butene**.

Worked example 5: Naming the alkenes

QUESTION

Draw the structural and molecular formula for the organic compound

3-methylbut-1-ene

SOLUTION

Step 1: Identify the functional group

The suffix -ene means that this compound is an alkene and there must be a double bond in the molecule. The number 1 immediately before the suffix means that the double bond must be at the first carbon in the chain (but-1-ene).

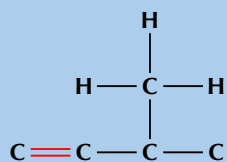
Step 2: Determine the number of carbon atoms in the longest chain containing the functional group

The prefix for the compound is but- so there must be four carbons in the longest chain containing the double bond.

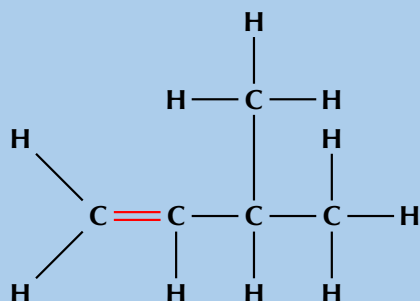


Step 3: Look for any branched groups

There is a methyl group at the third carbon atom in the chain. Count from the left so that the double bond carbon is the first carbon atom.



Step 4: Combine this information and add the hydrogen atoms



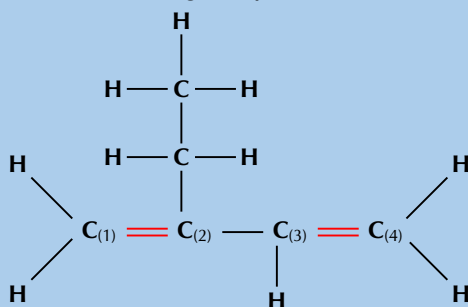
Step 5: Reduce the structural formula to the molecular formula

There are 5 carbon atoms and 10 hydrogen atoms so the molecular formula is C_5H_{10} . (Remember that there is no *structural* information given by the molecular formula)

Worked example 6: Naming the alkenes

QUESTION

Give the IUPAC name for the following compound:



SOLUTION

Step 1: Identify the functional group

The compound is an alkene and will have the suffix -ene. There is a double bond between the first and second carbons and also between the third and fourth carbons. The organic compound therefore contains '1,3-diene'.

Step 2: Find the longest carbon chain containing the functional group, and number the carbon atoms

Remember that the main carbon chain must contain both the double bonds. There are four carbon atoms in the longest chain containing the double bonds and so the prefix for this compound will be but-. The carbon atoms are already numbered 1 to 4 in the diagram.

Step 3: Look for any branched groups, name them and give their position on the carbon chain

There is an ethyl group on the second carbon.

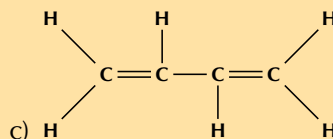
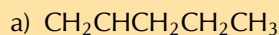
Note that if we had numbered from the right to left the suffix would still have been 1,3-diene, however the ethyl group would have been on the third carbon. So we had to number left to right.

Step 4: Combine the elements of the name into a single word in the following order: branched groups; prefix; name ending according to the functional group and its position along the longest carbon chain

The name of this compound is **2-ethylbut-1,3-diene**.

Exercise 4 – 9: Naming alkenes

1. Give the IUPAC name for each of the following alkenes:



2. Give the structural formula for each of the following alkenes:

a) ethene

b) hex-1-ene

c) hept-3-ene

d) 4-ethyloct-3-ene

3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27MP 2. 27MQ



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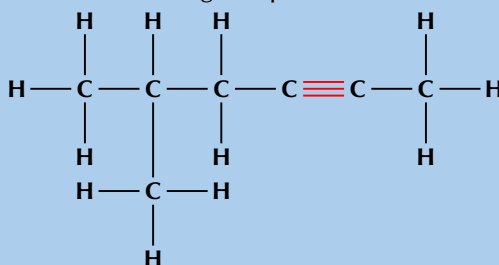
Naming alkynes

The suffix for an alkyne is -yne.

Worked example 7: Naming the alkynes

QUESTION

Give the IUPAC name for the following compound:



SOLUTION

Step 1: Identify the functional group

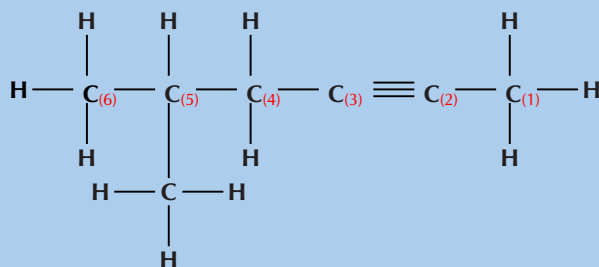
There is a triple bond between two of the carbon atoms, so this compound is an alkyne. The suffix will be -yne.

Step 2: Find the longest carbon chain containing the functional group

The functional group is a triple bond, so the longest chain must contain the triple bond. There are six carbon atoms in the longest chain. The prefix of the compound's name will be hex-.

Step 3: Number the carbons in the longest chain

In this example, you will need to number the carbons from right to left so that the triple bond is between carbon atoms with the lowest numbers (the suffix for the compound will therefore be -2-yne).



Step 4: Look for any branched groups, name them and assign the number of the carbon atom to which the group is attached

There is a methyl (CH_3) group attached to the fifth carbon (remember we have numbered the carbon atoms from right to left).

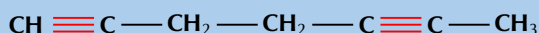
Step 5: Combine the elements of the name into a single word in the following order: branched groups; prefix; name ending according to the functional group and its position along the longest carbon chain

If we follow this order, the name of the compound is **5-methylhex-2-yne**.

Worked example 8: Naming the alkynes

QUESTION

Give the IUPAC name for the following compound:



SOLUTION

Step 1: Identify the functional group

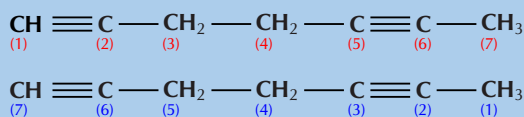
There are two triple bonds. The suffix will therefore be -diyne.

Step 2: Find the longest carbon chain containing the functional group

The functional group is a triple bond, so the longest chain must contain all triple bonds. The longest carbon chain contains seven carbon atoms, therefore the prefix will be hept-.

Step 3: Number the carbons in the longest chain

Numbering from left to right (shown in red) the first triple bond is on carbon 1 and the second is on carbon 5. The suffix will therefore be -1,5-diyne.



(Numbering from right to left (shown in blue) will give the suffix -2,6-diyne, and is incorrect).

Step 4: Look for any branched groups

There are no branched groups for this molecule.

Step 5: Combine the elements of the name into a single word in the following order: branched groups; prefix; name ending according to the functional group and its position along the longest carbon chain

The name of the compound is **hept-1,5-diyne**.

Worked example 9: Naming the alkynes

QUESTION

Draw the structural and condensed structural formula for the organic compound **6-methylhept-3-yne**

SOLUTION

Step 1: Identify the functional group

The suffix -3-yne means that this compound is an alkyne and there must be a triple bond located on carbon number 3.

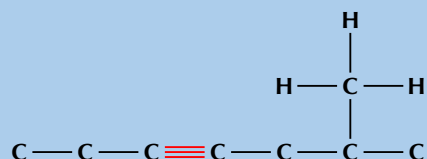
Step 2: Determine the number of carbon atoms in the longest chain containing the functional group

The prefix for the compound is hept- so there must be seven carbons in the longest chain.

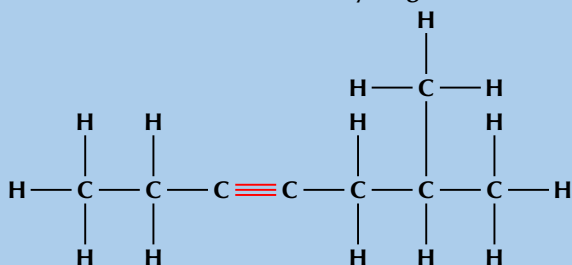


Step 3: Look for any branched groups

There is a methyl group located on carbon number 6.



Step 4: Combine this information and add the hydrogen atoms



Step 5: Condense the structural formula

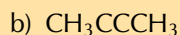
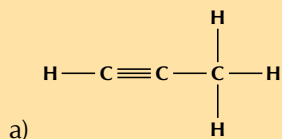
First condense the main chain: $\text{CH}_3\text{CH}_2\text{CCCH}_2\text{CHCH}_3$

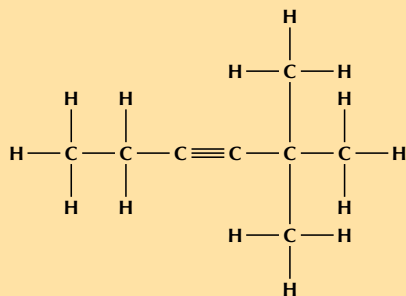
Then add the side chains (in brackets) on the relevant carbon atoms:



Exercise 4 – 10: Naming alkynes

- Give the structural formula for each of the following alkynes:
a) ethyne b) pent-1-yne c) 5-methylhept-3-yne
- Give the IUPAC names for the following alkynes:





3. More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.

1. 27MR 2. 27MS



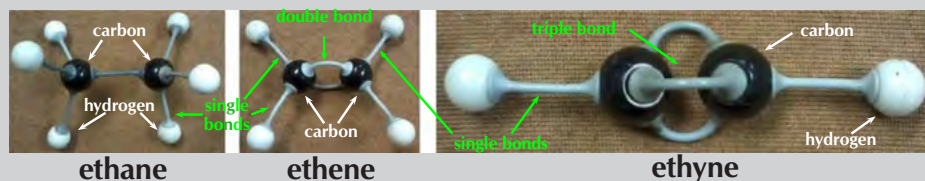
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Activity: Building hydrocarbons

An example of ethane, ethene and ethyne built with atomic model kits is given in the picture below:



- Using atomic model kits, build the molecules of methane, propane, butane, pentane and octane. If you don't have atomic model kits, jelly tots (or playdough) and toothpicks will work just as well. Use one colour jelly tot for the carbon atoms and one for the hydrogen atoms.
 - Remember that carbon atoms should have four bonds and hydrogen atoms can have only one. You should see that all these compounds have a similar formula, remember they all have the general formula C_nH_{2n+2} .
 - What is the name of the homologous series that all these molecules belong to?
- Build the molecules of prop-1-ene, but-1-ene, pent-1-ene and oct-1-ene. Use two toothpicks to represent a double bond. You should see that all these compounds have a similar formula. Remember, they all have the general formula C_nH_{2n} .
 - Try placing the double bond at different positions within the molecule. Does this make any difference to the total number of carbon and hydrogen atoms in the molecule?
 - What is the name of the homologous series that all these molecules belong to?
- Build the molecules of prop-1-yne, but-1-yne, pent-1-yne and oct-1-yne. Use three toothpicks to represent a triple bond. You should see that all these compounds have a similar formula. Remember, they are all alkynes.
 - Try placing the triple bond at different positions within the molecule. Does this make any difference to the total number of carbon and hydrogen atoms in the molecule?
 - What is the general formula for the alkynes?

with the lowest number. In this case you can number from either side.

Step 4: Name the halogen atom and assign the number for the carbon atom it is attached to

The halogen is a chlorine atom. It is attached to carbon number 2 and so will have the name 2-chloro.

Step 5: Look for branched groups

There are no branched groups in this compound.

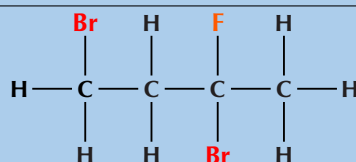
Step 6: Combine the elements of the name into a single word in the following order: halogen atoms; prefix; name ending according to functional group

The name of the compound is **2-chloropropane**.

Worked example 11: Naming the haloalkanes

QUESTION

Give the IUPAC name for the following compound:



SOLUTION

Step 1: Identify the functional group

There are three halogen atoms and no other functional groups. This compound is therefore a haloalkane, and will have the suffix -ane.

Step 2: Find the longest carbon chain containing the functional group

There are four carbons in the longest chain containing all the halogen atoms. The prefix for this compound will be but-.

Step 3: Number the carbon atoms in the longest chain

You need to number the carbon atoms so that the halogen atoms are on the carbon atoms with the lowest numbers. You must number from left to right here so that one halogen atom is on carbon 1 and two halogen atoms are on carbon 3.

Step 4: Name the halogen atoms and assign the number for the carbon atom attached to it

There are two halogen atoms that are bromine atoms and one that is fluorine. One bromine is attached to carbon 1 and one is attached to carbon 3. The fluorine atom is attached to carbon 3. So you have 1,3-dibromo- and 3-fluoro.

Step 5: Look for branched groups

There are no branched groups in this compound.

Step 6: Combine the elements of the name into a single word in the following order: halogen atoms in alphabetical order; prefix; name ending according to functional group

The name of the compound is **1,3-dibromo-3-fluorobutane**. Note that we place the halogens in alphabetical order: *bromo* (ignore the di/tri/tetra) is before *fluoro*.

Worked example 12: Naming the haloalkanes

QUESTION

Draw the structural and condensed structural formula for the organic compound **2-iodo-3-methylpentane**

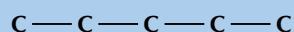
SOLUTION

Step 1: Identify the functional group

This compound has the suffix -ane, but also contains a halogen atom. It is therefore a haloalkane. Note that the methyl and iodo are written in alphabetical order.

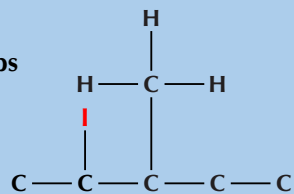
Step 2: Find the longest carbon chain containing the functional group

The prefix is pent- therefore there are 5 carbons in the longest chain.

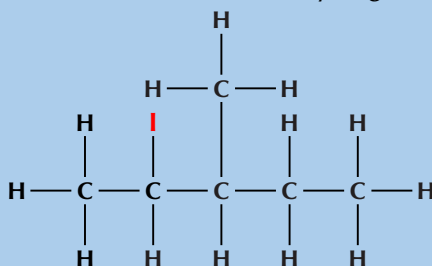


Step 3: Place the halogen atom(s) and any branched groups

There is an iodine atom on the second carbon atom, and a methyl branched group on the third carbon atom.



Step 4: Combine this information to and add the hydrogen atoms



Step 5: Condense the structural formula

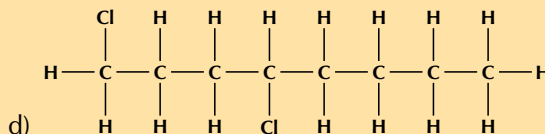
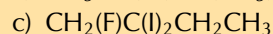
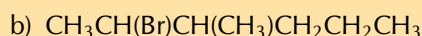
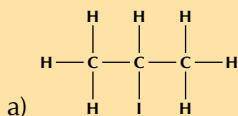
First condense the main chain: $\text{CH}_3\text{CHCHCH}_2\text{CH}_3$

Then add the side chains and halogen atoms (in brackets) on the relevant carbon atoms:



Exercise 4 – 12: Naming haloalkanes

- Give the structural representation for the following haloalkanes:
a) 2-chlorobutane b) 1-bromopropane c) 2,3-difluoropentane
- Give the IUPAC name for the following haloalkanes:



- More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.

1. 27MW 2. 27MX



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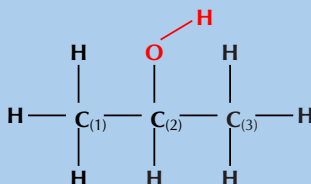
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The rules used to name the alcohols are similar to those already discussed for the hydrocarbons. The suffix of an alcohol is -ol (see Table 4.5).

Worked example 13: Naming the alcohols

QUESTION

Give the IUPAC name for the following organic compound



SOLUTION

Step 1: Identify the functional group

The compound has an -OH (hydroxyl) functional group and is therefore an alcohol. The compound will have the suffix -ol.

Step 2: Find the longest carbon chain containing the functional group

There are three carbon atoms in the longest chain that contains the functional group. The prefix for this compound will be prop-. As there are only single bonds between the carbon atoms, the prefix includes *an* to become propan-.

Step 3: Number the carbons in the carbon chain

In this case, it doesn't matter whether you start numbering from the left or right. The hydroxyl group will still be attached to second carbon atom (-2-ol).

Step 4: Look for branched groups

There are no branched groups in this compound.

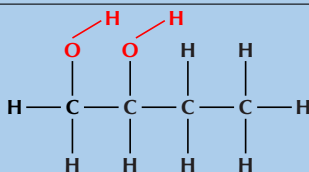
Step 5: Combine the elements of the compound's name into a single word in the order of branched groups; prefix; name ending according to the functional group

The compound's name is **propan-2-ol** or **2-propanol**.

Worked example 14: Naming the alcohols

QUESTION

Give the IUPAC name for the following compound:



SOLUTION

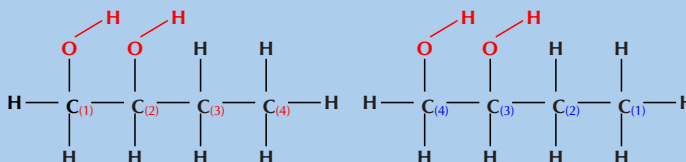
Step 1: Identify the functional group

The compound has an -OH (hydroxyl) functional group and is therefore an alcohol. There are two hydroxyl groups in the compound, so the suffix will be -diol.

Step 2: Find the longest carbon chain that contains the functional group

There are four carbon atoms in the longest chain that contains the functional group (but-) and only single bonds (an-). The prefix for this compound will be butan-.

Step 3: Number the carbons in the carbon chain



There are two hydroxyl groups attached to the main chain. If we number as shown in red (on the left) they are attached to the first and second carbon atoms. If we number as shown in blue (on the right) they are attached to the third and fourth carbon atoms.

The functional groups should have the lowest numbers possible. Therefore the red numbering is correct. The hydroxyl groups are attached to the first and second carbon atoms (1,2-diol).

Step 4: Look for branched groups

There are no branched groups in this compound.

Step 5: Combine the elements of the compound's name into a single word in the order of branched groups; prefix; name ending according to the functional group

The compound's name is **butan-1,2-diol**.

Worked example 15: Naming the alcohols

QUESTION

Draw the structural and condensed structural representations for the organic compound **4-ethyloctan-2,5-diol**

SOLUTION

Step 1: Identify the functional group

The compound has the suffix -ol. It is therefore an alcohol.

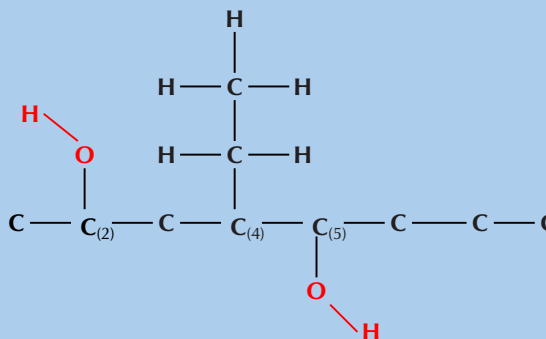
Step 2: Find the longest carbon chain that contains the functional group

The prefix is oct- therefore there are 8 carbons in the longest chain containing the functional group.

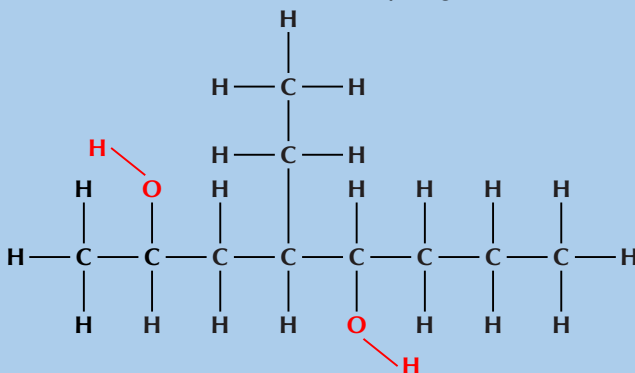


Step 3: Place the functional group as well as any branched groups

There is one -OH attached to carbon 2 and one attached to carbon 5. There is also an ethyl (-CH₂CH₃) branched group attached to carbon 4.



Step 4: Combine this information and add the hydrogen atoms



Step 5: Condense the structural formula

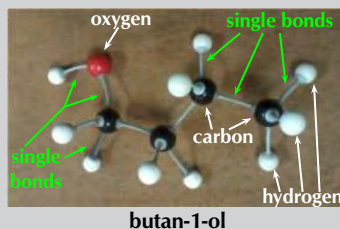
First condense the main chain: $\text{CH}_3\text{CHCH}_2\text{CHCHCH}_2\text{CH}_2\text{CH}_3$

Then add the side chains and alcohol functional groups (in brackets) on the relevant carbon atoms: **$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{CH}_2\text{CH}_3)\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$**

Activity: Building alcohols

The structural representation of butan-1-ol built using an atomic model kit is given below:

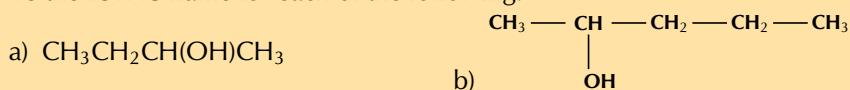
Using atomic model kits, build the molecules of methanol, ethanol, propan-1-ol, pentan-1-ol and octan-1-ol. If you don't have an atomic model kit remember that you can use jelly tots (or playdough) and toothpicks. Use different colour jelly tots to represent the different atoms.

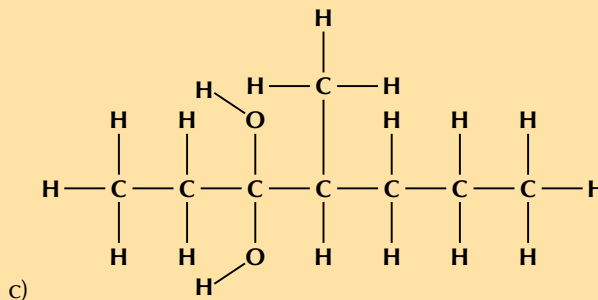


- You should see that all these compounds have a similar formula. Remember, they belong to the homologous series of the alcohols. What is the general formula for this series?
- Try placing the hydroxyl group at different positions within the molecule. Does this make any difference to the total number of carbon, hydrogen and oxygen atoms in the molecule?

Exercise 4 – 13: Naming alcohols

1. Give the structural representation of each of the following organic compounds:
a) pentan-3-ol b) butan-2,3-diol c) 2-methylpropan-1-ol
2. Give the IUPAC name for each of the following:





3. More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.

1. 27MY 2. 27MZ



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Naming carbonyl compounds

ESCKN

A carbonyl group consists of a carbon atom that is bonded to an oxygen atom through a double bond ($\text{C}=\text{O}$). There are many different functional groups that contain a carbonyl group.

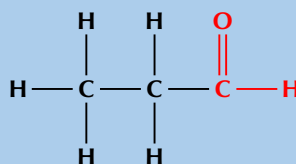
Naming aldehydes

If the carbonyl group is on the *end* of the carbon chain, the organic compound is called an *aldehyde*. An aldehyde has the suffix -al.

Worked example 16: Naming aldehydes

QUESTION

Give the IUPAC name and molecular formula for the following organic compound



SOLUTION

Step 1: Identify the functional group

The compound has a $\text{C}=\text{O}$ (carbonyl) group and no other functional groups. It is therefore either an aldehyde or a ketone. The carbonyl group is on the last (terminal) carbon in the main chain so the compound is an aldehyde. It will have the suffix -al.

Step 2: Find the longest carbon chain containing the functional group

There are three carbons in the longest chain that contains the functional group. The prefix for this compound will be prop-. As there are only single bonds between the carbon atoms, the prefix becomes propan-.

Step 3: Number the carbon atoms in the carbon chain

The carbon atoms will be numbered so that the carbon atom of the aldehyde group has the lowest number possible. In this case that is from right to left.

Step 4: Look for any branched groups

There are no branched groups in this compound.

Step 5: Combine the elements of the compound's name into a single word in the order of branched groups; prefix; name ending according to the functional group

The compound's name is **propanal** (there is no need to say propan-1-al as by definition all aldehydes are -1-al).

Step 6: Reduce the structural representation to the molecular formula

There are 3 carbon atoms, 6 hydrogen atoms and 1 oxygen atom so the molecular formula is **C₃H₆O**.

(Remember that there is no *structural* information given by the molecular formula)

TIP

Ketone is pronounced **keytone**. Therefore propanone is pronounced **propanown**.

FACT

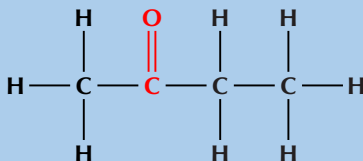
Note that butanone can only be butan-2-one. If the carbonyl group were on carbon 1 it would be an aldehyde, while if it were on carbon 3 we would simply count from the other side of the molecule. The 2 is still required for IUPAC naming however.

Naming ketones

If the carbonyl group is in the middle of the carbon chain, the compound is called a *ketone*. A ketone has the suffix *-one*.

Worked example 17: Naming ketones**QUESTION**

Give the IUPAC name and molecular formula for the following compound:

**SOLUTION****Step 1: Identify the functional group**

The compound has a C=O (carbonyl) group and no other functional groups. It is therefore either an aldehyde or a ketone. The carbonyl group is **not** at the end of the chain. Therefore the compound is a ketone and the suffix will be *-one*.

Step 2: Find the longest carbon chain containing the functional group

There are four carbons in the longest chain that contains the functional group, and only single carbon-carbon bonds. The prefix for this compound will be *butan-*.

Step 3: Number the carbon atoms in the carbon chain

The carbon atoms will be numbered from left to right so that the carbon atom of the ketone group has the lowest number possible (*-2-one*).

Step 4: Look for any branched groups

There are no branched groups in this compound.

Step 5: Combine the elements of the compound's name into a single word in the order of branched groups; prefix; name ending according to the functional group

The compound's name is **butan-2-one** or **2-butanone**.

Step 6: Reduce the structural representation to the molecular formula

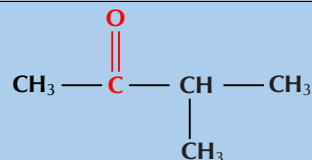
There are 4 carbon atoms, 8 hydrogen atoms and 1 oxygen atom so the molecular formula is **C₄H₈O**.

(Remember that there is no *structural* information given by the molecular formula)

Worked example 18: Naming carbonyl compounds

QUESTION

Give the IUPAC name for the following compound:



SOLUTION

Step 1: Identify the functional group

The compound has a C=O (carbonyl) group and no other functional groups. It is therefore either an aldehyde or a ketone. The carbonyl group is **not** at the end of the chain. Therefore the compound is a ketone and the suffix will be -one.

Step 2: Find the longest carbon chain that contains the functional group

The longest carbon chain that contains the functional group has four carbon atoms in it, and only single bonds. The prefix for this compound will be butan-.

Step 3: Number the carbon atoms in the carbon chain

The carbon atoms will be numbered from left to right so that the carbon atom of the ketone functional group has the lowest possible number. The suffix will be -2-one.

Step 4: Look for any branched groups

There is a branched group on carbon 3. This group has only one carbon atom. The branched group is attached to the third carbon atom (3-methyl).

Step 5: Combine the elements of the compound's name into a single word in the order of branched groups; prefix; name ending according to the functional group

The compound's name is **3-methylbutan-2-one**.

Worked example 19: Naming carbonyl compounds

QUESTION

Draw the structural and condensed structural representations for the organic compound **3-methylpentanal**.

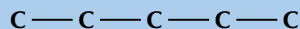
SOLUTION

Step 1: Identify the functional group

The compound has the suffix -al. It is therefore an aldehyde and has a C=O (carbonyl) group on the first carbon atom.

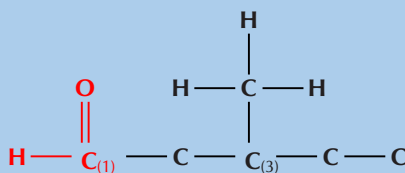
Step 2: Find the longest carbon chain that contains the functional group

The prefix is pent- so there are 5 carbon atoms in the longest chain.

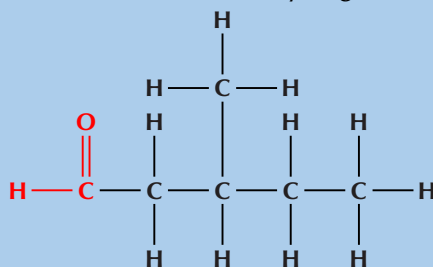


Step 3: Place the functional group as well as any branched groups

There is a C=O (carbonyl) group at the first carbon atom and a methyl group attached to the third carbon atom (3-methyl).



Step 4: Combine this information and add the hydrogen atoms



Step 5: Condense the structural representation

First condense the main chain including the carbonyl group oxygen atom: $\text{CHOCH}_2\text{CHCH}_2\text{CH}_3$

Then add the side chains on the relevant carbon atoms: $\text{CHOCH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$

Worked example 20: Naming carbonyl compounds

QUESTION

Draw the structural and condensed structural representations for the organic compound **3-methylpentan-2-one**.

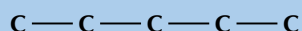
SOLUTION

Step 1: Identify the functional group

The compound has the suffix -one. It is therefore a ketone and has a $\text{C}=\text{O}$ (carbonyl) group. This group *cannot* be on the first carbon atom.

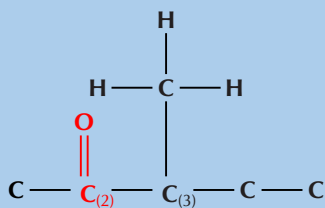
Step 2: Find the longest carbon chain that contains the functional group

The prefix is pent- therefore there are 5 carbon atoms in the longest chain.

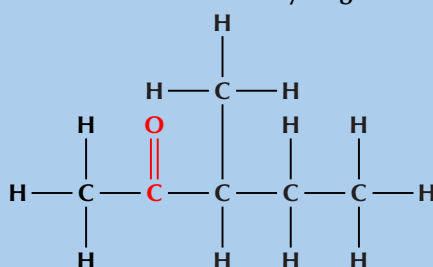


Step 3: Place the functional group as well as any branched groups

There is a $\text{C}=\text{O}$ (carbonyl) group at the second carbon atom (-2-one) and a methyl group attached to the third carbon atom (3-methyl).



Step 4: Combine this information and add the hydrogen atoms



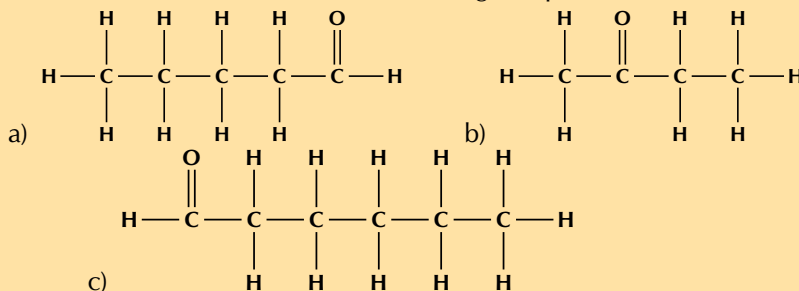
Step 5: Condense the structural representation

First condense the main chain including the carbonyl group oxygen atom:
 $\text{CH}_3\text{COCHCH}_2\text{CH}_3$

Then add the side chains on the relevant carbon atoms: $\text{CH}_3\text{COCH}(\text{CH}_3)\text{CH}_2\text{CH}_3$

Exercise 4 – 14: Naming aldehydes and ketones

1. Give the IUPAC name for each of the following compounds:



d) HCHO

e) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$

2. Give the structural representation for the following:

a) ethanal

b) propanone

c) heptan-3-one

3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27N2 2. 27N3



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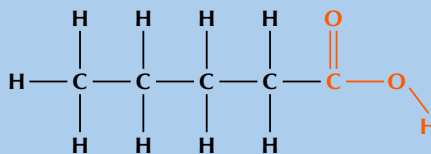
Naming carboxylic acids

Carboxylic acids are characterised by having a carboxyl group, which has the formula $-\text{COOH}$. In a carboxyl group a carbon atom is double-bonded to an oxygen atom (a carbonyl group), and is also bonded to a hydroxyl (alcohol) group. The IUPAC suffix for a carboxylic acid is -oic acid.

Worked example 21: Naming carboxylic acids

QUESTION

Give the IUPAC name and molecular formula for the following compound:



SOLUTION

Step 1: Identify the functional group

The compound has a $-\text{COOH}$ group and is therefore a carboxylic acid. The suffix will be -oic acid.

Step 2: Find the longest carbon chain that contains the functional group

There are five carbon atoms in the longest chain that contains the functional group, and only single bonds between carbon atoms. The prefix for this compound is pentan-.

Step 3: Number the carbon atoms in the carbon chain

The carbon atoms will be numbered from right to left so that the carboxylic acid functional group has the lowest numbered carbon atom.

Step 4: Look for any branched groups

There are no branched groups in this compound.

Step 5: Combine the elements of the compound's name into a single word in the order of branched groups; prefix, name ending according to the functional group

The compound's name is **pentanoic acid**.

Step 6: Reduce the structural representation to the molecular formula

There are 5 carbon atoms, 10 hydrogen atoms and 2 oxygen atoms so the molecular formula is **C₅H₁₀O₂**.

Worked example 22: Naming carboxylic acids**QUESTION**

Draw the structural and condensed structural representations for the organic compound **2-ethylhexanoic acid**.

SOLUTION**Step 1: Identify the functional group**

The compound has the suffix -oic acid. It is therefore a carboxylic acid and has a -COOH group. This group can only be on carbon 1 (at the end of the carbon chain).

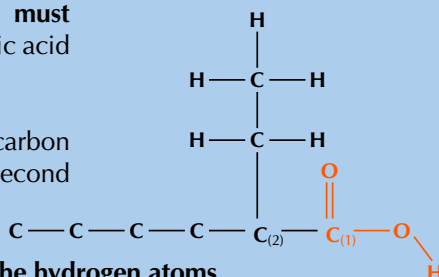
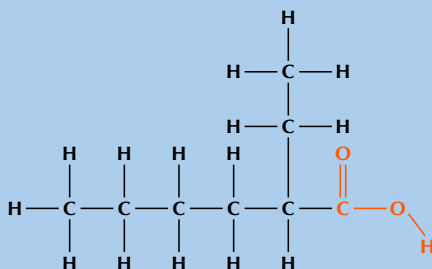
Step 2: Find the longest carbon chain that contains the functional group

The prefix is hex- therefore there are 6 carbons in the longest chain.

**Step 3: Place the functional group as well as any branched groups**

Remember that the main carbon chain **must** contain the functional group. This is hexanoic acid and **not** a substituted heptane.

The -COOH group contains the first carbon atom. The ethyl group is attached to the second carbon atom.

**Step 4: Combine this information and add the hydrogen atoms**

Step 5: Condense the structural representation

First condense the main chain, including the -COOH functional group:



Then add the side chains (in brackets) on the relevant carbon atoms:



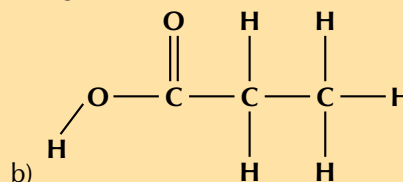
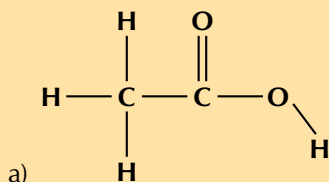
Activity: Building carboxylic acids

Using atomic model kits build the molecules of methanoic acid, ethanoic acid, butanoic acid, pentanoic acid and octanoic acid. If you don't have an atomic model kit remember that you can use jelly tots (or playdough) and toothpicks. Use different colour jelly tots to represent the different atoms and two toothpicks to represent double bonds.

- You should see that all these compounds have a similar formula. Remember, they belong to the carboxylic acid homologous series. What is the general formula for this series?
- Remember that carbon must have four bonds, oxygen must have two bonds and hydrogen can only have one bond. Thinking about this fact, is it possible to have the carboxylic acid (-COOH) group in a position other than the last (or first) carbon atom?

Exercise 4 – 15: Naming carboxylic acids

1. Give the structural representation for the following:
 - a) pentanoic acid
 - b) 4-ethyl-7-methyloctanoic acid
 - c) 4,4-diethylheptanoic acid
2. Give the IUPAC name for each of the following:



- c) $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_2\text{CH}_2\text{COOH}$
- d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_2\text{CH}_3)\text{CH}_2\text{CH}(\text{CH}_2\text{CH}_3)\text{COOH}$

3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27N4 2. 27N5



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Naming esters

When an alcohol reacts with a carboxylic acid an *ester* is formed. A new bond is formed between the oxygen atom of the hydroxyl group and the carbonyl carbon atom of the carboxylic acid. The suffix for an ester is -oate.

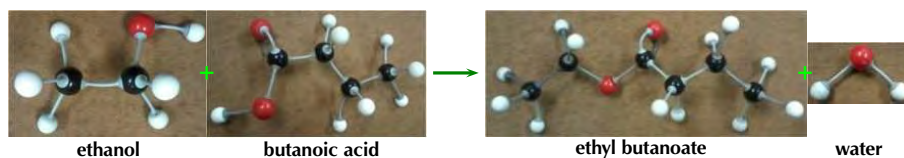


Figure 4.49: The esterification process of ethanol and butanoic acid to ethyl butanoate and water.

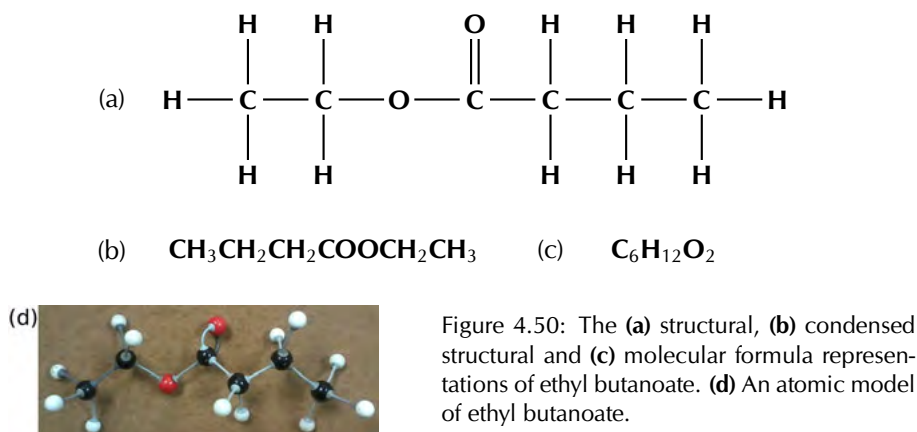


Figure 4.50: The (a) structural, (b) condensed structural and (c) molecular formula representations of ethyl butanoate. (d) An atomic model of ethyl butanoate.

Although the part of the ester from the alcohol (ethyl) is on the right, and the part from the carboxylic acid (butan-) is on the left in Figure 4.51, when naming the structure the part of the name from the alcohol is written first (ethyl butanoate). Don't forget to count the carbon atom in the carbonyl group when determining the number of carbon atoms in the chain.

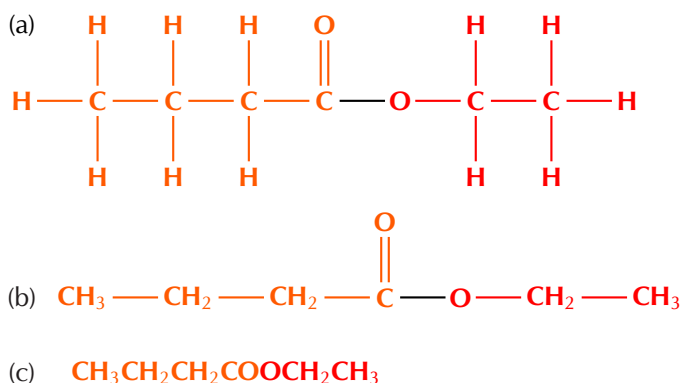
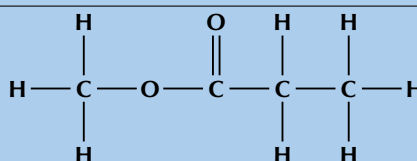


Figure 4.51: The (a) structural, (b) semi-structural structural and (c) condensed structural representations of **ethyl butanoate**.

Worked example 23: Naming esters

QUESTION

Give the IUPAC name for the following compound:



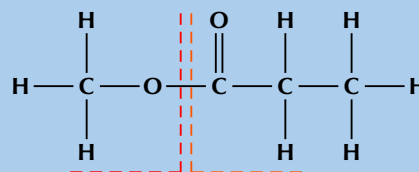
SOLUTION

Step 1: Identify the functional group

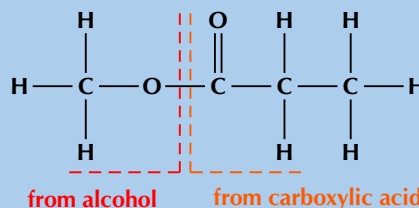
There is a -C=O (carbonyl) group as well as an oxygen atom bonded to the carbon atom of the carbonyl and another carbon atom. This is therefore an ester and the suffix is -oate.

Step 2: Determine which part is from the alcohol and which is from the carboxylic acid

An ester is a carboxylic acid derivative. Divide the molecule in two with the carbonyl group on one side and the oxygen bonded to two carbon atoms on the other.



The part containing the oxygen atom bonded to two different carbon atoms was formed from the alcohol and is on the left here. The part containing the carbonyl group was formed from the carboxylic acid and is on the right here.



Step 3: Number the carbon atoms on the carbon chains

There is only one carbon atom in the left-hand chain (from the alcohol). Therefore this will be methyl. There are three carbon atoms in the right-hand chain (from the carboxylic acid) therefore the prefix will be propan-.

Step 4: Combine the elements of the compound's name into a single word in the order of chain from the alcohol; prefix (from chain containing carbonyl functional group), name ending according to functional group

The compound's name is **methyl propanoate**.

Worked example 24: Naming esters

QUESTION

Draw the structural and condensed structural representations for the organic compound **ethyl hexanoate**.

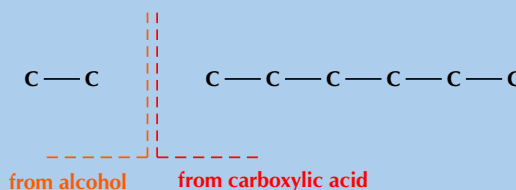
SOLUTION

Step 1: Identify the functional group

The compound has the suffix -oate. It is therefore an ester and has a $-\text{C}=\text{O}$ (carbonyl) group as well as an oxygen atom bonded to the carbon atom of the carbonyl and another carbon atom.

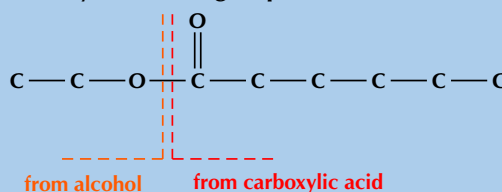
Step 2: Determine which part is from the alcohol and which is from the carboxylic acid

The ethyl tells us that there are two carbon atoms in the part of the chain from the alcohol. The prefix hex- tells us that there are six carbon atoms from the part of the chain from the carboxylic acid.

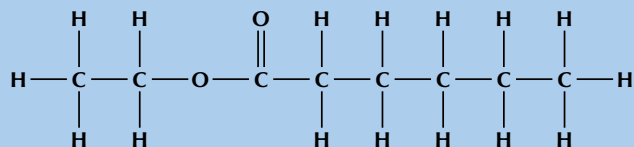


Step 3: Place the functional group as well as any branched groups

The oxygen atom bonded to two different carbon atoms is located between the two sections. The $-\text{C}=\text{O}$ (carbonyl) group is located at the first carbon atom of the carboxylic acid chain.



Step 4: Combine this information and add the hydrogen atoms



Step 5: Condense the structural representation

Condense the part of the compound that came from the carboxylic acid first, so start from the right here: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COO}$

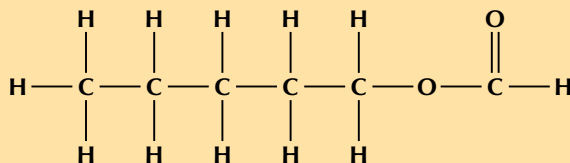
The first O is for the $-\text{C}=\text{O}$, the second is from the $-\text{O}-$. Now condense the part that came from the alcohol, starting from the $-\text{O}-$:



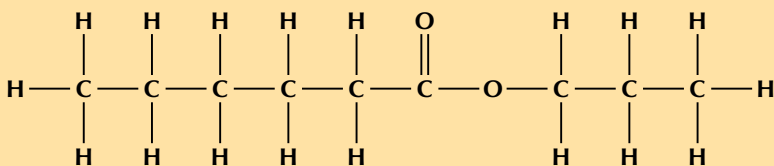
Exercise 4 – 16: Naming esters

1. Give the IUPAC name for each of the following compounds:

a) $\text{CH}_3\text{COOCH}_2\text{CH}_3$



b)



c)

2. Give the structural representations for the following esters:

a) heptyl propanoate b) methyl octanoate c) hexyl pentanoate

3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27N6 2. 27N7



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Worked example 25: Naming carbonyl compounds

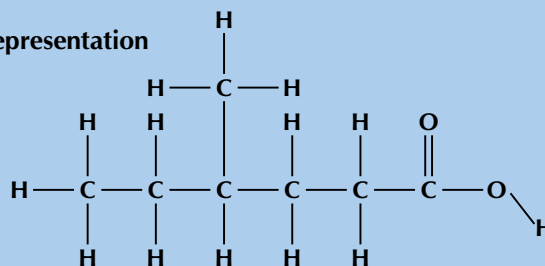
QUESTION

Give the IUPAC name for the following compound: $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{COOH}$

(Remember that the side groups are shown in brackets after the carbon atom to which they are attached)

SOLUTION

Step 1: Draw the structural representation



Step 2: Identify the functional group

The compound has a -COOH functional group. It is therefore a carboxylic acid and the suffix is -oic acid.

Step 3: Find the longest carbon chain containing the functional group

There are six carbon atoms in the longest chain containing the functional group. The prefix for this compound is hexan-.

Step 4: Number the carbon atoms in the carbon chain

The carbon atoms should be numbered from right to left so that the carboxylic acid functional group has the lowest numbered carbon atom.

Step 5: Look for any branched groups, name them and give their position on the carbon chain

There is a branched group attached to the fourth carbon atom. This group has only one carbon atom and is therefore a methyl group (4-methyl).

Step 6: Combine the elements of the compound's name into a single word in the order of branched groups; prefix; name ending according to the functional group

The compound's name is **4-methylhexanoic acid**.

Worked example 26: Naming carbonyl compounds

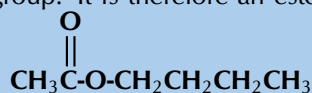
QUESTION

Give the IUPAC name for the following compound: $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

SOLUTION

Step 1: Identify the functional group

The compound has a -COO functional group. It is therefore an ester and the suffix will be -oate. This can also be shown as:



Step 2: Determine which part is from the alcohol and which is from the carboxylic acid

The left half of the compound contains the carbonyl group and is therefore from the carboxylic acid. The right half of the compound must be from the alcohol.



Step 3: Determine the number of carbon atoms in each chain

There are 2 carbon atoms in the part that contains the C=O group (from the carboxylic acid), therefore the prefix is ethan-. There are 4 carbon atoms in part of the chain without the carbonyl group (from the alcohol), which is therefore butyl.

Step 4: Look for any branched groups, name them and give their position on the carbon chain

There are no branched groups.

Step 5: Combine the elements of the compound's name into a single word in the order of name of chain from the alcohol; prefix (from chain containing carbonyl functional group); suffix

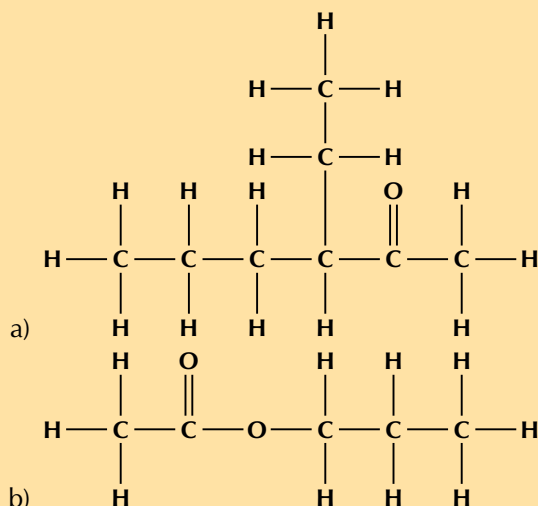
The compound's name is **butyl ethanoate**.

Exercise 4 – 17: Naming carbonyl compounds

1. Give the structural representation for the following compounds:.

- | | |
|--------------------------|---------------------|
| a) 3-methylpentanal | b) butyl pentanoate |
| c) 2-methylbutanoic acid | d) octan-4-one |

2. Give the IUPAC name for each of the following compounds:



c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$

d) HCOOH

3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27N8 2. 27N9



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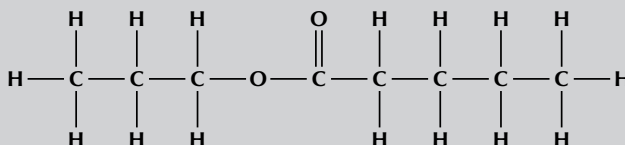
Activity: Building molecules

Using atomic model kits build molecules of butane, but-1-ene, but-1-yne, butan-1-ol, butanoic acid, butyl butanoate, butan-2-one, butanal. If you don't have an atomic model kit remember that you can use jelly tots (or playdough) and toothpicks. Use different colour jelly tots to represent the different atoms and extra toothpicks to represent double and triple bonds.

- Identify the functional group in each of these molecules. Move the functional group around the molecule. You should find that you can do so with but-1-ene,

but-1-yne and butan-1-ol. It will not be possible with butanoic acid and there is no functional group to move in butane.

- Can you see that if you move the carbonyl group in butan-2-one you will either still get butan-2-one or you will get butanal? Similarly, when moving the carbonyl group in butanal you can only get butanal or butan-2-one.
- Build a model of the molecule below:



Compare this to butyl butanoate. What differences are there between these two molecules? What similarities are there? What is the name of the new molecule?

- Continue by building different compounds for your friends. Make them identify the functional group, the number of carbon atoms and therefore the name of the compound.

Exercise 4 – 18: IUPAC naming

- Study the table below and answer the questions that follow:

	A	B	C
	Compound	Functional group	Number of carbon atoms
1	<i>e.g. methane</i>	<i>e.g. alkane</i>	<i>e.g. 1</i>
2	propanoic acid	alkyne	8
3	2-chloroethane	ketone	4
4	1-octanal	carboxylic acid	6
5	3-heptyne	aldehyde	2
6	butanone	ester	6
7	3-hexene	haloalkane	1 and 5
8	1-hexanol	alkene	3
9	methyl pentanoate	alcohol	7

- Match the compounds in column A with the correct functional group in column B. For example methane is an alkane: **A1, B1**.
 - Match the compounds in column A with the correct number of carbon atoms in column C. For example methane has one carbon atom in its longest chain: **A1, C1**.
- Match the structural representation in column A with the condensed structural representation (column B) and IUPAC name (column C).

	A	B	C
	Structure	Condensed	IUPAC name
1	$ \begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C} & -\text{C} & -\text{C}-\text{H} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array} $	CH ₃ CH(I)CH ₃	ethyl methanoate
2	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C}\equiv\text{C}-\text{H} \\ \\ \text{H} \end{array} $	CH ₃ CH ₂ CH ₃	propanone
3	$ \begin{array}{c} \text{H} & \text{O} & \text{H} \\ & & \\ \text{H}-\text{C} & -\text{C} & -\text{C}-\text{H} \\ & & \\ \text{H} & & \text{H} \end{array} $	CH ₃ COCH ₃	propane

	A	B	C
	Structure	Condensed	IUPAC name
4	<pre> H H O H - C - C - C - OH H H </pre>	$\text{HCOOCH}_2\text{CH}_3$	propyne
5	<pre> H I H H - C - C - C - H H H H </pre>	CH_3CCH	propanal
6	<pre> H H O H - C - C - O - C - H H H </pre>	CH_3CHCH_2	2-iodopropane
7	<pre> H H H H - C - C = C \ H H </pre>	$\text{CH}_3\text{CH}_2\text{CHO}$	propanoic acid
8	<pre> H H O H - C - C - C - H H H </pre>	$\text{CH}_3\text{CH}_2\text{COOH}$	propene

3. Fill in the gaps in the table below:

IUPAC name	Functional group	Condensed	Structural
		$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$	<pre> H H H O H - C - C - C - C - H H H H </pre>
ethanol			<pre> H O - H H - C - C - H H H </pre>
	carboxylic acid	$\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{COOH}$	
2-methyl pent-2-ene		$\text{CH}_3\text{C}(\text{CH}_3)\text{CHCH}_2\text{CH}_3$	
	alkane		<pre> H H - C - H H H H H - C - C - C - H H H H </pre>
	ester	$\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3$	
butanone	ketone		
1-pentyne			<pre> H H H H - C \equiv C - C - C - C - H H H H </pre>
	alkyl halide	$\text{CH}_2(\text{Cl})\text{CH}_2\text{CH}_3$	

4. More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.

1. 27NB 2. 27NC 3. 27ND



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The types of intermolecular forces that occur in a substance will affect its physical properties, such as its **phase**, **melting point** and **boiling point**. You should remember from the kinetic theory of matter (see [Grade 10](#)), that the *phase* of a substance is determined by how strong the forces are between its particles. The weaker the forces, the more likely the substance is to exist as a gas. This is because the particles are able to move far apart since they are not held together very strongly. If the forces are very strong, the particles are held closely together in a solid structure. Remember also that the *temperature* of a material affects the energy of its particles. The more energy the particles have, the more likely they are to be able to overcome the forces that are holding them together. This can cause a change in phase.

Figure 4.52 shows the three phases of water. Note that we are showing two-dimensional figures when in reality these are three-dimensional.

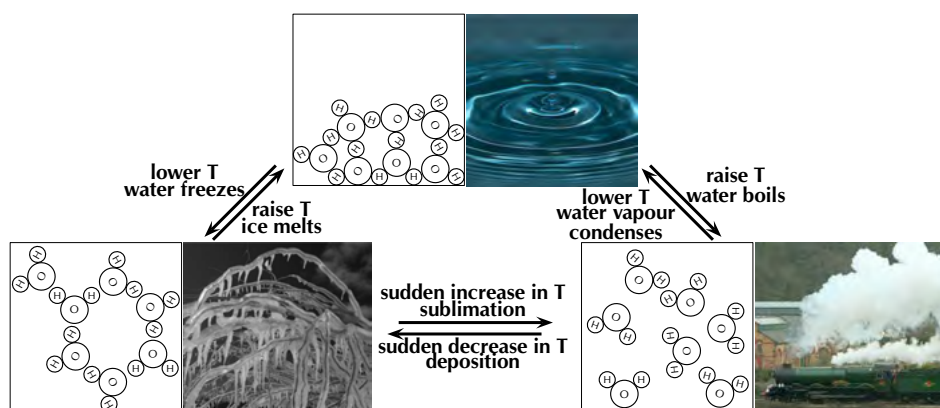


Figure 4.52: Submicroscopic diagrams of the three phases of water. (Photos by laszlo-photo, Fields of View and Capt' Gorgeous on Flickr)

The effects of intermolecular forces

The forces between molecules that bind them together are known as **intermolecular forces**. Intermolecular forces allow us to determine which substances are likely to dissolve in other substances, and what the melting and boiling points of substances are. Without intermolecular forces holding molecules together we would not exist.

DEFINITION: Intermolecular forces

Intermolecular forces are forces that act between molecules.

Remember from [Grade 11](#) that a dipole molecule is a molecule that has its charge unevenly distributed. One end of the molecule is slightly positive and the other is slightly negative. An overview of the different types of intermolecular forces that are discussed in this chapter are given below:

1. Dipole-dipole forces

When one dipole molecule comes into contact with another dipole molecule, the positive pole of the one molecule will be attracted to the negative pole of the other, and the molecules will be held together in this way.

One special case of this is hydrogen bonding:

- **Hydrogen bonds**

As the name implies, this type of intermolecular bond involves a hydrogen atom. When a molecule contains a hydrogen atom covalently bonded to a small, highly electronegative atom (e.g. O, N or F) this type of intermolecular force can occur. The highly electronegative atom on one molecule attracts the hydrogen atom on a nearby molecule (see Figure 4.53).

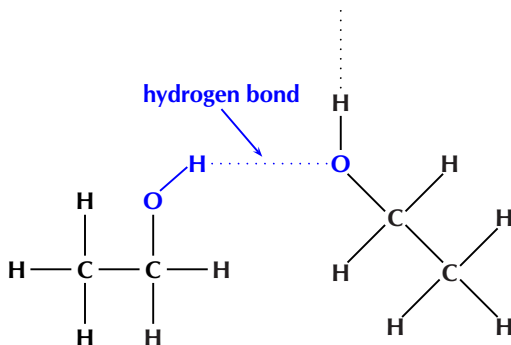


Figure 4.53: Hydrogen bonding between two molecules of ethanol.

2. van der Waals forces

- **Induced-dipole forces**

In non-polar molecules the electronic charge is usually evenly distributed but it is possible that at a particular moment in time, the electrons might not be evenly distributed (remember that the electrons are always moving in their orbitals). The molecule will have a *temporary dipole*. When this happens, molecules that are next to each other attract each other very weakly.

- **Dipole-induced-dipole forces**

These forces exist between dipoles and non-polar molecules. The dipole induces a dipole in the non-polar molecule leading to a weak, short lived force which holds the compounds together.

In this chapter we will focus on the effects of van der Waals forces and hydrogen bonding on the physical properties of organic molecules.

Viscosity

Viscosity is the resistance to flow of a liquid. Think how easy it is to pour water compared to syrup or honey. The water flows much faster than the syrup or honey.

DEFINITION: *Viscosity*

Viscosity is a measure of how much a liquid resists flowing. i.e. The higher the viscosity, the more viscous a substance is.



Figure 4.54: Pouring water versus pouring syrup.

You can see this if you take a cylinder filled with water and a cylinder filled with glycerol (propane-1,2,3-triol). Drop a small metal ball into each cylinder and note how easy it is for the ball to fall to the bottom (Figure 4.55). In the glycerol the ball falls slowly, while in the water it falls faster.

TIP

Do not confuse hydrogen bonds with *intramolecular* covalent bonds. Hydrogen bonding is an example of a scientist naming something, believing it to be one thing when in fact it was another. In this case the strength of the hydrogen bonds misled scientists into thinking this was an intramolecular bond when it is really just a strong intermolecular force.

FACT

Dipole-induced-dipole intermolecular forces are also sometimes called *London forces* or *dispersion forces*.

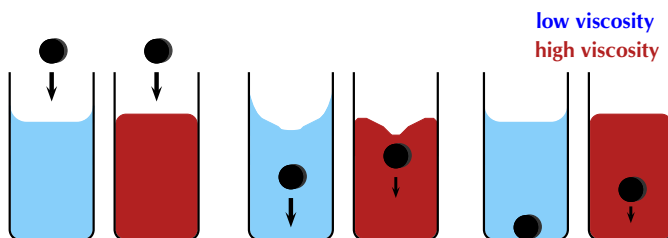


Figure 4.55: The higher the viscosity (red) the slower the ball moves through the liquid.

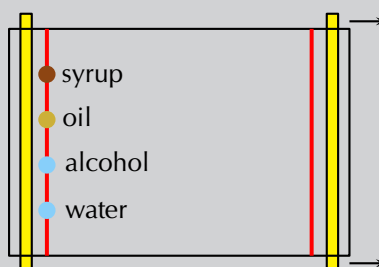
As implied by the definition, substances with stronger intermolecular forces are more viscous than substances with weaker intermolecular forces. The stronger the intermolecular forces the more the substance will resist flowing. The greater the internal friction the more a substance will slow down an object moving through it.

Activity: Resistance to flow

Take a sheet of glass at least 10 by 15 cm in size. Using a water-proof marker draw a straight line across the width of the glass about 2 cm from each end.

Place the glass flat on top of two pencils, then carefully put a drop of water on one end of the line. Leave at least 2 cm space next to the water drop, and carefully place a drop of alcohol. Repeat this with a drop of oil and a drop of syrup.

Slowly and carefully remove the pencil from the end opposite the drops (make sure you don't tilt the glass to either side in the process).



- Which drop moves fastest and reaches the end line first?
- Which drop moves slowest?

The fastest moving substance has the least resistance to flow, and therefore has the least viscosity (is the least viscous). The slowest moving substance has the most resistance to flow, and therefore has the most viscosity (is the most viscous).

Density

DEFINITION: *Density*

Density is a measure of the mass per unit of volume.

The solid phase is often the most dense phase (water is one noteworthy exception to this). This can be explained by the strong intermolecular forces found in a solid. These forces pull the molecules together, which results in more molecules in one unit of volume than in the liquid or gas phases. The more molecules in a unit volume the denser, and heavier that volume of the substance will be.

Density can be used to separate different liquids, with the more dense liquid settling

to the bottom of the container, while the less dense liquid floats on top. If you throw a leaf into a river or pond the leaf will float. If you instead throw a rock (with the same surface area and volume) into the river or pond the rock will sink. This is due to the different densities of the two substances: rocks are more dense than water while leaves are less dense than water.

Melting and boiling points

Intermolecular forces affect the boiling and melting points of substances. Substances with weak intermolecular forces will have low melting and boiling points as less energy (heat) is needed to overcome these forces. Those with strong intermolecular forces will have high melting and boiling points as more energy (heat) is required to overcome these forces. When the temperature of a substance is raised beyond its melting or boiling point the intermolecular forces are not weakened. Rather, the molecules have enough energy to overcome those forces.

FACT

An ether is a compound that contains two alkyl chains (e.g. methyl, ethyl) joined at an angle by an oxygen atom (—O—).

Name	Main intermolecular forces	Molecular Mass (g.mol ⁻¹)	Melting point (°C)	Boiling point (°C)	Phase (at 25 °C)
ethane	induced-dipole	30,06	-183	-89	gas
dimethyl ether	dipole-dipole	46,06	-141	-24	gas
chloroethane	dipole-dipole	64,5	-139	12,3	gas
pentane	induced-dipole	72,12	-130	36	liquid
propan-1-ol	hydrogen bonds	60,08	-126	97	liquid
ethanol	hydrogen bonds	46,06	-114	78,4	liquid
butan-1-ol	hydrogen bonds	74,1	-90	118	liquid
ethanoic acid	hydrogen bonds	60,04	16,5	118,5	liquid

Table 4.9: Relationship between intermolecular forces and melting point, boiling point and physical state.

As the intermolecular forces increase (from top to bottom in Table 4.9) the melting and boiling points increase. The stronger the intermolecular forces the more likely a substance is to be a liquid or a solid at room temperature.

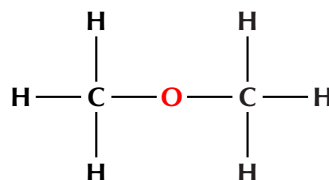


Figure 4.56: The structural formula of dimethyl ether.

Worked example 27: Comparing physical properties

QUESTION

Given:

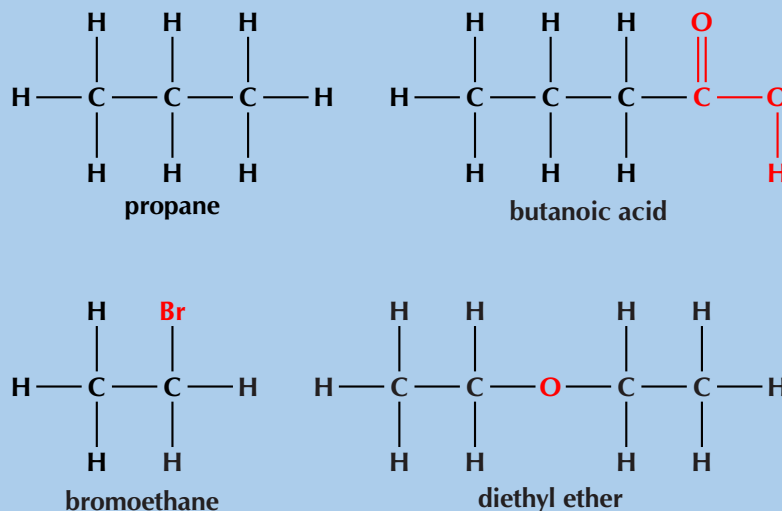
Fill in the table below:

	Melting point (°C)	Boiling point (°C)
propane	-188	-42
butanoic acid	-7,9	163,5
bromoethane	-118	38,5
diethyl ether	-116,3	34,6

Name	Main intermolecular forces	Molecular formula	Molecular mass (g.mol ⁻¹)	Phase (at 25 °C)
propane				
butanoic acid				
bromoethane				
diethyl ether				

SOLUTION

Step 1: Draw the structural representation of each molecule



Step 2: What are the intermolecular forces that each molecule will experience?

- **propane** has only single carbon-carbon bonds and no other functional group. It will therefore have induced-dipole forces only.
- **butanoic acid** has the carboxylic acid functional group. It can therefore form hydrogen bonds.
- **bromoethane** has the highly electronegative bromine atom. This means that it will form dipole-dipole interactions with neighbouring molecules.
- **diethyl ether** will have induced-dipole forces, however due to the flexible nature of the molecule it can also have dipole-induced-dipole (stronger van der Waals interactions) and dipole-dipole forces.

Step 3: Do these forces make sense with the melting and boiling points provided?

Propane has the lowest melting and boiling points and the weakest interactions. The next lowest melting and boiling points are for bromoethane and diethyl ether, which both have dipole-dipole interactions, the next strongest intermolecular forces. The highest melting and boiling points are for butanoic acid which has strong hydrogen bonds. Therefore these forces do make sense.

Step 4: Calculate the molecular mass of these molecules

- propane - C_3H_8 , molecular mass = $44,08 \text{ g}\cdot\text{mol}^{-1}$
- butanoic acid - $\text{C}_4\text{H}_8\text{O}_2$, molecular mass = $88,08 \text{ g}\cdot\text{mol}^{-1}$
- bromoethane - $\text{C}_2\text{H}_5\text{Br}$, molecular mass = $108,95 \text{ g}\cdot\text{mol}^{-1}$
- diethyl ether - $\text{C}_4\text{H}_{10}\text{O}$, molecular mass = $74,10 \text{ g}\cdot\text{mol}^{-1}$

Step 5: What will the phase of each compound be at 25°C ?

To determine the phase of a molecule at 25°C look at the melting and boiling points:

- The melting and boiling points of propane are both below 25°C , therefore the molecule will be a gas at 25°C .
- The melting points of butanoic acid, bromoethane and diethyl ether are below 25°C , however the boiling points of all three molecules are above 25°C . Therefore these molecules will be liquids at 25°C .

Step 6: Fill in the table

Name	Main intermolecular forces	Molecular formula	Molecular mass (g.mol ⁻¹)	Phase (at 25 °C)
propane	induced-dipole	C ₃ H ₈	44,08	gas
butanoic acid	hydrogen bonds	C ₄ H ₈ O ₂	88,08	liquid
bromoethane	dipole-dipole	C ₂ H ₅ Br	108,95	liquid
diethyl ether	dipole-dipole	C ₄ H ₁₀ O	74,10	liquid

Experiment: Investigation of boiling and melting points

Aim:

To investigate the relationship between boiling points and intermolecular forces

Apparatus:

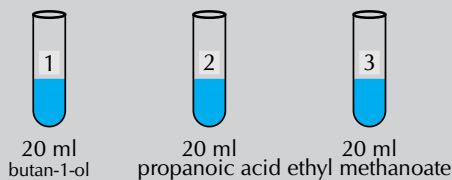
- butan-1-ol (CH₂(OH)CH₂CH₂CH₃), propanoic acid (CH₃CH₂COOH) and ethyl methanoate (HCOOCH₂CH₃), cooking oil
- three test tubes, a beaker, a thermometer, a hot plate

Method:

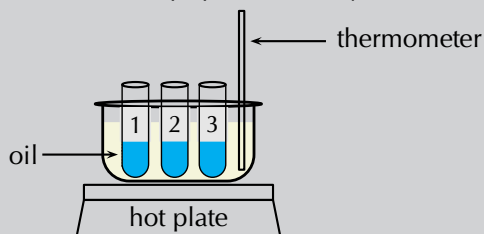
WARNING!

Ethyl methanoate can irritate your eyes, skin, nose and lungs. Keep open flames away from your experiment and make sure you work in a well ventilated area.

1. Label the test tubes **1**, **2** and **3**. Place 20 ml of butan-1-ol into test tube 1, 20 ml of propanoic acid into test tube 2, and 20 ml of ethyl methanoate into test tube 3.



2. Half-fill the beaker with cooking oil and place it on the hot plate.
3. Place the thermometer and the three test tubes in the beaker.
4. Make a note of the temperature when each substance starts to boil.



Results:

Fill in the gaps in the table below. Do the values you obtained match those reported in literature?

Compound	ethyl methanoate	butan-1-ol	propanoic acid
Molecular formula			
Molecular mass			
Main intermolecular forces			
Literature boiling point (°C)	54	118	141
Experimental boiling point (°C)			

Draw the structural representation of ethyl methanoate, butan-1-ol and propanoic acid.

Discussion and conclusion:

You should have found that the ethyl methanoate boiled first, then the butan-1-ol and then the propanoic acid. Ethyl methanoate has some dipole-dipole interactions, but cannot form a hydrogen bond. The alcohol (butan-1-ol) can form hydrogen bonds and so has a higher boiling point. This strong intermolecular force needs more energy to break and so the boiling point is higher.

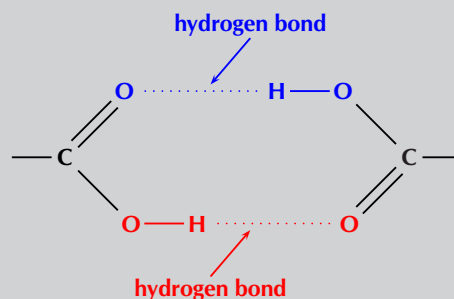


Figure 4.57: A carboxylic acid hydrogen bonding dimer.

For propanoic acid hydrogen bonds form between the carbonyl group on one acid and the hydroxyl group on another. This means that each molecule of propanoic acid can be part of two hydrogen bonds (this is called dimerisation, see Figure 4.57) and so the boiling point is even higher for propanoic acid than for butan-1-ol.

Flammability and vapour pressure

Flammability is a measure of how easy it would be for a substance to catch alight and burn. The *flash point* of a substance is the lowest temperature that is likely to form a gaseous mixture you could set alight. If a liquid has a low enough flash point it is considered flammable (able to be ignited easily) while those with higher flash points are considered nonflammable. A substance that is classified as nonflammable can still be forced to burn, but it will not ignite easily.

When a substance is in the liquid or solid state there will be some molecules in the gas state. These molecules have enough energy to overcome the intermolecular forces holding the majority of the substance in the liquid or solid phase. These gas molecules exert a pressure on the liquid or solid (and the container) and that pressure is the vapour pressure of that compound. The weaker the intermolecular forces within a substance the higher the vapour pressure will be. Compounds with higher vapour pressures have lower flash points and are therefore more flammable.

DEFINITION: Vapour pressure

The pressure exerted (at a specific temperature) on a solid or liquid compound by molecules of that compound that are in the gas phase.

Name	Main intermolecular forces	Vapour pressure (kPa at 20°C)	Flash point (°C)	Flammability
ethane	induced-dipole	3750	-135	very high
propane	induced-dipole	843	-104	very high
dimethyl ether	dipole-dipole	510	41	very high
butane	induced-dipole	204	-60	very high
chloroethane	dipole-dipole	132,4	-50	very high
pentane	induced-dipole	57,9	-49	very high
propanone	dipole-dipole	24,6	-17	high
ethanol	hydrogen bonds	5,8	17	high
water	hydrogen bonds	2,3	-	very low
propan-1-ol	hydrogen bonds	2	22	high
ethanoic acid	hydrogen bonds	1,6	40	moderate
butan-1-ol	hydrogen bonds	0,6	35	high

Table 4.10: Relationship between intermolecular forces and the flammability of a substance.

As the intermolecular forces increase (from top to bottom in Table 4.10) you can see a decrease in the vapour pressure. This corresponds with an increase in the flash point temperature and a decrease in the flammability of the substance. Figure 4.58 shows a few examples.

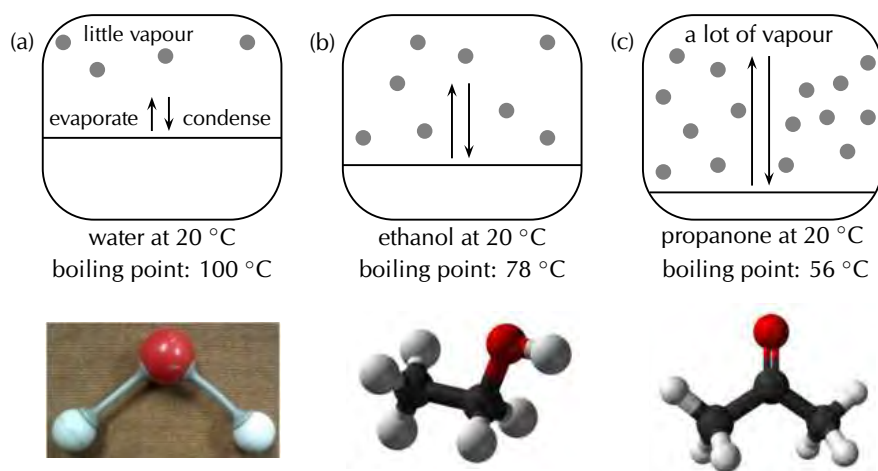


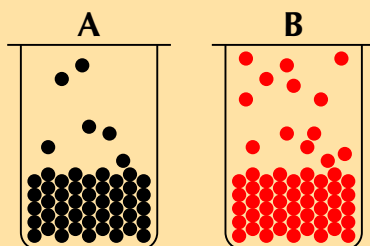
Figure 4.58: The vapour pressure of (a) water, (b) ethanol and (c) propanone at 20 °C. (Images by Duncan Watson)

Exercise 4 – 19: Types of intermolecular forces

- Use your knowledge of different types of intermolecular forces to explain the following statements:
 - The boiling point of hex-1-ene is much lower than the boiling point of propanoic acid.
 - Water evaporates more slowly than propanone.

	IUPAC name	Boiling point (°C)
A	ethane	−89,0
B	ethanol	78,4
C	ethanoic acid	118,5

- Which of the compounds listed in the table are gases at room temperature?
 - Name the main type of intermolecular forces for A, B and C.
 - Account for the difference in boiling point between A and B.
 - Account for the difference in boiling point between B and C.
 - Draw the structural representations of A, B and C.
- Which container (A or B) has the compound with higher vapour pressure in it? Explain your answer.



- Draw the condensed structural formula for each of these compounds:
 - propanoic acid
 - butan-1-ol
- Propanoic acid has a vapour pressure of 0,32 kPa at 20 °C

TIP

Vanilla has a sweet smell. The smell of ethene and dimethyl ether cling in your nose though, and the smell of ether can even stay on your skin up to 24 hours.

FACT

In humans, ethanol reduces the secretion of a hormone called antidiuretic hormone (ADH). The role of ADH is to control the amount of water that the body retains. When this hormone is not secreted in the right quantities, it can cause dehydration because too much water is lost from the body in the urine (ethanol is a diuretic). This is why people who drink too much alcohol can become dehydrated, and experience symptoms such as headaches, dry mouth, and lethargy. Part of the reason for the headaches is that dehydration causes the brain to shrink away from the skull slightly.

Butan-1-ol has a vapour pressure of 0,64 kPa at 20 °C

Explain the difference in vapour pressure.

4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 27NF 1b. 27NG 2. 27NH 3a. 27NJ 3b. 27NK 3c. 27NM



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Physical properties and functional groups

ESCKR

Compounds that contain very similar atoms can have very different properties depending on how those atoms are arranged. This is especially true when they have different functional groups. Table 4.11 shows some properties of different functional groups.

Functional group	Typical Smell	Example	Formula	Melting point (°C)	Boiling point (°C)	Phase (at 25°C)
alkane	odourless	ethane	C ₂ H ₆	-183	-89	gas
alkene	sweet/musky	ethene	C ₂ H ₄	-169,2	-103,7	gas
ether (-O-)	sweet	dimethyl ether	C ₂ H ₆ O	-141	-24	gas
haloalkane	almost odourless	chloro ethane	C ₂ H ₅ Cl	-139	12,3	gas
aldehyde	pungent fruity	ethanal	C ₂ H ₄ O	-123,4	20,2	gas
alcohol	sharp	ethanol	C ₂ H ₆ O	-114	78,4	liquid
ester	often fruity	methyl methanoate	C ₂ H ₄ O ₂	-100	32	liquid
alkyne	odourless	ethyne	C ₂ H ₂	-80,8	-84	gas
carboxylic acid	vinegar rancid butter	ethanoic acid	C ₂ H ₄ O ₂	16,5	118,5	liquid

Table 4.11: Some properties of compounds with different functional groups.

Listed in the table are the common smells and other physical properties found for common functional groups. Only one representative example from each homologous series is provided. This does not mean that all compounds in that series have exactly the same properties. For example, short-chain and long-chain alkanes are generally odourless, while those with moderate chain length (approximately 6 - 12 carbon atoms) smell like petrol (Table 4.15).

Some specific properties of a few functional groups will now be discussed in more detail.

Physical properties of the alcohols

The hydroxyl group (–OH) affects the **solubility** of the alcohols.

DEFINITION: Solubility

Solubility is a measure of the ability of a substance (solid, liquid or gas) to dissolve in another substance. The amount of the substance that can dissolve is the measure of its solubility.

The hydroxyl group generally makes the alcohol molecule *polar* and therefore more likely to be **soluble** in water. However, the carbon chain resists solubility, so there are two opposing trends in the alcohols. Alcohols with shorter carbon chains are usually more soluble in water than those with longer carbon chains.

Alcohols tend to have higher **boiling points** than the hydrocarbons because of the strong hydrogen bond between hydrogen atoms of one hydroxyl group and the oxygen atom of another hydroxyl group.

Physical properties of haloalkanes

There can be more than one halogen substituted for a hydrogen atom on one haloalkane. The more halogens are substituted the less **volatile** the haloalkane becomes. Take for example the haloalkane series shown in Figure 4.59.

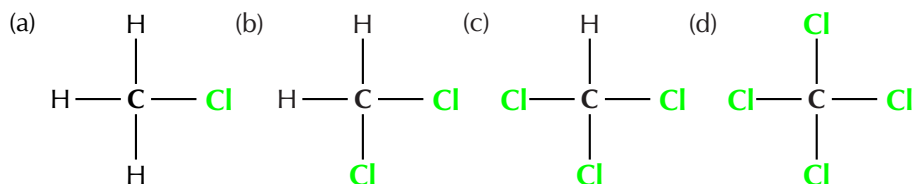


Figure 4.59: A series of haloalkanes with increasing numbers of chlorine atoms: (a) chloromethane, (b) dichloromethane, (c) trichloromethane and (d) tetrachloromethane.

DEFINITION: Volatility

The tendency of molecules at the surface of a compound to enter the gas phase. The more volatile a compound the more likely this is.

For every extra chlorine atom on the original methane molecule the volatility of the compound decreases. This can be seen by the increase in both the melting and boiling point (Table 4.12) as one goes from chloromethane through to tetrachloromethane. The more halogen atoms in the compound the stronger the intermolecular forces are, which leads to higher melting and boiling points.

Common Name	Number Cl atoms	Melting point (°C)	Boiling point (°C)
chloromethane	1	−97,4	−24,2
dichloromethane	2	−96,7	39,6
trichloromethane	3	−63,5	61,2
tetrachloromethane	4	−22,9	76,7

Table 4.12: Melting and boiling points of haloalkanes with increasing numbers of chlorine atoms.

Properties of carbonyl compounds

Carboxylic acids are **weak acids**, in other words they only dissociate partially. Why does the carboxyl group have acidic properties? In the carboxyl group, the hydrogen tends to separate itself (dissociate) from the oxygen atom. In other words, the carboxyl group becomes a source of positively-charged hydrogen ions (H^+). This is shown in Figure 4.60.

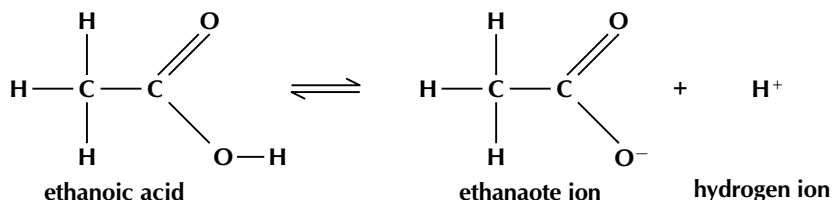
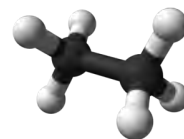


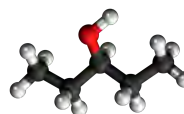
Figure 4.60: The dissociation of a carboxylic acid.

FACT

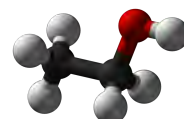
Ethane has a relatively low solubility in water.



Because hydroxyl ($-OH$) groups can hydrogen bond, all three pentanol molecules have a greater solubility in water than ethane.



Ethanol is completely soluble in water in any amount. This is because it contains a hydroxyl group and has a much shorter non-polar chain length than pentanol.



pentanol	solubility (g.dm ^{−3})
ethane	0,057
1-pentanol	22
2-pentanol	45
3-pentanol	59
ethanol	soluble

The carboxylic acid functional group is soluble in water. However, as the number of carbon atoms in the attached carbon chain increases the solubility decreases. This is discussed in greater detail in the next section.

Remember that carboxylic acids form hydrogen bonding dimers (the formation is called dimerisation) as shown in Figure 4.61.

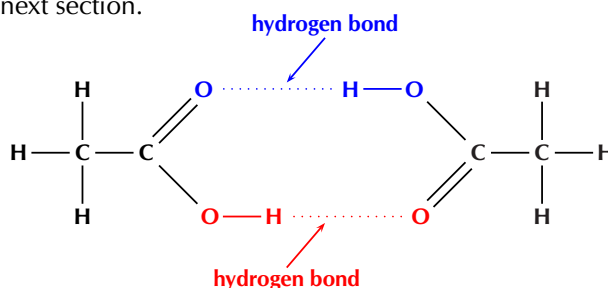


Figure 4.61: Ethanoic acid forming a carboxylic acid hydrogen bonding dimer.

The ability of a molecule to hydrogen bond leads to increased melting and boiling points when compared to a similar molecule that is unable to hydrogen bond. Similarly, the ability of a molecule to form a hydrogen bonding dimer leads to increased melting and boiling points when compared to similar molecules that can only form one hydrogen bond (Table 4.13).

Molecule	Hydrogen bonds per molecule	Melting point (°C)	Boiling point (°C)
ethane	0	-183	-89
ethanol	1	-114	78,4
ethanoic acid	2	16,5	118,5

Table 4.13: The melting and boiling points of similar organic compounds that can form different numbers of hydrogen bonds.

Physical properties of ketones

Hydrogen bonds are stronger than the van der Waals forces found in ketones. Therefore compounds with functional groups that can form hydrogen bonds are more likely to be soluble in water. This applies to aldehydes as well as to ketones.

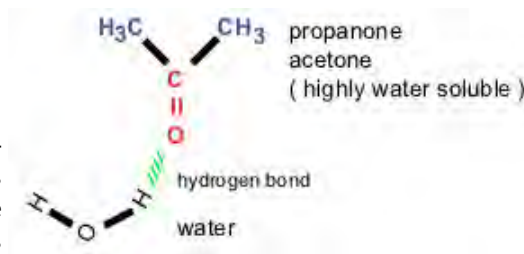


Figure 4.62: The hydrogen bond between water and a ketone (propanone).

Exercise 4 – 20: Physical properties and functional groups

1. Refer to the data table below which shows the melting point and boiling point for a number of organic compounds with different functional groups.

Formula	Name	Melting point (°C)	Boiling point (°C)
C ₄ H ₁₀	butane	-137	0
C ₄ H ₈	but-1-ene	-185	-6,5
C ₄ H ₁₀ O	butan-1-ol	-90	118
C ₄ H ₈ O ₂	butanoic acid	-7,9	163,5
C ₅ H ₁₂	pentane	-130	36
C ₅ H ₁₀	pent-1-ene	-165,2	30
C ₅ H ₁₂ O	pentan-1-ol	-78	138
C ₅ H ₁₀ O ₂	pentanoic acid	-34,5	186,5

- a) At room temperature (approx. 25°C), which of the organic compounds in the table are:

i. gases

ii. liquids

b) Look at an alkane, alkene, alcohol and carboxylic acid with the same number of carbon atoms:

i. How do their melting and boiling points compare?

ii. Explain why their melting points and boiling points are different?

2. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27NN



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Physical properties and chain length

ESCKS

Remember that the alkanes are a group of organic compounds that contain carbon and hydrogen atoms bonded together. The carbon atoms link together to form chains of varying lengths. We have already mentioned that the alkanes are relatively unreactive because of their stable C-C and C-H bonds. The boiling points and melting points of these molecules are determined by their molecular structure and their surface area.

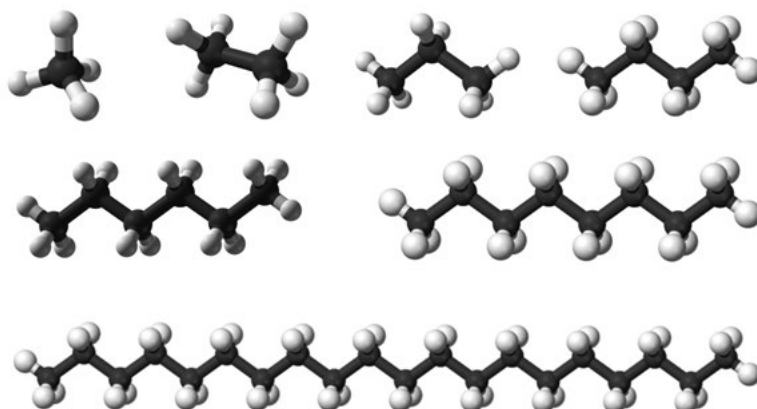


Figure 4.63: Some alkanes, from top left to bottom right: methane, ethane, propane, butane, hexane, octane and icosane.

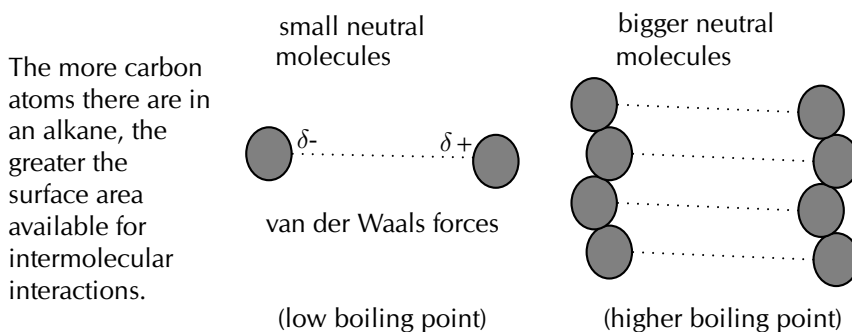


Figure 4.64: van der Waals intermolecular forces increase as chain length increases. (Image by Duncan Watson)

This increase in intermolecular attractions leads to higher melting and boiling points. This is shown in Table 4.14.

TIP

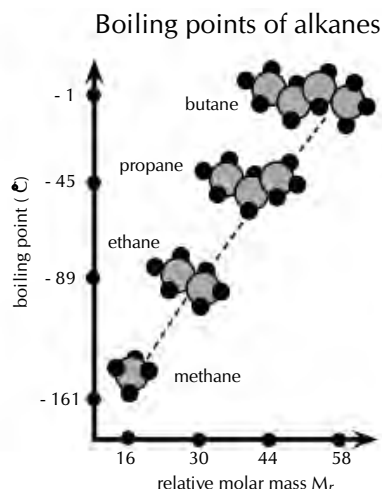
Be careful when comparing molecules with different types of intermolecular forces. For example, small molecule with hydrogen bonding can have stronger intermolecular forces than a large molecule with only van der Waals forces.

FACT

It is partly the stronger intermolecular forces that explain why petrol (mainly octane (C_8H_{18})) is a liquid, while candle wax ($\text{C}_{23}\text{H}_{48}$) is a solid. If these intermolecular forces did not increase with increasing molecular size we would not be able to put liquid fuel into our cars or use solid candles.

Formula	Name	Molecular mass (g.mol^{-1})	Melting point ($^{\circ}\text{C}$)	Boiling point ($^{\circ}\text{C}$)	Phase (at 25°C)
CH_4	methane	16,04	-182	-162	gas
C_2H_6	ethane	30,06	-183	-89	gas
C_3H_8	propane	44,08	-188	-42	gas
C_4H_{10}	butane	58,1	-137	0	gas
C_6H_{14}	hexane	86,14	-95	68,5	liquid
C_8H_{18}	octane	114,18	-57	125,5	liquid
$\text{C}_{20}\text{H}_{42}$	icosane	282,42	37	343	solid

Table 4.14: The physical properties of some alkanes.



Notice that when the molecular mass of the alkanes is low (i.e. there are few carbon atoms), the organic compounds are gases because the intermolecular forces are weak. As the number of carbon atoms and the molecular mass increases, the compounds are more likely to be liquids or solids because the intermolecular forces are stronger.

The larger a molecule is, the stronger the intermolecular forces are between the molecules. This is one of the reasons why methane (CH_4) is a gas at room temperature while hexane (C_6H_{14}) is a liquid and icosane ($\text{C}_{20}\text{H}_{42}$) is a solid.

Figure 4.65: As the molar mass of an alkane increases so does the boiling point. (Image by Duncan Watson)

Table 4.15 shows some other properties of the alkanes that also vary with chain length.

Name	Density (g.dm^{-3})	Flash point ($^{\circ}\text{C}$)	Smell	Phase (at 25°C)
methane	0,66	-188	odourless	gas
ethane	1,28	-135	odourless	gas
propane	2,01	-104	odourless	gas
butane	2,48	-60	like petrol	gas
hexane	650	-26	like petrol	liquid
octane	700	13	like petrol	liquid
icosane	790	>113	odourless	solid

Table 4.15: Properties of some of the alkanes.

Density increases with increasing molecular size. Note that compounds that are gaseous are much less dense than compounds that are liquid or solid.

Remember that the flash point of a volatile molecule is the lowest temperature at which that molecule can form a vapour mixture with air and be ignited. The flash point increases with increasing chain length meaning that the longer chains are less flammable (Table 4.10) although they can still be ignited.

The change in physical properties due to chain length does not only apply to the hydrocarbons. The solubility of ketones in water decreases as the chain length increases. The longer the chain length, the stronger the intermolecular interactions between the ketone molecules.

This means that more energy is required to overcome those interactions and the molecule is therefore less soluble in water. Long chains can also fold around the polar carbonyl group and stop water molecules from bonding with it.

This same idea can be applied to all the compounds with water-soluble functional groups. The longer the chain becomes, the less soluble the compound is in water.

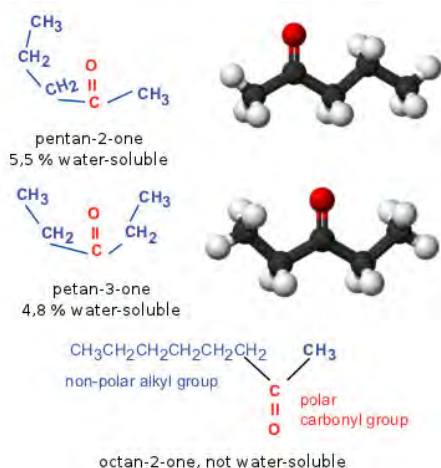


Figure 4.66: The water solubility of a ketone decreases as the chain length increases.

Exercise 4 – 21: Physical properties and chain length

1. Refer to the table below which gives information about a number of carboxylic acids, and then answer the questions that follow.

Condensed formula	Common name	Source	IUPAC name	Melting point (°C)	Boiling point (°C)
	formic acid	ants	methanoic acid	8,4	101
CH ₃ COOH		vinegar	ethanoic acid	16,5	118,5
	propionic acid	milk	propanoic acid	-20,8	141
CH ₃ (CH ₂) ₂ COOH	butyric acid	butter		-7,9	163,5
	valeric acid	valerian root	pentanoic acid	-34,5	186,5
CH ₃ (CH ₂) ₄ COOH	caproic acid	goat skin		-3,4	205,8
	enanthic acid	vines	heptanoic acid	-7,5	223
CH ₃ (CH ₂) ₆ COOH	caprylic acid	goat milk		16,7	239,7

- a) Fill in the missing spaces in the table by writing the formula, common name or IUPAC name.
 - b) Draw the structural representation of butyric acid.
 - c) Give the molecular formula for caprylic acid.
 - d)
 - i. Draw a graph to show the relationship between molecular mass (on the x-axis) and boiling point (on the y-axis)
 - ii. Describe the trend you see.
 - iii. Suggest a reason for this trend.
2. Refer to the data table below which shows the melting point and boiling point for a number of organic compounds with different functional groups.

Formula	Name	Melting point (°C)	Boiling point (°C)
C ₄ H ₁₀	Butane	-137	0
C ₅ H ₁₂	Pentane	-130	36
C ₆ H ₁₄	Hexane	-95	68,5
C ₄ H ₈	But-1-ene	-185	-6,5
C ₅ H ₁₀	Pent-1-ene	-165,2	30
C ₆ H ₁₂	Hex-1-ene	-140	63

- a) At room temperature (approx. 25 °C), which of the organic compounds in the table are: i. gases ii. liquids
- b) In the alkanes:
- Describe what happens to the melting point and boiling point as the number of carbon atoms in the compound increases.
 - Explain why this is the case.
3. Fill in the table below. Under boiling point put 1 for the compound with the lowest boiling point, 2 for the next lowest boiling point until you get to 6 for the compound with the highest boiling point. Do not use specific boiling point values, but rather use your knowledge of intermolecular forces.

Compound	Condensed Structure	Boiling point
propane	$\text{CH}_3\text{CH}_2\text{CH}_3$	
hexane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	
methane	CH_4	e.g. 1
octane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	
butane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$	
ethane	CH_3CH_3	

4. Draw the structural representations for each of the following compounds and answer the question that follows.
- but-2-yne
 - hex-2-yne
 - pent-2-yne
- d) Which of these compounds will have the highest viscosity?
5. More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.
1. 27NP 2. 27NQ 3. 27NR 4. 27NS



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Physical properties and branched groups

ESCKT

In a straight chain molecule, the carbon atoms are connected to *at most* two other carbon atoms. However, in a branched molecule some carbon atoms are connected to *three or four* other carbon atoms. This is the same principle that was discussed with primary, secondary and tertiary alcohols.

Straight chains always have higher boiling points than the equivalent molecule with branched chains. This is because the molecules with straight chains have a larger surface area that allows close contact. Branched chains have a lower boiling point due to a smaller area of contact. The molecules are more compact and cannot get too close together, resulting in fewer places for the van der Waals forces to act.

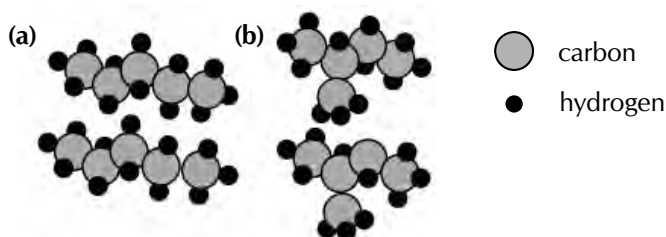


Figure 4.67: (a) Straight chains of pentane and (b) branched chains of 2-methylbutane. (Image by Duncan Watson)

Shown in Figure 4.68 are three isomers that all have the molecular formula C_5H_{12} . They are all alkanes with the only difference being branched groups leading to longer or shorter main chains.

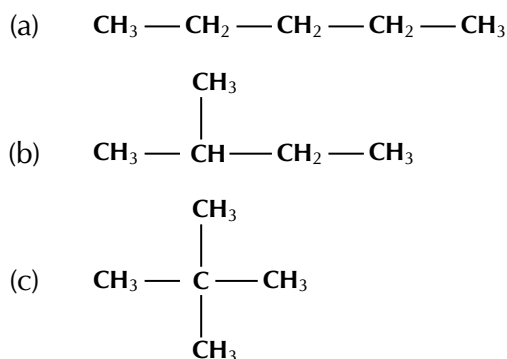


Figure 4.68: Three isomers of C_5H_{12} : (a) pentane, (b) 2-methylbutane and (c) 2,2-dimethylpropane.

As shown in Table 4.16 the properties of these three compounds are significantly different. The melting points have significant differences and the boiling points steadily decrease with an increasing number of branched groups.

Name	Melting point ($^{\circ}C$)	Boiling point ($^{\circ}C$)	Density ($g \cdot dm^{-3}$)	Vapour pressure (kPa at $20^{\circ}C$)	Flash point ($^{\circ}C$)
pentane	-130	36	621	57,9	-49
2-methylbutane	-160	27,7	616	77	-51
2,2-dimethylpropane	-16,6	9,5	586	147	<-7

Table 4.16: Properties of compounds with different numbers of branched groups.

Although the densities of all three compounds are very similar, there is a decrease in density with the increase of branched groups. The vapour pressure increases with increasing branched groups and the flash point of 2,2-dimethylpropane is significantly higher than those of pentane and 2-methylbutane.

It is interesting that the common trend of melting and boiling points is not followed here - 2,2-dimethylpropane has the highest melting point, lowest boiling point and is the least flammable although all three compounds are very flammable. **Symmetrical** molecules (such as 2,2-dimethylpropane) tend to have higher melting points than similar molecules with **less symmetry** due to their packing in the solid state. Once in the liquid state they follow the normal trends.

Exercise 4 – 22: Physical properties and branched chains

Structural Formula	Boiling point ($^{\circ}C$)
$\begin{array}{ccccccc} & H & H & H & H & & \\ & & & & & & \\ H & -C & -C & -C & -C & -Cl \\ & & & & & & \\ & H & H & H & H & & \end{array}$	78
1. $\begin{array}{c} H \\ \\ H-C-H \\ \quad \quad \\ H-C-C-C-H \\ \quad \quad \\ H \quad Cl \quad H \end{array}$	51

a) Give the IUPAC name for each compound

- b) Explain the difference in the melting points
2. There are five isomers with the molecular formula C_6H_{14} .
- Draw the structural representations of:
2-methylpentane and 2,2-dimethylbutane (two of the isomers)
 - What are the names of the other three isomers?
 - Draw the semi-structural representations of these three molecules.
 - The melting points of these three isomers are: -118 , -95 and -130 °C. Assign the correct melting points to the correct isomer. Give a reason for your answers.
3. More questions. Sign in at Everything Science online and click 'Practise Science'. Check answers online with the exercise code below or click on 'show me the answer'.
1. 27NT 2. 27NV



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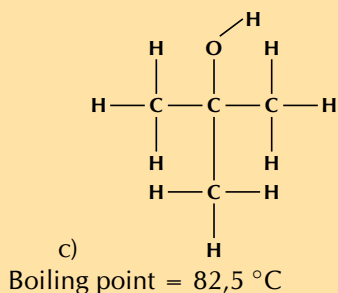
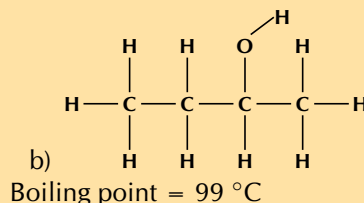
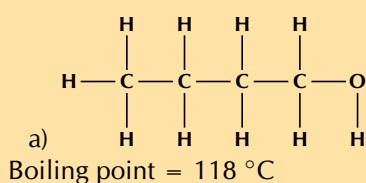
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Exercise 4 – 23: Physical properties of organic compounds

1. The table shows data collected for four organic compounds (A - C) during a practical investigation.

	Compound	Molecular Mass ($\text{g}\cdot\text{mol}^{-1}$)	Boiling point (°C)
A	$\text{CH}_3\text{CH}_2\text{CH}_3$	44,08	-42
B	CH_3CHO	44,04	20
C	$\text{CH}_3\text{CH}_2\text{OH}$	46,06	78

- Is compound **A** saturated or unsaturated? Give a reason for your answer.
 - To which homologous series does compound **B** belong?
 - Write down the IUPAC name for each of the following compounds:
 - B**
 - C**
 - Refer to intermolecular forces to explain the difference in boiling points between compounds **A** and **C**.
 - Which one of compounds **B** or **C** will have the highest vapour pressure at a specific temperature? Give a reason for your answer.
2. Give the IUPAC names for the following compounds and answer the questions that follow.



- d) Explain the difference in boiling points.

3. More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.

1. 27NW 2a. 27NX 2b. 27NY 2c. 27NZ 2d. 27P2



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4.5 Applications of organic chemistry

ESCKV

Alkanes as fossil fuels

ESCKW

Fossil fuels are fuels formed by the natural process of the decomposition of organisms under heat and pressure. They contain a high percentage of carbon and include fuels such as coal, petrol, and natural gases. They are also a **non-renewable**.

DEFINITION: *Hydrocarbon cracking*

Hydrocarbon cracking is the process of breaking carbon-carbon bonds in long-chain hydrocarbons to form simpler, shorter-chain hydrocarbons.

Hydrocarbon cracking is an important industrial process. Through this process long, bulky alkanes are broken up into smaller compounds. These compounds include shorter alkanes and alkenes. A few examples of cracking hydrocarbons are given in Figure 4.69.

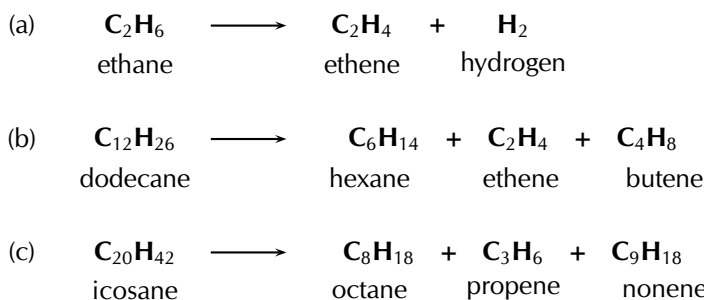


Figure 4.69: The cracking of various hydrocarbons to produce *alkanes* and *alkenes*.

There are two types of hydrocarbon cracking. **Thermal** cracking occurs under high pressures and temperatures without a catalyst, **catalytic** cracking occurs at lower pressures and temperatures in the presence of a catalyst. This process is a common source of shorter (more useful) alkanes as well as unsaturated alkenes. The alkanes are then used in combustion processes.

It is possible to separate the products of hydrocarbon cracking and obtain specific products from a crude oil mix through a process called *fractional distillation*. This is done using a *fractionating column*. Crude oil evaporates when heated to 700°C. The gas bubbles through a tray that is kept at a certain temperature. The alkanes and alkenes that condense at that temperature will then condense in the tray. For example if the tray is kept at 170°C the product will be paraffin oil (Figure 4.70).

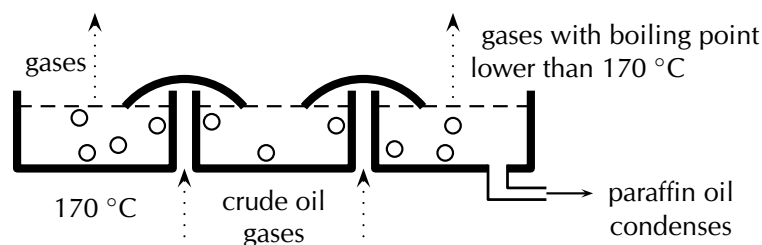


Figure 4.70: One layer of a fractionating column. (Image by Duncan Watson)

A fractionating column has a series of these trays (Figure 4.71), each at a constant temperature. This means that many compounds can be separated from the crude oil mix. The crude oil is heated to 700°C and the gas of the crude oil is passed through the column. Bitumen for tar roads is collected at the bottom of the fractionating column. These are all compounds with more than 70 carbon atoms. The temperature decreases as you move up the column. As the gases rise, compounds with different length carbon chains condense until only the chains with 1 - 4 carbon atoms are collected at the top of the column. These are used for liquid petroleum gas.

Note that this fits with what we learned in the previous section. The more carbon atoms in the chain, the greater the intermolecular forces and therefore the higher the boiling point. That means that these molecules will condense at higher temperatures.

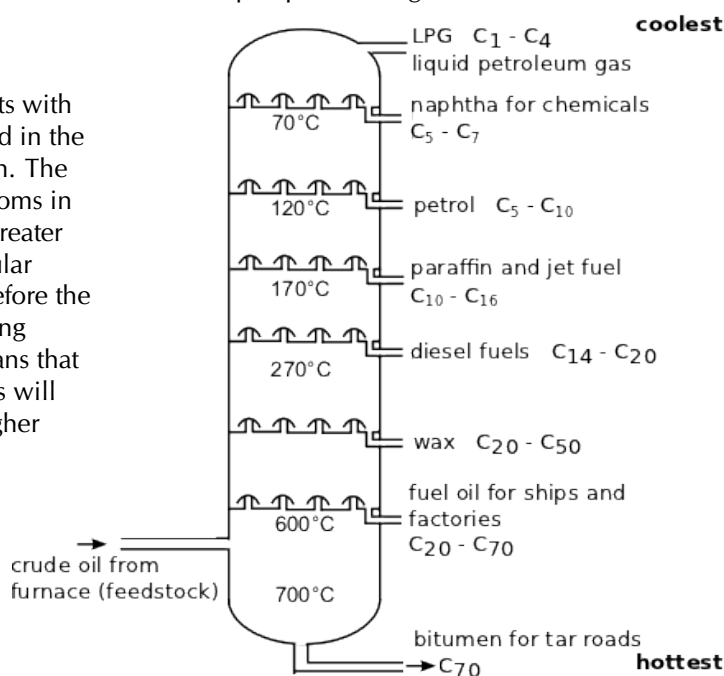


Figure 4.71: An example of a fractionating column.

For more information on the process of fractional distillation have a look at this animation:

▶ See simulation: [27P3](https://www.everythingscience.co.za/27P3) at www.everythingscience.co.za

Combustion of alkanes

Alkanes are our most important fossil fuels. The combustion (burning) of alkanes (also known as oxidation) is highly *exothermic*.

DEFINITION: Combustion

In a combustion reaction a substance reacts with an oxidising agent (e.g. oxygen), and heat and light are released.

In the complete combustion reaction of alkanes, carbon dioxide (CO_2) and water (H_2O) are released along with energy. Fossil fuels are burnt for the energy they re-

lease. The general reaction for the combustion of an alkane as a fossil fuel is given in Figure 4.72



Figure 4.72: The complete combustion of an alkane.

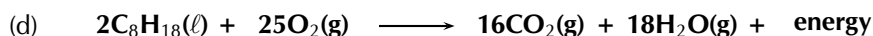
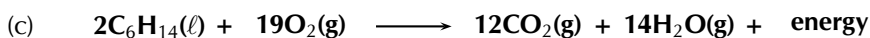
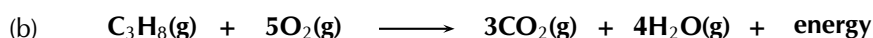
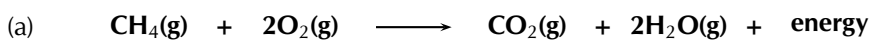


Figure 4.73: The complete combustion reactions of (a) methane, (b) propane, (c) hexane and (d) octane.

FACT

The complete combustion of alkanes produces only CO_2 and H_2O . Not all combustion process are complete though. An incomplete combustion will also produce carbon monoxide (CO).

TIP

Remember that an exothermic reaction releases energy ($\Delta H < 0$), while an endothermic reaction absorbs energy ($\Delta H > 0$). The fact that energy is released in a combustion reaction implies that $\Delta H < 0$ and the reaction is exothermic.

Worked example 28: Balancing equations

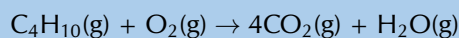
QUESTION

Balance the following equation: $\text{C}_4\text{H}_{10}(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$

SOLUTION

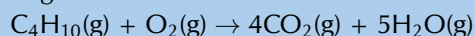
Step 1: Balance the carbon atoms

There are 4 carbon atoms on the left. There is 1 carbon atom on the right. Add a 4 in front of the CO_2 molecule on the right:



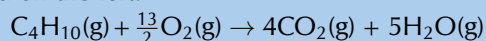
Step 2: Balance the hydrogen atoms

There are 10 hydrogen atoms on the left. There are 2 hydrogen atoms on the right. Add a 5 in front of the H_2O molecule on the right:



Step 3: Balance the oxygen atoms

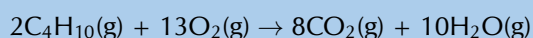
There are 2 oxygen atoms on the left. There are 13 oxygen atoms on the right (4×2 in the CO_2 and 5 in the H_2O). Divide the number of O atoms on the right by 2 to get $\frac{13}{2}$, this is the number of O_2 molecules required on the left:



This is acceptable but it is better for all numbers to be whole numbers.

Step 4: Make sure all numbers are whole numbers

There is $\frac{13}{2}$ in front of the O_2 while all other numbers are whole numbers. So multiply the entire equation by 2:



Worked example 29: Balancing equations

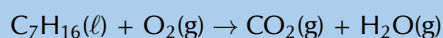
QUESTION

Balance the equation for the complete combustion of **heptane**

SOLUTION

Step 1: Write the unbalanced equation

The molecular formula for heptane is C_7H_{16} . Combustion always involves oxygen (O_2). The complete combustion of an alkane always produces carbon dioxide (CO_2) and water (H_2O):

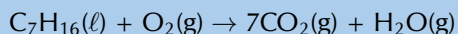


FACT

In the formation of an ester one oxygen atom comes from the alcohol molecule, while the carbonyl group comes from the carboxylic acid. This is known because it is possible to *label* (using radioactive nuclides) the atoms of the reactants and see where they end up in the products.

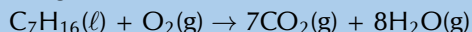
Step 2: Balance the carbon atoms

There are 7 carbon atoms on the left. There is 1 carbon atom on the right. Add a 7 in front of the CO_2 molecule on the right:



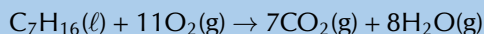
Step 3: Balance the hydrogen atoms

There are 16 hydrogen atoms on the left. There are 2 hydrogen atoms on the right. Add an 8 in front of the H_2O molecule on the right:



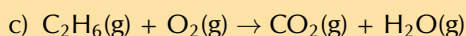
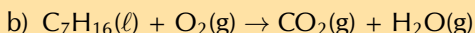
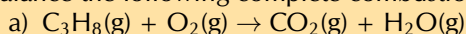
Step 4: Balance the oxygen atoms

There are 2 oxygen atoms on the left. There are 22 oxygen atoms on the right (7×2 in the CO_2 and 8 in the H_2O). Divide the number of O atoms on the right by 2 to get 11, this is the number of O_2 molecules required on the left:



Exercise 4 – 24: Alkanes as fossil fuels

1. Balance the following complete combustion equations:



2. Write the balanced equation for the complete combustion of:

a) octane

b) pentane

c) hexane

d) butane

3. Is the combustion of alkanes exothermic or endothermic? What does that mean?

4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 27P4 1b. 27P5 1c. 27P6 2a. 27P7 2b. 27P8 2c. 27P9

2d. 27PB 3. 27PC



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Esters

ESCKX

Production of esters

As was discussed earlier in this chapter one way to form an ester is through the reaction of an alcohol and a carboxylic acid. This process is called an **acid-catalysed condensation** or **esterification** of a carboxylic acid.

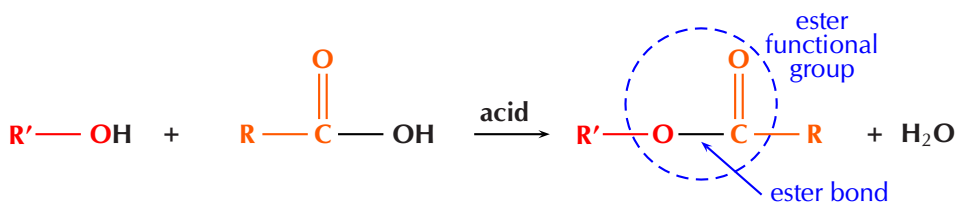


Figure 4.74: The acid-catalysed condensation of a carboxylic acid to form an ester.

In the general form above an alcohol (red) and a carboxylic acid (orange) combine to form an ester and water. A specific example is given in Figure 4.75 for the formation of butyl propanoate and water.

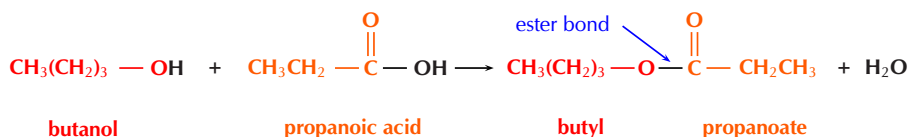
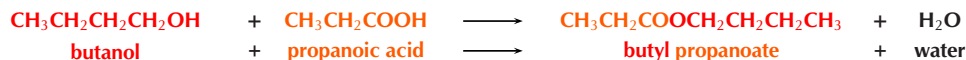


Figure 4.75: The esterification of butanol and propanoic acid to form butyl propanoate, water is also formed in this reaction.

This reaction can also be written as:



A more general example is:



It is important to be able to identify what ester a specific alcohol and carboxylic acid will form. Remember that the first part of the ester name takes its prefix from the alcohol with the suffix -yl. The second part of the ester takes its prefix from the carboxylic acid with the ester suffix -oate.

Worked example 30: Determining ester names

QUESTION

What is the name of the ester that will form from **hexanol** and **propanoic acid**

SOLUTION

Step 1: Which compound forms the first part of the ester name and which forms the second part of the ester name?

The alcohol forms the first part of the ester name and takes the suffix -yl. The carboxylic acid forms the second part of the ester name and takes the suffix -oate.

Step 2: Determine the first part of the ester name

The alcohol is *hexanol*, therefore there are 6 carbons and this will be hexyl.

Step 3: Determine the second part of the ester name

The carboxylic acid is *propanoic acid*, therefore there are 3 carbons and this will be propanoate.

Step 4: Combine the first and second parts of the ester name

The ester will be **hexyl propanoate**.

It is also important to be able to determine which compounds were used to form an ester.

Worked example 31: Determining starting materials of esters

QUESTION

What compounds did the ester **octyl heptanoate** come from?

SOLUTION

Step 1: What types of compounds are used to form esters?

Esters are formed from alcohols (which become the first part of the ester name) and carboxylic acids (which become the second part of the ester name).

Step 2: Determine the prefix for the alcohol

The first part of the ester name comes from the alcohol (-ol). Therefore the prefix is *oct-*.

Step 3: Determine the prefix for the carboxylic acid

The second part of the ester name comes from the carboxylic acid (-oic acid). Therefore the prefix is *hept-*.

Step 4: Determine the compounds used to form the ester

Octyl heptanoate was formed from **octanol** and **heptanoic acid**.

A few examples of esters are given in Table 4.17.

Ester name	Smell	Uses		
methyl methanoate	ether	quick-dry finishes	insecticide	pharmaceuticals
ethyl methanoate	rum	lacquers	safety glass	fumigating foods
propyl methanoate	pears	solvent	flavour	fragrance
methyl ethanoate	glue	glues	paints	nail polish remover
ethyl ethanoate	apple	glues etc	solvents	decaffeination
propyl ethanoate	pear	solvent	flavour	fragrance
butyl ethanoate	banana or apple	lacquers	flavour	
pentyl ethanoate	banana or apple	lacquers	solvents	resins
methyl butanoate	apple or pineapple	flavour	fragrance	
ethyl butanoate	pineapple	flavour	fragrance	plasticiser
propyl hexanoate	blackberry or cheese	flavour	solvent	fragrance

Table 4.17: The uses of some esters.

The following experiment will help you to prepare esters. Use what you have learned in this section to answer the questions that follow.

Experiment: Preparation of esters**Aim:**

To prepare and identify esters.

Apparatus:

- Methanol (CH_3OH), ethanol ($\text{CH}_3\text{CH}_2(\text{OH})$), pentan-1-ol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2(\text{OH})$), methanoic acid (HCOOH), ethanoic acid (CH_3COOH), sulfuric acid (H_2SO_4)
- Marble chips (or small, clean stones)

- Five tall test tubes or beakers, a water bath, a hot-plate (or bunsen burner), a thermometer, rubber bands, paper towel, five bowls of cold water

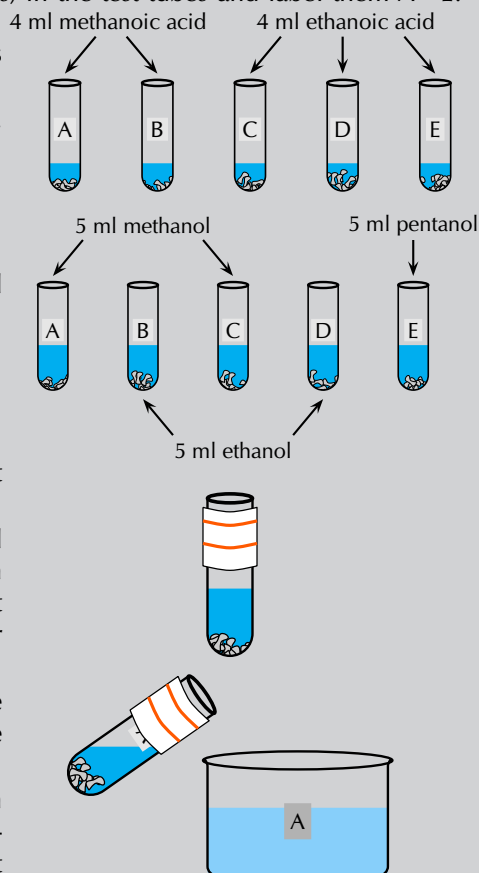
Method:

WARNING!

Concentrated acids can cause serious burns. We suggest using gloves and safety glasses whenever you work with an acid. Always add the acid to the water and avoid sniffing the acid.

Remember that all alcohols are toxic, methanol is particularly toxic and can cause blindness, coma or death. Handle all chemicals with care.

1. Place the marble chips (or clean stones) in the test tubes and label them A - E.
2. Add 4 ml methanoic acid to test tubes A and B.
3. Add 4 ml ethanoic acid to test tubes C, D and E.
4. Add 5 ml methanol to test tubes A and C.
5. Add 5 ml ethanol to test tubes B and D.
6. Add 5 ml pentanol to test tube E.
7. **Slowly** add 2 ml H_2SO_4 to each test tube.
8. Soak the paper towel in cold water and attach it to the sides near the top of each test tube with a rubber band (do not close the test tube, just wrap the paper around the sides near the top).
9. Heat the water bath to 60 °C (using the hot-plate or bunsen burner) and place the test tubes in it for 10-15 min.
10. After 10-15 min cool each test tube in cold water. Label the bowls of cold water A - E and pour the contents of test tube A into bowl A, etc.
11. Observe the surface of the water and note the smell in each bowl.



Questions:

1. What is the purpose of the H_2SO_4 ?
2. What is the purpose of the wet paper towel?
3. What did you observe on the top of the water in each bowl at the end?
4. Fill in the details in a table like this one:

Carboxylic acid	Alcohol	Product	Smell
methanoic acid	methanol		
methanoic acid	ethanol		
ethanoic acid	methanol		
ethanoic acid	ethanol		
ethanoic acid	pentanol		

Discussion and Conclusion:

An ester is the product of an acid-catalysed condensation between an alcohol and a carboxylic acid. Esters have identifiable aromas (like the fragrant smell and odour of fruit) and are used in perfumes. If you didn't smell anything it is likely that your water bath was set at too high a temperature and the ester degraded.

Esters are less water soluble than the carboxylic acid they were formed from and appear as an oily substance on water. Before being cooled however some of the ester would have been a vapour, so the wet paper towel would help to prevent a loss of the product.

Refer to Table 4.17 and see if the smells match those listed there. Remember to waft the smell towards you using your hand, do not sniff the fumes directly!

Exercise 4 – 25: Esters

1. Give the IUPAC name for the product in the esterification of ethanoic acid with:
a) methanol b) octanol c) hexanol d) propanol
2. What is another name for the type of reaction in the question above?
3. Give the IUPAC name for the product in the reaction of butanol with:
a) ethanoic acid b) pentanoic acid c) heptanoic acid d) methanoic acid
4. Fill in the missing reactant or part of the product name in the reactions below:
a) octanol + _____ $\rightarrow$ _____ hexanoate
b) _____ + propanoic acid $\rightarrow$ hexyl _____
c) _____ + butanol $\rightarrow$ _____ pentanoate
5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 27PD 1b. 27PF 1c. 27PG 1d. 27PH 2. 27PJ 3a. 27PK
3b. 27PM 3c. 27PN 3d. 27PP 4a. 27PQ 4b. 27PR 4c. 27PS



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4.6 Addition, elimination and substitution reactions ESKY

We will study three main types of reactions - addition, elimination and substitution.

- An **addition reaction** occurs when two or more reactants combine to form a single product. This product will contain *all* the atoms that were present in the reactants. Addition reactions occur with **unsaturated** compounds.

The general equation for an **addition** reaction: $A + B \rightarrow C$

Notice that C is the final product with no A or B remaining as a residue.

- An **elimination reaction** occurs when a reactant is broken up into two products. Elimination reactions occur with **saturated** compounds.

The general equation for an **elimination** reaction: $A \rightarrow B + C$

- A **substitution reaction** occurs when an exchange of elements in the reactants takes place. The initial reactants are transformed or *swopped around* to give a final product.

The general equation for a **substitution** reaction: $AB + CD \rightarrow AD + BC$

Some specific examples of these reactions are given in the following pages.

Addition reactions

ESCKZ

1. Hydrohalogenation

Hydrohalogenation involves the addition of a **hydrogen atom** and a **halogen atom** to an unsaturated compound (containing a carbon-carbon double bond). An example is given in Figure 4.76. X can be fluorine (F), chlorine (Cl), bromine (Br) or iodine (I).

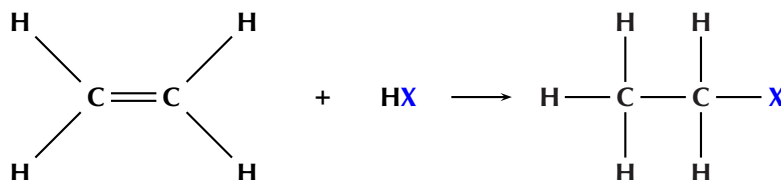


Figure 4.76: The hydrohalogenation of *ethene* to a *haloethane*.

If more than one product is possible the **major product** will be the compound where:

- the **hydrogen atom** is added to **least substituted carbon atom**
- i.e. the carbon atom with the least number of carbon atoms bonded to it
- the **halogen atom** is added to the **more substituted carbon atom**
- i.e. the carbon atom with the most number of carbon atoms bonded to it

This is shown in Figure 4.77.

DEFINITION: *Major and minor products*

The major product of a reaction is the product that is most likely to form. Minor products are those that are less likely to form.

FACT

Remember from the section on saturated and unsaturated structures that you can perform tests to determine if a compound is saturated or not. If bromine water or potassium permanganate are decolourised by a compound the compound is unsaturated, if they are not decolourised the compound is saturated.

TIP

Remember that a **tertiary carbon atom** is bonded to three other carbon atoms, a **secondary carbon atom** is bonded to two other carbon atoms, and a **primary carbon atom** is bonded to one other carbon atom. So tertiary is the most substituted, secondary is less substituted than tertiary and more substituted than primary, and primary is the least substituted.

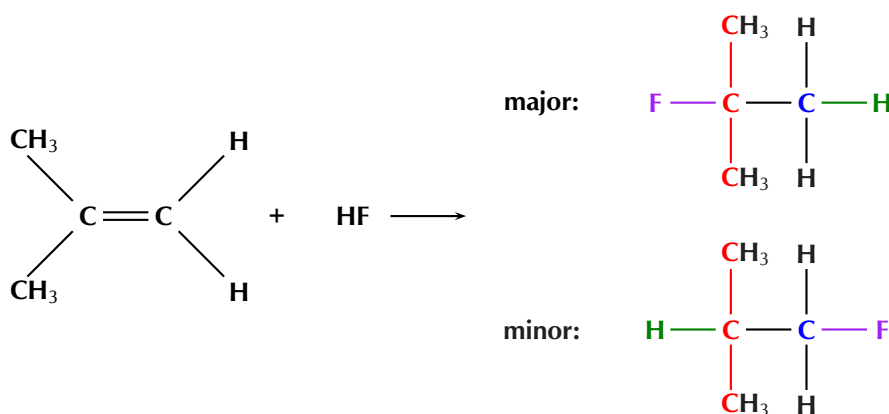


Figure 4.77: The hydrohalogenation of *2-methylpropene* to form **2-fluoro-2-methylpropane** (major product) and **1-fluoro-2-methylpropane** (minor product).

Reaction conditions:

- **no water** present in the reaction

2. Halogenation

Halogenation is very similar to hydrohalogenation but a **diatomic halogen molecule** is added across the double bond. An example is given in Figure 4.78.

FACT

Fermentation can refer to the conversion of sugar to alcohol using yeast (a fungus). The process of fermentation produces items such as wine, beer and yogurt. To make wine, grape juice is fermented to produce alcohol. This reaction is shown below:

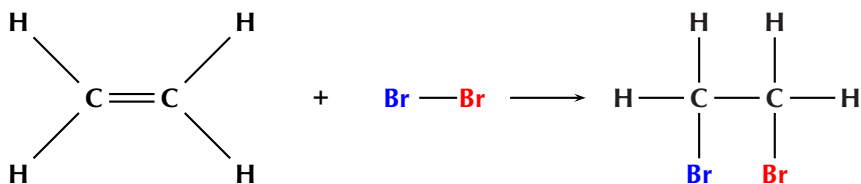
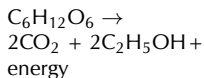


Figure 4.78: The reaction between *ethene* and *bromine* to form *1,2-dibromoethane*.

3. Hydration

A hydration reaction involves the addition of **water** (H_2O) to an **unsaturated compound**. This is one way of preparing an alcohol from the corresponding alkene (Figure 4.79).

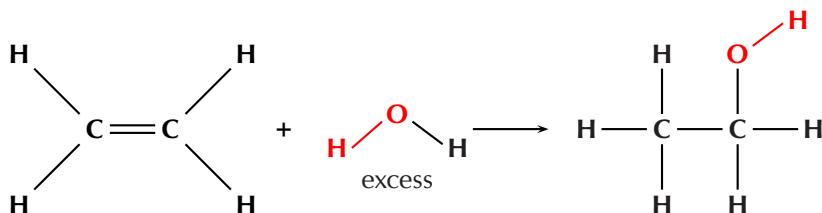


Figure 4.79: The hydration of *ethene* to *ethanol*.

If more than one product is possible the **major product** will be the compound where:

- the **hydrogen atom** is added to the **least substituted carbon atom**
- the **hydroxyl anion** (OH^-) is added to the **more substituted carbon atom**

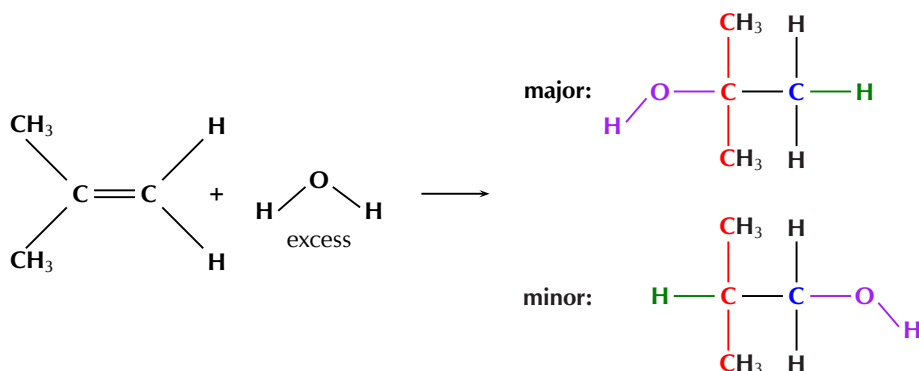


Figure 4.80: The hydration of *2-methylpropene* to form **2-methylpropan-2-ol** (major product) and **2-methylpropan-1-ol** (minor product).

Reaction conditions:

- water must be present in **excess**
- an **acid catalyst** is needed for this reaction to take place
- the catalyst that is most commonly used is phosphoric acid (H_3PO_4)

Activity: Formation of alcohol through fermentation

Ethanol can also be formed through the process of fermentation. In almost every country there is a local product that people make by fermenting local fruits. For example in Malawi people ferment potatoes, sugar cane or maize to form *katchasu*. Here in South Africa people ferment marula fruit to make *mampoer*, grapes to make *witblits* and sorghum to make beer.

- Research the chemistry behind the fermentation process.

- What other plants do people ferment?
- Fermentation leads to acid formation, research why this occurs.
- Sorghum is also used to make sour porridge. How is it made?

4. Hydrogenation

Hydrogenation involves adding **hydrogen** (H_2) to an **alkene**. During hydrogenation the double bond is broken (as with hydrohalogenation and halogenation) and more hydrogen atoms are added to the molecule. A specific example is shown in Figure 4.81.

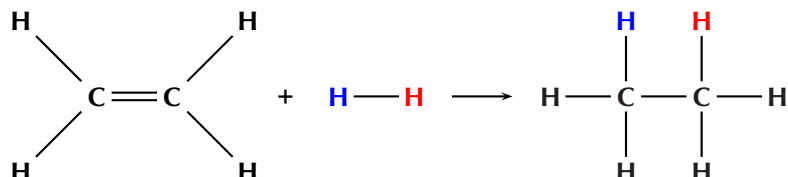


Figure 4.81: The hydrogenation of *ethene* to *ethane*.

Reaction conditions:

- a **catalyst** such as platinum (Pt), palladium (Pd) or nickel (Ni) is needed for these reactions
- **heating** is required
- the reaction must be done under an **inert atmosphere**, not air (e.g. $\text{N}_2(\text{g})$ atmosphere)

The hydrogenation of vegetable oils to form margarine is another example of this addition reaction (see Figure 4.82).

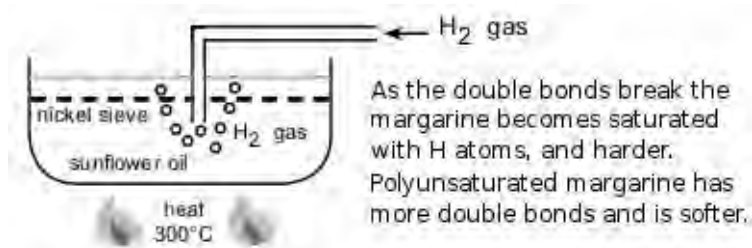


Figure 4.82: The hydrogenation of sunflower oil to make margarine.

5. Polymerisation reactions

A polymer is made up of lots of smaller units called *monomers*. When these monomers are added together, they form a polymer. One way for *polymerisation* to occur is through an **addition** reaction. More details are given later in this chapter in *Plastics and polymers* (Section 4.7).

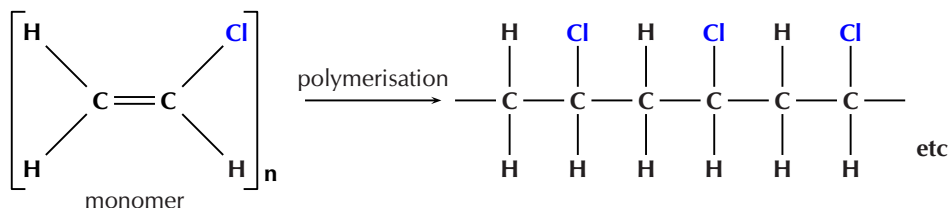
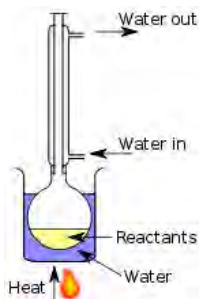


Figure 4.83: The polymerisation of **vinyl chloride** monomers to form a **polyvinyl chloride** polymer.

Polyvinyl chloride (Figure 4.101) is used in construction and in clothing, as well as having many other uses.

FACT

Reflux is a technique whereby a reaction solution is placed in a container with a small opening at the top. A tube that is constantly being cooled is connected to the opening. The solution is heated to approximately its boiling point, but any gases produced are condensed in the tube and fall back into the container. This way the reaction rate can be increased without the loss of material.

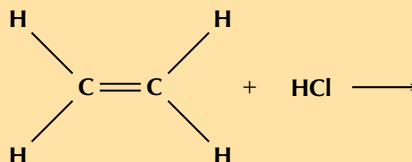


TIP

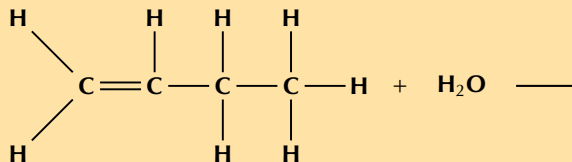
See hydrolysis in the substitution reactions section for more information on what happens when different reaction conditions are used.

Exercise 4 – 26: Addition reactions

1. The following diagram shows the reactants in an addition reaction.



- a) Draw the structural representation of the final product in this reaction.
b) What is the condensed structural representation of the product?
2. The following diagram shows the reactants in an addition reaction.



- a) Draw the structural representation of the major product in this reaction and give reasons.
b) Draw the structural representation of the minor product in this reaction and give reasons.
c) What type of addition reaction is this reaction?
3. a) Which types of homologous series can undergo addition reactions?
b) Is this a saturated or unsaturated series?
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27PT 2. 27PV 3. 27PW



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Elimination reactions

ESCM2

1. Dehydrohalogenation

In dehydrohalogenation a **haloalkane** is exposed to a **base**, the base then helps the elimination of the halogen and a hydrogen atom. A double bond is formed (alkane $\rightarrow$ alkene). Dehydrohalogenation is considered the opposite of hydrohalogenation. The elimination of iodine from iodoethane is an example of dehydrohalogenation (Figure 4.84).

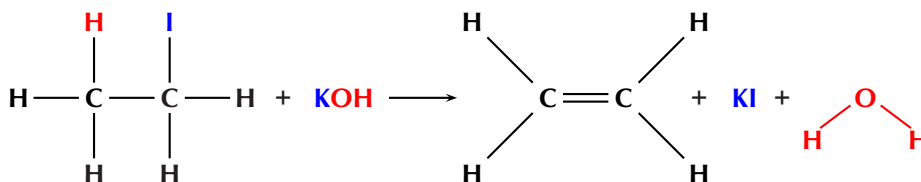


Figure 4.84: The dehydrohalogenation of iodoethane with potassium hydroxide (KOH) to form ethene, potassium iodide (KI) and water.

In order for elimination to occur the following reaction conditions must be used:

- heat under reflux (approximately 70°C)
- a concentrated, strong base (e.g. NaOH, KOH)
- the base must be dissolved in pure ethanol (hot ethanolic base)

Dihalogenated saturated compounds can also undergo elimination reactions to become unsaturated, losing one halogen and one hydrogen atom. In the example below (Figure 4.85), an atom of hydrogen and chlorine are **eliminated** from the original compound to form an unsaturated haloalkene and hydrochloric acid.

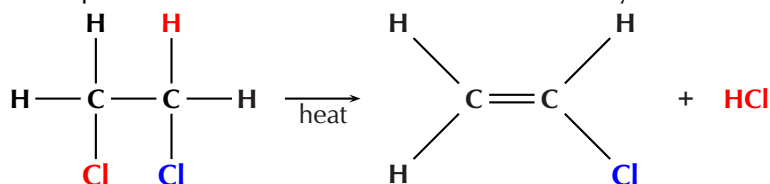


Figure 4.85: The elimination of hydrochloric acid (HCl) from 1,2-dichloroethane to form chloroethene.

If more than one product is possible the major product will be the compound where:

- the hydrogen atom is removed from the **more substituted carbon atom**
- i.e. the carbon that is bonded to the most number of carbon atoms

See Figure 4.86.

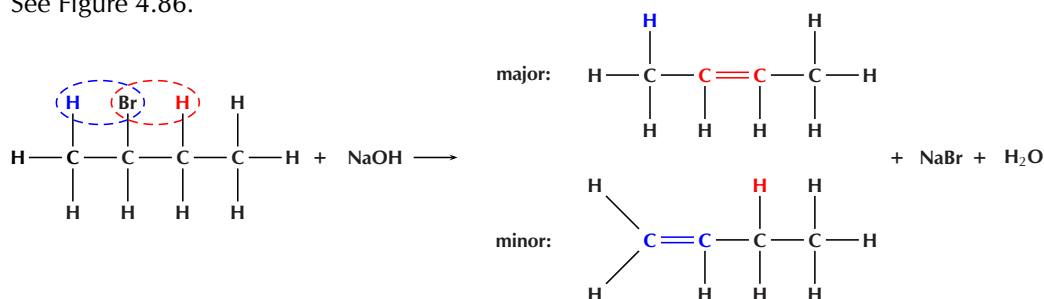


Figure 4.86: The dehydrohalogenation of 2-bromobutane to form **but-2-ene** (major product) and **but-1-ene** (minor product).

2. Dehydration of an alcohol

During the dehydration of an alcohol the **hydroxyl** (-OH) group and a **hydrogen atom** are **eliminated** from the reactant. A molecule of water is formed as a product in the reaction, along with an alkene (Figure 4.87). This can be thought of as the reverse of a hydration (addition) reaction.

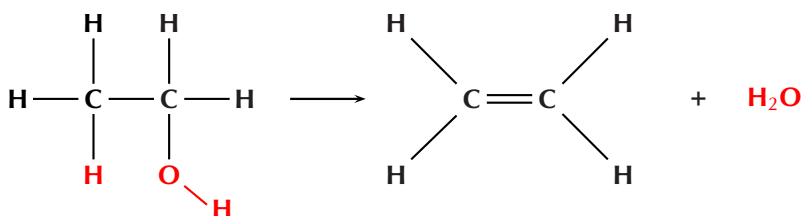


Figure 4.87: The dehydration of *ethanol* to form *ethene* and H_2O .

Reaction conditions:

- an **excess** of a **strong acid catalyst** (generally H_2SO_4 or H_3PO_4)
- high temperature (approximately 180°C)

If more than one elimination product is possible the major product will be the compound where:

- the hydrogen atom is removed from the carbon atom bonded to the most number of carbon atoms (i.e., the **more substituted carbon atom**)

This is shown in Figure 4.88.

FACT

Sulfuric acid (H_2SO_4) is a dehydrating agent and is often used to prepare dried fruits.



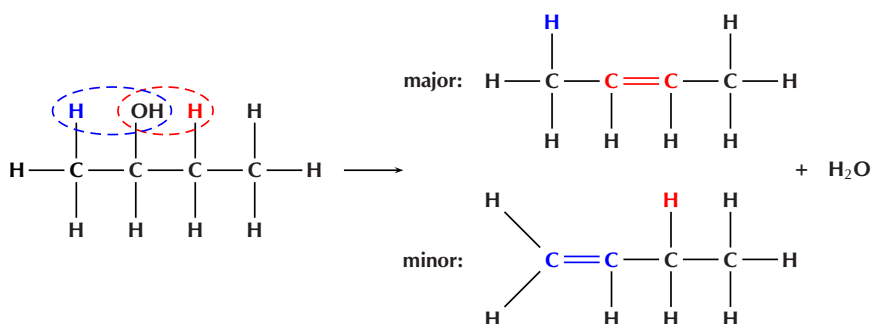


Figure 4.88: The dehydration of *butan-2-ol* to form **but-2-ene** (major product) and **but-1-ene** (minor product).

Exercise 4 – 27: Elimination reactions

- Answer the following questions about elimination reactions:
 - What two compounds do you need for a dehydrohalogenation reaction?
 - What carbon atom will the hydrogen atom be removed from in a dehydration reaction (major product)?
 - Are the reactants in an elimination reaction saturated or unsaturated?
 - Name the elimination reaction that is considered the reverse of a hydration addition reaction
 - More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.
1. 27PX 2. 27PY



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Substitution reactions

ESCM3

Some simple examples of substitution reactions are shown below:

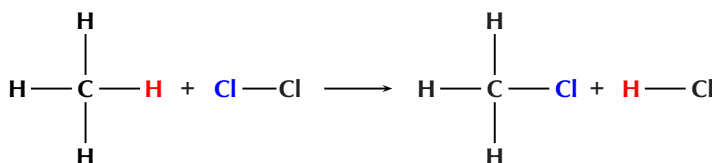
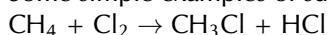


Figure 4.89: A chlorine atom (from Cl_2) and a hydrogen atom (from CH_4) are exchanged to create new products (CH_3Cl and HCl).

1. Formation of haloalkanes

Haloalkanes can be formed when the hydroxyl ($-\text{OH}$) group of an **alcohol** is replaced by a **halogen atom** ($\text{X} = \text{Cl}, \text{Br}$). This reaction works best with tertiary alcohols where it can occur at room temperature (Figure 4.90)

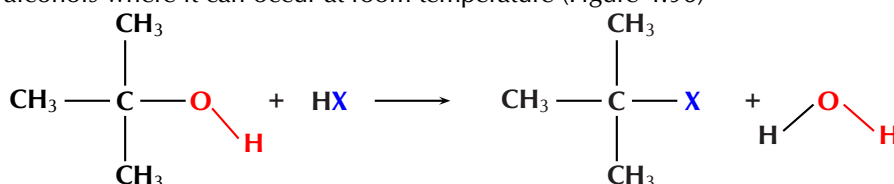


Figure 4.90: The formation of a *tertiary haloalkane* from a *tertiary alcohol* by a substitution with HX ($\text{X} = \text{Cl}, \text{Br}$).

A *tertiary* carbon atom is attached to three other carbon atoms. A *secondary* carbon atom is attached to two other carbon atoms while a *primary* carbon atom is attached to one other carbon atom. In Figure 4.90 the carbon is attached to the functional group as well as three other carbon atoms. This makes it a tertiary alcohol on the left of the arrow and a tertiary haloalkane on the right.

It is possible for a secondary or primary alcohol to form a secondary or primary haloalkane as well (see Figure 4.91) but this requires high temperatures and the reaction is very slow.

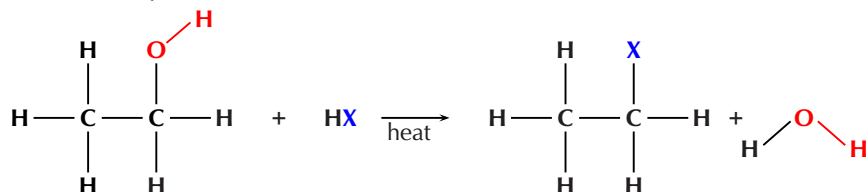


Figure 4.91: The formation of a *primary haloethane* from *ethanol* by a substitution with HX (X = Cl, Br).

2. Hydrolysis

In a previous section we saw the hydration of an ethene to form an alcohol. That is an addition reaction. Alcohols can also be formed through a substitution reaction with a **haloalkane**. In the example given in Figure 4.92 the haloalkane would be dissolved in water.

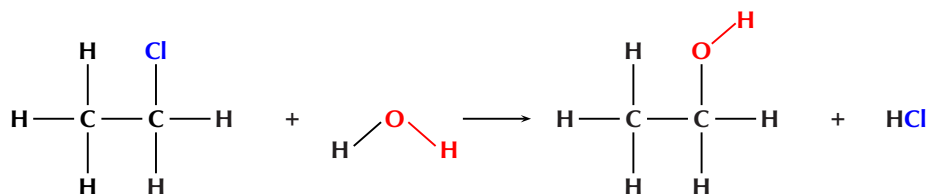


Figure 4.92: The formation of *ethanol* from *chloroethane* through a substitution reaction with water.

Using a base (e.g. KOH, NaOH) dissolved in water and warming the solution would increase the rate of the reaction (Figure 4.93). However, in order for substitution to occur the following reaction conditions must be used:

- low temperatures (around room temperature)
- a dilute solution of a strong base (e.g. NaOH)
- the solution must be aqueous (in water)

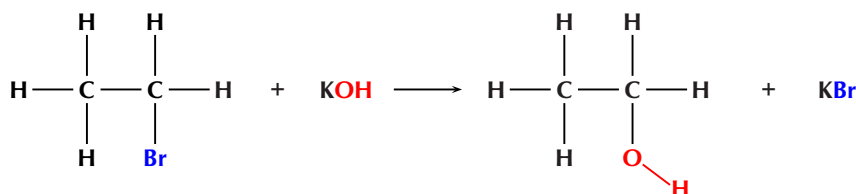


Figure 4.93: The formation of *ethanol* from *bromoethane* through a substitution reaction with potassium hydroxide (KOH).

If the concentration of the base is too high then the halogen atom and a hydrogen atom could be eliminated (dehydrohalogenation).

3. Formation of haloalkanes from alkanes

Another way of forming a haloalkane involves the removal of a hydrogen atom from a **saturated compound**. The hydrogen atom is replaced by a **halogen** (F, Cl, Br or I) to form a haloalkane (Figure 4.94). As alkanes are not very reactive light is needed for this reaction to take place.

TIP

See dehydrohalogenation in the elimination reactions section for more information on what happens when different reaction conditions are used.

FACT

Refer to the section on saturated and unsaturated structures for more information on why alkanes are not very reactive.

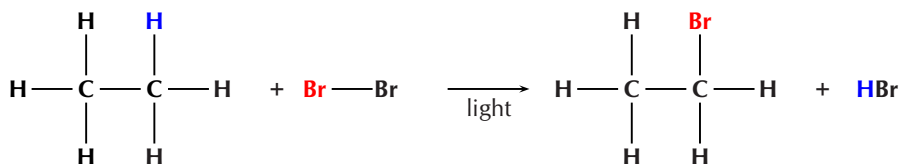


Figure 4.94: The formation of *bromoethane* from *ethane* through a substitution reaction with Br_2 .

Reaction conditions:

- energy in the form of light

Exercise 4 – 28: Substitution reactions

- When forming haloalkanes through a substitution reaction with alcohols is it best to use a tertiary, secondary or primary alcohol? Give a reason for your answer.
 - What is a substitution reaction?
 - Write the general equation for a substitution reaction using the reactants WX and YZ
 - More questions. Sign in at Everything Science online and click 'Practise Science'.
- Check answers online with the exercise code below or click on 'show me the answer'.

1. 27PZ 2. 27Q2



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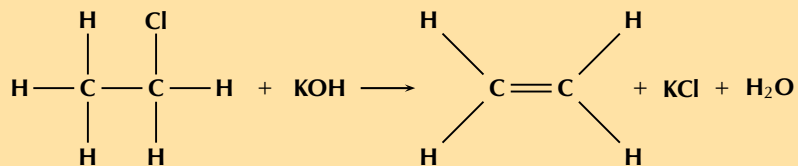
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Exercise 4 – 29: Addition, elimination and substitution reactions

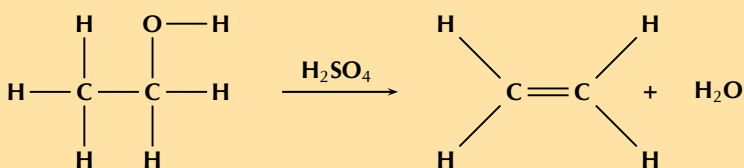
- Answer the following questions about hydration reactions:
 - What type of reaction is a hydration reaction?
 - What is the most common catalyst used in this reaction?
 - Name one of the most common catalysts used in a hydrogenation reaction.
 - Answer the following questions about alkane reactions
 - Are alkanes reactive compared to alkenes and alkynes?
 - In a haloalkane reaction an alkane and a halogen react. What condition is necessary for this reaction to take place?
 - Look at the general equations given below and answer the questions that follow:
 - haloalkane + base $\rightarrow$ A + B
 - Name A and B for a hydrolysis reaction.
 - What type of reaction is a hydrolysis reaction?
 - haloalkane + base $\rightarrow$ A + B + C
 - Name A, B and C for a dehydrohalogenation reaction.
 - What type of reaction is a dehydrohalogenation reaction?
 - Fill in the reaction conditions required for dehydrohalogenation and hydrolysis reactions in the table below:

	Dehydrohalogenation	Hydrolysis
Temperature		
Solvent		
Concentration of base		

- Refer to the diagram and then answer the questions that follow:



- What type of reaction is this reaction?
 - Give a reason for your answer above.
 - What are the required reaction conditions?
3. Write the equation for the following reactions using the condensed structural representation. Say what type of reaction they are:
- The dehydration of hexan-2-ol ($\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$)
 - The hydration of pent-1-ene ($\text{CH}_2\text{CHCH}_2\text{CH}_2\text{CH}_3$)
 - The hydrolysis of 2-fluorobutane ($\text{CH}_3\text{CH}(\text{F})\text{CH}_2\text{CH}_3$) with water (no base).
4. The following reaction takes place:



- What type of reaction is this?
 - Give a reason for your answer.
5. Write the equations for the following reactions using condensed structural representations. State the type of reaction as well.
- The hydrogenation of propene (CH_2CHCH_3)
 - The formation of a haloalkane from:
 - butan-1-ol ($\text{CH}_2(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$) and hydrochloric acid (HCl)
 - 2-methylpent-2-ene ($\text{CH}_3\text{C}(\text{CH}_3)\text{CHCH}_2\text{CH}_3$) and hydroiodic acid (HI).
Indicate the major and minor products and give reasons.
 - Write balanced equations for the following reactions involving hept-3-ene ($\text{CH}_3\text{CH}_2\text{CHCHCH}_2\text{CH}_2\text{CH}_3$) and name the product.
 - Hydrohalogenation, choose any halogen
 - The dehydrohalogenation with potassium hydroxide (KOH) of the product from c) i
 - Halogenation, choose any halogen
 - The dehydrohalogenation of the product from c) iii
6. More questions. Sign in at Everything Science online and click 'Practise Science'.
- Check answers online with the exercise code below or click on 'show me the answer'.
- 1a. [27Q3](#) 1b. [27Q4](#) 1c. [27Q5](#) 1d. [27Q6](#) 2. [27Q7](#) 3. [27Q8](#)
 4. [27Q9](#) 5. [27QB](#)



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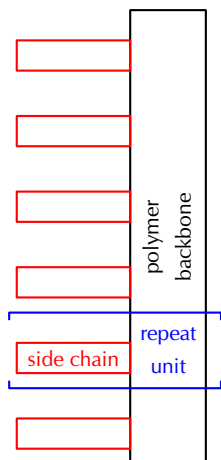
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FACT

Mono means one, while **poly** means many. So a monomer is the single unit, and a polymer is made from many monomers.

TIP

A polymer consists of a backbone, made up of repeating units.



FACT

A thermoplastic is a plastic that can be heated and molded multiple times without changing chemically.



4.7 Plastics and polymers

ESCM4

What is a polymer?

ESCM5

Polymers are large molecules (macromolecules) that are made up of many repeating structural units called **monomers** which have various functional groups. To put it more simply, a monomer is like a building block.

DEFINITION: Monomer

A monomer is a small molecule that can be combined through a reaction to form a polymer. It is a repeating unit in the polymer.

When lots of monomers are joined together by covalent bonds, they form a **polymer**.

DEFINITION: Polymer

Polymer is a term used to describe large molecules consisting of repeating structural units (monomers) connected by covalent chemical bonds.

In an **organic polymer**, the monomers are joined by the *carbon* atoms of the polymer backbone or *chain*. A polymer can also be **inorganic**, in which case there may be atoms such as *silicon* in the place of carbon atoms. We will look solely at organic polymers. Polymers are a specific group of *macromolecules*. A macromolecule is any compound with a large number of atoms. A *biological macromolecule* is one that is found in living organisms. Biological macromolecules include molecules such as carbohydrates, proteins, nucleic acids and lipids. They are essential for all known forms of life to survive.

DEFINITION: Macromolecule

A macromolecule is any molecule containing a large number of atoms.

The key feature that makes a polymer different from other macromolecules is the repetition of identical or similar monomers in the polymer chain. Polymers will contain chains of the same type of functional group and that functional group is dependent on the monomer used. The examples given over the following pages will help to make these concepts clearer. *Plastics* are a group of polymers that can be molded during manufacture. They can be one polymer or a blend (mixture) of polymers and may contain other substances as well. These other substances can be inorganic (e.g. used for electronic packaging) or stabilising (e.g. used for increasing fire resistance).

DEFINITION: Plastic

A subgroup of organic polymers that can be molded. Plastics may contain more than one organic polymer as well as other additives.

Activity: What is a polymer?

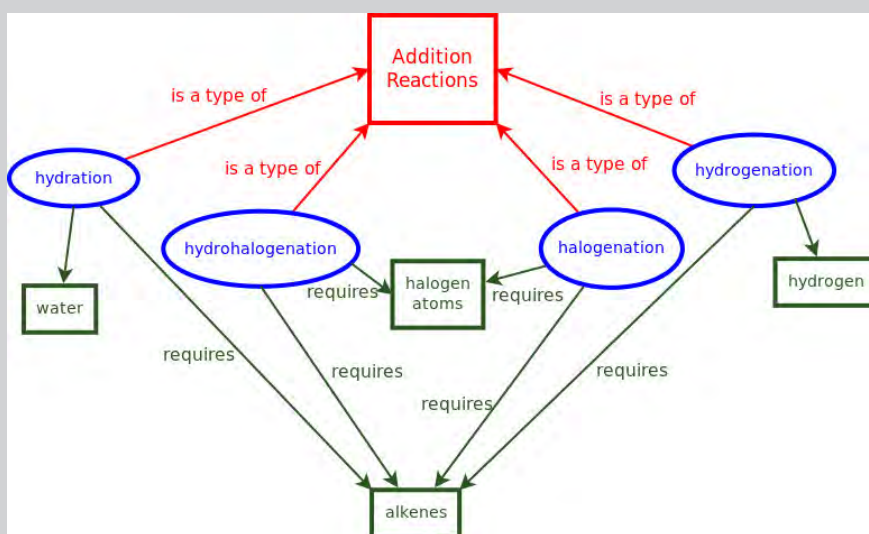
1. SASOL polymers is the largest division of the Johannesburg based companies' chemical business. Divide the class into 4 groups and have each group research the production of polymers in South Africa, including investigations into one of the products listed below:

- Kevlar and Mylar
- Break pads
- Windscreens
- Polymers to replace glass

Research the functions of the materials, who invented or discovered them, and what they are made of. Present your findings to the rest of the class in a five minute discussion.

2. Read this section (What is a polymer?) carefully. Research any concepts you don't understand and draw up a concept map (mindmap) describing what a polymer is.

A concept map is a way of showing concepts and linking knowledge using graphics. An example is given below:



How do polymers form?

ESCM6

Polymers are formed through a process called **polymerisation**, where monomers react together to form a polymer chain. Two of the types of polymerisation reactions are **addition polymerisation** and **condensation polymerisation**.

DEFINITION: Polymerisation

Polymerisation is a process of bonding monomers, or *single units* together to form longer chains called polymers.

Addition polymerisation

In this type of reaction, **monomer molecules** are added to a growing **polymer chain** one at a time. (No *small molecules* are eliminated in the process).

Four major examples of addition polymers are polyethylene, polypropylene, polyvinylchloride (PVC) and polystyrene (Table 4.18). All four of these organic polymers are also plastics.

Polymer name	Production (metric tons)	Some uses		
polyethylene	>80 million	plastic bags	bottles	plastic films
polypropylene	>45 million	labelling	textiles	stationery
polyvinyl chloride	>30 million	construction	clothing	insulation
polystyrene	>2 million	packaging	molds	cutlery

Table 4.18: Common uses of four major polymers formed through addition reactions.

1. Polyethene (polyethylene)

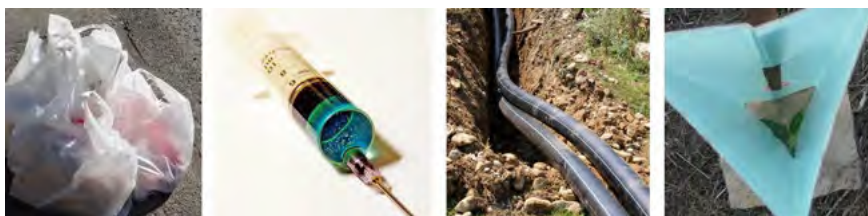


Figure 4.95: A collection of products made from polyethylene: plastic bags, a syringe, pipes, a plastic tree guard.

Earlier in this chapter we looked at the structure of a group of hydrocarbons called the *alkenes*. One example is the molecule **ethene**. The structural formula of ethene is shown in Figure 4.96. When lots of ethene molecules bond together, a polymer called **polyethene** (commonly called polyethylene) is formed. Ethene is the *unsaturated monomer* which, when joined to other ethene molecules through an addition reaction, forms the *saturated polymer* **polyethene**.

Polyethene is the most common plastic with over 80 million metric tons produced each year. It is commonly known as polyethylene. It is cheap and is used to make squeeze bottles, plastic bags, films, toys and molded objects as well as electric insulation. It has a recycling number 4 which means that it is easy to process, has strength, toughness, flexibility, is easy to seal and has a barrier to moisture.

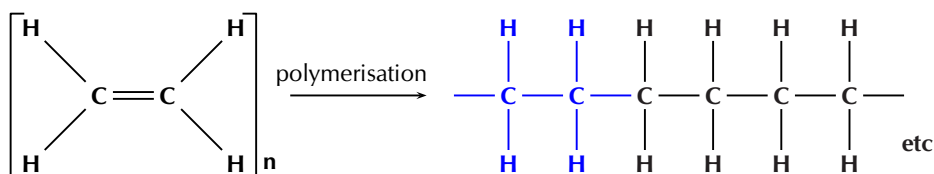


Figure 4.96: The polymerisation of an **ethene** monomer to form a **polyethene** polymer. The repeat unit is highlighted in blue.

A polymer may be a chain of thousands of monomers, and so it is impossible to draw the entire polymer. Rather, the structure of a polymer can be condensed and represented as shown in Figure 4.97.

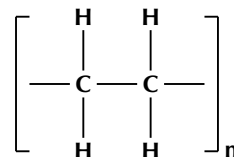


Figure 4.97: A simplified representation of polyethene.

The monomer is enclosed in brackets and the n represents the number of **repeating units** (the saturated form of the monomer) in the polymer, where n is any whole number. What this shows is that the monomer is repeated an indefinite number of times in a molecule of polyethene.

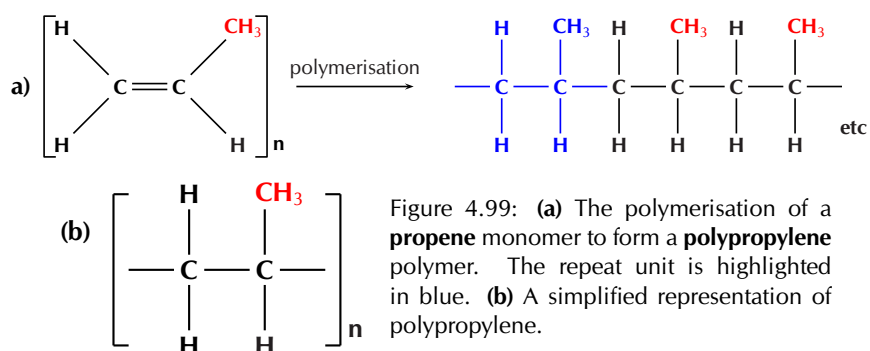
► See video: 27QC at www.everythingscience.co.za

2. Polypropylene (polypropylene)



Figure 4.98: A collection of products made from polypropylene: a lamp cover, computer parts, and shopping trolleys.

Another example of a polymer is *polypropylene* (Figure 2). Polypropylene (commonly known as polypropylene) is also a plastic, but is stronger than polyethene and is used to make crates, fibres and ropes as well as being used in textiles, stationery and car parts. In this polymer, the monomer is the alkene called **propene**.



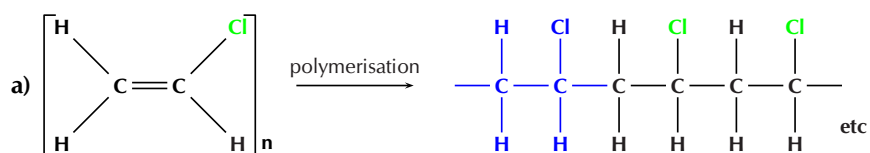
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3. Polyvinyl chloride (PVC)



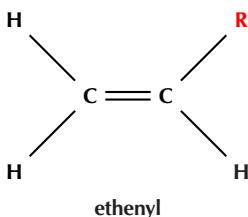
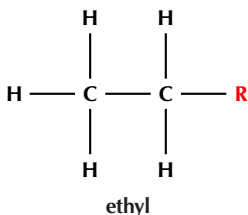
Figure 4.100: A collection of products made from polyvinyl chloride: pipes, electrical tape, and car parts.

Polyvinyl chloride or *PVC* (Figure 4.101) is formed from the monomer chloroethene, which is commonly known as vinyl chloride. PVC is used in construction, especially plastic piping. With the addition of a plasticiser it is also used in clothing and upholstery and to replace rubber. The role of the plasticiser is to increase the ability of a material to change shape without breaking.

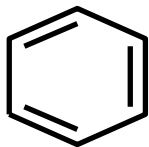


TIP

Ethenyl is not the same as ethyl. Ethyl is an ethane molecule bonded to another compound (R). Ethenyl is an ethene molecule bonded to another compound (R).

**FACT**

Benzene can be represented with three double bonds:



or with a circle inside the ring:

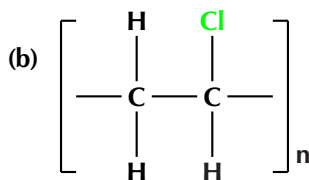
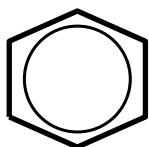


Figure 4.101: **(a)** The polymerisation of a **chloroethane** monomer to form a **polyvinyl chloride** polymer. The repeat unit is highlighted in blue. **(b)** A simplified representation of polyvinyl chloride.

See video: 27QF at www.everythingscience.co.za

4. Polyvinyl acetate

Polyvinyl acetate or *PVA* (Figure 4.103) is formed from the monomer ethenyl ethanoate, which is commonly known as vinyl acetate. PVA is used in various glues and adhesives (such as wood glue).

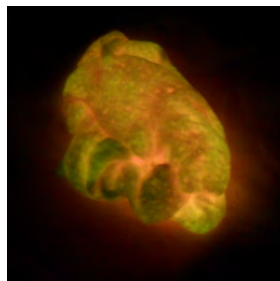


Figure 4.102: Glowing slime made with polyvinyl acetate.

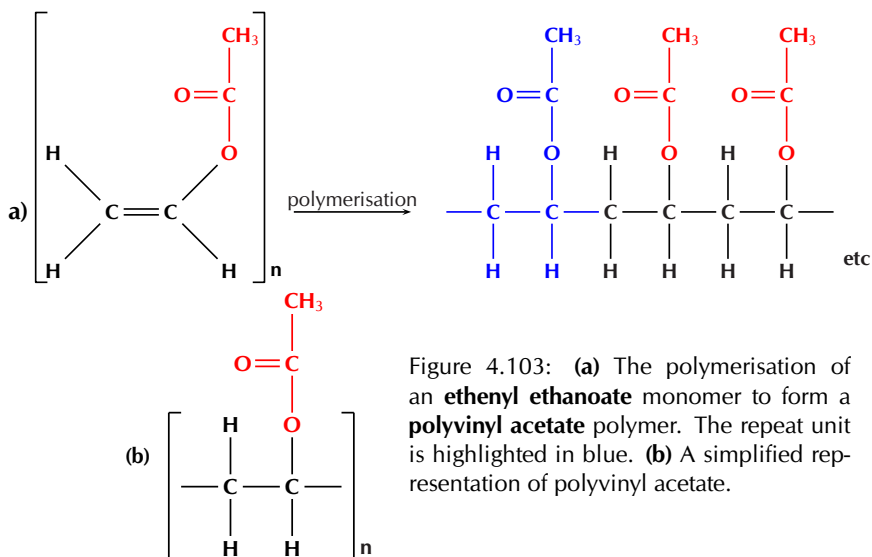


Figure 4.103: **(a)** The polymerisation of an **ethenyl ethanoate** monomer to form a **polyvinyl acetate** polymer. The repeat unit is highlighted in blue. **(b)** A simplified representation of polyvinyl acetate.

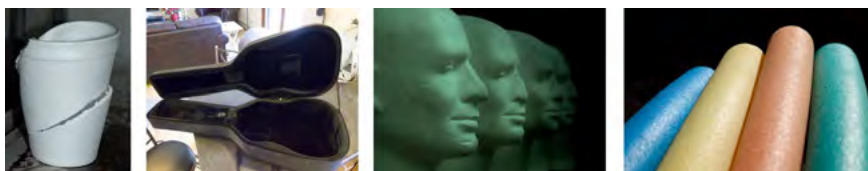
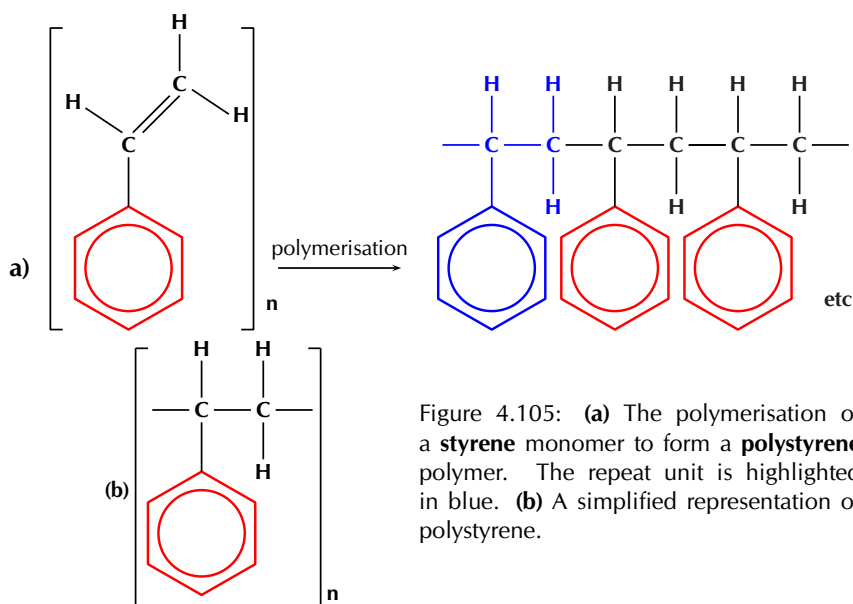
5. Polystyrene

Figure 4.104: A collection of products made from polystyrene: a cup, a guitar case, polystyrene carvings, and floating pool noodles.

Polystyrene is made from the monomer *styrene* which is a liquid petrochemical. Styrene consists of a benzene ring (a six membered ring with three double bonds) bonded to an ethene chain. Polystyrene is an aromatic polymer and has many uses including protective packaging, in trays, as plastic lids and bottles.



FACT

Remember that the IUPAC name of:

- ethylene is ethene
- propylene is propene
- vinyl chloride is chloroethene

Figure 4.105: (a) The polymerisation of a **styrene** monomer to form a **polystyrene** polymer. The repeat unit is highlighted in blue. (b) A simplified representation of polystyrene.

► See video: 27QG at www.everythingscience.co.za

It is interesting to note that the polymerisation of a monomer leads to different physical properties of the polymer (Table 4.19). For the polymers the melting points are dependent on the polymer grade.

Name	Melting point (°C)	Boiling point (°C)	Phase (at 25°C)
ethylene	-169,2	-103,7	gas
polyethylene	105 - 130	does not boil	solid
propylene	-185,2	-47,6	gas
polypropylene	130 - 171	does not boil	solid
vinyl chloride	-153,8	-13,4	gas
polyvinyl chloride	100 - 260	does not boil	solid
vinyl acetate	-93	72,7	liquid
polyvinyl acetate	60	does not boil	solid
styrene	-30	145	liquid
polystyrene	240	does not boil	solid

Table 4.19: Different physical properties of addition monomers and polymers.

Activity: Development of addition polymers

Polyethene was discovered by accident, twice. Look into the history of this commonly used plastic as well as other addition polymers.

- Who discovered them?
- How were they discovered?
- What difference have they made to modern life?

Activity: Building polymers

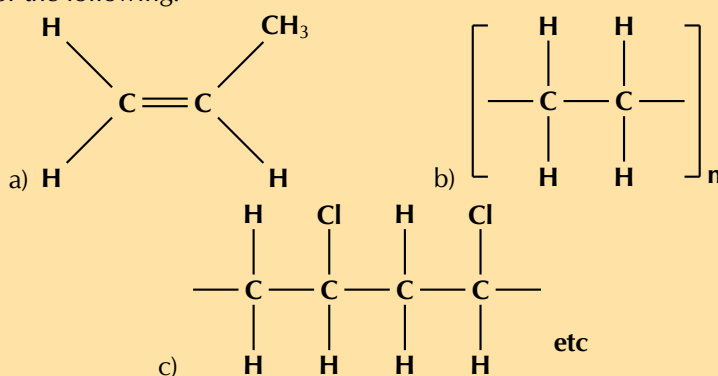
Using atomic model kits, jelly tots, or playdough and toothpicks build four ethene monomers, four propene monomers and four vinyl chloride monomers.

- Join three of each type of monomer to form polyethene, polypropene and polyvinyl chloride.
- Identify the repeating unit in each polymer.
- List the differences and similarities between the monomers and the repeating unit.

Note that the monomers all contain double bonds, while the repeating unit has only a single bond.

Exercise 4 – 30: Addition polymers

1. What trend is visible when comparing the melting points of monomers and their polymers?
2. Name the monomer (where polymer is given) or polymer (where monomer is given) for the following:



3. More questions. Sign in at Everything Science online and click 'Practise Science'.
Check answers online with the exercise code below or click on 'show me the answer'.

1. 27QH 2. 27QJ



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Condensation polymerisation

In this type of reaction, two monomer molecules combine by means of a covalent bond, and a small molecule such as water is lost in the bonding process. Nearly all biological macromolecules are formed using this process. **Polyesters** are polymers that form through condensation polymerisation.

Polyesters have a number of characteristics which make them very useful. They are resistant to stretching and shrinking, they are easily washed and dry quickly, and they are resistant to mildew. It is for these reasons that polyesters are being used more and more in **textiles**. Polyesters are stretched out into fibres and can then be made into fabric and articles of clothing. In the home, polyesters are used to make clothing, carpets, curtains, sheets, pillows and upholstery.

Polyesters are a group of polymers that contain the **ester** functional group in their main chain. This bond is called an **ester linkage**. For a polyester however there needs to be continuation of the chain. This requires a *-diol* (two alcohol functional groups) and a *diacid* (two carboxylic acid functional groups) as shown in Figure 4.106, or a monomer that contains both a hydroxyl group and a carboxylic acid.

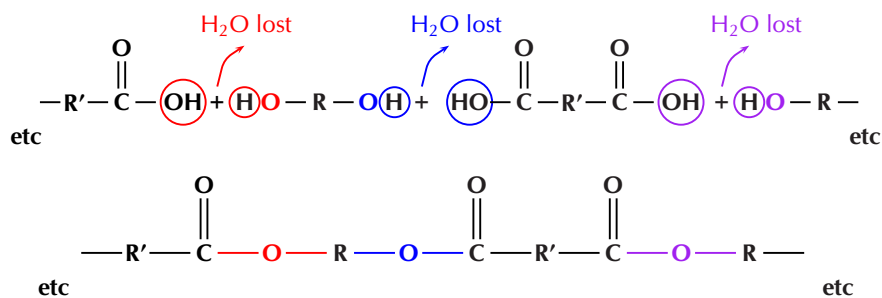


Figure 4.106: The propagation of the **ester linkage** in a **polyester**.

FACT

Polyethylene terephthalate (PET) is not just a textile. PET is in fact a plastic which can also be used to make plastic drink bottles. Many drink bottles are recycled by being reheated and turned into polyester fibres. This type of recycling helps to reduce disposal problems.

1. Polyethylene terephthalate (PET)



Figure 4.107: A collection of products made from polyethylene terephthalate: sailcloth and plastic squeeze bottles.

Although there are many forms of polyesters, the term *polyester* usually refers to *polyethylene terephthalate (PET)*. PET is made from ethan-1,2-diol (ethylene glycol, an alcohol) and terephthalic acid (a carboxylic acid). In the reaction, a hydrogen atom is lost from the alcohol, and a hydroxyl group is lost from the carboxylic acid. Together these form one water molecule which is lost during condensation reactions. A new bond is formed between an oxygen and a carbon atom. This bond is called an **ester linkage**. The reaction is shown in Figure 4.108.

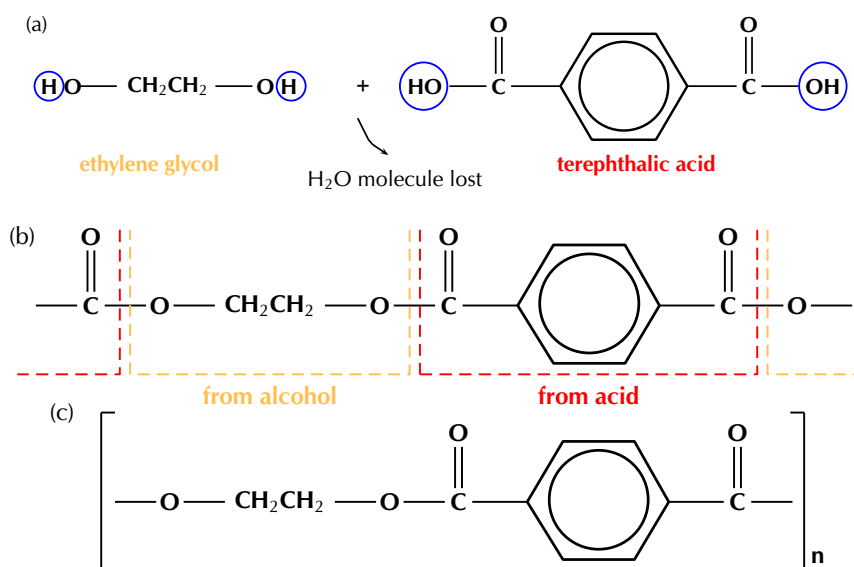


Figure 4.108: (a) Ethylene glycol (alcohol) and terephthalic acid (carboxylic acid) monomers react to form (b) the polymer poly(ethylene terephthalate) (PET). (c) A simplified representation of PET.

► See video: 27QK at www.everythingscience.co.za

FACT

3D printing is the process of printing a solid, three-dimensional object. If you have a 3D printer it is possible to print objects ranging from fashion items (e.g. cell phone covers), to human organs, although organs require living cells as the 'ink'.

2. Polylactic acid (PLA)

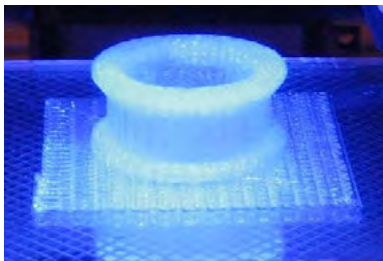


Figure 4.109: 3-D printing using polylactic acid as the ink.

Despite the name, polylactic acid (PLA) is actually a polyester (look for the ester linkage we spoke about before). It is an interesting polymer because the monomer used for this polymer (Figure 4.110) comes from the biological fermentation of plant materials, while most monomers used in plastics come from petroleum. As a result PLA is biodegradable and has low carbon dioxide (CO₂) emissions.

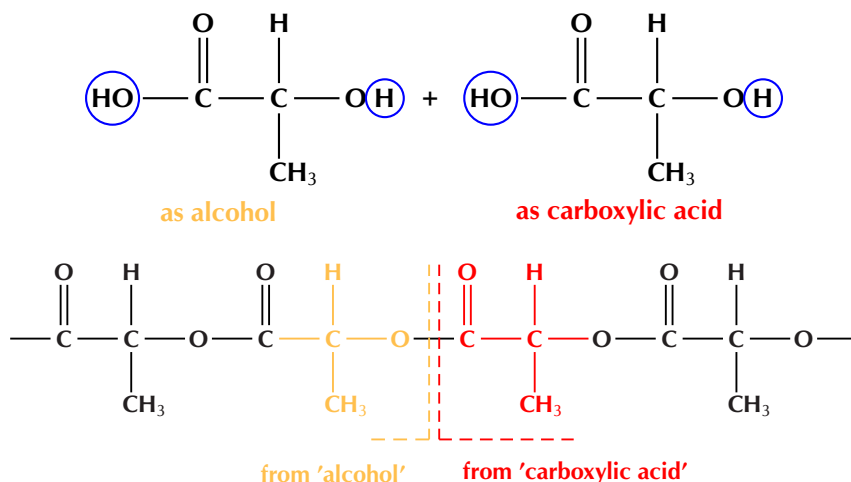


Figure 4.110: Ester linkages between lactic acid monomers.

PLA is mostly used for packaging material and, because it is biodegradable, it has the potential to alleviate land-fill disposal problems. The PLA polymer structural repeat unit is given in Figure 4.111.

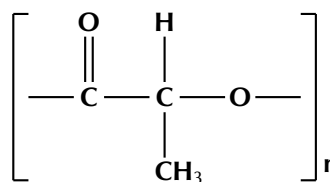


Figure 4.111: The polyester polylactic acid.

▶ See video: 27QM at www.everythingscience.co.za

Polymer name	Some uses		
polyethylene terephthalate	synthetic fibre	plastic containers	resins
polylactic acid	medical implants	packaging	textiles

Table 4.20: Some common uses of polymers formed through condensation reactions.

For polyethylene terephthalate there are two monomers *ethylene glycol* and *terephthalic acid* that join to form the polymer. Each monomer has its own unique properties (Table 4.21).

Name	Monomer/ polymer	Melting point (°C)	Boiling point (°C)	Phase (at 25°C)
terephthalic acid	monomer	300	sublimes (solid → gas)	solid
ethylene glycol	monomer	-12,9	197,3	liquid
polyethylene terephthalate	polymer	>250	decomposes	solid
lactic acid	monomer	16,8	122	liquid
polylactic acid	polymer	150 - 160	does not boil	solid

Table 4.21: Different physical properties of condensation monomers and polymers.

Exercise 4 – 31: Condensation polymers

1. What is the main difference between the reactants used in an addition polymerisation and those used in a condensation polymerisation?
2. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27QN



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Extension

Initiation, propagation and termination

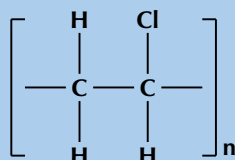
There are three stages in the process of addition polymerisation. **Initiation** refers to a chemical reaction that triggers the first reaction. In other words, initiation is the starting point of the polymerisation reaction. **Chain propagation** is the part where monomers are continually added to form a longer and longer polymer chain. During chain propagation it is the reactive end groups of the polymer chain that react in each propagation step to add a new monomer to the chain. Once a monomer has been added the reactive part of the polymer is now in this last monomer unit so that propagation will continue. **Termination** refers to a chemical reaction that destroys the reactive part of the polymer chain so that propagation stops.

It is important to be able to identify the monomer used to produce a polymer from the repeat unit of the chain.

Worked example 32: Identifying monomers

QUESTION

Which monomer was used to make this polymer (give the name and draw the structure):



SOLUTION

Step 1: Identify the polymer through functional groups

This polymer contains a chlorine atom. Which polymers contain chlorine? Polyvinyl chloride contains chlorine.

Step 2: Consider what you know of this polymer

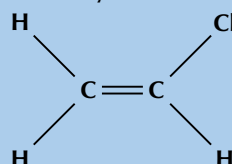
PVC forms through an addition reaction. There are two carbon atoms in the repeat unit (ethane) and the polymer contains chlorine atoms.

Step 3: Apply this knowledge to the monomer

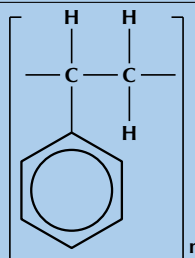
An addition reaction indicates that the monomer must contain a double bond. As the polymer contains units of ethane the monomer must be an ethene. The monomer must contain a chlorine atom.

Step 4: Name the monomer

The monomer must be **chloroethene** also known as **vinyl chloride**

Step 5: Draw the structure of this monomer**Worked example 33: Identifying monomers****QUESTION**

Which monomer was used to make this polymer (give the name and draw the structure):

**SOLUTION****Step 1: Identify the polymer through functional groups**

This polymer contains a benzene ring connected to two carbon atoms. Which polymers contain a benzene ring? Polystyrene contains a benzene ring connected to two carbon atoms.

Step 2: Consider what you know of this polymer

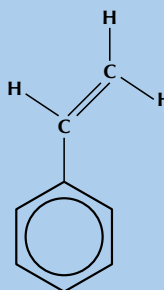
Polystyrene forms through an addition reaction. There are two carbon atoms attached to the benzene ring.

Step 3: Apply this knowledge to the monomer

An addition reaction means that the monomer must contain a double bond. The polymer contains two carbon atoms attached to the benzene ring which must be connected by a double bond in the monomer.

Step 4: Name the monomer

The monomer must be **styrene**.

Step 5: Draw the structure of this monomer

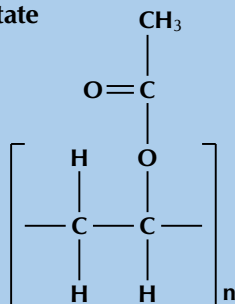
Worked example 34: Identifying monomers

QUESTION

Give the structure of the monomer used to make polyvinyl acetate:

SOLUTION

Step 1: Draw the repeat unit structure of polyvinyl acetate

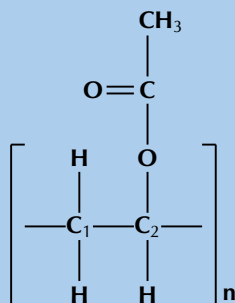


Step 2: Consider what you know of this polymer

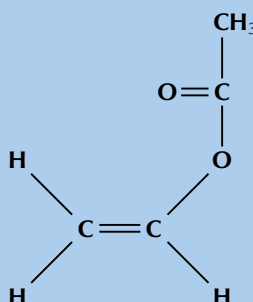
Polyvinyl acetate must form from vinyl acetate.
The IUPAC name of vinyl acetate is ethenyl ethanoate.
The polymer forms through an addition reaction.

Step 3: Apply this knowledge to the monomer

An addition reaction means that the monomer must contain a double bond. The only place this bond can occur is between carbon 1 and carbon 2:



Step 4: Draw the structure of the monomer



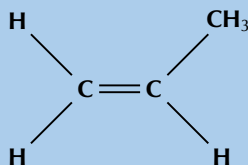
Worked example 35: Identifying polymers

QUESTION

Which polymer is formed from propene? Give the name and repeat structural unit.

SOLUTION

Step 1: Draw the structural representation of the monomer



Step 2: Consider what you about this monomer

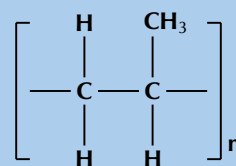
Propene contains a double bond between the first and second carbon atoms.

Step 3: Apply this knowledge to the polymer

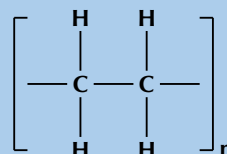
The polymer must form through an addition reaction by breaking the double bond.
There must be a methyl group on every second carbon atom.

Step 4: Name the polymer

The polymer must be **polypropene** (polypropylene).

Step 5: Draw the repeat structural unit of this polymer**Worked example 36: Identifying types of polymerisation reactions****QUESTION**

Was an addition or a condensation reaction used to make this polymer?

**SOLUTION****Step 1: Look at the types of bonds and functional groups in the polymer repeat unit**

There are only single carbon-carbon bonds, there are no functional groups.

Step 2: What type of monomer could this polymer have formed from?

As there are no oxygen atoms in the repeat unit, this polymer could only have formed from a monomer containing a double bond.

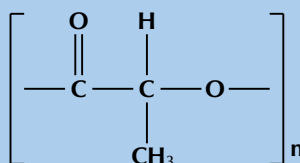
Step 3: Apply this knowledge to determine the type of reaction

This polymer must have formed through an **addition** reaction.

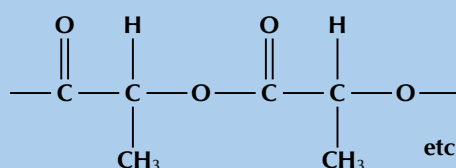
(This is polyethene and was formed by the addition of ethene monomers)

Worked example 37: Identifying types of polymerisation reactions**QUESTION**

Was an addition or a condensation reaction used to make this polymer?

**SOLUTION****Step 1: Look at the types of bonds in the polymer repeat unit**

There are carbon-carbon single bonds. If you expand the repeating unit you can see that there is an ester linkage in the structural chain.

**Step 2: What type of monomer could this polymer have formed from?**

The ester linkage requires an alcohol and a carboxylic acid monomer or a monomer that contains both a hydroxyl and carboxyl functional group.

Step 3: Apply this knowledge to determine the type of reaction

The reaction of an alcohol and carboxylic acid to form an ester linkage (and resulting in the loss of water) is a **condensation** reaction.

(This is polylactic acid and was formed by the condensation reaction of lactic acid monomers)

Experiment: Making polymer slime

Aim:

To compare the differences in slime made from polyvinyl alcohol and polyvinyl acetate.

Apparatus:

- 4% aqueous polyvinyl alcohol solution, 4% aqueous borax (hydrated sodium tetraborate) solution, white glue (polyvinyl acetate), food colourant
- a beaker, two disposable plastic cups, a metal spatula, disposable plastic gloves

Method:

1. Place 40 cm³ of the polyvinyl alcohol solution in one plastic cup and label it A.
2. Place 20 cm³ of the white glue in the second cup. Add 20 cm³ of water and stir. Label this cup B.
3. Add one drop of food colourant to each cup and stir well.
4. Measure out 10 cm³ of the borax solution into the beaker, then add to cup A.
5. Repeat step 4 with cup B.
6. Stir until gelling is complete (i.e. the slime).
7. Put on the disposable plastic gloves, take the slime out of cup A and roll it around in your hand to remove air bubbles, repeat with the slime in cup B.

Questions:

1. Pull each slime apart slowly, what happens?
2. Pull each slime apart quickly, what happens?
3. Roll each slime into a ball and drop it onto the bench, what happens?
4. Place a small bit of each slime on the bench and hit it hard with your hand, what happens?
5. What are the similarities and differences between the two slimes?

Discussion and Conclusions:

You should find that the slime does not break under slow stress, but will break if you pull it apart too quickly. This is because the slime can expand under stress. It should even bounce slightly if you throw it onto a hard surface.

Impact of polymers and plastics

ESCM7

Plastics and the environment

Although plastics have had a huge impact globally, there is also an environmental price that has to be paid for their use. The following are just some of the ways in which plastics can cause damage to the environment.

FACT
The Great Pacific Garbage Patch is a large area of waste that has collected in the central North Pacific Ocean. It is estimated to cover at least 700 000 km².



- 1. Waste disposal**
Plastics are not easily broken down by micro-organisms and therefore most are not easily biodegradable. This leads to waste disposal problems.
- 2. Air pollution**
When plastics burn, they can produce toxic gases such as carbon monoxide, hydrogen cyanide and hydrogen chloride (particularly from PVC and other plastics that contain chlorine or nitrogen).
- 3. Recycling**
It is difficult to recycle plastics because each type of plastic has different properties and so different recycling methods may be needed for each plastic. Some plastics can be remelted and re-used, while others can be ground up and used as a filler. One of the problems with recycling plastics is that they have to be sorted according to plastic type. A list of some of the different types and their identification codes are given in Table 4.22. Alternatively, plastics should be re-used. In many countries, including South Africa, shoppers must now pay for plastic bags. This encourages people to collect and re-use the bags they already have.

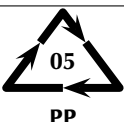
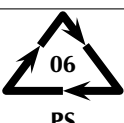
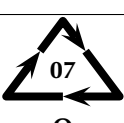
PIC	Plastic type	Properties	Uses	Recycling
	polyethylene terephthalate	very clear high strength barrier to gas barrier to moisture	water bottles peanut butter jars mouthwash bottles	can recycle most places
	high-density polyethylene	very stiff high strength barrier to moisture allows gas through	yoghurt tubs detergent bottles cereal box liners	can recycle most places
	polyvinyl chloride	easily blended high strength very toughness	detergent bottles medical equipment piping	very rarely recycled
	low-density polyethylene	very flexible high strength barrier to moisture	clothing squeeze bottles bread bags	not often recycled
	polypropylene	high strength resistance to heat barrier to moisture	syrup bottles plastic caps straws	fairly easy to recycle
	polystyrene	many uses very clear is easily formed	egg cartons disposable cups take-away boxes	fairly easy to recycle
	other	dependent on combination of polymers	large containers sunglasses computer cases	often not recycled

Table 4.22: The plastic identification code (PIC), type, properties and uses of various commonly used plastics.

Case study: Plastic pollution in South Africa

Read the following extract, taken from *Planet Ark* (September 2003), and then answer the questions that follow.

'South Africa launches a programme this week to exterminate its national flower - the millions of used plastic bags that litter the landscape.

Beginning on Friday, plastic shopping bags used in the country must be both thicker and more recyclable, a move officials hope will stop people from simply tossing them away. 'Government has targeted plastic bags because they are the most visible kind of waste,' said Phindile Makwakwa, spokeswoman for the Department of Environmental Affairs and Tourism. 'But this is mostly about changing people's mindsets about the environment.'

South Africa is awash in plastic pollution. Plastic bags are such a common eyesore that they are dubbed roadside daisies and referred to as the national flower. Bill Naude of the Plastics Federation of South Africa said the country used about eight billion plastic bags annually, a figure which could drop by 50 percent if the new law works.'

It is difficult sometimes to imagine exactly how much waste is produced in our country every year. Where does all of this go to? You are going to do some simple calculations to try to estimate the volume of plastic packets that is produced in South Africa every year.

1. Take a plastic shopping packet and fold it until it forms a tightly compressed cube.
 - a) Measure the approximate length, breadth and depth of your compressed plastic bag.
 - b) Calculate the approximate volume that is occupied by the packet.
 - c) Now calculate the approximate volume of your classroom by measuring its length, breadth and height.
 - d) Calculate the number of compressed plastic packets that would fit into a classroom of this volume.
 - e) If South Africa produces an average of 8 billion plastic bags each year, how many classrooms would be filled if all of these bags were thrown away and not re-used?
2. What has South Africa done to try to reduce the number of plastic bags that are produced?
3. Do you think this has helped the situation?
4. What can *you* do to reduce the amount of plastic that you throw away?
5. Take a survey of your classmates, family and friends on the quantities and types of solid waste they generate. What methods could they use to reduce that waste?
6. It has now been over *ten years* since shops started charging for plastic bags. Research the effects this law has had on South African plastic waste.

Case study: Biodegradable plastics

Read the following extract, taken from *Nova: Science in the news* (July 2006), and then answer the questions that follow.

Our whole world seems to be wrapped in plastic. Almost every product we buy, most of the food we eat and many of the liquids we drink come

encased in plastic. Plastic packaging provides excellent protection for the product, it is cheap to manufacture and seems to last forever. Lasting forever, however, is proving to be a major environmental problem. Another problem is that traditional plastics are manufactured from non-renewable resources - oil, coal and natural gas. In an effort to overcome these problems, researchers and engineers have been trying to develop biodegradable plastics that are made from renewable resources, such as plants.

The term *biodegradable* means that a substance can be broken down into simpler substances by the activities of living organisms, and therefore is unlikely to remain in the environment. The reason most plastics are not biodegradable is because their long polymer molecules are too large and too tightly bonded together to be broken apart and used by decomposer organisms. However, plastics based on natural plant polymers that come from wheat or corn starch have molecules that can be more easily broken down by microbes.

Starch is a natural polymer. It is a white, granular carbohydrate produced by plants during photosynthesis and it serves as the plant's energy store. Many plants contain large amounts of starch. Starch can be processed directly into a bioplastic but, because it is soluble in water, articles made from starch will swell and deform when exposed to moisture, and this limits its use. This problem can be overcome by changing starch into a different polymer. First, starch is harvested from corn, wheat or potatoes, then micro-organisms transform it into lactic acid, a monomer. Finally, the lactic acid is chemically treated to cause the molecules of lactic acid to link up into long chains or polymers, which bond together to form a plastic called polylactide (PLA).

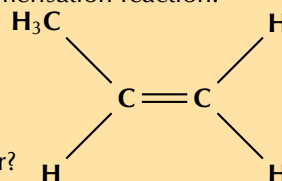
PLA can be used for products such as plant pots and disposable nappies. It has been commercially available in some countries since 1990, and certain blends have proved successful in medical implants, sutures and drug delivery systems because they are able to dissolve away over time. However, because PLA is much more expensive than normal plastics, it has not become as popular as one would have hoped.

Questions:

1. In your own words, explain what is meant by a *biodegradable plastic*.
2. Using your knowledge of chemical bonding, explain why some polymers are biodegradable and others are not.
3. Explain why lactic acid is a more useful monomer than starch, when making a biodegradable plastic.
4. If you were a consumer (shopper), would you choose to buy a biodegradable plastic rather than another? Explain your answer.
5. What do you think could be done to make biodegradable plastics more popular with consumers?

Exercise 4 – 32: Polymers

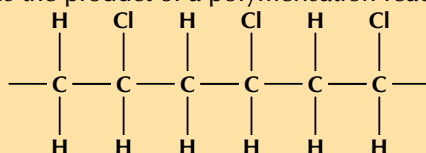
1. The following monomer is a reactant in a polymerisation reaction:



- a) What is the IUPAC name of this monomer?

- b) Give the structural representation of the polymer that is formed in this polymerisation reaction.
 c) Is the reaction an addition or condensation reaction?

2. The polymer below is the product of a polymerisation reaction.



- a) Give the structural formula of the monomer used to form this polymer.
 b) What is the name of the monomer?
 c) Draw the abbreviated structural formula for the polymer (the repeat unit).
 d) Has this polymer been formed through an addition or condensation polymerisation reaction?
3. A polymerisation reaction takes place between two lactic acid monomers.
- a) Give the structural representation for:
- lactic acid
 - the repeat unit of the polymer product
- b) What is the name of the product?
 c) Was this polymer formed through an addition or a condensation reaction?
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27QP 2. 27QQ 3. 27QR



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4.8 Chapter summary

ESCM8

► See presentation: 27QS at www.everythingscience.co.za

► See presentation: 27QT at www.everythingscience.co.za

- **Organic chemistry** is the branch of chemistry that deals with organic molecules. An **organic molecule** is one that contains carbon.
- All **living organisms** contain carbon. Plants use sunlight to convert carbon dioxide in the air into organic compounds through the process of **photosynthesis**. Animals and other organisms then feed on plants to obtain their own organic compounds. **Fossil fuels** are another important source of carbon.
- It is the **unique properties of the carbon atom** that give organic compounds certain properties.
- The carbon atom has **four valence electrons**, so it can bond with many other atoms, often resulting in long chain structures. It also forms mostly **covalent bonds** with the atoms that it bonds to.
- An organic compound can be represented in different ways, using its **molecular formula**, **structural formula**, **semi-structural formula** or **condensed structural formula**.
- A **functional group** is a particular group of atoms within a molecule, which give it certain reaction characteristics. Organic compounds can be grouped according to their functional group.
- The **hydrocarbons** are organic compounds that contain only carbon and hydro-

gen. They can be further divided into the alkanes, alkenes and alkynes, based on the type of bonds between the carbon atoms.

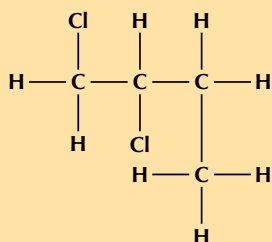
- The **alkanes** have only **single bonds** between their carbon atoms and are unreactive. The **alkenes** have at least one **double bond** between two of their carbon atoms and the **alkynes** have at least one **triple bond** between two of their carbon atoms. They are both more reactive than the alkanes.
- An **alcohol** is an organic compound that contains a **hydroxyl group** ($-\text{OH}$).
- The alcohols have a number of different uses including their use as a solvent, for medicinal purposes and in alcoholic drinks.
- The alcohols share a number of **properties** because of the hydroxyl group. The hydroxyl group affects the **solubility** of the alcohols (in water). Those with shorter carbon chains are generally more soluble, and those with longer chains are less soluble. The strong hydrogen bond between the hydrogen and oxygen atoms in the hydroxyl group gives alcohols a higher melting point and boiling point than the hydrocarbons.
- A **carbonyl group** consists of an oxygen atom that is double-bonded to a carbon atom. In a **ketone** the carbonyl group is in the middle of the carbon chain. In an **aldehyde** the carbonyl group is at the end of the chain.
- The **carboxylic acids** are organic acids that contain a **carboxyl group** with the formula $-\text{COOH}$. In a carboxyl group a carbon atom is double-bonded to an oxygen atom, and it is also bonded to a hydroxyl group.
- The carboxylic acids have weak **acidic properties** because the hydrogen atom is able to dissociate from the carboxyl group.
- An **ester** is formed when an alcohol reacts with a carboxylic acid. Esters contain a carbonyl functional group as well as an oxygen atom bonded to the carbonyl carbon atom and another carbon atom.
- A molecule is said to be **saturated** if it contains the maximum possible number of hydrogen (or other) atoms for that molecule. The alkanes are all saturated compounds.
- A molecule is **unsaturated** if it does not contain the maximum number of hydrogen atoms for that molecule. The alkenes and alkynes are examples of unsaturated molecules. If a double or triple bond is broken, more hydrogen (or other) atoms can be added to the molecule.
- If two compounds are **isomers**, it means that they have the same molecular formulae but different structural formulae.
- Organic compounds are **named** according to their functional group and its position in the molecule, the number of carbon atoms in the molecule and the position of any double and triple bonds. The IUPAC rules for nomenclature are used in the naming of organic molecules.
- Many of the **properties** of the hydrocarbons are determined by their **molecular structure**, the **bonds** between atoms and molecules, and their **surface area**.
- The **melting points** and **boiling points** of the hydrocarbons increases as their number of carbon atoms increases.
- The **molecular mass** of the hydrocarbons determines whether they will be in the gaseous, liquid or solid phase at specific temperatures.
- The weaker the **intermolecular forces** between molecules the more **volatile** and the higher the **vapour pressure** of that compound. Haloalkanes become less volatile the more halogen atoms they contain. This is due to increased intermolecular forces.
- Three types of reactions that occur are **addition**, **elimination** and **substitution**. The alkenes undergo **addition** reactions because they are unsaturated.
- A **polymer** is a macromolecule that is made up of many repeating structural units

called **monomers**, which are joined by covalent bonds.

- Polymers that contain carbon atoms in the main chain are called **organic polymers**.
- Organic polymers can be divided into **natural organic polymers** (e.g. natural rubber) or **synthetic organic polymers** (e.g. polystyrene).
- The polymer **polyethene** for example, is made up of many ethene monomers that have been joined into a polymer chain.
- Polymers form through a process called **polymerisation**. Two examples of polymerisation reactions are **addition** and **condensation** reactions.
- An **addition polymerisation reaction** occurs when unsaturated monomers (e.g. alkenes) are added to each other one by one. The breaking of a double bond between carbon atoms in the monomer means that a bond can form with the next monomer.
- In a **condensation polymerisation reaction** a molecule of water is released as a product of the reaction. The water molecule is made up of atoms that have been lost from each of the monomers.
- The **chemical properties** of polymers (e.g. ability to withstand stress and melting point) are determined by the types of atoms in the polymer, and by the strength of the bonds between adjacent polymer chains. The stronger the bonds, the greater the strength of the polymer, and the higher its melting point.
- **Polyethene, polypropene, polyvinyl chloride, polyvinyl acetate and polystyrene** are all polymers formed through addition polymerisation.
- **Polyethylene terephthalate and polylactic acid** are polymers formed through condensation polymerisation.
- **Plastics** are a subgroup of organic polymers that can be molded. They can contain more than one polymer.
- It is not easy to **recycle** all plastics and so they create environmental problems.
- Some of these **environmental problems** include issues of waste disposal, air pollution and recycling.
- **Polylactic acid** is an example of a biodegradable polymer made from renewable resources.

Exercise 4 – 33:

1. Give **one word or phrase** for each of the following descriptions:
 - a) The name of the homologous series to which 2-methylpropene belongs.
 - b) The name of the functional group that gives alcohols their properties.
 - c) The group of organic compounds that have acidic properties.
 - d) The IUPAC name of the organic compound that is found in vinegar.
 - e) The name of the organic compound that is found in alcoholic beverages.
2. When 1-propanol is oxidised by acidified potassium permanganate, the possible product formed is:
 - propane • propanoic acid • methyl propanol • propyl methanoate(IEB 2004)
3. What is the IUPAC name for the compound represented by the following structure?



- 1,2-dichlorobutane
- 2,2-dichlorobutane
- 1,2-dichloro-3-methylpropane
- 1,1-dichloro-3-methylpropane

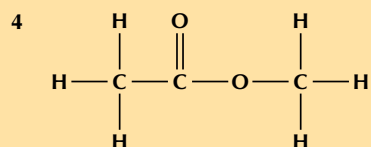
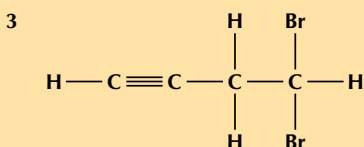
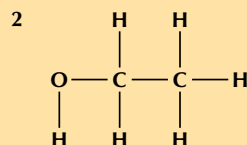
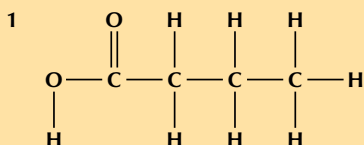
(IEB 2003)

4. Give the structural representation and IUPAC name of all possible isomers for C_6H_{14} (hint: there are 5)
5. Write balanced equations for the following reactions:
 - a) Ethene reacts with bromine (use condensed structural representations)
 - b) The complete combustion of ethyne gas (used molecular formula)
 - c) Ethanoic acid ionises in water
6. The table gives the boiling point of ten organic compounds.

	Compound	Molecular formula	Boiling Point ($^{\circ}C$)
1	methane	CH_4	-162
2	ethane	C_2H_6	-89
3	propane	C_3H_8	-42
4	butane	C_4H_{10}	0
5	pentane	C_5H_{12}	36
6	methanol	CH_4O	64,7
7	ethanol	C_2H_6O	78,4
8	propan-1-ol	C_3H_8O	97
9	propan-1,2-diol	$C_3H_8O_2$	188
10	propan-1,2,3-triol	$C_3H_8O_3$	290

The following questions refer to the compounds shown in the above table.

- a) To which homologous series do the following compounds belong?
 - i. Compounds 1 - 5
 - ii. Compounds 6 - 10
 - b) Which of the above compounds are gases at room temperature?
 - c) What is the reason for the trend of increasing boiling points seen in compounds 1 to 5?
 - d) Despite the fact that the length of the carbon chain in compounds 8,9 and 10 is the same, the boiling point of propan-1,2,3-triol is much higher than the boiling point of propan-1-ol. What is responsible for this large difference in boiling point?
 - e) Give the IUPAC name and the structural representation of an isomer of butane.
 - f) Which **one** of the compounds is used as a reactant in the preparation of the ester ethyl methanoate?
 - g) Using structural representation, write an equation for the reaction which produces ethyl methanoate.
- (IEB 2004)
7. Refer to the numbered diagrams below and then answer the questions that follow.



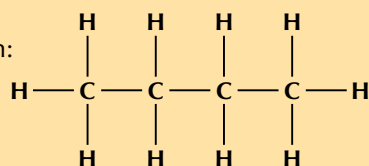
- a) Which one of the above compounds is produced from the fermentation of starches and sugars in plant matter?

- b) To which one of the following homologous series does compound 1 belong?
- i. esters ii. alcohols iii. aldehydes iv. carboxylic acids
- c) The correct IUPAC name for compound 3 is...
- i. 1,1-dibromobut-3-yne ii. 4,4-dibromobut-1-yne
iii. 2,4-dibromobut-1-yne iv. 4,4-dibromoprop-1-yne
- d) What is the correct IUPAC name for compound 4?
- i. propanoic acid ii. ethyl methanoate
iii. methyl ethanoate iv. methyl propanoate
- IEB 2005

8. Answer the following questions:

- a) Explain the term homologous series.
- b) A mixture of ethanoic acid and methanol is warmed in the presence of concentrated sulfuric acid.
- i. Using structural representations, give an equation for the reaction which takes place.
- ii. What is the IUPAC name of the organic compound formed in this reaction?

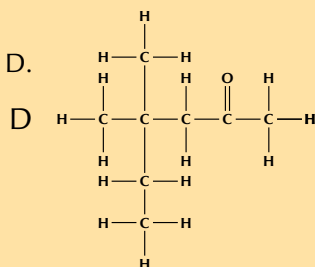
c) Consider the following hydrocarbon:



- i. Give the IUPAC name for this compound.
- ii. Give the balanced equation for the complete combustion of this compound in excess oxygen (use molecular formulae).
- (IEB Paper 2, 2003)

9. Consider the organic compounds labelled A to D.

- A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
B. C_6H_6
C. $\text{CH}_3 - \text{Cl}$



- a) Write a balanced chemical equation for the preparation of compound C using an alkane as one of the reactants.
- b) Write down the IUPAC name for compound D.
- c) Write down the structural representation of an isomer of compound A that has only FOUR carbon atoms in the longest chain.
- d) Write down the structural representation of compound B.
10. Chlorine and ethane react to form chloroethane and hydrochloric acid.
- a) Write a balanced chemical equation for this reaction, using molecular formulae.
- b) Give the structural representation of chloroethane.
- c) What type of reaction has taken place in this example?
11. Petrol is in fact not pure C_8H_{18} but a mixture of various *alkanes*. The *octane rating* of petrol refers to the percentage of the petrol which is C_8H_{18} . For example, 93 octane fuel contains 93% C_8H_{18} and 7% other alkanes. The *isomer* of C_8H_{18} referred to in the *octane rating* is in fact not octane but 2,2,4-trimethylpentane.
- a) Write a balanced equation for the chemical reaction which takes place when C_8H_{18} burns in excess oxygen (complete combustion).
- b) Give the general formula of the *alkanes*.

- c) Define the term *isomer*.
d) Use the information given in this question and your knowledge of naming organic compounds to deduce and draw the full structural formula for 2,2,4-trimethylpentane.(I.E.B.)

12. More questions. Sign in at Everything Science online and click 'Practise Science'. Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. [27QV](#) 1b. [27QW](#) 1c. [27QX](#) 1d. [27QY](#) 1e. [27QZ](#) 2. [27R2](#)
3. [27R3](#) 4. [27R4](#) 5a. [27R5](#) 5b. [27R6](#) 5c. [27R7](#) 6. [27R8](#)
7. [27R9](#) 8. [27RB](#) 9. [27RC](#) 10. [27RD](#) 11. [27RF](#)



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